

Boundary, Terminal, Diagonal, and Cycle Limit Laws for Luce Permutations

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Abstract

The stationary distribution of the Tsetlin library is the Luce distribution, equivalently the order induced by independent exponential clocks with prescribed rates. Chatterjee, Diaconis, and Kim initiated an enumerative theory for this non-uniform distribution, and Borga, Chatterjee, and Diaconis developed its permuton and local-limit theory. This paper proves boundary, terminal, diagonal, and cycle limit laws for Luce permutations.

The boundary, diagonal, and endpoint results are proved by local-entry analysis in exponential-clock and shifted-Gumbel coordinates. For top windows we prove exact total-variation and one-sided relative L^∞ identities; for bottom windows we identify the precise terminal leakage when the limiting integral is defective. Writing $C_r = C_r(\sigma_n)$ for the number of r -cycles, we prove, for uniformly positive continuous profiles, joint Poisson limits for every fixed vector $(C_1, \dots, C_R)$, together with sparse deterministic-entry and overlap laws. The limiting cycle means λ_r satisfy

$$\lambda_r = \frac{1}{r} + O_f(q^r/r)$$

for some $q = q(f) \in (0, 1)$. Under the weaker assumption that the largest and smallest weights are uniformly comparable, a complementary clock-slab mixing and regeneration argument proves that the ranked macroscopic cycle lengths converge to PD(1).

For the singular Sukhatme weights and, more generally, endpoint-regular profiles with regular-variation index $\gamma > 0$, fixed points form a local Poisson point process on logarithmic terminal scale with intensity

$$c_\gamma dx, \quad c_\gamma = \gamma K_\gamma e^{-K_\gamma}, \quad K_\gamma = \Gamma(1 + 1/\gamma)^\gamma.$$

With $F_n = C_1(\sigma_n)$, we prove the global approximation

$$d_{\text{TV}}(\mathcal{L}(F_n), \text{Pois}(\mathbb{E}F_n)) = O\left(\frac{\log \log n}{\sqrt{\log n}}\right).$$

For every fixed collection of endpoint cycle lengths we determine all mixed factorial-moment asymptotics; in particular,

$$\mathbb{E}C_r \sim \frac{k_\gamma^{*r}(0)}{r} \log n, \quad k_\gamma(x) = \gamma K_\gamma e^{-\gamma x} e^{-K_\gamma e^{-\gamma x}},$$

and the normalized fixed-length cycle counts satisfy a joint law of large numbers. Finally, on every mesoscopic scale $b_n \rightarrow \infty$, $b_n = o(n)$, the normalized endpoint graph converges to the deterministic bistochastic tangent kernel

$$\mathcal{K}_\gamma(x, y) = y^{-1} k_\gamma(\log(y/x)).$$

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1 Introduction and scope

Let $\theta_1, \dots, \theta_n$ be positive weights. The Luce distribution, originating in Luce's choice model [11], is generated on S_n by sampling labels without replacement, choosing each remaining label with probability proportional to its weight. Equivalently, if

$$X_i \sim \text{Exp}(\theta_i), \quad 1 \leq i \leq n,$$

are independent, then the labels in increasing order of the X_i 's have the Luce law. This is the stationary distribution of the Tsetlin library; see Chatterjee, Diaconis, and Kim [8]. The Tsetlin library also belongs to the hyperplane-walk framework of Bidigare, Hanlon, and Rockmore [3] and Brown and Diaconis [7]. Borga, Chatterjee, and Diaconis developed the permuton and local-limit theory of Luce permutations [6]; for general permutation-limit background, see Hoppen, Kohayakawa, Moreira, Ráth, and Sampaio [10].

Mukherjee [12, Theorems 1.4, 1.6, and 1.9] derives fixed-point, overlap, and fixed short-cycle Poisson limits from a global uniform approximation of finite point probabilities for convergent random permutations. In the regular Luce setting, Proposition 5.7 proves the corresponding multipoint approximation only on separated bulk configurations, with microscopic entry density equal to the Luce permuton density identified in [6, Theorem 2.1 and Proposition 2.2]. Because the estimate is not uniform through endpoint and collision regimes, we do not invoke Mukherjee's theorems directly. Instead, endpoint and collision envelopes are combined with the separated estimate in a self-contained factorial-moment argument. The Sukhatme and endpoint-regular profiles lie outside the uniformly positive regime and require a distinct terminal analysis.

We use *terminal-condition flux* as an organizing description for the local-entry mechanism in most of the proofs. A statistic is controlled by the asymptotic mass entering a relevant rank, boundary, diagonal, or terminal set. The exponential-clock representation makes these entries explicit: top-window statistics are governed by sampling without replacement, bottom-window laws by leakage through a terminal endpoint, regular fixed points and short cycles by microscopic

point-entry estimates, and singular profiles by logarithmic terminal saddles. The equivalent shifted-Gumbel representation

$$Y_i = -\log X_i = \log \theta_i + G_i$$

turns the same events into threshold crossings. For local rank events, the density of the specified priority and the derivative of the Bernoulli threshold mean contain a common factor that cancels. The terminology introduces no additional assumption; all proofs are stated in standard probabilistic terms.

The results fall into four regimes. First, at the boundary, we prove exact top-window distance identities and identify the precise leakage of candidate infinite bottom-window laws. The shifted-Gumbel threshold formulas and the vanishing-block construction in the preliminaries show, respectively, how local rank probabilities are calculated and why sparse cycle statistics cannot be recovered from permuton convergence alone.

Second, for uniformly positive continuous profiles, we prove joint Poisson limits for every fixed vector of cycle counts, sparse-entry and deterministic-overlap laws, and exponentially accurate long-trace universality. Third, under the weaker assumption of uniformly comparable weights, the clock-slab and permuton-matching argument of Section 6 yields PD(1) macroscopic-cycle universality. A common minorization gives a spectral gap depending only on the weight-comparability constant; quantitative prefix balance, pointwise rooted smoothing, and regular-history deletion provide the regeneration step.

Fourth, for the singular Sukhatme profile, we establish the logarithmic fixed-point mean, a local terminal Poisson point process, and all fixed factorial moments. The endpoint-regular theory extends these conclusions under regular variation, proves a global total-variation Poisson approximation by an approximate-Palm rank transposition, identifies finite-dimensional endpoint displacements, and constructs microscopic and mesoscopic endpoint limits. It also determines the logarithmic means and all fixed mixed factorial moments of the endpoint cycle counts of every fixed length, yielding a joint law of large numbers.

These results answer, under explicit microscopic regularity hypotheses, the fixed-point question and the joint-law question for every fixed vector of short cycle counts raised by Borga, Chatterjee, and Diaconis [6, Section 5.2]. The asymptotic law of the total number of cycles remains open; see Conjecture 13.2. Proposition 2.6 explains why permuton convergence alone cannot imply such microscopic conclusions.

Most results use local-entry analysis: identify the relevant event, write an exact integral or sequential estimate, and evaluate it using a lattice local central limit theorem, saddle analysis, or product bounds. The macroscopic PD(1) theorem instead uses a complementary clock-slab mixing and regenerative-cycle argument.

Conventions

The notation $\text{Exp}(\theta)$ denotes the exponential distribution with rate θ . Weight vectors may form triangular arrays and therefore depend on n . The phrase *lattice local central limit theorem* below refers to point probabilities for lattice-valued sums and is distinct from permutation local-limit notions. All exponential clocks are almost surely distinct. If π is the one-line ordering of labels, then $\pi(r)$ is the label drawn at position r . Let $\sigma = \pi^{-1}$, so $\sigma(i)$ is the position of label i . Fixed points and cycle counts are invariant under inversion. In fixed-point and cycle arguments we use σ , since

$$\sigma(i) = i \iff \#\{j : X_j < X_i\} = i - 1.$$

2 Preliminaries

2.1 The exponential race

Let $\theta_1, \dots, \theta_n > 0$, and let $X_i \sim \text{Exp}(\theta_i)$ be independent. For distinct labels $i_1, \dots, i_n$,

$$\mathbb{P}(X_{i_1} < \dots < X_{i_n}) = \prod_{r=1}^n \frac{\theta_{i_r}}{\theta_{i_r} + \theta_{i_{r+1}} + \dots + \theta_{i_n}}.$$

This is the standard exponential race identity and follows from the memoryless property.

2.2 Gumbel priorities

A standard Gumbel random variable has distribution

$$\mathbb{P}(G \leq y) = \exp(-e^{-y}), \quad y \in \mathbb{R}.$$

If $E_i \sim \text{Exp}(1)$ are independent, then $-\log E_i$ are independent standard Gumbels. Since

$$X_i \stackrel{d}{=} E_i / \theta_i,$$

the exponential race may equivalently be generated by the shifted priorities

$$Y_i = -\log X_i = \log \theta_i + G_i, \quad 1 \leq i \leq n,$$

ordered in decreasing order. Thus

$$\sigma(i) = 1 + \#\{j : Y_j > Y_i\}.$$

For later reference we record the exact threshold formulas. The density and upper-tail function of Y_i are

$$f_i(y) = \theta_i e^{-y} \exp(-\theta_i e^{-y}), \quad \mathbb{P}(Y_i > y) = 1 - \exp(-\theta_i e^{-y}).$$

Lemma 2.1 (Gumbel threshold identities). *For a single specified label p ,*

$$\mathbb{P}(\sigma(p) = q) = \int_{-\infty}^{\infty} f_p(y) \mathbb{P}\left(\sum_{j \neq p} \mathbf{1}_{\{Y_j > y\}} = q - 1\right) dy.$$

More generally, let $p_1, \dots, p_\ell$ be distinct labels and let $q_1 < \dots < q_\ell$ be distinct target positions. Put $P = \{p_1, \dots, p_\ell\}$ and

$$S_a(y) = \sum_{j \notin P} \mathbf{1}_{\{Y_j > y_a\}}, \quad 1 \leq a \leq \ell.$$

Then

$$\begin{aligned} \mathbb{P}(\sigma(p_a) = q_a, \ 1 \leq a \leq \ell) \\ = \int_{y_1 > \dots > y_\ell} \prod_{a=1}^{\ell} f_{p_a}(y_a) \mathbb{P}(S_a(y) = q_a - a, \ 1 \leq a \leq \ell) dy_1 \dots dy_\ell. \end{aligned}$$

Proof. Condition on the specified priorities. For one label, the event $\sigma(p) = q$ is exactly the event that precisely $q - 1$ nonspecified priorities exceed the observed value of Y_p . This gives the first identity.

For the multi-point identity, the ordering of the target positions forces the specified priorities to satisfy $Y_{p_1} > \dots > Y_{p_\ell}$. Conditional on $Y_{p_a} = y_a$ in this chamber, exactly $a - 1$ specified priorities are above y_a . Hence $\sigma(p_a) = q_a$ is equivalent to $S_a(y) = q_a - a$, simultaneously for $1 \leq a \leq \ell$. Integrating against the independent specified-priority densities gives the formula. $\square$

Remark 2.2 (The derivative cancellation). The one-point identity explains the local-entry density used later. Let

$$M_{n,P}(y) = \sum_{j \notin P} \mathbb{P}(Y_j > y) = \sum_{j \notin P} (1 - e^{-\theta_j e^{-y}}).$$

At a rank saddle y_* , where $M_{n,P}(y_*)$ equals the target rank up to $O(1)$, a one-dimensional local CLT gives formally

$$\mathbb{P}(\sigma(p) = q) \approx \frac{f_p(y_*)}{|M'_{n,P}(y_*)|}.$$

Here

$$|M'_{n,P}(y_*)| = \sum_{j \notin P} \theta_j e^{-y_*} e^{-\theta_j e^{-y_*}},$$

while

$$f_p(y_*) = \theta_p e^{-y_*} e^{-\theta_p e^{-y_*}}.$$

Thus the common factor e^{-y_*} cancels. In the regular-profile notation of Section 5, the exact finite- n saddle uses $A_{n,P}(t) = (q - 1)/n$. Replacing $(q - 1)/n$ by q/n changes only a lower-order term, and putting $t = e^{-y_*}$ gives

$$\frac{f_p(y_*)}{|M'_{n,P}(y_*)|} = \frac{1}{n} \frac{\theta_p e^{-\theta_p t}}{B_{n,P}(t)} \longrightarrow \frac{1}{n} \rho(p/n, q/n).$$

This is the Gumbel form of the exponential-clock saddle calculation in Proposition 5.7.

Remark 2.3 (Hyperplane-walk priorities). The same representation applies to the face-sampling step in the Brown–Diaconis stationary construction for a finite hyperplane walk whose face measure is separating. If the faces in the support have positive weights w_F , then sampling them without replacement with probabilities proportional to the remaining weights is equivalent to ordering independent priorities

$$\log w_F + G_F$$

in decreasing order and taking their ordered Tits product. The resulting chamber is therefore a deterministic function of a Gumbel-priority order. This is an exact probabilistic representation, although it does not by itself give a closed formula for arbitrary arrangement statistics; see [3, 7] for the face-semigroup construction.

2.3 Analytic input

We use standard estimates for sums of independent Bernoulli variables. They are stated explicitly to make clear what is invoked.

Lemma 2.4 (Point probability and Bernstein bounds). *Let $Y = \sum_{j=1}^N \xi_j$, where the ξ_j are independent, not necessarily identically distributed, Bernoulli variables, and set $\mu = \mathbb{E}Y$, $v = \text{Var} Y$. There is an absolute constant C such that*

$$\sup_m \mathbb{P}(Y = m) \leq \frac{C}{\sqrt{1+v}}.$$

Moreover, there is an absolute constant $c > 0$ such that, for every $u \geq 0$,

$$\mathbb{P}(|Y - \mu| \geq u) \leq 2 \exp \left[-c \min \left(\frac{u^2}{v+1}, u \right) \right].$$

Proof. The point bound is the classical concentration-function estimate for sums of Bernoulli variables. It follows from Fourier inversion and the standard inequality

$$|1 - p + pe^{it}| \leq \exp \left[-\frac{2}{\pi^2} p(1-p) \text{dist}(t, 2\pi\mathbb{Z})^2 \right],$$

where $\text{dist}(t, 2\pi\mathbb{Z})$ denotes distance from t to the nearest multiple of 2π . Bernstein's inequality gives the second bound for bounded independent summands. See Petrov [14, Chapter VII] for both estimates. $\square$

Lemma 2.5 (Uniform lattice local CLT with a tilted local bound). *Fix a dimension d and a support bound $B < \infty$. Let*

$$Y_{n,m} = \sum_{j=1}^{N_n} \zeta_{n,m,j}, \quad m \in \mathcal{M}_n,$$

be sums of independent $\mathbb{Z}^d$ -valued random vectors satisfying $\|\zeta_{n,m,j}\| \leq B$. Write

$$\mu_{n,m} = \mathbb{E}Y_{n,m}, \quad \Sigma_{n,m} = \text{Var}(Y_{n,m}),$$

and suppose that, uniformly in m ,

$$cs_n I_d \leq \Sigma_{n,m} \leq Cs_n I_d, \quad s_n \rightarrow \infty.$$

For $\lambda \in \mathbb{R}^d$, let $\mathbb{P}_\lambda$ denote the product exponential tilt

$$\mathbb{P}_\lambda(\zeta_{n,m,j} = x) = \frac{e^{\lambda \cdot x} \mathbb{P}(\zeta_{n,m,j} = x)}{\mathbb{E}e^{\lambda \cdot \zeta_{n,m,j}}}.$$

Assume that there is $\lambda_0 > 0$ for which the following two properties hold uniformly for $\|\lambda\| \leq \lambda_0$:

(i) the tilted covariance matrices satisfy

$$c_0 s_n I_d \leq \text{Var}_\lambda(Y_{n,m}) \leq C_0 s_n I_d;$$

(ii) for every $\varepsilon > 0$ there are $\kappa_\varepsilon > 0$ and n_ε such that

$$\left| \prod_{j=1}^{N_n} \mathbb{E}_\lambda e^{iu \cdot \zeta_{n,m,j}} \right| \leq e^{-\kappa_\varepsilon s_n}$$

whenever $n \geq n_\varepsilon$ and $u \in [-\pi, \pi]^d$ has distance at least ε from $2\pi\mathbb{Z}^d$.

Then, uniformly in m and $z \in \mathbb{Z}^d$,

$$\begin{aligned} \mathbb{P}(Y_{n,m} = z) &= (2\pi)^{-d/2} (\det \Sigma_{n,m})^{-1/2} \\ &\times \exp \left[-\frac{1}{2} (z - \mu_{n,m})^T \Sigma_{n,m}^{-1} (z - \mu_{n,m}) \right] + o(s_n^{-d/2}). \end{aligned}$$

Moreover,

$$\sup_z \mathbb{P}(Y_{n,m} = z) \leq C_1 s_n^{-d/2}.$$

For every fixed $K < \infty$ and $\gamma < 1/2$, there are $c_1, C_2 > 0$ such that, uniformly when

$$\|z - \mu_{n,m}\| \leq K s_n^{1/2+\gamma},$$

one has

$$\mathbb{P}(Y_{n,m} = z) \leq C_2 s_n^{-d/2} \exp \left[-c_1 \frac{\|z - \mu_{n,m}\|^2}{s_n} \right].$$

All constants depend only on the fixed dimension, the support bound, and the uniform constants in the hypotheses.

Proof. We suppress n, m and write Y, μ, Σ, s . Let $X_j = \zeta_j - \mathbb{E}\zeta_j$ and

$$\Psi(u) = \mathbb{E} e^{iu \cdot (Y - \mu)}.$$

Since $\|X_j\| \leq 2B$ and $\sum_j \mathbb{E}\|X_j\|^2 = \text{tr } \Sigma = O(s)$,

$$\sum_j \mathbb{E}\|X_j\|^3 = O(s).$$

Consequently, uniformly for $\|u\|$ sufficiently small,

$$\log \Psi(u) = -\frac{1}{2} u^T \Sigma u + O(s\|u\|^3). \quad (2.1)$$

A symmetrization argument also gives, on a fixed neighborhood of the origin,

$$|\Psi(u)| \leq e^{-cs\|u\|^2}. \quad (2.2)$$

Indeed, if X'_j is an independent copy of X_j , then $1 - |\mathbb{E} e^{iu \cdot X_j}|^2 = \mathbb{E}[1 - \cos(u \cdot (X_j - X'_j))]$; for small u the right-hand side is bounded below by a constant multiple of $\text{Var}(u \cdot X_j)$, and multiplication over j yields (2.2).

Fourier inversion on $[-\pi, \pi]^d$ now proves the local expansion. Choose $L_s \rightarrow \infty$ with $L_s^3/\sqrt{s} \rightarrow 0$ and split the integral into

$$\|u\| \leq L_s/\sqrt{s}, \quad L_s/\sqrt{s} < \|u\| \leq \varepsilon, \quad \|u\| > \varepsilon.$$

On the first region, (2.1), followed by the change of variables $v = \Sigma^{1/2}u$, gives the Gaussian Fourier integral with an $o(s^{-d/2})$ error. The estimate (2.2) makes the second region $o(s^{-d/2})$, and tilted aperiodicity with $\lambda = 0$ makes the third region exponentially small. The oscillatory factor $e^{-iu \cdot (z - \mu)}$ has modulus one, so every error is uniform in z . Taking absolute values in the same inversion formula proves the concentration bound. This is the usual fixed-dimensional Fourier proof of the lattice local central limit theorem; compare Petrov [14, Chapter VII].

It remains to prove the Gaussian local upper bound. Let $h = z - \mu$ and define the centered cumulant generating function

$$K(\lambda) = \log \mathbb{E} e^{\lambda \cdot (Y - \mu)}.$$

For $\|\lambda\| \leq \lambda_0$, its Hessian is $\nabla^2 K(\lambda) = \text{Var}_\lambda(Y)$ and is therefore between $c_0 s I_d$ and $C_0 s I_d$. If $\|h\| = o(s)$, strong convexity shows that the minimum of $K(\lambda) - h \cdot \lambda$ over the ball $\|\lambda\| \leq \lambda_0/2$ is attained in its interior for all large n . Thus there is a unique λ_h in that ball satisfying

$$\nabla K(\lambda_h) = h, \quad \|\lambda_h\| \leq C\|h\|/s.$$

The associated Legendre transform $I(h) = \lambda_h \cdot h - K(\lambda_h)$ satisfies

$$I(h) \geq c\|h\|^2/s.$$

For example, the last inequality follows by testing the variational formula $I(h) = \sup_\lambda \{\lambda \cdot h - K(\lambda)\}$ at a sufficiently small multiple of h/s and using the tilted covariance upper bound.

Under $\mathbb{P}_{\lambda_h}$, the mean of Y is z . Repeating the absolute-value Fourier estimate above, now using the two tilted hypotheses, gives

$$\sup_w \mathbb{P}_{\lambda_h}(Y = w) \leq C s^{-d/2}.$$

The change-of-measure identity therefore yields

$$\mathbb{P}(Y = z) = e^{-I(h)} \mathbb{P}_{\lambda_h}(Y = z) \leq C s^{-d/2} e^{-c\|h\|^2/s}.$$

The stated displacement range is $o(s)$ because $\gamma < 1/2$, completing the proof. $\square$

2.4 A permuton-level obstruction

The regularity assumptions in the cycle theorems below cannot be replaced by permuton convergence alone. A vanishing exceptional set of labels can carry an arbitrary sublinear permutation while leaving the limiting profile unchanged almost everywhere.

Proposition 2.6 (Vanishing-block implantation). *Let $m_n \rightarrow \infty$ with $m_n = o(n)$, and let $\pi_n \in S_{m_n}$ be arbitrary. There are Luce weights $\theta_{n,1}, \dots, \theta_{n,n}$ such that*

$$\mathbb{P}(\sigma_n(i) = \pi_n(i) \text{ for every } 1 \leq i \leq m_n) \rightarrow 1,$$

while the step profiles

$$f_n(x) = \theta_{n, \lceil nx \rceil}, \quad 0 < x \leq 1,$$

converge almost everywhere to the constant profile $f \equiv 1$. Consequently, σ_n has the uniform permuton limit, but its cycle structure on $m_n = o(n)$ prescribed labels can be chosen arbitrarily.

Proof. Write $\tau_r = \pi_n^{-1}(r)$, $1 \leq r \leq m_n$, and choose R_n so that $R_n/n \rightarrow \infty$. Set

$$\theta_{n, \tau_r} = R_n^{m_n - r + 1}, \quad 1 \leq r \leq m_n, \quad \theta_{n, i} = 1, \quad i > m_n.$$

Let $X_{n,i} \sim \text{Exp}(\theta_{n,i})$ be independent, and let $Y_n = \min_{i > m_n} X_{n,i} \sim \text{Exp}(n - m_n)$. For $1 \leq r < m_n$,

$$\mathbb{P}(X_{n, \tau_r} \geq X_{n, \tau_{r+1}}) = \frac{1}{R_n + 1},$$

while

$$\mathbb{P}(X_{n,\tau_{m_n}} \geq Y_n) = \frac{n - m_n}{R_n + n - m_n}.$$

A union bound therefore gives

$$\begin{aligned} \mathbb{P}(X_{n,\tau_1} < \dots < X_{n,\tau_{m_n}} < Y_n) \\ \geq 1 - \frac{m_n - 1}{R_n + 1} - \frac{n - m_n}{R_n + n - m_n} \longrightarrow 1, \end{aligned}$$

because $n/R_n \rightarrow 0$ and $m_n/R_n = (m_n/n)(n/R_n) \rightarrow 0$. On this event, the first m_n labels in the Luce draw order are $\tau_1, \dots, \tau_{m_n}$, so $\sigma_n(\tau_r) = r = \pi_n(\tau_r)$ for every r .

For every fixed $x > 0$, the index $\lceil nx \rceil$ exceeds m_n for all large n , and hence $f_n(x) = 1$. Thus $f_n \rightarrow 1$ almost everywhere. On the high-probability event above, the remaining $n - m_n$ labels all have equal weight and therefore appear in a uniformly random order after the implanted block. The implanted points carry permuton mass $m_n/n = o(1)$, while a uniform permutation of the remaining labels has the uniform permuton limit. Hence the whole permutation has the uniform permuton limit. The final assertion follows by prescribing the cycle decomposition of π_n . $\square$

Remark 2.7. Taking π_n to be the identity produces at least m_n fixed points with probability tending to one, despite the uniform permuton limit. More generally, one may implant $\lfloor m_n/r \rfloor$ cycles of any fixed length r , or one cycle of length m_n . Thus fixed points, fixed short-cycle counts, and the total number of cycles on sublinear scales are not continuous statistics of the permuton topology. The uniformly positive continuous-profile assumptions in Section 5 exclude this type of microscopic exceptional block.

3 Top windows: exact boundary identities

Normalize $\sum_i \theta_i = 1$. Let P_k be the law of the first k labels under Luce sampling without replacement, and let Q_k be the product law of k independent draws from $[n]$ with probabilities θ_i . Define

$$\mathcal{D}_k = \{(i_1, \dots, i_k) \in [n]^k : i_1, \dots, i_k \text{ are pairwise distinct}\}$$

and

$$e_k(\theta_1, \dots, \theta_n) = \sum_{1 \leq j_1 < \dots < j_k \leq n} \theta_{j_1} \dots \theta_{j_k}.$$

Theorem 3.1 (Exact top-window total variation). *For every $k \leq n$,*

$$\|P_k - Q_k\|_{\text{TV}} = Q_k(\mathcal{D}_k^c) = 1 - k!e_k(\theta_1, \dots, \theta_n).$$

Proof. The measure P_k is supported on $\mathcal{D}_k$. If $i = (i_1, \dots, i_k) \in \mathcal{D}_k$, then

$$P_k(i) = \prod_{r=1}^k \frac{\theta_{i_r}}{1 - \theta_{i_1} - \dots - \theta_{i_{r-1}}},$$

with the empty sum equal to zero. Hence $P_k(i) \geq Q_k(i)$ on $\mathcal{D}_k$, while $P_k(i) = 0$ on $\mathcal{D}_k^c$. The positive and negative parts therefore have equal mass, and

$$\begin{aligned} 2\|P_k - Q_k\|_{\text{TV}} &= \sum_{i \in \mathcal{D}_k} (P_k(i) - Q_k(i)) + \sum_{i \notin \mathcal{D}_k} Q_k(i) \\ &= 2Q_k(\mathcal{D}_k^c). \end{aligned}$$

Thus $\|P_k - Q_k\|_{\text{TV}} = 1 - Q_k(\mathcal{D}_k)$. Finally,

$$Q_k(\mathcal{D}_k) = \sum_{\{j_1, \dots, j_k\}} k! \theta_{j_1} \cdots \theta_{j_k} = k! e_k(\theta_1, \dots, \theta_n).$$

□

Define the one-sided relative defect

$$d_{\infty}^+(P_k, Q_k) = \left\| 1 - \frac{d(Q_k|_{\mathcal{D}_k})}{dP_k} \right\|_{L^{\infty}(P_k)} = \max_{i \in \mathcal{D}_k} \left(1 - \frac{Q_k(i)}{P_k(i)} \right).$$

This is not a metric. Let $\theta_{(1)} \geq \cdots \geq \theta_{(n)}$ be the decreasing rearrangement.

Theorem 3.2 (Exact one-sided relative L^{∞} defect). *For $k \leq n$,*

$$d_{\infty}^+(P_k, Q_k) = 1 - \prod_{r=1}^{k-1} \left(1 - \sum_{j=1}^r \theta_{(j)} \right),$$

where the empty product is one when $k = 1$.

Proof. For $i = (i_1, \dots, i_k) \in \mathcal{D}_k$,

$$\frac{Q_k(i)}{P_k(i)} = \prod_{r=1}^{k-1} \left(1 - \sum_{j=1}^r \theta_{i_j} \right).$$

For every r , $\sum_{j=1}^r \theta_{i_j} \leq \sum_{j=1}^r \theta_{(j)}$. Since each factor $1 - x$ is decreasing in x , the product is minimized by taking the first $k-1$ selected weights to be $\theta_{(1)}, \dots, \theta_{(k-1)}$. The final selected label does not enter the product. This proves the formula. □

Corollary 3.3 (Birthday transfer). *Let $\xi_1, \dots, \xi_k$ be independent with law θ , and let*

$$W_k = \sum_{1 \leq r < s \leq k} \mathbf{1}_{\{\xi_r = \xi_s\}}.$$

Then

$$\|P_k - Q_k\|_{\text{TV}} = \mathbb{P}(W_k \geq 1).$$

Consequently, every asymptotic estimate for the no-collision probability $\mathbb{P}(W_k = 0)$, equivalently for the collision probability $\mathbb{P}(W_k \geq 1)$, transfers immediately to the top-window total-variation distance.

Proof. Under Q_k , the event $\{W_k = 0\}$ is precisely $\mathcal{D}_k$. Hence

$$Q_k(W_k = 0) = Q_k(\mathcal{D}_k) = k! e_k(\theta_1, \dots, \theta_n).$$

Theorem 3.1 therefore gives

$$\|P_k - Q_k\|_{\text{TV}} = 1 - Q_k(\mathcal{D}_k) = Q_k(W_k \geq 1),$$

which is the claimed identity. □

4 Bottom windows and terminal leakage

Let $(\theta_i)_{i \geq 1}$ be positive rates and let $X_i \sim \text{Exp}(\theta_i)$ be independent. Define

$$f(x) = \sum_{i=1}^{\infty} e^{-\theta_i x}, \quad x_0 = \inf\{x : f(x) < \infty\}.$$

The function f is nonincreasing. For a finite set $A \subset \mathbb{N}$ and $x > 0$, define

$$\prod_{i \notin A} (1 - e^{-\theta_i x}) := \lim_{N \rightarrow \infty} \prod_{\substack{i \leq N \\ i \notin A}} (1 - e^{-\theta_i x}).$$

The limit always exists. It is positive exactly when $\sum_{i \notin A} e^{-\theta_i x} < \infty$, and otherwise it equals zero. Indeed, for $u_i = e^{-\theta_i x} \in (0, 1)$, the partial products are decreasing, and $\log(1 - u_i) \sim -u_i$ once $u_i \rightarrow 0$; if u_i does not tend to zero, the product plainly vanishes.

For distinct $a_1, \dots, a_k$, define

$$\mu_k(a_1, \dots, a_k) := \int_{x_1 > \dots > x_k > 0} \prod_{j=1}^k \theta_{a_j} e^{-\theta_{a_j} x_j} \prod_{i \notin \{a_1, \dots, a_k\}} (1 - e^{-\theta_i x_k}) dx_1 \cdots dx_k.$$

The following finite-volume statement identifies μ_k as an actual limiting subprobability law rather than only a formal candidate.

Proposition 4.1 (Finite bottom-window convergence). *For $N \geq \max_j a_j$, let $\mu_k^{(N)}(a_1, \dots, a_k)$ be the probability that the k largest values among $X_1, \dots, X_N$, listed in decreasing order, have labels $a_1, \dots, a_k$. Then, for every fixed ordered distinct tuple,*

$$\mu_k^{(N)}(a_1, \dots, a_k) \longrightarrow \mu_k(a_1, \dots, a_k).$$

Thus μ_k is the coordinatewise limit of the finite bottom-window laws.

Proof. For $N \geq \max_j a_j$,

$$\begin{aligned} \mu_k^{(N)}(a_1, \dots, a_k) &= \int_{x_1 > \dots > x_k > 0} \prod_{j=1}^k \theta_{a_j} e^{-\theta_{a_j} x_j} \\ &\quad \times \prod_{\substack{1 \leq i \leq N \\ i \notin \{a_1, \dots, a_k\}}} (1 - e^{-\theta_i x_k}) dx_1 \cdots dx_k. \end{aligned}$$

The finite products decrease pointwise to the defining infinite product and are bounded by one. The remaining factor is the joint density of the specified independent clocks and is integrable over the chamber. Dominated convergence proves the assertion. $\square$

Theorem 4.2 (Terminal leakage). *Assume $x_0 < \infty$. Let*

$$N_0 = \sum_{i=1}^{\infty} \mathbf{1}_{\{X_i > x_0\}}.$$

Then

$$\sum_{a_1, \dots, a_k}^* \mu_k(a_1, \dots, a_k) = \mathbb{P}(N_0 \geq k),$$

where $\sum^*$ denotes summation over distinct ordered k -tuples. In particular, if $f(x_0) = \infty$, then the mass is one. If $f(x_0) < \infty$, the escaped mass is $\mathbb{P}(N_0 < k) > 0$, so the limiting law is strictly defective.

Proof. For every $x > x_0$, $f(x) < \infty$. Hence

$$\sum_i \mathbb{P}(X_i > x) = \sum_i e^{-\theta_i x} < \infty,$$

so Borel–Cantelli implies that only finitely many of the events $\{X_i > x\}$ occur almost surely. Applying this with $x = x_0 + 1/m$, $m \geq 1$, shows that the point set $\{X_i : X_i > x_0\}$ is locally finite in (x_0, ∞) . Thus its largest, second largest, and subsequent points are well-defined whenever $N_0 \geq k$; possible accumulation can occur only at the endpoint x_0 , not above it. If $x_0 = 0$, then $N_0 = \infty$ almost surely, since $X_i > 0$ almost surely for every i , and the same argument applies with boundary $x_k = 0$.

For a fixed ordered distinct k -tuple $a = (a_1, \dots, a_k)$, the event that the largest k clock values above x_0 , equivalently the ordered bottom-window labels, are $a_1, \dots, a_k$ and have heights in $dx_1 \cdots dx_k$, with $x_1 > \cdots > x_k > x_0$, has density

$$\prod_{j=1}^k \theta_{a_j} e^{-\theta_{a_j} x_j} \prod_{i \notin \{a_1, \dots, a_k\}} (1 - e^{-\theta_i x_k}).$$

The first factor fixes the specified heights and the product says that every other variable is below x_k . Conversely, if this density contributes with $x_k > x_0$, then these are exactly the largest k clock values above x_0 . If $x_k < x_0$, the infinite product is zero because $f(x_k) = \infty$. The hyperplane $x_k = x_0$ has Lebesgue measure zero. Therefore integrating and summing over all ordered distinct a 's partitions the event $\{N_0 \geq k\}$, giving the identity.

If $f(x_0) = \infty$, then $\sum_i \mathbb{P}(X_i > x_0) = \infty$, so independence and the second Borel–Cantelli lemma imply $N_0 = \infty$ almost surely. If $f(x_0) < \infty$, then necessarily $x_0 > 0$, and N_0 is a finite Poisson-binomial variable with parameters $e^{-\theta_i x_0}$. Since these parameters are all strictly less than one and have finite sum,

$$\mathbb{P}(N_0 = 0) = \prod_{i=1}^{\infty} (1 - e^{-\theta_i x_0}) > 0.$$

Hence $\mathbb{P}(N_0 < k) > 0$. □

Remark 4.3 (Complete leakage when $x_0 = \infty$). If $x_0 = \infty$, then $f(x) = \infty$ for every finite x . The infinite product in the definition of μ_k therefore vanishes for every point of the integration chamber, so $\mu_k \equiv 0$. In this case all finite-label coordinate mass escapes.

Corollary 4.4 (A log-log phase diagram). *Suppose, as $i \rightarrow \infty$,*

$$\theta_i = \beta \log i + \alpha \log \log i + O(1), \quad \beta > 0.$$

Then $x_0 = 1/\beta$, and the bottom fixed-window law is non-defective if and only if $\alpha \leq \beta$.

Proof. For large i ,

$$e^{-\theta_i x} \asymp i^{-\beta x} (\log i)^{-\alpha x}.$$

Thus $\sum_i e^{-\theta_i x}$ converges for $x > 1/\beta$ and diverges for $x < 1/\beta$, so $x_0 = 1/\beta$. At x_0 ,

$$f(x_0) \asymp \sum_i \frac{1}{i(\log i)^{\alpha/\beta}},$$

which diverges exactly when $\alpha/\beta \leq 1$. Apply Theorem 4.2. $\square$

5 Regular positive profiles: fixed points and short cycles

Let σ_n be the label-to-position Luce permutation generated by independent exponentials

$$X_{n,i} \sim \text{Exp}(\theta_{n,i}), \quad 1 \leq i \leq n,$$

so that

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}.$$

Throughout this section assume

$$\theta_{n,i} = a_n(f(i/n) + \varepsilon_{n,i}), \quad \max_{1 \leq i \leq n} |\varepsilon_{n,i}| \rightarrow 0,$$

where

$$f \in C[0, 1], \quad 0 < m_f \leq f(x) \leq M_f < \infty.$$

Multiplying all rates by a_n only rescales the exponential-clock time axis and does not change their order, so take $a_n = 1$. Fix constants

$$0 < \underline{\theta} < \bar{\theta} < \infty$$

such that, for every sufficiently large n , $\underline{\theta} \leq \theta_{n,i} \leq \bar{\theta}$ for all i ; for example, one may take $\underline{\theta} = m_f/2$ and $\bar{\theta} = 2M_f$.

Define

$$A(t) = \int_0^1 (1 - e^{-f(u)t}) du, \quad B(t) = A'(t) = \int_0^1 f(u) e^{-f(u)t} du.$$

Then

$$A(0) = 0, \quad A'(t) = B(t) > 0, \quad A(t) \uparrow 1 \quad (t \rightarrow \infty),$$

so $\tau = A^{-1} : (0, 1) \rightarrow (0, \infty)$ is well defined. The density

$$\rho(x, y) = \frac{f(x) e^{-f(x)\tau(y)}}{B(\tau(y))}, \quad 0 \leq x \leq 1, \quad 0 < y < 1,$$

is the Luce permuton density identified by Borga, Chatterjee, and Diaconis [6, Proposition 2.2]. Here it will also arise as the microscopic point-entry density. It is strictly positive and continuous, and is bounded on every strip $[0, 1] \times [\delta, 1 - \delta]$.

For $r \geq 1$ define

$$\lambda_r = \lim_{\delta \downarrow 0} \frac{1}{r} \int_{[\delta, 1-\delta]^r} \rho(x_1, x_2) \rho(x_2, x_3) \cdots \rho(x_r, x_1) dx_1 \cdots dx_r,$$

where $x_{r+1} = x_1$. The proof below shows that these improper limits exist and are finite.

Theorem 5.1 (Regular-profile fixed points and short cycles). *For every fixed R ,*

$$(C_1(\sigma_n), \dots, C_R(\sigma_n)) \Rightarrow (Z_1, \dots, Z_R),$$

where $Z_1, \dots, Z_R$ are independent Poisson variables with means $\lambda_1, \dots, \lambda_R$. In particular,

$$C_1(\sigma_n) \Rightarrow \text{Pois}(\lambda_1), \quad \lambda_1 = \lim_{\delta \downarrow 0} \int_{\delta}^{1-\delta} \rho(x, x) dx.$$

The proof has three parts. First, we prove a uniform local estimate for finitely many point constraints $\sigma_n(p_a) = q_a$ when the target positions lie in a separated bulk region. Second, we prove endpoint and collision envelopes showing that the complement of this separated bulk region has negligible total contribution in fixed-order factorial-moment sums. Third, we sum the separated estimate over fixed points and over disjoint directed cycles. Thus the argument uses only fixed-order factorial moments; no global total-variation approximation is needed.

5.1 Separated bulk local estimate

For a finite set of specified labels $P \subset [n]$, define empirical functions

$$A_{n,P}(t) = \frac{1}{n} \sum_{j \notin P} (1 - e^{-\theta_{n,j}t}), \quad B_{n,P}(t) = A'_{n,P}(t) = \frac{1}{n} \sum_{j \notin P} \theta_{n,j} e^{-\theta_{n,j}t}.$$

If $|P|$ is fixed, then $A_{n,P} \rightarrow A$ and $B_{n,P} \rightarrow B$ uniformly on compact subsets of $[0, \infty)$.

Lemma 5.2 (Uniform saddle geometry). *Fix $\ell \geq 1$, $\delta > 0$, and $\eta > 0$. Uniformly over sets $P \subset [n]$ of cardinality at most ℓ and over integers*

$$q_1 < \dots < q_\ell, \quad q_a/n \in [\delta, 1 - \delta], \quad q_{a+1} - q_a \geq \eta n,$$

put $y_a = (q_a - a)/n$. For all large n , the equations

$$A_{n,P}(t_{n,a}^*) = y_a, \quad 1 \leq a \leq \ell,$$

have unique solutions. These solutions lie in a compact interval $[c_\delta, C_\delta] \subset (0, \infty)$, satisfy

$$t_{n,a}^* = \tau(q_a/n) + o(1), \quad B_{n,P}(t_{n,a}^*) = B(\tau(q_a/n)) + o(1),$$

uniformly, and are separated:

$$t_{n,a+1}^* - t_{n,a}^* \geq c_{\delta,\eta} > 0, \quad 1 \leq a < \ell.$$

Proof. Because A is continuous, strictly increasing, $A(0) = 0$, and $A(t) \uparrow 1$, choose $0 < c_\delta < C_\delta < \infty$ so that

$$A(c_\delta) < \delta/3, \quad A(C_\delta) > 1 - \delta/3.$$

For large n , uniformly over bounded-size P , we have

$$\sup_{0 \leq t \leq C_\delta} |A_{n,P}(t) - A(t)| \rightarrow 0.$$

Also $y_a \in [\delta/2, 1 - \delta/2]$ for all large n , uniformly in the allowed positions. Hence $A_{n,P}(c_\delta) < y_a < A_{n,P}(C_\delta)$. Since $|P| \leq \ell$ and the rates are uniformly bounded in $[\underline{\theta}, \bar{\theta}]$, for all large n ,

$$B_{n,P}(t) = \frac{1}{n} \sum_{j \notin P} \theta_{n,j} e^{-\theta_{n,j} t} \geq \frac{\underline{\theta}}{2} e^{-\bar{\theta} t} > 0$$

on bounded intervals. Thus $A_{n,P}$ is strictly increasing, and the solution exists uniquely in $[c_\delta, C_\delta]$.

Uniform convergence of $A_{n,P}$ to A , together with the uniform continuity of $\tau = A^{-1}$ on $[\delta/2, 1 - \delta/2]$, gives

$$t_{n,a}^* = \tau(y_a) + o(1) = \tau(q_a/n) + o(1),$$

uniformly. Uniform convergence of $B_{n,P}$ to B on $[c_\delta, C_\delta]$ gives the displayed convergence of derivatives.

Finally, $\tau'(y) = 1/A'(\tau(y))$. Since A' is bounded above on $[c_\delta, C_\delta]$, τ' is bounded below on $[\delta/2, 1 - \delta/2]$. Moreover,

$$y_{a+1} - y_a = \frac{q_{a+1} - q_a - 1}{n} \geq \eta/2$$

for all large n . Hence the limiting saddle points $\tau(y_a)$ are separated by a positive constant depending only on δ and η . The uniform convergence just proved transfers this lower bound to $t_{n,a}^*$ for all large n . $\square$

Lemma 5.3 (Uniform LLT for Luce cumulative counts). *Fix $\ell \geq 1$, $0 < c < C < \infty$, $\eta > 0$, and $K < \infty$. Let $P \subset [n]$ satisfy $|P| \leq K$, and let*

$$c \leq t_1 < \dots < t_\ell \leq C, \quad t_{a+1} - t_a \geq \eta.$$

For independent $X_{n,j} \sim \text{Exp}(\theta_{n,j})$, define

$$S_a(t) = \sum_{j \notin P} \mathbf{1}_{\{X_{n,j} \leq t_a\}}, \quad 1 \leq a \leq \ell,$$

and $S(t) = (S_1(t), \dots, S_\ell(t))$. Then $S(t)$ satisfies the fixed-dimensional local CLT of Lemma 2.5 uniformly over the above parameters with scale $s_n = n$. In particular, if $\Sigma_{n,P}(t) = n^{-1} \text{Var } S(t)$, then

$$c_0 I_\ell \leq \Sigma_{n,P}(t) \leq C_0 I_\ell$$

for constants depending only on $\ell, c, C, \eta, K, \underline{\theta}, \bar{\theta}$, and uniformly for $\|z - \mathbb{E}S(t)\| = O(n^{1/2+\gamma})$, $\gamma < 1/6$,

$$\begin{aligned} \mathbb{P}(S(t) = z) &= (2\pi n)^{-\ell/2} (\det \Sigma_{n,P}(t))^{-1/2} \\ &\times \exp \left[-\frac{1}{2n} (z - \mathbb{E}S(t))^T \Sigma_{n,P}(t)^{-1} (z - \mathbb{E}S(t)) \right] + o(n^{-\ell/2}). \end{aligned}$$

Moreover, in the same range,

$$\mathbb{P}(S(t) = z) \leq C n^{-\ell/2} \exp \left[-c \frac{\|z - \mathbb{E}S(t)\|^2}{n} \right].$$

Proof. For a single nonspecified label j , the increment vector

$$V_j(t) = (\mathbf{1}_{\{X_{n,j} \leq t_1\}}, \dots, \mathbf{1}_{\{X_{n,j} \leq t_\ell\}})$$

takes the $\ell + 1$ adjacent cumulative values

$$(1, \dots, 1), (0, 1, \dots, 1), \dots, (0, \dots, 0, 1), (0, \dots, 0).$$

The corresponding categories are

$$(0, t_1], (t_1, t_2], \dots, (t_{\ell-1}, t_\ell], (t_\ell, \infty).$$

Because the rates lie in $[\underline{\theta}, \bar{\theta}]$, the times stay in $[c, C]$, and successive times are separated by at least η , every category probability is bounded below by a constant $c_* > 0$, uniformly in all parameters.

The adjacent support differences are the coordinate vectors. For any $u \in \mathbb{R}^\ell$, the variance identity

$$\text{Var}(u \cdot V_j) = \frac{1}{2} \sum_{v,w} p_v p_w \{u \cdot (v - w)\}^2$$

therefore gives

$$\text{Var}(u \cdot V_j) \geq c_*^2 \sum_{a=1}^{\ell} u_a^2.$$

Summing over $n - |P|$ labels gives the covariance lower bound, and bounded increments give the upper bound.

Fix $\lambda_0 < \infty$. Under any exponential tilt $\|\lambda\| \leq \lambda_0$, the probabilities of the finitely many atoms are multiplied by factors lying between positive constants depending only on ℓ and λ_0 . Thus every tilted category probability is still bounded below uniformly, and the same covariance comparison holds under all such tilts. The same observation gives tilted strong aperiodicity. Indeed, for $u \in [-\pi, \pi]^\ell$ away from $2\pi\mathbb{Z}^\ell$, at least one adjacent phase difference u_a is bounded away from zero. The two corresponding adjacent atoms have uniformly positive tilted mass, so compactness gives

$$|\mathbb{E}_\lambda e^{iu \cdot V_j(t)}| \leq 1 - \kappa$$

for some $\kappa > 0$. Multiplication over $n - |P|$ labels yields the exponential Fourier bound required in Lemma 2.5. All hypotheses of that lemma are therefore satisfied with scale $s_n = n$, proving the stated local expansion and Gaussian local upper bound. $\square$

Lemma 5.4 (Off-saddle exclusion). *Fix $\ell \geq 1$, $\delta > 0$, $\eta > 0$, and $\gamma \in (0, 1/6)$. In the setting of Lemma 5.2, let*

$$K_\delta = [c_\delta, C_\delta]$$

be a compact interval containing all saddles. In the integral representation for finitely many point constraints below, the contribution from

$$\max_a |t_a - t_{n,a}^*| > n^{-1/2+\gamma}$$

is $o(n^{-\ell})$, uniformly over the allowed labels and positions.

Proof. Enlarge K_δ slightly, if necessary, so that outside K_δ the limiting mean $A(t)$ is separated from $[\delta/2, 1 - \delta/2]$. Uniform convergence of $A_{n,P}$ to A then implies that if some $t_a \notin K_\delta$, the corresponding Bernoulli sum has mean separated from its target by order n . Bernstein's inequality gives an exponentially small bound for the marginal event $\{S_a(t) = q_a - a\}$. The full joint point-constraint event is contained in this marginal event, so the same bound applies to the integrand. Since the specified-clock density integrates to at most one over the full positive chamber, this part is $O(e^{-cn})$.

It remains to consider $t \in K_\delta^\ell$ with $\max_a |t_a - t_{n,a}^*| > n^{-1/2+\gamma}$. On K_δ , the derivatives $B_{n,P}$ are uniformly bounded below. Hence, for at least one coordinate a ,

$$|nA_{n,P}(t_a) - nA_{n,P}(t_{n,a}^*)| \geq cn^{1/2+\gamma}.$$

Since $nA_{n,P}(t_{n,a}^*) = q_a - a$, Bernstein's inequality applied to the one-dimensional count $S_a(t)$ gives

$$\mathbb{P}(S_a(t) = q_a - a) \leq \exp(-cn^{2\gamma}).$$

The full joint point-constraint event is contained in this one-dimensional marginal event, so the same bound applies to the integrand. The specified-clock density is uniformly bounded on K_δ^ℓ , and the volume is bounded. Thus the contribution of this region is exponentially small, hence $o(n^{-\ell})$. $\square$

Lemma 5.5 (Gaussian-scale saddle tail). *Fix $\ell \geq 1$, $\delta > 0$, $\eta > 0$, and $\gamma \in (0, 1/6)$. Let $p_1, \dots, p_\ell$ be distinct labels and let $q_1 < \dots < q_\ell$ be target positions satisfying*

$$q_a/n \in [\delta, 1 - \delta], \quad q_{a+1} - q_a \geq \eta n.$$

Let $t_n^ = (t_{n,1}^*, \dots, t_{n,\ell}^*)$ be the saddle vector from Lemma 5.2. In the corresponding point-constraint integral, put*

$$S(t) = (S_1(t), \dots, S_\ell(t)), \quad z_n = (q_1 - 1, \dots, q_\ell - \ell).$$

For

$$t = t_n^* + h/\sqrt{n}, \quad \|h\| \leq n^\gamma,$$

the integrand

$$\prod_{a=1}^{\ell} \theta_{n,p_a} e^{-\theta_{n,p_a} t_a} \mathbb{P}(S(t) = z_n)$$

is bounded by

$$Cn^{-\ell/2} e^{-c\|h\|^2},$$

with constants independent of the allowed labels and positions. Consequently, uniformly in those labels and positions, the part of the saddle window with $\|h\| > R$ contributes at most

$$Cn^{-\ell} \int_{\|h\| > R} e^{-c\|h\|^2} dh.$$

Proof. For $\|h\| \leq n^\gamma$, the saddle separation from Lemma 5.2 implies that the coordinates of t remain ordered and separated for all large n . Moreover, t remains in a fixed compact subinterval of $(0, \infty)$. Hence Lemma 5.3, including the Gaussian local upper bound inherited from Lemma 2.5, applies uniformly.

Let

$$A_n(t) = (A_{n,P}(t_1), \dots, A_{n,P}(t_\ell)), \quad z_n = (q_1 - 1, \dots, q_\ell - \ell).$$

Since $A_n(t_n^*) = z_n/n$, Taylor's theorem and the uniform C^2 bounds for $A_{n,P}$ on the compact saddle region give

$$nA_n(t) - z_n = \sqrt{n} DA_n(t_n^*)h + O(\|h\|^2).$$

The diagonal matrix $DA_n(t_n^*)$ has entries $B_{n,P}(t_{n,a}^*)$, uniformly bounded above and below. Since $\gamma < 1/2$, the quadratic remainder is $o(\sqrt{n}\|h\|)$ uniformly when $\|h\| \leq n^\gamma$ and $\|h\| \rightarrow \infty$. For bounded $\|h\|$, the desired Gaussian upper bound follows from the uniform local bound $Cn^{-\ell/2}$, after decreasing c . Thus, in the range relevant for Gaussian decay,

$$\|nA_n(t) - z_n\| \geq c\sqrt{n}\|h\|.$$

The Gaussian local upper bound gives

$$\mathbb{P}(S(t) = z_n) \leq Cn^{-\ell/2}e^{-c\|h\|^2}.$$

The specified-clock density $\prod_a \theta_{n,pa} e^{-\theta_{n,pa} t_a}$ is uniformly bounded on the compact region, which proves the integrand bound. Since $dt = n^{-\ell/2} dh$, the displayed tail estimate follows. $\square$

Lemma 5.6 (Uniform integrated lattice saddle). *Fix $\ell \geq 1$, $\gamma \in (0, 1/6)$, and an open set $U \subset \mathbb{R}^\ell$. Let $K \Subset U$ be compact. Suppose $t_n^* \in K$, $z_n \in \mathbb{Z}^\ell$, and $S_n(t)$ is an integer-valued random vector such that*

$$\mathbb{E}S_n(t) = nA_n(t), \quad A_n(t_n^*) = z_n/n.$$

Assume, uniformly for $t_n^ \in K$, that:*

1. A_n is C^2 in a fixed neighborhood of K , with uniformly bounded second derivatives;
2. $D_n = DA_n(t_n^*)$ is nonsingular and $\|D_n^{-1}\|$ is uniformly bounded;
3. the covariance matrices $\Sigma_n(t) = n^{-1} \text{Var } S_n(t)$ are uniformly positive definite and uniformly continuous on a fixed neighborhood of K ;
4. a uniform local CLT holds for $S_n(t)$ when $t = t_n^* + O(n^{-1/2+\gamma})$;
5. the contribution outside $\|t - t_n^*\| \leq n^{-1/2+\gamma}$ is $o(n^{-\ell})$, and inside this window the Gaussian tail bound

$$|g_n(t)|\mathbb{P}(S_n(t) = z_n) \leq Cn^{-\ell/2}e^{-c\|h\|^2}, \quad t = t_n^* + h/\sqrt{n},$$

holds for some $C, c > 0$.

If g_n is uniformly bounded and equicontinuous near K , then

$$\int_U g_n(t)\mathbb{P}(S_n(t) = z_n) dt = n^{-\ell}g_n(t_n^*)|\det D_n|^{-1} + o(n^{-\ell}),$$

uniformly over the family.

Proof. The off-saddle hypothesis reduces the integral to $t = t_n^* + h/\sqrt{n}$ with $\|h\| \leq n^\gamma$. By the Gaussian tail hypothesis, the part with $\|h\| > R$ is bounded by

$$Cn^{-\ell} \int_{\|h\| > R} e^{-c\|h\|^2} dh,$$

which is $o_R(n^{-\ell})$ uniformly as $R \rightarrow \infty$. It therefore suffices to evaluate the integral on each fixed ball $\|h\| \leq R$.

On such a ball, Taylor's theorem gives

$$z_n - nA_n(t_n^* + h/\sqrt{n}) = -\sqrt{n} D_n h + O(1),$$

uniformly. The covariance matrices satisfy

$$\Sigma_n(t_n^* + h/\sqrt{n}) = \Sigma_n(t_n^*) + o(1)$$

uniformly on $\|h\| \leq R$, and $g_n(t_n^* + h/\sqrt{n}) = g_n(t_n^*) + o(1)$ uniformly. The local CLT yields

$$\begin{aligned} \mathbb{P}(S_n(t_n^* + h/\sqrt{n}) = z_n) &= (2\pi n)^{-\ell/2} (\det \Sigma_n(t_n^*))^{-1/2} \\ &\quad \times \exp \left[-\frac{1}{2} h^T D_n^T \Sigma_n(t_n^*)^{-1} D_n h \right] + o(n^{-\ell/2}), \end{aligned}$$

uniformly on the fixed ball. All $o(1)$ terms in this fixed-ball argument are uniform over the family and over $\|h\| \leq R$. Since $dt = n^{-\ell/2} dh$, the contribution from $\|h\| \leq R$ equals

$$\begin{aligned} n^{-\ell} g_n(t_n^*) (2\pi)^{-\ell/2} (\det \Sigma_n(t_n^*))^{-1/2} \\ \times \int_{\|h\| \leq R} \exp \left[-\frac{1}{2} h^T D_n^T \Sigma_n(t_n^*)^{-1} D_n h \right] dh + o_R(n^{-\ell}). \end{aligned}$$

Letting $R \rightarrow \infty$, uniformly in the family, the Gaussian integral is

$$(2\pi)^{\ell/2} (\det \Sigma_n(t_n^*))^{1/2} |\det D_n|^{-1}.$$

Indeed,

$$\det(D_n^T \Sigma_n(t_n^*)^{-1} D_n) = (\det D_n)^2 (\det \Sigma_n(t_n^*))^{-1},$$

so the covariance determinant cancels and the Gaussian integral contributes $|\det D_n|^{-1}$. Combining this with the preceding tail estimate proves the claim. $\square$

Proposition 5.7 (Separated bulk local estimate). *Fix $\ell \geq 1$, $\delta > 0$, and $\eta > 0$. Uniformly over distinct labels $p_1, \dots, p_\ell$ and distinct positions $q_1, \dots, q_\ell$ satisfying*

$$q_a/n \in [\delta, 1 - \delta], \quad |q_a - q_b| \geq \eta n \quad (a \neq b),$$

we have

$$\mathbb{P}(\sigma_n(p_a) = q_a, 1 \leq a \leq \ell) = n^{-\ell} \prod_{a=1}^{\ell} \rho(p_a/n, q_a/n) + o(n^{-\ell}).$$

Proof. Reorder the pairs so that $q_1 < \dots < q_\ell$, relabeling the corresponding p_a 's. Conditional on $X_{n,p_a} = t_a$ with $0 < t_1 < \dots < t_\ell$, define

$$S_a(t) = \sum_{j \notin \{p_1, \dots, p_\ell\}} \mathbf{1}_{\{X_{n,j} \leq t_a\}}.$$

The event $\sigma_n(p_a) = q_a$ for all a is equivalent to

$$S_a(t) = q_a - a, \quad 1 \leq a \leq \ell,$$

because exactly $a - 1$ specified clocks occur before t_a . Thus

$$\begin{aligned} \mathbb{P}(\sigma_n(p) = q) &= \int_{0 < t_1 < \dots < t_\ell} \prod_{a=1}^{\ell} \theta_{n,p_a} e^{-\theta_{n,p_a} t_a} \\ &\quad \times \mathbb{P}(S_a(t) = q_a - a, 1 \leq a \leq \ell) dt_1 \dots dt_\ell. \end{aligned}$$

Put $P = \{p_1, \dots, p_\ell\}$, $y_a = (q_a - a)/n$, and let $t_{n,a}^*$ be the saddle point from Lemma 5.2. The ordered integration domain contains a fixed neighborhood of $t_n^* = (t_{n,1}^*, \dots, t_{n,\ell}^*)$, because the saddle coordinates are uniformly separated.

For fixed t , the vector $S(t) = (S_1(t), \dots, S_\ell(t))$ has mean

$$\mathbb{E}S_a(t) = nA_{n,P}(t_a),$$

and covariance $n\Sigma_{n,P}(t)$, where

$$\Sigma_{n,P,ab}(t) = \frac{1}{n} \sum_{j \notin P} (1 - e^{-\theta_{n,j} t_a \wedge b}) e^{-\theta_{n,j} t_a \vee b}.$$

In the saddle neighborhood, Lemma 5.3 gives the required uniform local CLT. Lemma 5.5 supplies the Gaussian-scale tail bound, and Lemma 5.4 removes the complementary region. Apply Lemma 5.6 with

$$A_n(t) = (A_{n,P}(t_1), \dots, A_{n,P}(t_\ell)), \quad z_n = (q_1 - 1, \dots, q_\ell - \ell),$$

and

$$g_n(t) = \prod_{a=1}^{\ell} \theta_{n,p_a} e^{-\theta_{n,p_a} t_a}.$$

The functions g_n are uniformly bounded and equicontinuous in the saddle neighborhood, since the specified rates are uniformly bounded and the neighborhood is compact. Thus the hypotheses of Lemma 5.6 are satisfied uniformly over the separated bulk configurations. Here

$$DA_n(t_n^*) = \text{diag}(B_{n,P}(t_{n,1}^*), \dots, B_{n,P}(t_{n,\ell}^*)).$$

Therefore the integral equals

$$n^{-\ell} \prod_{a=1}^{\ell} \frac{\theta_{n,p_a} e^{-\theta_{n,p_a} t_{n,a}^*}}{B_{n,P}(t_{n,a}^*)} + o(n^{-\ell}).$$

Finally, Lemma 5.2 and the uniform convergence of the rates give

$$\theta_{n,p_a} = f(p_a/n) + o(1), \quad t_{n,a}^* = \tau(q_a/n) + o(1), \quad B_{n,P}(t_{n,a}^*) = B(\tau(q_a/n)) + o(1),$$

uniformly. This is exactly

$$n^{-\ell} \prod_{a=1}^{\ell} \rho(p_a/n, q_a/n) + o(n^{-\ell}).$$

□

5.2 Endpoint and collision domination

We next prove global domination estimates sufficient for the factorial-moment sums. Let a Luce permutation of N labels have all rates in $[\underline{\theta}, \bar{\theta}]$, and let R_i be the rank of label i . Put

$$\alpha = \frac{\bar{\theta}}{\underline{\theta}} \in (0, 1].$$

For $1 \leq q \leq N$, define

$$w_N(q) = \begin{cases} N^{-1}, & 1 \leq q \leq N/2, \\ N^{-\alpha}(N - q + 1)^{\alpha-1}, & N/2 < q \leq N. \end{cases}$$

Lemma 5.8 (One-point rank envelope). *There is $C = C(\underline{\theta}, \bar{\theta})$ such that, for all N , all rate vectors in $[\underline{\theta}, \bar{\theta}]^N$, all labels i , and all q ,*

$$\mathbb{P}(R_i = q) \leq C w_N(q).$$

Proof. First suppose $q \leq N/2$. Reveal the Luce order sequentially. Conditional on label i not having appeared before position q , at position q there are at least $N - q + 1 \geq N/2$ labels remaining. The total remaining weight is at least $m(N - q + 1)$, and the weight of label i is at most M . Hence

$$\mathbb{P}(R_i = q) \leq \frac{\bar{\theta}}{\underline{\theta}(N - q + 1)} \leq \frac{2\bar{\theta}}{\underline{\theta}N}.$$

Thus $\mathbb{P}(R_i = q) \leq CN^{-1}$ for $q \leq N/2$.

Now suppose $q > N/2$. Put $s = N - q + 1$. The event $R_i = q$ means that label i survives until there are s labels remaining and is then selected. When there are r labels remaining and i is among them, the conditional probability that i is selected at the next draw is at least

$$\frac{\underline{\theta}}{\bar{\theta}r} = \frac{\alpha}{r}.$$

Therefore the conditional probability that i is not selected at that draw is at most $1 - \alpha/r$. Hence the probability that i survives from N labels down to s labels is at most

$$\prod_{r=s+1}^N \left(1 - \frac{\alpha}{r}\right).$$

Indeed,

$$\sum_{r=s+1}^N \frac{1}{r} \geq \log(N/s) - O(1),$$

and therefore

$$\prod_{r=s+1}^N \left(1 - \frac{\alpha}{r}\right) \leq \exp\left(-\alpha \sum_{r=s+1}^N \frac{1}{r}\right) \leq C_\alpha \left(\frac{s}{N}\right)^\alpha.$$

Given survival to s labels, the conditional probability that i is selected next is at most

$$\min\left(1, \frac{\bar{\theta}}{\underline{\theta}s}\right) \leq \frac{1}{\alpha s}.$$

Therefore

$$\mathbb{P}(R_i = q) \leq C \left(\frac{s}{N} \right)^\alpha \frac{1}{s} = CN^{-\alpha} s^{\alpha-1}.$$

Since $s = N - q + 1$, this is the desired bound. $\square$

Lemma 5.9 (Gap envelope). *Fix $\ell \geq 1$. There is C_ℓ such that the following holds. Let $p_1, \dots, p_\ell$ be distinct labels and $q_1, \dots, q_\ell$ distinct positions in a Luce permutation of $[n]$ with rates in $[\underline{\theta}, \bar{\theta}]$. Sort the target positions:*

$$q_{(1)} < q_{(2)} < \dots < q_{(\ell)}.$$

Set $q_{(0)} = 0$, $r_a = q_{(a)} - q_{(a-1)}$, and $N_a = n - q_{(a-1)}$. Then

$$\mathbb{P}(\sigma_n(p_a) = q_a, 1 \leq a \leq \ell) \leq C_\ell \prod_{a=1}^{\ell} w_{N_a}(r_a).$$

Proof. Let $p_{(a)}$ denote the specified label paired with the sorted target position $q_{(a)}$. Reveal the Luce order from left to right, stopping at the specified target positions. At the first specified position $q_{(1)}$, the label $p_{(1)}$ must appear at relative rank $r_1 = q_{(1)}$ in a Luce order of $N_1 = n$ labels. By Lemma 5.8, this has probability at most $Cw_{N_1}(r_1)$.

At step a , condition on the full history revealed up to $q_{(a-1)}$. If $p_{(a)}$ has already appeared, the conditional probability of the joint event is zero. If $p_{(a)}$ is still alive, the Plackett–Luce deletion property says that the unexposed tail is again a Luce order on the remaining labels, with the remaining rates. This property follows immediately from the exponential-clock representation and memorylessness: after the revealed clocks are removed, the residual order of the unrevealed labels is generated by independent exponential clocks with their original rates. In this tail there are $N_a = n - q_{(a-1)}$ labels, and $p_{(a)}$ must appear at relative rank

$$r_a = q_{(a)} - q_{(a-1)}.$$

Lemma 5.8 gives a conditional probability at most $Cw_{N_a}(r_a)$. Iterating over $a = 1, \dots, \ell$ gives the stated bound, with $C_\ell = C^\ell$. $\square$

Lemma 5.10 (Summability of the gap envelope). *There is $C < \infty$ such that*

$$\sum_{r=1}^N w_N(r) \leq C$$

for every N . Also, for $1 \leq h \leq N/2$,

$$\sum_{r > N-h} w_N(r) \leq C \left(\frac{h}{N} \right)^\alpha.$$

For $\ell \geq 1$, define

$$H_\ell(N, h) = \sum_{\substack{r_1, \dots, r_\ell \geq 1 \\ r_1 + \dots + r_\ell \leq N \\ N - r_1 - \dots - r_\ell \leq h}} \prod_{a=1}^{\ell} w_{N-r_1-\dots-r_{a-1}}(r_a),$$

where the empty partial sum is zero. Then, for every $0 < \beta < \alpha$,

$$H_\ell(N, h) \leq C_{\ell, \beta} \min \left\{ 1, \left(\frac{h}{N} \right)^\beta \right\}, \quad N \geq 1, h \geq 1.$$

Proof. The first estimate follows from

$$\sum_{r \leq N/2} N^{-1} \leq \frac{1}{2}$$

and, after writing $s = N - r + 1$,

$$\sum_{r > N/2} N^{-\alpha} (N - r + 1)^{\alpha-1} = N^{-\alpha} \sum_{s \leq N/2} s^{\alpha-1} \leq C.$$

The same substitution gives the terminal-tail estimate.

We prove the final assertion by induction on ℓ . If $h \geq N/2$, repeated use of the first estimate gives $H_\ell(N, h) \leq C^\ell$, which is bounded by the asserted right-hand side after increasing the constant. It therefore suffices to assume $h < N/2$.

For $\ell = 1$, the defining condition implies $r > N - h - 1$, so the terminal-tail bound, with a harmless endpoint term, gives $H_1(N, h) \leq C(h/N)^\alpha \leq C(h/N)^\beta$. Assume the result for $\ell - 1$. After the first gap r , put $s = N - r$. The induction hypothesis gives

$$H_\ell(N, h) \leq C \sum_{r=1}^{N-1} w_N(r) \min \left\{ 1, \left(\frac{h}{N-r} \right)^\beta \right\}.$$

For $s \leq h$, the contribution is at most $C(h/N)^\alpha$. For $h < s < N/2$, one has $r > N/2$ and hence

$$\begin{aligned} \sum_{h < s < N/2} w_N(N-s) \left(\frac{h}{s} \right)^\beta &\leq C N^{-\alpha} h^\beta \sum_{h < s < N/2} s^{\alpha-1-\beta} \\ &\leq C(h/N)^\beta. \end{aligned}$$

Finally, for $s \geq N/2$ one has $r \leq N/2$, so $w_N(r) = N^{-1}$ and each summand is at most $C N^{-1} (h/N)^\beta$. Summing over at most N values completes the induction. $\square$

Let

$$E_{\delta, n} = \{1, \dots, \lfloor \delta n \rfloor\} \cup \{\lceil (1 - \delta)n \rceil, \dots, n\}.$$

Lemma 5.11 (Endpoint domination). *Fix $\ell \geq 1$, $0 < \beta < \alpha$, and $0 < \delta < 1/2$. Then, for all sufficiently large n ,*

$$\sum_{\substack{q_1, \dots, q_\ell \text{ distinct} \\ q_a \in E_{\delta, n} \text{ for some } a}} \sup_{p_1, \dots, p_\ell} \mathbb{P}(\sigma_n(p_a) = q_a, 1 \leq a \leq \ell) \leq C_{\ell, \beta} (\delta + \delta^\beta),$$

where $C_{\ell, \beta}$ is independent of n and δ .

Proof. Sort the target positions. There are at most $\ell!$ possible orderings, so it suffices to sum over ordered target positions and use the gap envelope.

If some target position lies in the first endpoint block, then $q_{(1)} \leq \delta n$, equivalently $r_1 \leq \delta n$. For $\delta < 1/2$,

$$\sum_{r_1 \leq \delta n} w_n(r_1) \leq \sum_{r_1 \leq \delta n} \frac{1}{n} \leq \delta.$$

The remaining $\ell - 1$ gaps have total envelope mass bounded by a constant by Lemma 5.10. Thus this endpoint block contributes at most $C_\ell \delta$.

If some target position lies in the last endpoint block, then $q_{(\ell)} \geq (1 - \delta)n$. Equivalently, after the last specified target position, at most δn labels remain. For fixed $\delta > 0$ and all sufficiently large n , we have $1 \leq \delta n \leq n/2$, so Lemma 5.10 applies with $h = \delta n$. The final estimate in that lemma gives total envelope mass at most $C_{\ell,\beta} \delta^\beta$. Multiplying by the finite number of orderings proves the lemma. $\square$

Lemma 5.12 (Bulk collision domination). *Fix $\ell \geq 1$ and $\delta > 0$. Let $B_{\delta,\eta,n}$ be the set of distinct ℓ -tuples $q_1, \dots, q_\ell$ such that*

$$q_a/n \in [\delta, 1 - \delta]$$

for all a , but $|q_a - q_b| \leq \eta n$ for some $a \neq b$. Then

$$\sum_{q \in B_{\delta,\eta,n}} \sup_{p_1, \dots, p_\ell} \mathbb{P}(\sigma_n(p_a) = q_a, 1 \leq a \leq \ell) \leq C_{\ell,\delta} \eta.$$

Proof. Sort the positions. If two target positions are within ηn , then some adjacent ordered pair has gap $r_j = q_{(j)} - q_{(j-1)} \leq \eta n$ with $j \geq 2$. Since all target positions are in $[\delta n, (1 - \delta)n]$, before that gap the remaining deck size satisfies $N_j \geq \delta n$. If $\eta < \delta/2$, then $r_j \leq N_j/2$ and

$$\sum_{r_j \leq \eta n} w_{N_j}(r_j) \leq \frac{\eta n}{N_j} \leq \frac{\eta}{\delta}.$$

For all other gaps, we sum the gap envelope iteratively and use the uniform bound

$$\sum_{r=1}^N w_N(r) \leq C$$

from Lemma 5.10. This gives a factor depending only on ℓ , uniformly in the intermediate deck sizes. Summing over the possible adjacent pair and orderings gives the result. The case $\eta \geq \delta/2$ is absorbed by increasing the constant. $\square$

Corollary 5.13 (Endpoint tightness and improper integrability). *Fix $\ell \geq 1$. For every $0 < \beta < \alpha$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \sum_{\substack{q_1, \dots, q_\ell \text{ distinct} \\ q_a \in E_{\delta,n} \text{ for some } a}} \sup_{p_1, \dots, p_\ell} \mathbb{P}(\sigma_n(p_a) = q_a, 1 \leq a \leq \ell) = 0.$$

Moreover, all limiting integrals obtained from fixed-order fixed-point and cycle factorial moments by Proposition 5.7 exist as finite improper integrals.

Proof. The displayed endpoint-tightness assertion follows immediately from Lemma 5.11 by first taking $\limsup_{n \rightarrow \infty}$ and then letting $\delta \downarrow 0$.

For the integrability assertion, fix $\delta, \eta > 0$. On the truncated region $[\delta, 1 - \delta]^\ell$ with pairwise target separation greater than η , Proposition 5.7 identifies the relevant finite sums with Riemann sums for the corresponding product of ρ -factors. Lemma 5.12 bounds the omitted finite- n collision contribution by $O_{\ell,\delta}(\eta)$. Since ρ is bounded and continuous on the truncated strip, the limiting integral over the collision neighborhood also vanishes as $\eta \downarrow 0$. Thus the full truncated integral is obtained by first taking $n \rightarrow \infty$ and then $\eta \downarrow 0$.

The gap envelope and Lemma 5.10 bound the complete fixed-order finite sums uniformly in n . Endpoint tightness shows that the portion lost outside $[\delta, 1 - \delta]^\ell$ tends to zero as $\delta \downarrow 0$. Consequently the nonnegative truncated integrals are uniformly bounded and increase to finite limits. This applies in particular to

$$\int_{\delta}^{1-\delta} \rho(x, x) dx$$

and to every cyclic trace integral

$$\int_{[\delta, 1-\delta]^r} \rho(x_1, x_2) \cdots \rho(x_r, x_1) dx_1 \cdots dx_r,$$

as well as to their finite products in mixed factorial moments. $\square$

5.3 Factorial moments

Proposition 5.14 (Fixed-point factorial moments). *For every fixed $k \geq 1$,*

$$\mathbb{E}(C_1(\sigma_n))_k \rightarrow \lambda_1^k.$$

Proof. Write

$$C_1(\sigma_n) = \sum_{i=1}^n \mathbf{1}_{\{\sigma_n(i)=i\}}.$$

For fixed k ,

$$\mathbb{E}(C_1(\sigma_n))_k = \sum_{i_1, \dots, i_k}^* \mathbb{P}(\sigma_n(i_a) = i_a, 1 \leq a \leq k),$$

where $*$ means that the indices are distinct. Fix $0 < \delta < 1/2$ and $\eta > 0$. Split the sum into endpoint terms, non-separated bulk terms, and separated bulk terms. Lemma 5.11 bounds the endpoint terms by $O_k(\delta + \delta^\beta)$, for any fixed $0 < \beta < \alpha$. Lemma 5.12 bounds the non-separated bulk terms by $O_{k,\delta}(\eta)$. On the separated bulk part, Proposition 5.7 gives

$$\mathbb{P}(\sigma_n(i_a) = i_a, 1 \leq a \leq k) = n^{-k} \prod_{a=1}^k \rho(i_a/n, i_a/n) + o(n^{-k}),$$

uniformly. The error is uniform over the separated bulk configurations, and there are $O(n^k)$ such configurations, so its total contribution is $o(1)$. Hence, as $n \rightarrow \infty$, the separated bulk contribution tends to

$$\left(\int_{\delta}^{1-\delta} \rho(x, x) dx \right)^k,$$

up to an error that vanishes as $\eta \downarrow 0$. Indeed, on $[\delta, 1 - \delta]$ the function $x \mapsto \rho(x, x)$ is bounded and continuous, so the contribution of the η -neighborhood of the diagonals in $[\delta, 1 - \delta]^k$ is $O_{k,\delta}(\eta)$. Letting $\eta \downarrow 0$ and then $\delta \downarrow 0$, and using Corollary 5.13, gives

$$\mathbb{E}(C_1(\sigma_n))_k \rightarrow \lambda_1^k.$$

$\square$

Proposition 5.15 (Mixed short-cycle factorial moments). *For every fixed $R \geq 1$ and fixed nonnegative integers $k_1, \dots, k_R$,*

$$\mathbb{E} \prod_{r=1}^R (C_r(\sigma_n))_{k_r} \rightarrow \prod_{r=1}^R \lambda_r^{k_r}.$$

Proof. Let $\mathcal{C}_{n,r}$ be the set of directed r -cycles on $[n]$, counted up to cyclic rotation. For $\gamma = (i_1, \dots, i_r) \in \mathcal{C}_{n,r}$, let

$$E_\gamma = \{\sigma_n(i_1) = i_2, \sigma_n(i_2) = i_3, \dots, \sigma_n(i_r) = i_1\}.$$

Then $C_r(\sigma_n) = \sum_{\gamma \in \mathcal{C}_{n,r}} \mathbf{1}_{E_\gamma}$. Put $L = \sum_{r=1}^R r k_r$. Expanding the mixed factorial moment gives a sum over ordered collections of distinct cycles. For k_r cycles of the same length r , the factorial moment $(C_r(\sigma_n))_{k_r}$ keeps those cycles ordered, so no additional $k_r!$ denominator appears; the only normalization inside a single directed r -cycle is the factor $1/r$ for cyclic rotations. If two distinct cycles share a label, their events are incompatible; hence only disjoint collections contribute.

For a disjoint collection, the event imposes L distinct point constraints $\sigma_n(p_a) = q_a$. The labels p_a are distinct, and the target positions q_a are the same labels cyclically shifted within each cycle, hence are also distinct. The separated local estimate requires separation of the target positions; no separate metric separation of the labels is needed.

Fix $0 < \delta < 1/2$ and $\eta > 0$. Decompose the sum according to whether some target position lies in $E_{\delta,n}$, whether all target positions lie in the bulk but two are within ηn , or whether the target positions are separated in the bulk. For fixed $R, k_1, \dots, k_R$, the number of ways to arrange a given ordered L -tuple of distinct point constraints into the prescribed ordered cycle collection is bounded by a constant depending only on R and the k_r 's. Therefore Lemmas 5.11 and 5.12 bound the endpoint terms by $O_{R,k}(\delta + \delta^\beta)$ and the non-separated bulk terms by $O_{R,k,\delta}(\eta)$. On separated bulk configurations, Proposition 5.7 applies.

Consider first a single r -cycle. On the separated bulk, the local estimate gives the Riemann sum

$$\frac{1}{r} \int_{[\delta, 1-\delta]^r} \rho(x_1, x_2) \rho(x_2, x_3) \cdots \rho(x_r, x_1) dx_1 \cdots dx_r,$$

where the factor $1/r$ appears because an ordered r -tuple represents the same directed cycle under r cyclic rotations. For an ordered collection of disjoint cycles, the local estimate gives

$$n^{-L} \prod_{\gamma} \prod_{a=1}^{|\gamma|} \rho(x_{\gamma,a}, x_{\gamma,a+1}) + o(n^{-L}), \quad x_{\gamma,|\gamma|+1} = x_{\gamma,1},$$

uniformly on the separated bulk. The disjointness restriction only removes configurations lying on equality diagonals among the L labels. There are $O(n^{L-1})$ such grid configurations, and on the truncated bulk the limiting product density is bounded, so their Riemann-sum weight is $O(n^{L-1}) \cdot O(n^{-L}) = O(n^{-1})$. Thus the corresponding limiting integral factors over the cycle components. After $n \rightarrow \infty$ and $\eta \downarrow 0$, the contribution from separated bulk collections tends to the product of the truncated cycle means. Finally $\delta \downarrow 0$ is justified by Corollary 5.13. We obtain

$$\mathbb{E} \prod_{r=1}^R (C_r(\sigma_n))_{k_r} \rightarrow \prod_{r=1}^R \lambda_r^{k_r}.$$

□

Proof of Theorem 5.1. Propositions 5.14 and 5.15 give convergence of all fixed mixed factorial moments:

$$\mathbb{E} \prod_{r=1}^R (C_r(\sigma_n))_{k_r} \rightarrow \prod_{r=1}^R \lambda_r^{k_r}.$$

These are exactly the mixed factorial moments of independent Poisson variables $Z_r \sim \text{Pois}(\lambda_r)$. Since ordinary mixed moments are finite linear combinations of mixed factorial moments, all fixed mixed ordinary moments also converge to those of $(Z_1, \dots, Z_R)$. The limiting vector has finite exponential moments and is therefore moment-determinate, so the multivariate method of moments gives

$$(C_1(\sigma_n), \dots, C_R(\sigma_n)) \Rightarrow (Z_1, \dots, Z_R),$$

with independent $Z_r \sim \text{Pois}(\lambda_r)$. □

Corollary 5.16 (Total-variation convergence for fixed cycle vectors). *For every fixed R ,*

$$d_{\text{TV}} \left(\mathcal{L}(C_1(\sigma_n), \dots, C_R(\sigma_n)), \bigotimes_{r=1}^R \text{Pois}(\lambda_r) \right) \rightarrow 0.$$

Proof. Let p_n and p be the probability mass functions of the two laws on the countable discrete space $\mathbb{Z}_{\geq 0}^R$. Theorem 5.1 implies $p_n(z) \rightarrow p(z)$ for every $z \in \mathbb{Z}_{\geq 0}^R$. Given $\varepsilon > 0$, choose a finite set $A \subset \mathbb{Z}_{\geq 0}^R$ with $p(A) > 1 - \varepsilon$. Then $\sum_{z \in A} |p_n(z) - p(z)| \rightarrow 0$ and $p_n(A^c) \rightarrow p(A^c)$. Hence

$$\sum_{z \in \mathbb{Z}_{\geq 0}^R} |p_n(z) - p(z)| \leq \sum_{z \in A} |p_n(z) - p(z)| + p_n(A^c) + p(A^c) \leq 3\varepsilon + o(1).$$

Letting $\varepsilon \downarrow 0$ proves the claim. □

5.4 Bistochasticity and long trace universality

The fixed-length cycle means admit an operator interpretation that also determines their large-length behavior. Define the integral operator K on $L^2([0, 1])$ by

$$(Kh)(x) = \int_0^1 \rho(x, y) h(y) dy,$$

and let Π be the rank-one projection onto the constants,

$$(\Pi h)(x) = \int_0^1 h(y) dy.$$

Lemma 5.17 (Bistochastic Hilbert–Schmidt kernel). *The kernel ρ is bistochastic:*

$$\int_0^1 \rho(x, y) dy = 1 \quad \text{for every } x \in [0, 1], \quad \int_0^1 \rho(x, y) dx = 1 \quad \text{for every } y \in (0, 1).$$

Moreover, K is a positivity-preserving Hilbert–Schmidt Markov contraction on $L^2([0, 1])$ and

$$\|K\|_{\text{HS}}^2 = \int_{[0, 1]^2} \rho(x, y)^2 dx dy \leq \frac{M_f}{m_f}.$$

In particular, K is compact, $K\mathbf{1} = \mathbf{1}$, $K^*\mathbf{1} = \mathbf{1}$, and $K\Pi = \Pi K = \Pi$.

Proof. Put $y = A(t)$, so that $dy = B(t) dt$. Then

$$\begin{aligned}\int_0^1 \rho(x, y) dy &= \int_0^\infty f(x) e^{-f(x)t} dt = 1, \\ \int_0^1 \rho(x, y) dx &= \frac{\int_0^1 f(x) e^{-f(x)\tau(y)} dx}{B(\tau(y))} = 1.\end{aligned}$$

Thus $K\mathbf{1} = K^*\mathbf{1} = \mathbf{1}$. Jensen's inequality and the two marginal identities give

$$|Kh(x)|^2 \leq \int_0^1 \rho(x, y) |h(y)|^2 dy, \quad \|Kh\|_2^2 \leq \|h\|_2^2,$$

so K is a contraction.

For the Hilbert–Schmidt norm, again use $y = A(t)$:

$$\begin{aligned}\|K\|_{\text{HS}}^2 &= \int_0^\infty \frac{\int_0^1 f(x)^2 e^{-2f(x)t} dx}{B(t)} dt \\ &\leq M_f \int_0^\infty \frac{B(2t)}{B(t)} dt.\end{aligned}$$

Since $f \geq m_f$,

$$B(2t) = \int_0^1 f(x) e^{-f(x)t} e^{-f(x)t} dx \leq e^{-m_f t} B(t),$$

and hence $\|K\|_{\text{HS}}^2 \leq M_f/m_f$. Hilbert–Schmidt operators are compact. The identities involving Π follow from bistochasticity. $\square$

Lemma 5.18 (Cyclic trace formula). *Let K be a Hilbert–Schmidt integral operator on $L^2([0, 1])$ with nonnegative kernel $\rho \in L^2([0, 1]^2)$. For every $r \geq 2$, K^r is trace class and*

$$\text{Tr}(K^r) = \int_{[0, 1]^r} \rho(x_1, x_2) \rho(x_2, x_3) \cdots \rho(x_r, x_1) dx_1 \cdots dx_r.$$

Proof. Approximate ρ in L^2 by nonnegative step kernels ρ_m , and let K_m be the corresponding finite-rank operators. The formula is the ordinary matrix trace identity for K_m . Since $K_m \rightarrow K$ in Hilbert–Schmidt norm and the Hilbert–Schmidt norms remain bounded, the Schatten ideal inequality

$$\|AB\|_1 \leq \|A\|_2 \|B\|_2$$

and a telescoping expansion give $K_m^r \rightarrow K^r$ in trace norm. Hence $\text{Tr}(K_m^r) \rightarrow \text{Tr}(K^r)$. Applying the same Cauchy–Schwarz inequalities to the cyclic products shows that their integrals converge to the displayed integral. This proves both its finiteness and the identity. $\square$

Theorem 5.19 (Long trace universality). *There are constants $C < \infty$ and $q \in (0, 1)$, depending only on the limiting profile f , such that, for every $r \geq 2$,*

$$\lambda_r = \frac{1}{r} + O\left(\frac{q^r}{r}\right).$$

Equivalently,

$$r\lambda_r = 1 + O(q^r).$$

Consequently,

$$\lim_{r \rightarrow \infty} \lim_{n \rightarrow \infty} r \mathbb{E} C_r(\sigma_n) = 1, \quad \sum_{r=2}^{\infty} \left| \lambda_r - \frac{1}{r} \right| < \infty.$$

Proof. We first identify the peripheral spectrum of K . Suppose $Kh = \zeta h$ with $|\zeta| = 1$. Since K is a contraction,

$$\|h\|_2 = \|Kh\|_2.$$

Equality must therefore hold in the pointwise Jensen inequality used in the proof of Lemma 5.17 for almost every x . Because $\rho(x, y) > 0$ for every $x \in [0, 1]$ and every $y \in (0, 1)$, equality in Jensen's inequality forces $h(y)$ to be almost everywhere constant. It follows that $\zeta = 1$. Thus the only eigenvectors with eigenvalue of modulus one are the constants. The eigenvalue 1 is algebraically simple as well. Indeed, if $(K - I)g = c\mathbf{1}$ with $c \neq 0$, then $K^n g = g + nc\mathbf{1}$, contradicting $\|K^n\| \leq 1$.

Since K is compact, its nonzero spectrum consists of isolated eigenvalues of finite algebraic multiplicity, with possible accumulation only at zero. Hence there is $q \in (0, 1)$ such that every eigenvalue of K other than 1 has modulus at most q . Put $T = K - \Pi$. Then T is Hilbert–Schmidt, $T\Pi = \Pi T = 0$, and

$$K^r = \Pi + T^r, \quad r \geq 1.$$

For $r \geq 2$, Lemma 5.18 gives

$$\mathrm{Tr}(K^r) = \int_{[0,1]^r} \rho(x_1, x_2) \rho(x_2, x_3) \cdots \rho(x_r, x_1) dx_1 \cdots dx_r = r\lambda_r.$$

The equality with the improper integral defining λ_r follows by monotone convergence and Corollary 5.13.

Let $(\zeta_j)_{j \geq 2}$ be the nontrivial eigenvalues of K , repeated according to algebraic multiplicity. Lidskii's theorem [17, Theorem 3.7] gives

$$r\lambda_r = \mathrm{Tr}(K^r) = 1 + \mathrm{Tr}(T^r) = 1 + \sum_{j \geq 2} \zeta_j^r.$$

Weyl's eigenvalue inequality for Schatten classes [17, Theorem 2.3] gives

$$\sum_{j \geq 2} |\zeta_j|^2 \leq \|T\|_{\mathrm{HS}}^2.$$

Therefore

$$|r\lambda_r - 1| \leq q^{r-2} \sum_{j \geq 2} |\zeta_j|^2 \leq q^{r-2} \|T\|_{\mathrm{HS}}^2.$$

After changing the constant, this is $O(q^r)$. The two consequences follow immediately. $\square$

5.5 Diagonal bands and transverse curves

The separated local estimate has several consequences for sparse deterministic entry sets. We first record the fixed-width diagonal-band law, and then prove a master sparse-entry theorem. The latter removes the transversality and smoothness assumptions from the thin-curve statement and yields deterministic overlap laws as a further corollary.

Theorem 5.20 (Fixed-width diagonal bands). *Under the hypotheses of Theorem 5.1, fix an integer $h \geq 0$, and define*

$$N_h = \#\{i : 1 \leq i \leq n, |\sigma_n(i) - i| \leq h\}.$$

Then

$$N_h \Rightarrow \text{Pois}((2h+1)\lambda_1), \quad \lambda_1 = \lim_{\delta \downarrow 0} \int_{\delta}^{1-\delta} \rho(x, x) dx.$$

Proof. Write

$$N_h = \sum_{d=-h}^h \sum_{\substack{1 \leq i \leq n \\ 1 \leq i+d \leq n}} \mathbf{1}_{\{\sigma_n(i)=i+d\}}.$$

Fix $k \geq 1$. Expanding $\mathbb{E}(N_h)_k$ gives a sum over ordered k -tuples (i_a, d_a) . Terms with repeated labels or repeated target positions are zero unless they are identical summands, which are excluded by the factorial expansion. Thus the contributing terms impose k distinct point constraints $\sigma_n(i_a) = i_a + d_a$.

Fix $0 < \delta < 1/2$ and $\eta > 0$. Endpoint target positions contribute $O_{k,h}(\delta + \delta^\beta)$, for every $0 < \beta < \alpha$, by Lemma 5.11; the finite number of offsets only changes the constant. Among the remaining bulk terms, the contribution from target positions within ηn of one another is $O_{k,h,\delta}(\eta)$ by Lemma 5.12. On the separated bulk part, Proposition 5.7 gives, uniformly,

$$\mathbb{P}(\sigma_n(i_a) = i_a + d_a, 1 \leq a \leq k) = n^{-k} \prod_{a=1}^k \rho(i_a/n, (i_a + d_a)/n) + o(n^{-k}).$$

For each fixed offset d , uniformly on the truncated bulk,

$$\rho(i/n, (i+d)/n) = \rho(i/n, i/n) + o(1),$$

because ρ is continuous on compact subintervals of $[0, 1] \times (0, 1)$. Hence, after summing over the $(2h+1)^k$ offset choices and passing to Riemann sums, the separated bulk contribution tends to

$$\left((2h+1) \int_{\delta}^{1-\delta} \rho(x, x) dx \right)^k,$$

up to an error vanishing as $\eta \downarrow 0$. Letting $\eta \downarrow 0$ and then $\delta \downarrow 0$, using Corollary 5.13, gives

$$\mathbb{E}(N_h)_k \rightarrow ((2h+1)\lambda_1)^k.$$

These are the factorial moments of $\text{Pois}((2h+1)\lambda_1)$, which proves the claim by the method of moments. $\square$

5.6 Sparse entry sets, thin graphs, and overlaps

The preceding diagonal arguments are instances of a single sparse-entry theorem. For a deterministic set $E_n \subset [n]^2$, put

$$N(E_n) = \sum_{(i,j) \in E_n} \mathbf{1}_{\{\sigma_n(i)=j\}}, \quad \nu_n = \frac{1}{n} \sum_{(i,j) \in E_n} \delta_{(i/n, j/n)}.$$

The measures ν_n need not be probability measures.

Theorem 5.21 (Sparse bulk entry sets). *Assume the hypotheses of Theorem 5.1. Fix $\delta \in (0, 1/2)$, and suppose that $E_n \subset [n] \times [\delta n, (1 - \delta)n]$ satisfies*

$$|E_n| = O(n), \quad \max_{1 \leq i \leq n} \#\{j : (i, j) \in E_n\} = o(n),$$

and

$$\lim_{\eta \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \sup_{q \in [n]} \#\{(i, j) \in E_n : |j - q| \leq \eta n\} = 0.$$

If $\nu_n \Rightarrow \nu$ weakly as finite measures on $[0, 1]^2$, then

$$N(E_n) \Rightarrow \text{Pois} \left(\int_{[0, 1]^2} \rho(x, y) \nu(dx, dy) \right).$$

Proof. Fix $k \geq 1$. Expanding the factorial moment gives

$$\mathbb{E}(N(E_n))_k = \sum_{e_1, \dots, e_k \in E_n}^* \mathbb{P}(\sigma_n(i_a) = j_a, 1 \leq a \leq k), \quad e_a = (i_a, j_a),$$

where the star excludes repeated edges. If two distinct selected edges have the same row or the same target column, the corresponding event is impossible.

We first record a uniform bulk bound. If $j_1, \dots, j_k$ are distinct and all belong to $[\delta n, (1 - \delta)n]$, then Lemma 5.9 gives

$$\sup_{i_1, \dots, i_k} \mathbb{P}(\sigma_n(i_a) = j_a, 1 \leq a \leq k) \leq C_{k, \delta} n^{-k}.$$

Indeed, after sorting the target positions, every remaining deck size is at least δn , and every gap in Lemma 5.9 is separated from the terminal end of its remaining deck by at least δn ; hence each gap factor is $O_\delta(n^{-1})$.

Let

$$D_n = \max_i \#\{j : (i, j) \in E_n\}, \quad M_n(\eta) = \sup_q \#\{(i, j) \in E_n : |j - q| \leq \eta n\}.$$

The number of ordered k -tuples containing two distinct edges in the same row is at most

$$C_k D_n |E_n|^{k-1} = o(n^k).$$

The number containing a pair of target columns at distance at most ηn is at most

$$C_k |E_n|^{k-1} M_n(\eta).$$

Consequently, by the preceding $O(n^{-k})$ probability bound, the total factorial-moment contribution from repeated rows or ηn -close target columns is

$$o(1) + O_k(M_n(\eta)/n).$$

After $n \rightarrow \infty$ and then $\eta \downarrow 0$, this contribution vanishes.

On the remaining tuples the rows are distinct and the target columns are pairwise separated by more than ηn . Proposition 5.7 gives, uniformly,

$$\begin{aligned} \mathbb{P}(\sigma_n(i_a) = j_a, 1 \leq a \leq k) \\ = n^{-k} \prod_{a=1}^k \rho(i_a/n, j_a/n) + o(n^{-k}). \end{aligned}$$

There are $O(n^k)$ tuples, so the accumulated error is $o(1)$. Since ρ is bounded and continuous on $[0, 1] \times [\delta, 1 - \delta]$, the removal of the repeated-row and close-target tuples also changes the deterministic product sum by $o(1)$ after $n \rightarrow \infty$ and $\eta \downarrow 0$. Hence

$$\begin{aligned} \mathbb{E}(N(E_n))_k &= \left(\frac{1}{n} \sum_{(i,j) \in E_n} \rho(i/n, j/n) \right)^k + o(1) \\ &\longrightarrow \left(\int \rho(x, y) \nu(dx, dy) \right)^k. \end{aligned}$$

These are the factorial moments of the stated Poisson law. The method of moments proves the theorem. $\square$

Corollary 5.22 (General thin-graph Poisson law). *Assume the hypotheses of Theorem 5.1. Let $q_{n,i} \in [\delta n, (1 - \delta)n]$ be deterministic and suppose*

$$\frac{1}{n} \sum_{i=1}^n \delta_{(i/n, q_{n,i}/n)} \Rightarrow \nu,$$

where ν is a probability measure on $[0, 1] \times [\delta, 1 - \delta]$. If

$$\lim_{\eta \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \sup_q \#\{i : |q_{n,i} - q| \leq \eta n\} = 0,$$

then

$$\#\{i : \sigma_n(i) = q_{n,i}\} \Rightarrow \text{Pois} \left(\int \rho(x, y) \nu(dx, dy) \right).$$

If, in addition to the preceding no-crowding assumption, $q_{n,i}/n \rightarrow g(i/n)$ uniformly for a continuous function g with values in $[\delta, 1 - \delta]$, then the limiting mean is

$$\int_0^1 \rho(x, g(x)) dx.$$

Proof. Apply Theorem 5.21 to $E_n = \{(i, q_{n,i}) : 1 \leq i \leq n\}$. Every row degree is one, and the remaining hypotheses are exactly those in the statement. The final assertion follows from convergence of the empirical graph measures to the pushforward of Lebesgue measure under $x \mapsto (x, g(x))$. $\square$

Corollary 5.23 (Overlaps with deterministic permutations). *Assume the hypotheses of Theorem 5.1. Let $\tau_n \in S_n$ be deterministic and suppose its graph permutoins*

$$\mu_{\tau_n} = \frac{1}{n} \sum_{i=1}^n \delta_{(i/n, \tau_n(i)/n)}$$

converge weakly to a permuton μ . Then

$$\#\{i : \sigma_n(i) = \tau_n(i)\} \Rightarrow \text{Pois}(\Lambda_\mu),$$

where

$$\Lambda_\mu = \lim_{\delta \downarrow 0} \int_{[0,1] \times [\delta, 1-\delta]} \rho(x, y) \mu(dx, dy) < \infty.$$

Proof. Fix $\delta \in (0, 1/2)$ and retain only the edges $(i, \tau_n(i))$ whose target belongs to $[\delta n, (1-\delta)n]$. The row degrees and column degrees are one, and

$$\sup_q \#\{i : |\tau_n(i) - q| \leq \eta n\} \leq 2\eta n + 2.$$

Since the second marginal of every permuton is Lebesgue measure, the boundary lines $y = \delta$ and $y = 1 - \delta$ have μ -measure zero. Theorem 5.21 therefore gives a Poisson limit with mean

$$\Lambda_\mu(\delta) = \int_{[0,1] \times [\delta, 1-\delta]} \rho(x, y) \mu(dx, dy).$$

The expected number of omitted overlaps is bounded by

$$\sum_{q \in E_{\delta, n}} \sup_i \mathbb{P}(\sigma_n(i) = q),$$

which tends to zero as $\delta \downarrow 0$, uniformly in n along the limit, by Corollary 5.13 with $\ell = 1$. Hence the omitted count tends to zero in probability after first taking $n \rightarrow \infty$ and then $\delta \downarrow 0$. To see that the limiting mean is finite, fix $0 < \delta' < \delta$. Applying Theorem 5.21 to the graph edges with target in $[\delta', \delta] \cup [1 - \delta, 1 - \delta']$ identifies the corresponding limiting integral with the limit of their expected overlap count. The latter is bounded by the endpoint estimate above and is therefore $O(\delta + \delta^\beta)$. Thus $\Lambda_\mu(\delta)$ is Cauchy as $\delta \downarrow 0$ and has a finite monotone limit. The converging-together lemma now yields the stated Poisson limit. $\square$

Theorem 5.24 (Transverse thin-curve Poisson law). *Assume the hypotheses of Theorem 5.1. Let $g \in C^1[0, 1]$ satisfy*

$$0 < \delta_g \leq g(x) \leq 1 - \delta_g < 1, \quad \inf_{0 \leq x \leq 1} |g'(x)| > 0.$$

For all large n , put

$$q_{n,i} = \lfloor ng(i/n) \rfloor, \quad N_g = \#\{i : \sigma_n(i) = q_{n,i}\}.$$

Then

$$N_g \Rightarrow \text{Pois} \left(\int_0^1 \rho(x, g(x)) dx \right).$$

Proof. The derivative condition makes g monotone and bi-Lipschitz on its image. Hence

$$\sup_q \#\{i : |q_{n,i} - q| \leq \eta n\} \leq C_g(\eta n + 1),$$

while $q_{n,i}/n \rightarrow g(i/n)$ uniformly. Corollary 5.22 applies. $\square$

6 Macroscopic cycles under uniformly comparable weights

The preceding results concern cycles whose lengths remain fixed as $n \rightarrow \infty$. We now consider cycles of order n . Under a uniform bound on the ratio of the largest to the smallest weight, the macroscopic cycle partition is universal. We write $\text{GEM}(1)$ for the residual-allocation sequence

$$U_1, \quad (1 - U_1)U_2, \quad (1 - U_1)(1 - U_2)U_3, \dots,$$

where the U_j are independent uniform random variables, and $\text{PD}(1)$ for its decreasing rearrangement; see Pitman [15, Chapter 3]. Let

$$\mathcal{S}^\downarrow = \left\{ x_1 \geq x_2 \geq \dots \geq 0 : \sum_{j \geq 1} x_j = 1 \right\},$$

equipped with the ℓ^1 metric.

Theorem 6.1 (Macroscopic cycle universality). *Let σ_n be a Luce permutation with positive weights $\theta_{n,1}, \dots, \theta_{n,n}$ and assume*

$$\sup_n \frac{\max_i \theta_{n,i}}{\min_i \theta_{n,i}} < \infty.$$

If $L_{n,1} \geq L_{n,2} \geq \dots$ are the cycle lengths of σ_n , then

$$\left(\frac{L_{n,1}}{n}, \frac{L_{n,2}}{n}, \dots \right) \Rightarrow \text{PD}(1)$$

in $\mathcal{S}^\downarrow$. Moreover, if $\tilde{L}_{n,1}, \tilde{L}_{n,2}, \dots$ are obtained by choosing a uniform vertex, listing the length of its cycle, deleting that cycle, and repeating on the residual vertex set, then

$$\left(\frac{\tilde{L}_{n,1}}{n}, \frac{\tilde{L}_{n,2}}{n}, \dots \right) \Rightarrow \text{GEM}(1)$$

in ℓ^1 . In particular, if I_n is a uniformly chosen label and $\mathcal{L}_n(I_n)$ is the length of its cycle, then

$$\frac{\mathcal{L}_n(I_n)}{n} \Rightarrow \text{Unif}(0, 1).$$

The proof uses one fixed finite partition of clock time. Conditional on the clock slabs, each slab row is a permanent matching whose entries are nearly constant. A common minorization gives a spectral gap depending only on the weight-comparability constant. A relative-prefix theorem then controls every target category under arbitrary adaptive exploration. We formulate all concentration statements uniformly over an explicit class of regular residual histories, and prove that deleting an interior macroscopic cycle preserves that class. This supplies the regeneration needed for the $\text{GEM}(1)$ induction.

Multiplying all weights by a common positive factor does not change the Luce order. We therefore normalize throughout the proof so that, for a fixed $\kappa < \infty$,

$$1 \leq \theta_{n,i} \leq \kappa, \quad 1 \leq i \leq n. \tag{6.0}$$

6.1 Subsequential marked kernels and a uniform contraction

Fix an arbitrary subsequence. Passing to a further subsequence, compactness gives

$$\eta_n := \frac{1}{n} \sum_{i=1}^n \delta_{(i/n, \theta_{n,i})} \Longrightarrow \eta$$

on $[0, 1] \times [1, \kappa]$. The first marginal of η is Lebesgue measure. Choose a disintegration

$$\eta(dx, d\vartheta) = dx \eta_x(d\vartheta),$$

defined for almost every x . Put

$$A(t) = \int (1 - e^{-\vartheta t}) \eta(dx, d\vartheta), \quad B(t) = A'(t) = \int \vartheta e^{-\vartheta t} \eta(dx, d\vartheta),$$

let $\tau = A^{-1}$, and define, for almost every x ,

$$g_x(t) = \int \vartheta e^{-\vartheta t} \eta_x(d\vartheta), \quad \rho(x, y) = \frac{g_x(\tau(y))}{B(\tau(y))}.$$

Let K be the associated Markov operator on $L^2[0, 1]$.

Lemma 6.2 (Bistochasticity and common minorization). *The kernel ρ is bistochastic up to null sets and*

$$\iint_{[0,1]^2} \rho(x, y)^2 dx dy \leq \kappa.$$

On the mean-zero subspace,

$$\|K\|_{L_0^2 \rightarrow L_0^2} \leq q_\kappa := \sqrt{1 - \kappa^{-1}} < 1,$$

and the same bound holds for K^ .*

Proof. Using $y = A(t)$ and $dy = B(t) dt$,

$$\int_0^1 \rho(x, y) dy = \int_0^\infty g_x(t) dt = 1$$

for almost every x , while

$$\int_0^1 \rho(x, y) dx = \frac{\int_0^1 g_x(\tau(y)) dx}{B(\tau(y))} = 1.$$

Jensen's inequality and $\vartheta \leq \kappa$ give

$$\int_0^1 g_x(t)^2 dx \leq \int \vartheta^2 e^{-2\vartheta t} \eta(dx, d\vartheta) \leq \kappa B(2t).$$

Since $\vartheta \geq 1$, $B(2t) \leq e^{-t} B(t)$, and hence

$$\iint \rho^2 = \int_0^\infty \frac{\int g_x(t)^2 dx}{B(t)} dt \leq \kappa.$$

For the contraction, regard $g_x(t) dt$ as the conditional law of a clock time T given x . From (6.0), $g_x(t) \geq e^{-\kappa t}$. Let

$$\alpha = \kappa^{-1}, \quad \nu(dt) = \kappa e^{-\kappa t} dt.$$

There are probability kernels R_x such that

$$g_x(t) dt = \alpha \nu(dt) + (1 - \alpha) R_x(dt).$$

The marginal law of T is $\mu = \alpha \nu + (1 - \alpha) \bar{R}$, where $\bar{R} = \int_0^1 R_x dx$. If $f \in L^2(\mu)$ and $\mu(f) = 0$, then

$$\int f(t) g_x(t) dt = (1 - \alpha) \{R_x(f) - \bar{R}(f)\}.$$

Jensen's inequality therefore yields

$$\begin{aligned} \int_0^1 \left| \int f(t) g_x(t) dt \right|^2 dx &\leq (1 - \alpha)^2 \bar{R}(f^2) \\ &\leq (1 - \alpha) \mu(f^2). \end{aligned}$$

The change of variables $y = A(t)$ is an isometry from $L^2(dy)$ to $L^2(\mu)$, and under this isometry the displayed conditional-expectation operator is K . This proves the bound; the adjoint has the same norm. $\square$

6.2 Clock slabs and residual permanental matchings

Fix $T > 0$ and a partition

$$0 = t_0 < t_1 < \cdots < t_d = T, \quad \max_a (t_a - t_{a-1}) \leq h,$$

and append the tail interval $I_\star = (T, \infty)$. Put

$$I_a = (t_{a-1}, t_a] \quad (a \leq d), \quad H_{n,a} = \{i : X_{n,i} \in I_a\}, \quad m_{n,a} = |H_{n,a}|.$$

If $M_{n,a} = \sum_{c \leq a} m_{n,c}$, then the ranks occupied by $H_{n,a}$ are

$$Q_{n,a} = \{M_{n,a-1} + 1, \dots, M_{n,a}\};$$

the tail row is defined analogously. Set

$$N_{n,ab} = |Q_{n,a} \cap H_{n,b}|, \quad p_{n,a} = m_{n,a}/n.$$

When $m_{n,a} > 0$, put $P_{n,ab} = N_{n,ab}/m_{n,a}$; if $m_{n,a} = 0$, define that row of P_n arbitrarily. On the good events used below every row is nonempty. Thus P_n is stochastic and $p_n P_n = p_n$.

Let

$$J_a = (A(t_{a-1}), A(t_a)] \quad (a \leq d), \quad J_\star = (A(T), 1], \quad p_a = |J_a|,$$

and define

$$q_{ab} = \int_{J_a \times J_b} \rho(x, y) dx dy, \quad P_{ab} = q_{ab}/p_a.$$

Every p_a and q_{ab} is strictly positive.

Lemma 6.3 (Slab law and deterministic good events). *For every fixed partition,*

$$p_{n,a} \xrightarrow{\mathbb{P}} p_a, \quad N_{n,ab}/n \xrightarrow{\mathbb{P}} q_{ab}$$

simultaneously in a, b . There are deterministic $\varepsilon_n \downarrow 0$ and events $\mathcal{G}_n$ with $\mathbb{P}(\mathcal{G}_n) \rightarrow 1$ such that, on $\mathcal{G}_n$:

(i) *all of the preceding convergences hold with error at most ε_n^2 , and, for all sufficiently large n ,*

$$p_{n,a} \geq \frac{1}{2}p_a, \quad N_{n,ab} \geq \frac{1}{2}q_{ab}n$$

for every a, b ;

(ii) *for a constant $q_0 \in (q_\kappa, 1)$,*

$$\|P_n\|_{L_0^2(p_n) \rightarrow L_0^2(p_n)} \leq q_0;$$

(iii) *if S is a partition endpoint and $J_b \subset [0, A(S)]$, then*

$$P_{n,ab} \leq 2M_S p_{n,b} \quad \text{for every } a, \quad M_S = \kappa e^{(\kappa-1)S}; \quad (6.1)$$

(iv) *for every union D of slab states,*

$$\sum_{a \in D} \sum_b p_{n,a} \frac{P_{n,ab}^2}{p_{n,b}} \leq 2\kappa; \quad (6.2)$$

(v) *if U is a partition endpoint and $D(U)$ is the union of rows whose clock intervals lie in (U, ∞) , then*

$$\sum_{a \in D(U)} p_{n,a} \leq 2e^{-U} + \varepsilon_n^2. \quad (6.3)$$

The constants may depend on the fixed partition and on S , but not on n .

Proof. For a fixed clock cell $I_b = (u, v]$, the variables $\mathbf{1}_{\{X_{n,i} \in I_b\}}$ are independent and have means $e^{-\theta_{n,i}u} - e^{-\theta_{n,i}v}$. Kolmogorov's maximal inequality gives

$$\max_{k \leq n} \left| \sum_{i \leq k} \left(\mathbf{1}_{\{X_{n,i} \in I_b\}} - \mathbb{E} \mathbf{1}_{\{X_{n,i} \in I_b\}} \right) \right| = o_{\mathbb{P}}(n).$$

Marked empirical convergence and the Lebesgue first marginal imply, uniformly in $x \in [0, 1]$,

$$\frac{1}{n} \sum_{i \leq nx} \mathbb{E} \mathbf{1}_{\{X_{n,i} \in I_b\}} \longrightarrow \int_0^x \int (e^{-\vartheta u} - e^{-\vartheta v}) \eta_z(d\vartheta) dz.$$

This proves the slab-size convergence and convergence of the endpoints of $Q_{n,a}/n$ to those of J_a . Applying the same uniform partial-sum estimate at those random endpoints gives

$$\frac{N_{n,ab}}{n} \longrightarrow \int_{J_a} \int (e^{-\vartheta t_{b-1}} - e^{-\vartheta t_b}) \eta_x(d\vartheta) dx = q_{ab}.$$

The last equality follows by integrating $\rho(x, y)$ over J_b and using $y = A(t)$.

The matrix P is the compression of K to functions that are constant on the sets J_a : explicitly, it is $\Pi_{\mathcal{J}} K \Pi_{\mathcal{J}}$. Conditional expectation is a contraction, so Lemma 6.2 gives $\|P\|_{L^2_0(p)} \leq q_\kappa$. Entrywise convergence of the conjugated matrices $D_{p_n}^{1/2} P_n D_{p_n}^{-1/2}$ proves (ii).

If $y \leq A(S)$, then $\tau(y) \leq S$ and

$$\rho(x, y) = \frac{g_x(\tau(y))}{B(\tau(y))} \leq \frac{\kappa e^{-\tau(y)}}{e^{-\kappa\tau(y)}} \leq \kappa e^{(\kappa-1)S} =: M_S.$$

Hence $P_{ab} \leq M_S p_b$, which yields (6.1) by entrywise convergence. On every rectangle, Cauchy's inequality gives

$$p_a \frac{P_{ab}^2}{p_b} = \frac{(\int_{J_a \times J_b} \rho)^2}{p_a p_b} \leq \int_{J_a \times J_b} \rho^2.$$

Summing and using Lemma 6.2 proves the limiting version of (6.2), and entrywise convergence gives its finite- n form. Finally, the expected proportion of clocks exceeding U is at most e^{-U} because all rates are at least one; the slab law gives (6.3). Since the partition is finite, choose deterministic $\delta_n \downarrow 0$ so that all convergence assertions hold with error at most δ_n on a single event of probability tending to one, and set $\varepsilon_n = \sqrt{\delta_n}$. This gives the stated event $\mathcal{G}_n$ and the errors ε_n^2 in (i) and (6.3). $\square$

Condition on the complete slab data: the sets $H_{n,a}$, the rank intervals $Q_{n,a}$, and the ordered clock values

$$s_{a,1} < \dots < s_{a,m_{n,a}}$$

in every slab. Write $Q_{n,a} = \{q_{a,1} < \dots < q_{a,m_{n,a}}\}$, and let $\phi_a : H_{n,a} \rightarrow [m_{n,a}]$ be defined by $\sigma_n(i) = q_{a,\phi_a(i)}$. Then

$$\mathbb{P}(\phi_a = \phi \mid \text{slab data}) = \frac{\prod_{i \in H_{n,a}} e^{-\theta_{n,i} s_{a,\phi(i)}}}{\text{per}(e^{-\theta_{n,i} s_{a,j}})_{i,j}}. \quad (6.3a)$$

The row matchings are conditionally independent. The factors $\prod_i \theta_{n,i}$ cancel from the conditional density.

Lemma 6.4 (Residual permanental factorization). *Condition on the slab data and on any compatible partial collection of exposed matching edges. In every row, delete the exposed source and target vertices. The remaining row matchings are conditionally independent, and each has the permanental law (6.3a) on its remaining submatrix.*

Proof. The conditional density of the complete matching is a product over rows and, within each row, a product over its edges. Conditioning on fixed exposed edges removes their constant factors and restricts the sum to bijections of the remaining rows and columns. The normalizing sum is the permanent of the remaining submatrix, and the factorization over rows is unchanged. $\square$

Lemma 6.5 (Hereditary flatness). *Let $A = (a_{ij})$ be a positive $m \times m$ matrix satisfying*

$$e^{-\epsilon} \leq a_{ij}/a_{ik} \leq e^\epsilon$$

for every row i and every pair of columns j, k . Under the permanental matching law proportional to $\prod_i a_{i,\phi(i)}$, after conditioning on an arbitrary partial matching, every remaining edge probability satisfies

$$\frac{e^{-2\epsilon}}{s} \leq \mathbb{P}(\phi(i) = j \mid \text{partial matching}) \leq \frac{e^{2\epsilon}}{s},$$

where s rows and columns remain. For a Luce slab of width at most h , this holds with $\epsilon = \kappa h$; write

$$\beta = 2\kappa h. \quad (6.4)$$

Proof. It suffices to work with the current remaining submatrix. For two remaining columns j, k ,

$$\frac{\mathbb{P}(\phi(i) = j)}{\mathbb{P}(\phi(i) = k)} = \frac{a_{ij} \text{ per } A_{-i,-j}}{a_{ik} \text{ per } A_{-i,-k}}.$$

Map a term of the first minor to a term of the second by locating the row sent to k and sending that row to j instead. Corresponding products differ by a factor in $[e^{-\epsilon}, e^{\epsilon}]$. Including the factor a_{ij}/a_{ik} shows that every pair of edge probabilities has ratio in $[e^{-2\epsilon}, e^{2\epsilon}]$. Since the s probabilities sum to one, the individual bounds follow. In a clock slab,

$$e^{-\kappa h} \leq e^{-\theta_i(s_j - s_k)} \leq e^{\kappa h}.$$

□

Lemma 6.6 (Predictable row exposure). *Condition on the slab data and on a compatible partial collection of exposed matching edges. Let $(\mathcal{F}_r)$ be an exploration filtration such that, before the $(r+1)$ st edge is exposed, its source is an unexposed source that is $\mathcal{F}_r$ -measurable. Then, conditional on $\mathcal{F}_r$, the target of that source has the one-edge marginal of the residual permanental matching in its row. In particular, if that row has s targets remaining and has width at most h , every individual target has conditional probability in*

$$[e^{-\beta}/s, e^{\beta}/s], \quad \beta = 2\kappa h.$$

If a distinguished target v in that row is forbidden and one conditions the next target not to equal v , then every one of the other $s-1$ targets has conditional probability in

$$[e^{-\beta}/(s-1), e^{\beta}/(s-1)].$$

The conclusions remain true when the next source is selected adaptively using all previously exposed edges, including edges from the same row.

Proof. By Lemma 6.4, conditional on $\mathcal{F}_r$ the unexposed part of the relevant row has the permanental law on its current submatrix. Lemma 6.5 therefore gives the first interval of probabilities. If v is forbidden, conditioning away from v merely renormalizes the probabilities of the other targets. Their pairwise ratios are unchanged and lie in $[e^{-\beta}, e^{\beta}]$; since their conditional probabilities sum to one, each lies in the second displayed interval. The measurability assumption is precisely what is needed to apply the residual-factorization statement before the new edge is revealed. No independence between the chosen source and the previously exposed edges is required. □

6.3 A quantitative relative-prefix theorem

Lemma 6.7 (Relative prefix law). *Fix $d \geq 1$ and $c_* > 0$. There are absolute constants $\beta_0 > 0$ and $C < \infty$, and deterministic sequences $\chi_m \downarrow 0$ and $\pi_m \downarrow 0$, depending only on d and c_* , with the following property. Let m targets be partitioned into d categories of sizes $m_1, \dots, m_d$ satisfying $m_b \geq c_* m$. Targets are selected sequentially without replacement in an arbitrary predictable,*

adaptive source order. Suppose that, whenever s targets remain, every individual remaining target has conditional selection probability in

$$[e^{-\beta}/s, e^{\beta}/s], \quad 0 \leq \beta \leq \beta_0.$$

If $X_b(r)$ is the number of category- b targets among the first r selections, then

$$\mathbb{P} \left(\max_{b \leq d} \max_{0 \leq r \leq m} \frac{|X_b(r) - rm_b/m|}{m_b} > C\beta + \chi_m \right) \leq \pi_m. \quad (6.5)$$

The estimate is uniform over all adaptive source rules satisfying the displayed one-target bounds. It remains valid after deleting any fixed number of distinguished targets before the procedure, or after forbidding those targets throughout the procedure, after changing c_* and the two deterministic sequences harmlessly.

Proof. Fix one category and write

$$N_r = m_b - X_b(r), \quad s_r = m - r.$$

Let Y_{r+1} be the indicator that the next selected target belongs to this category. By summing the individual-target bounds over the N_r remaining targets in the category,

$$\pi_r := \mathbb{E}(Y_{r+1} \mid \mathcal{F}_r) \in \left[e^{-\beta} \frac{N_r}{s_r}, e^{\beta} \frac{N_r}{s_r} \right]. \quad (6.6)$$

Set $a_m = m_b^{2/3}$ and stop at

$$\tau = \inf\{r : N_r < a_m \text{ or } s_r < a_m\}.$$

For $r < \tau$, put $Z_r = \log(N_r/s_r)$. Since

$$Z_{r+1} - Z_r = -\log(1 - s_r^{-1}) + Y_{r+1} \log(1 - N_r^{-1}),$$

Taylor's formula and (6.6) give

$$|\mathbb{E}(Z_{r+1} - Z_r \mid \mathcal{F}_r)| \leq C_1 \frac{\beta}{s_r} + C_1 \left(\frac{1}{s_r^2} + \frac{1}{s_r N_r} \right), \quad (6.7)$$

and the conditional variance of the centered increment is at most $C_1/(s_r N_r)$. Because $m_b \geq c_* m$,

$$\sum_{r < \tau} \frac{1}{s_r N_r} \leq a_m^{-1} \sum_{s \geq a_m} \frac{1}{s} \leq C_{c_*} \frac{1 + \log m}{m^{2/3}} =: v_m, \quad (6.8)$$

and the sum of the two second-order drift terms in (6.7) is $O(v_m)$.

Let M_r be the stopped martingale obtained by summing the centered increments. Doob's L^2 inequality and (6.8) imply

$$\mathbb{P} \left(\max_{r \leq \tau} |M_r| > v_m^{1/4} \right) \leq C v_m^{1/2}. \quad (6.9)$$

On the complementary event, for every $r < \tau$,

$$\left| \log \frac{N_r/m_b}{s_r/m} \right| \leq C_2 \beta \log \frac{m}{s_r} + C_2 (v_m^{1/4} + v_m). \quad (6.10)$$

Choose β_0 so that $C_2\beta_0 \leq 1/2$. Writing $u = s_r/m$ and $\eta_m = C_2(v_m^{1/4} + v_m)$, (6.10) yields

$$e^{-\eta_m} u^{1+C_2\beta} \leq \frac{N_r}{m_b} \leq e^{\eta_m} u^{1-C_2\beta}. \quad (6.11)$$

For $|a| \leq 1/2$,

$$\sup_{0 \leq u \leq 1} |u^{1+a} - u| \leq C|a|,$$

which follows from the mean-value theorem in the exponent and $\sup_{0 < u \leq 1} u |\log u| < \infty$. Hence, before τ ,

$$\left| \frac{N_r}{m_b} - \frac{s_r}{m} \right| \leq C_3\beta + C_3\eta_m. \quad (6.12)$$

At the stopping time, if $s_\tau < a_m$, then $s_\tau/m = O(m^{-1/3})$ and the upper bound in (6.11), applied one step earlier, gives $N_\tau/m_b = o(1)$. If $N_\tau < a_m$, then the lower bound in (6.11) gives $s_\tau/m = o(1)$. In either case, after τ both normalized remaining counts are nonincreasing and their difference is bounded by $C_3\beta + o(1)$. Thus (6.12) extends to all $0 \leq r \leq m$.

For example, one may take

$$\chi_m = C_{d,c_*} \left(v_m^{1/4} + m^{-1/12} \right), \quad \pi_m = C_{d,c_*} v_m^{1/2}.$$

A union bound over the fixed number of categories proves (6.5). Deleting or forbidding finitely many targets changes every relevant category size and the total size by $O(1)$; for all large m the positive-proportion hypothesis and the same one-target ratio bounds remain valid after changing c_* by a fixed factor. $\square$

In every application below the mesh is chosen sufficiently fine that $\beta = 2\kappa h \leq \beta_0$.

6.4 Regular residual histories and scale-sensitive orbit balance

A residual history $\mathcal{H}$ consists of the complete slab data, all roots selected so far, and the matching edges in a collection of fully exposed and deleted cycles; it contains no additional information about unexposed row matchings. After such a history, let $R \subset [n]$ be the remaining vertex set and $N = |R|$. Put

$$m_a^R = |H_{n,a} \cap R|, \quad N_{ab}^R = |Q_{n,a} \cap H_{n,b} \cap R|, \quad r = N/n.$$

For every deleted cycle C and every slab a ,

$$|C \cap H_{n,a}| = |C \cap Q_{n,a}|, \quad (1)$$

because σ_n maps $C \cap H_{n,a}$ bijectively onto $C \cap Q_{n,a}$. Consequently $m_a^R = |Q_{n,a} \cap R|$.

Let U be a master-partition endpoint. A residual history is called (U, δ, ξ) -regular if $N \geq \xi n$, $0 \leq \delta < 1/4$, and, for every row whose clock interval is contained in $[0, U]$ and every target category b ,

$$\left| \frac{m_a^R}{rm_{n,a}} - 1 \right| \leq \delta, \quad \left| \frac{N_{ab}^R}{rN_{n,ab}} - 1 \right| \leq \delta. \quad (2)$$

The initial history is $(U, 0, 1)$ -regular. By Lemma 6.4, conditional on any residual history, the unexposed row matchings are independent permanental matchings on their remaining submatrices.

An *admissible exploration* starts from a residual history and reveals one outgoing matching edge at a time. Before an edge is revealed, its source must be measurable with respect to the filtration generated by the slab data, the residual history, and all previously exposed edges. We also permit a fixed number of distinguished targets to be forbidden, with the next edge in a row containing such a target sampled conditionally on avoiding it. Lemma 6.6 shows that every actual cycle exploration and every root-avoiding exploration used below is admissible.

Consider an injective orbit segment $v_0, v_1, \dots, v_\ell$, where $v_{j+1} = \sigma_n(v_j)$ for the actual exploration, and allow $v_\ell = v_0$ only at a terminal closing step. For an auxiliary exploration, v_{j+1} denotes the target selected under its conditional avoid-target law. Put

$$R_a(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a}\},$$

and

$$R_{ab}(\ell) = \#\{0 \leq j < \ell : v_j \in H_{n,a}, v_{j+1} \in H_{n,b}\}.$$

Whenever no ambiguity is possible we suppress the argument ℓ . The deterministic flow identity is

$$R_b - \sum_a R_{ab} = \mathbf{1}_{\{v_0 \in H_{n,b}\}} - \mathbf{1}_{\{v_\ell \in H_{n,b}\}}. \quad (3)$$

Lemma 6.8 (Scale-sensitive two-scale orbit balance). *Fix master-partition endpoints $0 < S < U \leq T$, $\xi > 0$, and $0 \leq \delta < 1/4$. There are deterministic sequences $\chi_n \downarrow 0$ and $\pi_n \downarrow 0$, depending on the fixed partition, U , and ξ but not on the residual history, with the following property. On $\mathcal{G}_n$, uniformly over every (U, δ, ξ) -regular history and every admissible exploration, with conditional probability at least $1 - \pi_n$ the following assertions hold simultaneously at every defined prefix length $\ell \geq 1$.*

For every row a whose clock interval is contained in $[0, U]$ and every target category b ,

$$\left| R_{ab}(\ell) - \frac{R_a(\ell)}{m_a^R} N_{ab}^R \right| \leq (C\beta + \chi_n) N_{ab}^R, \quad (4)$$

where C is absolute and $\beta = 2\kappa h$. Define

$$\Delta_n(U, \delta, h) = \delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}.$$

Then, for every inner state b with $J_b \subset [0, A(S)]$,

$$\left| \frac{R_b(\ell)}{\ell p_{n,b}} - 1 \right| \leq C_{\kappa,S} \frac{n}{\ell} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{\ell}. \quad (5)$$

Here $C_{\kappa,S}$ is independent of the mesh and of ℓ ; the finite constant $C_{\mathcal{J}}$ may depend on the fixed master partition. In particular, for every fixed $\lambda > 0$, all prefixes with $\ell \geq \lambda n$ satisfy

$$\max_{b: J_b \subset [0, A(S)]} \left| \frac{R_b(\ell)}{\ell p_{n,b}} - 1 \right| \leq C_{\kappa,S,\lambda} \Delta_n(U, \delta, h) + o(1).$$

Proof. On $\mathcal{G}_n$, every $m_{n,a}$ and every $N_{n,ab}$ is bounded below by a positive constant times n . If the history is regular, the same is true for m_a^R and N_{ab}^R , uniformly over the stated histories. Consequently, for each row below U , the residual target categories have proportions bounded below by a positive constant depending only on the fixed partition. Apply Lemmas 6.6 and 6.7 to the adaptive sequence of queried sources in that row, extending it predictably after the

orbit segment ends until all ordinary targets in that row have been selected. In a row with a forbidden target, this means a complete order of the allowable targets. A union bound over the finitely many rows and categories gives deterministic sequences $\chi_n, \pi_n \downarrow 0$ for which (4) holds simultaneously for every prefix. If finitely many distinguished targets are forbidden, remove them from the corresponding category counts; the resulting $O(1)$ changes are absorbed into χ_n .

Write $p = p_n$, $P = P_n$, $q_a = R_a/\ell$, and

$$E_{ab} = R_{ab} - R_a P_{ab}.$$

For a row below U , regularity gives

$$\frac{N_{ab}^R}{m_a^R} = P_{ab}\{1 + O(\delta)\},$$

uniformly, and $R_a \leq m_a^R$. Hence

$$|E_{ab}| \leq C(\delta + \beta + \chi_n)N_{n,ab} \quad (a \notin D(U)), \quad (6)$$

where $D(U)$ is the union of rows whose clock intervals lie in (U, ∞) . For an outer row we use only injectivity of the explored path: $R_{ab} \leq N_{ab}^R \leq N_{n,ab}$ and $R_a P_{ab} \leq m_a^R P_{ab} \leq N_{n,ab}$. Thus

$$|E_{ab}| \leq 2N_{n,ab} \quad (a \in D(U)). \quad (7)$$

Let

$$G_b = \sum_{a \notin D(U)} E_{ab}, \quad H_b = \sum_{a \in D(U)} E_{ab}.$$

Since $\sum_a N_{n,ab} = m_{n,b} = np_b$, equation (6) gives

$$\left\| \frac{G}{\ell p} \right\|_{L^2(p)} \leq C \frac{n}{\ell} (\delta + \beta + \chi_n). \quad (8)$$

For the outer rows, Cauchy's inequality and the Hilbert–Schmidt block estimate on $\mathcal{G}_n$ give

$$\begin{aligned} \sum_b \frac{H_b^2}{m_{n,b}} &\leq 4 \sum_b \frac{(\sum_{a \in D(U)} N_{n,ab})^2}{m_{n,b}} \\ &\leq 4m_{n,D(U)} \sum_{a \in D(U)} \sum_b \frac{m_{n,a} P_{n,ab}^2}{m_{n,b}} \leq C m_{n,D(U)}. \end{aligned} \quad (9)$$

Therefore, using the tail-mass estimate on $\mathcal{G}_n$,

$$\left\| \frac{H}{\ell p} \right\|_{L^2(p)} \leq C \frac{n}{\ell} \left(e^{-U/2} + \varepsilon_n \right). \quad (10)$$

Let d_b denote the right-hand side of (3). It has at most two nonzero coordinates, each of absolute value at most one. Dividing (3) by ℓ gives

$$q - qP = e, \quad e_b = \frac{d_b + G_b + H_b}{\ell}.$$

Since the partition is fixed,

$$\left\| \frac{e}{p} \right\|_{L^2(p)} \leq C \frac{n}{\ell} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{\ell}. \quad (11)$$

Put $u_a = q_a/p_a - 1$. Because $pP = p$ and $\sum_a p_a u_a = 0$,

$$u - P^*u = e/p.$$

The spectral-gap event in $\mathcal{G}_n$ yields

$$\|u\|_{L^2(p)} \leq (1 - q_0)^{-1} \left\| \frac{e}{p} \right\|_{L^2(p)}. \quad (12)$$

It remains to recover a coordinatewise estimate for an inner state b . The inner-column bound and stationarity imply

$$\sum_a p_a \left(\frac{P_{ab}}{p_b} \right)^2 \leq \left(\max_a \frac{P_{ab}}{p_b} \right) \sum_a \frac{p_a P_{ab}}{p_b} \leq C_{\kappa, S}.$$

Hence

$$|(P^*u)_b| \leq C_{\kappa, S} \|u\|_{L^2(p)}.$$

For the coordinate $(e/p)_b$, equations (6) and (7), together with $P_{ab} \leq C_{\kappa, S} p_b$ for an inner column, give

$$\left| \frac{e_b}{p_b} \right| \leq C_{\kappa, S} \frac{n}{\ell} (\delta + \beta + \chi_n + e^{-U} + \varepsilon_n^2) + \frac{C_{\mathcal{J}}}{\ell}.$$

Combining this estimate with $u_b = (e/p)_b + (P^*u)_b$, (11), and (12) proves (5). Every probabilistic input was the conditional relative-prefix event, so the failure probability π_n is uniform over the entire stated class of histories and explorations. $\square$

6.5 The cycle of a uniform residual root

For a vertex v , let $c(v)$ be the unique row index for which $v \in Q_{n, c(v)}$. An orbit can hit v only when its current source lies in $H_{n, c(v)}$.

Lemma 6.9 (Root-avoiding change of measure). *Fix a residual history and a distinguished remaining target v . Explore the path from v . At the j th opportunity to hit v , let λ_j be the conditional probability, under the original residual matching law and given that v has not yet been selected, that the next target is v . Let $\mathbb{P}^\circ$ be the auxiliary law obtained by conditioning each such choice notto select v . If $S(k)$ is the number of closure opportunities among the first k completed edges, then*

$$\mathbb{P}(|\mathcal{C}(v)| > k \mid \mathcal{H}, v) = \mathbb{E}^\circ \prod_{j=0}^{S(k)-1} (1 - \lambda_j), \quad (13)$$

where the product is interpreted as zero if the auxiliary exploration is forced to stop before completing k edges. If the closure row initially has m available targets, including v , then at its j th closure opportunity,

$$\frac{e^{-\beta}}{m-j} \leq \lambda_j \leq \frac{e^{\beta}}{m-j}, \quad \beta = 2\kappa h. \quad (14)$$

Proof. At a closure opportunity, the original transition law is the mixture of an atom of mass λ_j at v and the conditional avoid- v law of mass $1 - \lambda_j$. A non-root previously visited vertex cannot be selected, because its target column has already been used; thus the only possible first repeat is the root. Multiplying the sequential Radon–Nikodym factors gives (13). Before the j th opportunity, exactly j non- v targets have been removed from the closure row, so $m - j$ targets remain. Equation (14) follows from hereditary flatness. If only v remains, the next closure probability is one, and the survival product is zero, which agrees with the stated convention. $\square$

Lemma 6.10 (Harmonic-product approximation). *Fix $\zeta \in (0, 1)$. Uniformly for $c \in [1/2, 2]$, integers $m \geq 1$, and $0 \leq s \leq (1 - \zeta)m$ for which all factors are nonnegative,*

$$\prod_{j=0}^{s-1} \left(1 - \frac{c}{m-j}\right) = (1 - s/m)^c \{1 + O_\zeta(m^{-1})\}.$$

Moreover, uniformly for $x \in [\zeta, 1 - \zeta]$ and $|a| \leq 1/2$,

$$|(1-x)^{1+a} - (1-x)| \leq C_\zeta |a|.$$

Proof. Since $m - j \geq \zeta m$,

$$\sum_{j < s} \log \left(1 - \frac{c}{m-j}\right) = -c \sum_{j < s} \frac{1}{m-j} + O_\zeta(m^{-1}).$$

The harmonic sum equals $\log(m/(m-s)) + O_\zeta(m^{-1})$, which proves the first assertion after exponentiation. The second follows from the mean-value theorem in the exponent and the boundedness of $|(1-x)^b \log(1-x)|$ for $x \in [\zeta, 1 - \zeta]$ and $b \in [1/2, 3/2]$. $\square$

Proposition 6.11 (Uniform conditional residual fraction). *Fix master-partition endpoints $0 < S < U \leq T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \leq \delta < 1/4$. On $\mathcal{G}_n$, let $\mathcal{H}$ be a (U, δ, ξ) -regular history. Choose a root uniformly from its N remaining vertices, independently of the unexposed matchings, and let L be the root-cycle length. With χ_n, π_n as in Lemma 6.8, uniformly over all such histories,*

$$\begin{aligned} & \sup_{x \in [\zeta, 1-\zeta]} \left| \mathbb{P} \left(\frac{L}{N} \leq x \mid \mathcal{H} \right) - x \right| \\ & \leq C_{\kappa, S, \xi, \zeta} (\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}) + C_\xi (e^{-S} + \varepsilon_n^2) + \pi_n + \frac{C_{\mathcal{J}, \xi, \zeta}}{n}. \end{aligned} \quad (15)$$

The constants multiplying the principal error are independent of the mesh; $C_{\mathcal{J}, \xi, \zeta}$ may depend on the fixed master partition.

Proof. Let v be the uniform root and put $C = c(v)$. Complete-cycle deletion gives $|Q_{n,a} \cap R| = m_a^R$, so, conditional on the history,

$$\mathbb{P}(C = a \mid \mathcal{H}) = m_a^R / N.$$

Consequently, using $N \geq \xi n$ and the tail bound on $\mathcal{G}_n$,

$$\mathbb{P}(C \in D(S) \mid \mathcal{H}) \leq C_\xi (e^{-S} + \varepsilon_n^2). \quad (16)$$

We henceforth work on the event $C \notin D(S)$. Put $m = m_C^R$ and $p_C = p_{n,C}$. Regularity gives

$$\frac{m}{Np_C} = 1 + O(\delta). \quad (17)$$

For the fixed master partition, p_C is bounded below over the finitely many rows below S ; thus $m \geq c_{\mathcal{J},S} \xi n$ for all large n .

We first rule out premature termination of the root-avoiding exploration. Let r be the number of completed edges if that exploration stops because only v remains in the closure row. Then exactly $m - 1$ closure-row targets have been used, so

$$R_C(r) = m - 1, \quad r \geq m - 1.$$

On the simultaneous prefix event of Lemma 6.8, its scale-sensitive estimate at this terminal prefix gives

$$\left| \frac{m - 1}{rp_C} - 1 \right| \leq C_{\kappa,S} \frac{n}{r} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{r}.$$

Writing $\theta = r/N$ and multiplying by θ yields

$$\left| \frac{m - 1}{Np_C} - \theta \right| \leq C_{\kappa,S,\xi} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}}}{N}.$$

Together with (17),

$$|1 - r/N| \leq C_{\kappa,S,\xi} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},S,\xi}}{n}. \quad (18)$$

If the right-hand side is smaller than $\zeta/2$, the auxiliary exploration cannot stop before $(1 - \zeta)N$ completed edges. If it is not smaller than $\zeta/2$, the right-hand side of (15), after increasing its constant, is at least one and the proposition is trivial. Thus we may assume that the auxiliary exploration is defined through every $k \leq (1 - \zeta)N$.

Fix $x \in [\zeta, 1 - \zeta]$ and put $k = \lfloor xN \rfloor$. The number $S(k)$ of closure opportunities is exactly $R_C(k)$ under the avoid- v law. Applying (5) at $\ell = k$, and using $k \geq \zeta \xi n$, gives, on the simultaneous prefix event,

$$\left| \frac{S(k)}{kp_C} - 1 \right| \leq C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}.$$

Combining this with (17) and $k/N = x + O(n^{-1})$ gives

$$\sup_{x \in [\zeta, 1 - \zeta]} \left| \frac{S(\lfloor xN \rfloor)}{m} - x \right| \leq C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}. \quad (19)$$

The exceptional conditional probability is at most π_n .

On the event in (19), the ratio $S(k)/m$ lies in $[\zeta/2, 1 - \zeta/2]$ once the displayed error is small; in the complementary parameter regime the final estimate is again trivial. Equations (13) and (14) squeeze the survival product between

$$\prod_{j=0}^{S(k)-1} \left(1 - \frac{e^\beta}{m-j} \right) \quad \text{and} \quad \prod_{j=0}^{S(k)-1} \left(1 - \frac{e^{-\beta}}{m-j} \right).$$

Lemma 6.10, followed by (19), shows that both products differ from $1 - x$ by at most

$$C_{\kappa,S,\xi,\zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J},\xi,\zeta}}{n}.$$

On the exceptional prefix event we use the trivial bound by one. Averaging under $\mathbb{P}^\circ$, then adding the closure-tail probability (16), gives

$$\sup_{x \in [\zeta, 1-\zeta]} |\mathbb{P}(L > \lfloor xN \rfloor \mid \mathcal{H}) - (1-x)|$$

bounded by the right-hand side of (15). Since $\{L/N \leq x\} = \{L \leq \lfloor xN \rfloor\}$, this is the claimed CDF estimate. $\square$

6.6 Deletion stability and regeneration

Lemma 6.12 (Deletion of an interior macroscopic cycle). *Fix master-partition endpoints $0 < V < U \leq T$, $\xi > 0$, $\zeta \in (0, 1/2)$, and $0 \leq \delta < 1/4$. On $\mathcal{G}_n$, start from a (U, δ, ξ) -regular history and expose the cycle of a uniform residual root. Let N and L be the residual size and the exposed cycle length. With χ_n, π_n as in Lemma 6.8, there is a constant $C = C(\kappa, V, \xi, \zeta)$ such that*

$$\mathbb{P}\left(\zeta N \leq L \leq (1-\zeta)N, \text{ the post-deletion history is not } (V, \delta', \zeta\xi)\text{-regular} \mid \mathcal{H}\right) \leq \pi_n, \quad (20)$$

where

$$\delta' = C(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2}) + \frac{C_{\mathcal{J}, \xi, \zeta}}{n}. \quad (21)$$

The conclusion is asserted whenever the displayed δ' is smaller than $1/4$, which is the only regime used below. This is a joint-event estimate; it does not condition on the exact random value of L .

Proof. Let d_a be the number of vertices of the exposed cycle in $H_{n,a}$, and let d_{ab} be the number of its directed edges from $H_{n,a}$ to $H_{n,b}$. The conditional prefix event in Lemma 6.8 is simultaneous over all prefix lengths. It may therefore be evaluated at the random closing time without conditioning on L .

On the event $\zeta N \leq L \leq (1-\zeta)N$, we have $L \geq \zeta\xi n$. Apply the scale-sensitive estimate (5) with inner cutoff V and $\ell = L$. For every row below V ,

$$\left| \frac{d_a}{Lp_{n,a}} - 1 \right| \leq C_{\kappa, V, \xi, \zeta} \Delta_n(U, \delta, h) + \frac{C_{\mathcal{J}, \xi, \zeta}}{n}.$$

Regularity gives $m_a^R = Np_{n,a}(1 + O(\delta))$, and hence

$$\frac{d_a}{m_a^R} = \frac{L}{N} + O\left(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2} + \frac{C_{\mathcal{J}, \xi, \zeta}}{n}\right) \quad (22)$$

uniformly over rows below V .

The row-prefix estimate (4), evaluated at the same random closing time, gives

$$d_{ab} = \frac{d_a}{m_a^R} N_{ab}^R + O((\beta + \chi_n) N_{ab}^R).$$

Combining this with (22),

$$d_{ab} = \frac{L}{N} N_{ab}^R + O\left(\delta + \beta + \chi_n + \varepsilon_n + e^{-U/2} + \frac{C_{\mathcal{J}, \xi, \zeta}}{n}\right) N_{ab}^R. \quad (23)$$

The quantity d_{ab} is exactly the number of vertices removed from $Q_{n,a} \cap H_{n,b}$, because every cycle edge from row a to category b uses one such target.

Write $x = L/N$ and let R' be the post-deletion vertex set. Equations (22)–(23) give

$$m_a^{R'} = (1-x)m_a^R + O(\Delta_n^* m_a^R), \quad N_{ab}^{R'} = (1-x)N_{ab}^R + O(\Delta_n^* N_{ab}^R),$$

where Δ_n^* denotes the parenthesized error in (22). Since $1-x \geq \zeta$ and $|R'| = (1-x)N \geq \zeta \xi n$, division by the new total fraction preserves the regularity inequalities with error at most the quantity in (21). Residual permanental factorization and hereditary flatness remain valid after deleting the exposed edges. The only conditional failure event is the simultaneous prefix failure, whose probability is at most π_n . This proves (20). $\square$

6.7 Independent residual fractions

Order the cycles size-biased by choosing a uniform root from the current residual vertex set, deleting its cycle, and repeating. Let

$$V_{n,j} = \frac{\tilde{L}_{n,j}}{n - \tilde{L}_{n,1} - \cdots - \tilde{L}_{n,j-1}}$$

whenever the denominator is positive, and set $V_{n,j} = 0$ after exhaustion.

Lemma 6.13 (Conditional test functions from an interior CDF bound). *Let $W \in [0, 1]$ and let $\mathcal{F}$ be a sigma-field. Suppose that, almost surely,*

$$\sup_{x \in [a,b]} |\mathbb{P}(W \leq x \mid \mathcal{F}) - x| \leq \eta, \quad 0 < a < b < 1.$$

Then

$$\mathbb{P}(W \notin [a, b] \mid \mathcal{F}) \leq a + (1-b) + 2\eta.$$

If f is absolutely continuous, vanishes outside $[a, b]$, and satisfies $f(a) = f(b) = 0$, then

$$\left| \mathbb{E}(f(W) \mid \mathcal{F}) - \int_0^1 f(u) du \right| \leq \eta \int_a^b |f'(u)| du.$$

Proof. The endpoint estimate follows by evaluating the CDF bound at a and b . For the second assertion, conditional Stieltjes integration by parts and the vanishing boundary values give

$$\mathbb{E}(f(W) \mid \mathcal{F}) = - \int_a^b \mathbb{P}(W \leq u \mid \mathcal{F}) f'(u) du,$$

whereas

$$\int_a^b f(u) du = - \int_a^b u f'(u) du.$$

Subtracting proves the claim. $\square$

Proposition 6.14 (Finite-dimensional GEM(1) limits). *For every fixed k ,*

$$(V_{n,1}, \dots, V_{n,k}) \Rightarrow (U_1, \dots, U_k),$$

where $U_1, \dots, U_k$ are independent $\text{Unif}(0, 1)$ variables.

Proof. Fix k and $\zeta \in (0, 1/4)$, and put $\xi_* = \zeta^{k-1}$. We first record a noncircular parameter hierarchy. Let $\eta > 0$ be a target error.

Choose T_0 so large that the closure-tail term $C_{\xi_*} e^{-T_0}$ in Proposition 6.11 is below a prescribed small fraction of η . Suppose $T_0, \dots, T_j$ have been chosen. Let R_j and D_j be, respectively, upper bounds for the principal-error coefficients in the root law and deletion lemma with inner cutoff T_j , residual fraction at least ξ_* , and truncation parameter ζ . These constants are now fixed. An error created at this level can subsequently be amplified only by $D_{j-1}, \dots, D_0$, all of which have already been fixed. Put

$$A_j = (1 + R_j + D_j) \prod_{i=0}^{j-1} (1 + D_i),$$

with the empty product equal to one, and choose $T_{j+1} > T_j$ so large that

$$A_j e^{-T_{j+1}/2} \leq 2^{-j-3} \eta.$$

Continue until T_k has been chosen.

Use one master partition of $[0, T_k]$ containing $T_0, \dots, T_k$, append the tail interval, and let its mesh be at most h . After the cutoffs are fixed, all coefficients in the finitely many root and deletion estimates are finite. Choose h so small that the errors involving $\beta = 2\kappa h$ remain below their assigned fractions of η after the finite deletion recursion. More explicitly, if D_j denotes the coefficient in (21) with inner cutoff T_j , the deterministic recursion

$$d_0 = 0, \quad d_{r+1} = D_{k-r-1}(d_r + \beta + e^{-T_{k-r}/2})$$

has $\max_{r \leq k} d_r \rightarrow 0$ as all displayed input errors tend to zero. The preceding choices make this maximum smaller than a prescribed fraction of η , and in particular smaller than $1/8$. Finally, for this fixed master partition take the maxima of the finitely many sequences χ_n and π_n that occur at the different cutoff levels, and choose n so large that

$$\varepsilon_n + \chi_n + \pi_n + C_{\mathcal{J}}/n$$

is below its assigned fraction of η and $\mathbb{P}(\mathcal{G}_n^c) \leq \eta$. This order of choices is important: the inner cutoff is fixed before its outer cutoff, and the mesh is chosen only after all cutoff-dependent constants are known.

On the event that

$$\zeta \leq V_{n,j} \leq 1 - \zeta, \quad 1 \leq j < r, \quad (6.20)$$

the residual size before stage r is at least $\zeta^{r-1} n \geq \xi_* n$. At stage r , use outer cutoff T_{k-r+1} and inner cutoff T_{k-r} . Inductively, the current history is regular at the outer cutoff with a regularity error bounded by the deterministic recursion above, except on a conditional event of probability $O_k(\eta)$. Proposition 6.11 therefore gives, uniformly over every such current history,

$$\sup_{x \in [\zeta, 1-\zeta]} |\mathbb{P}(V_{n,r} \leq x \mid \mathcal{F}_{r-1}) - x| \leq C_k \eta. \quad (24)$$

On the event $V_{n,r} \in [\zeta, 1 - \zeta]$, Lemma 6.12 makes the next history regular at cutoff T_{k-r} , except on a conditional event of probability at most $C_k \eta$. All these estimates are uniform over the preceding regular history.

We now turn the conditional estimates into joint convergence. Let $f_1, \dots, f_k$ be bounded Lipschitz functions on $[0, 1]$. Choose a piecewise linear cutoff φ_ζ which is zero outside $[\zeta, 1 - \zeta]$ and equal to one on $[2\zeta, 1 - 2\zeta]$, and put $g_j = f_j \varphi_\zeta$. Lemma 6.13, applied to (24), gives

$$\left| \mathbb{E}(g_r(V_{n,r}) \mid \mathcal{F}_{r-1}) - \int_0^1 g_r(u) du \right| \leq C_{f_r, \zeta} \eta$$

on every regular history. Let $\mathcal{R}_r$ be the event that, after the first r deletions, the history is regular at cutoff T_{k-r} with the prescribed error; put $\mathcal{R}_0 = \mathcal{G}_n$. Because g_r vanishes outside $[\zeta, 1 - \zeta]$, the joint-event deletion lemma also gives, on $\mathcal{R}_{r-1}$,

$$\left| \mathbb{E}(g_r(V_{n,r}) \mathbf{1}_{\mathcal{R}_r} \mid \mathcal{F}_{r-1}) - \int_0^1 g_r(u) du \right| \leq C_{f_r, \zeta} \eta.$$

Multiplying by $\prod_{j < r} g_j(V_{n,j}) \mathbf{1}_{\mathcal{R}_{r-1}}$, taking expectations, and inducting on r yields

$$\left| \mathbb{E} \left(\prod_{j=1}^k g_j(V_{n,j}) \mathbf{1}_{\mathcal{R}_k} \right) - \prod_{j=1}^k \int_0^1 g_j(u) du \right| \leq C_{f,k,\zeta} \eta + o_n(1).$$

On the event that every $g_j(V_{n,j})$ is nonzero, every residual fraction is in the interior range, so a union bound over the conditional deletion failures shows that removing $\mathbf{1}_{\mathcal{R}_k}$ costs another $C_{f,k,\zeta} \eta + o_n(1)$. Consequently

$$\left| \mathbb{E} \prod_{j=1}^k g_j(V_{n,j}) - \prod_{j=1}^k \int_0^1 g_j(u) du \right| \leq C_{f,k,\zeta} \eta + o_n(1). \quad (25)$$

Here the $o_n(1)$ includes $\mathbb{P}(\mathcal{G}_n^c)$.

Evaluating (24) at 2ζ and $1 - 2\zeta$, and adding the regularity-failure probabilities, gives

$$\limsup_{n \rightarrow \infty} \mathbb{P}(V_{n,j} \notin [2\zeta, 1 - 2\zeta] \text{ for some } j \leq k) \leq 4k\zeta + C_k \eta. \quad (26)$$

Consequently, replacing g_j by f_j in (25) costs at most $C_{f,k}(\zeta + \eta)$, and the same is true for replacing $\int g_j$ by $\int f_j$. Let first $n \rightarrow \infty$, then $\eta \downarrow 0$, and finally $\zeta \downarrow 0$. We obtain

$$\mathbb{E} \prod_{j=1}^k f_j(V_{n,j}) \longrightarrow \prod_{j=1}^k \int_0^1 f_j(u) du.$$

Products of bounded Lipschitz functions form a convergence-determining class on $[0, 1]^k$. This proves convergence to k independent uniform variables. The convention $V_{n,j} = 0$ after exhaustion causes no difficulty, because exhaustion before stage k is contained in the endpoint event in (26). $\square$

Proof of Theorem 6.1. Put

$$X_{n,j} = \frac{\tilde{L}_{n,j}}{n} = V_{n,j} \prod_{i < j} (1 - V_{n,i}), \quad R_{n,k} = 1 - \sum_{j=1}^k X_{n,j} = \prod_{i=1}^k (1 - V_{n,i}).$$

Proposition 6.14 and the continuous mapping theorem give, for every fixed k ,

$$(X_{n,1}, \dots, X_{n,k}, R_{n,k}) \Rightarrow \left(U_1, (1 - U_1)U_2, \dots, U_k \prod_{i < k} (1 - U_i), \prod_{i=1}^k (1 - U_i) \right). \quad (27)$$

Because $0 \leq R_{n,k} \leq 1$,

$$\mathbb{E} R_{n,k} \longrightarrow 2^{-k}.$$

Thus, for every $\epsilon > 0$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(R_{n,k} > \epsilon) \leq \epsilon^{-1} 2^{-k}.$$

For a bounded Lipschitz functional on ℓ^1 , truncate after k coordinates, use (27), and bound the truncation error by the residual tail mass. Equation (6.7), followed by $k \rightarrow \infty$, proves

$$(X_{n,1}, X_{n,2}, \dots) \Rightarrow \text{GEM}(1)$$

in ℓ^1 .

For finite nonnegative sequences, the usual two-point uncrossing inequality shows that matching coordinates in decreasing order minimizes the sum of absolute differences. Passing to truncations gives

$$\|x^\downarrow - y^\downarrow\|_1 \leq \|x - y\|_1$$

for nonnegative ℓ^1 sequences. The continuous mapping theorem therefore gives the PD(1) limit. The first residual fraction is the cycle length of a uniform root divided by n , which proves the final assertion of the theorem.

The argument was carried out along an arbitrary subsequence after extracting a marked-profile limit. The resulting law does not depend on that profile. Every subsequence therefore has a further subsequence with the same limit, so the full sequence converges. $\square$

Corollary 6.15 (Power sums of macroscopic cycles). *Under the hypotheses of Theorem 6.1, for every $q > 1$,*

$$\sum_{j \geq 1} \left(\frac{L_{n,j}}{n} \right)^q \Rightarrow \sum_{j \geq 1} P_j^q, \quad (P_j) \sim \text{PD}(1),$$

and

$$\mathbb{E} \sum_{j \geq 1} \left(\frac{L_{n,j}}{n} \right)^q \longrightarrow \frac{1}{q}.$$

Proof. The map $x \mapsto \sum_j x_j^q$ is continuous and bounded by one on $\mathcal{S}^\downarrow$. If $\tilde{P}$ is a size-biased pick from a partition (P_j) , then, conditionally on the partition,

$$\mathbb{E}(\tilde{P}^{q-1} \mid (P_j)) = \sum_j P_j^q.$$

For a PD(1) partition, its size-biased ordering has the GEM(1) law, so $\tilde{P}$ is uniform on $(0, 1)$. Hence

$$\mathbb{E} \sum_j P_j^q = \mathbb{E} \tilde{P}^{q-1} = \int_0^1 u^{q-1} du = 1/q.$$

Bounded convergence transfers the expectation limit. $\square$

7 Sukhatme fixed points

We now prove the singular Sukhatme fixed-point mean, a local terminal Poisson point-process limit, and fixed-order factorial moment asymptotics. Throughout this section

$$\theta_i = n + 1 - i.$$

We call this the Sukhatme profile because the same decreasing rates $n, n-1, \dots, 1$ occur in the classical exponential-spacing representation associated with Sukhatme and Rényi [19, 16]. No external result from that theory is used below; the citations record only the historical source of the rate pattern and its standard order-statistic interpretation. It is convenient to use the terminal rate coordinate

$$a = n + 1 - i.$$

Thus rate coordinate a is the actual exponential rate of the corresponding clock, and it corresponds to label $i = n + 1 - a$. We write

$$T_a \sim \text{Exp}(a), \quad 1 \leq a \leq n,$$

for these independent clocks. This is a change of notation from the regular-profile section, where the rates were denoted $\theta_{n,i}$. The fixed-point indicator in rate coordinates is

$$I_{n,a} = \mathbf{1} \left\{ \sum_{b \neq a} \mathbf{1}_{\{T_b > T_a\}} = a - 1 \right\},$$

and the total number of fixed points is

$$F_n = \sum_{a=1}^n I_{n,a}.$$

Thus F_n is the fixed-point count $C_1(\sigma_n)$ specialized to the singular Sukhatme profile, written in terminal rate coordinates.

The scale in this section is terminal rather than bulk. The relevant rates satisfy $a \rightarrow \infty$, $a = o(n)$, and their clock times are of order $1/a$. A single rate a contributes asymptotically e^{-1}/a . Summing over a logarithmic range gives the mean $e^{-1} \log n$. Joint probabilities factor for rates separated on logarithmic scale, which gives a local Poisson point process on the coordinate $\log a$. The proof parallels the regular-profile argument, but with terminal rather than bulk geometry: a saddle computation and local central limit estimate again produce the local entry probability, while the integrated saddle is anisotropic and the residual domination comes from the terminal death process rather than the bulk endpoint and collision envelopes. The proof has three ingredients: finite-tail saddle geometry, an anisotropic terminal local central limit estimate, and a death-process product bound for non-separated tuples.

Lemma 7.1 (Death-process product bound). *For every finite set $A \subset [n]$,*

$$\mathbb{P}(I_{n,a} = 1 \text{ for all } a \in A) \leq \prod_{a \in A} \frac{2}{a+1} \leq 2^{|A|} \prod_{a \in A} \frac{1}{a}.$$

Proof. Expose the exponential race as a pure death process. When the alive set is S , the next deleted rate is $b \in S$ with conditional probability

$$\frac{b}{\sum_{c \in S} c}.$$

The event $I_{n,a} = 1$ says that rate a is deleted at the unique transition at which the number of alive particles drops from a to $a - 1$. Equivalently, immediately before T_a there are a alive rates, including a , and immediately after T_a there are $a - 1$. Thus $I_{n,a} = 1$ is exactly the event that rate a is deleted at this death transition. Condition on the entire history before this transition. If rate a is already dead, the conditional probability is zero. If it is alive, the alive set has size a and contains a . The smallest possible total rate of such a set is

$$1 + 2 + \cdots + a = \frac{a(a+1)}{2},$$

so the conditional probability that rate a is next deleted is at most

$$\frac{a}{a(a+1)/2} = \frac{2}{a+1}.$$

Order the elements of A decreasingly and condition successively on the histories before the corresponding death levels. Multiplying the conditional bounds gives the result. $\square$

7.1 The one-point terminal saddle

We first record the elementary finite-tail estimates used throughout the terminal analysis. They are stated in relative form because this is the form needed below: when $x = o(n)$, the omitted infinite tail is negligible relative to x , but it need not be $O(1)$.

Lemma 7.2 (Finite-tail geometric estimates). *Let $x = x_n \rightarrow \infty$ and $x = o(n)$. Uniformly for s in compact subsets of $(0, \infty)$,*

$$\sum_{b=1}^n e^{-bs/x} = \frac{x}{s}(1 + o(1)),$$

$$\sum_{b=1}^n \frac{b}{x} e^{-bs/x} = \frac{x}{s^2}(1 + o(1)), \quad \sum_{b=1}^n e^{-2bs/x} = \frac{x}{2s}(1 + o(1)).$$

Also

$$\sum_{b=1}^n b^2 e^{-bs/x} = O(x^3),$$

uniformly for the same range of s . The same estimates remain true if a bounded number of summands is omitted.

Proof. For example,

$$\sum_{b=1}^n e^{-bs/x} = \frac{e^{-s/x}(1 - e^{-sn/x})}{1 - e^{-s/x}}.$$

Since s is bounded away from zero and infinity, $e^{-sn/x} = o(1)$ uniformly and $(1 - e^{-s/x})^{-1} = x/s + O(1)$. This gives the first estimate in relative form.

For the weighted sum, put $q = e^{-s/x}$. Then

$$\sum_{b=1}^n bq^b = \frac{q(1 - (n+1)q^n + nq^{n+1})}{(1-q)^2}.$$

Since $n/x \rightarrow \infty$, the truncation terms are exponentially small on the n/x scale, while

$$\frac{q}{(1-q)^2} = \frac{x^2}{s^2}(1 + o(1))$$

uniformly on compact s -intervals. Dividing by x gives the second estimate. The third estimate is the first estimate with s replaced by $2s$. For the final bound, compare with the infinite tail:

$$\sum_{b=1}^n b^2 e^{-bs/x} \leq \sum_{b=1}^{\infty} b^2 e^{-bs/x} = O(x^3),$$

uniformly on compact s -intervals bounded away from zero. Omitting a bounded number of summands changes the first, second, and third displayed sums by $O(1)$, $O(1)$, and $O(1)$, respectively, and changes the final sum by at most $O(x^2)$. Indeed,

$$\sup_{b \geq 1} \frac{b}{x} e^{-bs/x} = O(1), \quad \sup_{b \geq 1} b^2 e^{-bs/x} = O(x^2),$$

uniformly for s in compact subsets of $(0, \infty)$. These errors are negligible relative to the corresponding main scales, or harmless for the displayed $O(x^3)$ bound. $\square$

Lemma 7.3 (One-point terminal saddle geometry). *Assume $a \rightarrow \infty$ and $a = o(n)$. Define*

$$\mu_{n,a}(s) = \sum_{b \neq a} e^{-bs/a}, \quad v_{n,a}(s) = \sum_{b \neq a} e^{-bs/a} (1 - e^{-bs/a}).$$

Let $s_{n,a}^*$ be the unique solution of

$$\mu_{n,a}(s_{n,a}^*) = a - 1.$$

Then, uniformly in this range,

$$s_{n,a}^* \rightarrow 1, \quad -\mu'_{n,a}(s_{n,a}^*) \sim a, \quad v_{n,a}(s_{n,a}^*) \sim \frac{a}{2}.$$

Moreover, for all sufficiently large n , uniformly for $s \in [1/2, 2]$,

$$|\mu_{n,a}(s) - (a - 1)| \geq ca|s - s_{n,a}^*|$$

whenever $|s - s_{n,a}^*| \leq 1/4$, with an absolute constant $c > 0$.

Proof. The function $\mu_{n,a}$ is continuous and strictly decreasing, tends to $n - 1$ as $s \downarrow 0$, and tends to zero as $s \rightarrow \infty$. Hence the saddle exists and is unique. By Lemma 7.2, uniformly for s in compact subsets of $(0, \infty)$,

$$\mu_{n,a}(s) = \frac{a}{s}(1 + o(1)), \quad -\mu'_{n,a}(s) = \sum_{b \neq a} \frac{b}{a} e^{-bs/a} = \frac{a}{s^2}(1 + o(1)),$$

and

$$v_{n,a}(s) = \sum_{b \neq a} e^{-bs/a} - \sum_{b \neq a} e^{-2bs/a} = \frac{a}{2s}(1 + o(1)).$$

These estimates also locate the saddle. Fix $\varepsilon \in (0, 1)$. Uniformly,

$$\mu_{n,a}(1 - \varepsilon) = \frac{a}{1 - \varepsilon}(1 + o(1)) > a - 1,$$

whereas

$$\mu_{n,a}(1 + \varepsilon) = \frac{a}{1 + \varepsilon}(1 + o(1)) < a - 1$$

for all sufficiently large n . Since $\mu_{n,a}$ is strictly decreasing, this implies $1 - \varepsilon < s_{n,a}^* < 1 + \varepsilon$, and hence $s_{n,a}^* \rightarrow 1$. The displayed derivative and variance asymptotics follow by substitution.

Finally, on $[1/2, 2]$ the derivative satisfies $-\mu'_{n,a}(s) \geq ca$ for all large n . The mean-value theorem gives the stated linear separation from the saddle. $\square$

Lemma 7.4 (Sukhatme one-point estimate). *Let $p_{n,a} = \mathbb{P}(I_{n,a} = 1)$. Uniformly for $a \rightarrow \infty$ and $a = o(n)$,*

$$p_{n,a} \sim \frac{1}{ea}.$$

Moreover, for all $1 \leq a \leq n$,

$$p_{n,a} \leq \frac{2}{a + 1}.$$

Proof. The upper bound is Lemma 7.1 with $A = \{a\}$. It remains to prove the asymptotic.

Conditioning on $T_a = t$ and then putting $t = s/a$ gives

$$p_{n,a} = \int_0^\infty e^{-s} \mathbb{P}(Y_{n,a}(s) = a - 1) ds,$$

where

$$Y_{n,a}(s) = \sum_{b \neq a} \xi_b(s), \quad \xi_b(s) \sim \text{Bernoulli}(e^{-bs/a})$$

are independent. Let $s_{n,a}^*$ be the finite saddle from Lemma 7.3.

In the saddle window write

$$s = s_{n,a}^* + \frac{u}{\sqrt{a}}, \quad |u| \leq a^{1/10}.$$

For $s \in [1/2, 2]$, a positive proportion of the indices $b \asymp a$ have Bernoulli parameters $e^{-bs/a}$ in a fixed compact subinterval of $(0, 1)$. Under every bounded scalar exponential tilt these parameters remain uniformly nondegenerate. The total tilted variance is comparable to a , and the product characteristic function has exponential decay away from $2\pi\mathbb{Z}$. Thus all tilted hypotheses of Lemma 2.5 hold uniformly with scale $s_n = a$. We use its local estimate in two forms. First, on each fixed u -ball, Lemma 2.5, together with Lemma 7.3, gives

$$\mathbb{P}(Y_{n,a}(s) = a - 1) = \frac{1}{\sqrt{\pi a}} e^{-u^2} (1 + o(1)).$$

The exponent is obtained from

$$a - 1 - \mu_{n,a}(s) = -\mu'_{n,a}(s_{n,a}^*)(s - s_{n,a}^*) + o(\sqrt{a}) = u\sqrt{a}(1 + o(1)),$$

and $v_{n,a}(s_{n,a}^*) \sim a/2$. Second, the Gaussian upper bound in the same one-dimensional form of Lemma 2.5 gives

$$\mathbb{P}(Y_{n,a}(s) = a - 1) \leq C a^{-1/2} e^{-cu^2}, \quad |u| \leq a^{1/10}.$$

This bound justifies extending the fixed u -ball to the whole growing saddle window.

The contribution away from the saddle is negligible. If $s \notin [1/2, 2]$, the mean of $Y_{n,a}(s)$ is separated from $a - 1$ by a positive multiple of a , so Bernstein's inequality gives an exponentially small contribution. If $s \in [1/2, 2]$ but $|s - s_{n,a}^*| > a^{-2/5}$, Lemma 7.3 gives

$$|\mu_{n,a}(s) - (a - 1)| \geq ca|s - s_{n,a}^*|.$$

Since $v_{n,a}(s) = O(a)$ uniformly on $[1/2, 2]$, Bernstein's inequality gives

$$\mathbb{P}(Y_{n,a}(s) = a - 1) \leq C \exp[-ca(s - s_{n,a}^*)^2].$$

The integral of this bound over $|s - s_{n,a}^*| > a^{-2/5}$ is $o(a^{-1})$. Also $e^{-s} = e^{-s_{n,a}^*}(1 + o(1))$ uniformly for $|u| \leq a^{1/10}$. Therefore

$$\begin{aligned} p_{n,a} &\sim \int_{-\infty}^{\infty} e^{-s_{n,a}^*} \frac{1}{\sqrt{\pi a}} e^{-u^2} \frac{du}{\sqrt{a}} \\ &\sim \frac{e^{-1}}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} du = \frac{1}{ea}. \end{aligned}$$

□

Theorem 7.5 (Sukhatme fixed-point mean). *For Sukhatme weights $\theta_i = n + 1 - i$,*

$$\mathbb{E}F_n \sim e^{-1} \log n.$$

Proof. Choose integer cutoffs

$$L_n = \lceil \log n \rceil, \quad U_n = \lfloor n / \log n \rfloor.$$

On this range $L_n \rightarrow \infty$ and $U_n = o(n)$. Hence the uniform estimate in Lemma 7.4 gives

$$\sup_{L_n \leq a \leq U_n} |ea p_{n,a} - 1| \rightarrow 0.$$

Therefore

$$\sum_{L_n \leq a \leq U_n} p_{n,a} = e^{-1}(1 + o(1)) \sum_{L_n \leq a \leq U_n} \frac{1}{a} \sim e^{-1} \log n.$$

The lower range contributes

$$\sum_{a < L_n} p_{n,a} \leq 2 \sum_{a < L_n} \frac{1}{a + 1} = O(\log \log n) = o(\log n),$$

and the upper range contributes

$$\sum_{a > U_n} p_{n,a} \leq 2 \sum_{a > U_n} \frac{1}{a + 1} = O(\log \log n) = o(\log n).$$

Thus $\mathbb{E}F_n = \sum_a p_{n,a} \sim e^{-1} \log n$.

□

7.2 Separated terminal joint estimates

The joint estimate is anisotropic: the covariance scale in the j -th terminal coordinate is a_j , not a common scalar scale. We therefore use a terminal local CLT adapted to logarithmically separated rates. The one-point saddle above differentiated with respect to the scaled variable $s = at$; in the joint saddle below we differentiate with respect to the clock time t , which accounts for the change from derivative scale a to derivative scale a^2 .

Lemma 7.6 (Joint terminal saddle geometry). *Fix $r \geq 1$ and $\eta > 0$. Uniformly over integers*

$$1 \ll a_1 < \cdots < a_r \ll n, \quad a_{j+1}/a_j \geq 1 + \eta,$$

where the ratio-separation condition is empty when $r = 1$. Let $A = \{a_1, \dots, a_r\}$ and define

$$\mu_j(t) = \sum_{b \notin A} e^{-bt}, \quad 1 \leq j \leq r.$$

There is a unique $t_j^ > 0$ satisfying*

$$\mu_j(t_j^*) = a_j - j.$$

Moreover

$$a_j t_j^* \rightarrow 1, \quad -\mu_j'(t_j^*) \sim a_j^2, \quad \mu_j''(t_j^*) = O(a_j^3),$$

uniformly in j . Finally, for all large n ,

$$t_1^* > t_2^* > \cdots > t_r^*,$$

and the ratios t_j^/t_{j+1}^* are bounded below by a constant greater than one depending only on η .*

Proof. Fix j . For $t = s/a_j$, Lemma 7.2 gives, uniformly for s in compact subsets of $(0, \infty)$,

$$\mu_j(s/a_j) = \frac{a_j}{s}(1 + o(1)), \quad -\mu_j'(s/a_j) = \frac{a_j^2}{s^2}(1 + o(1)).$$

The omitted set A has fixed size r . It changes the first sum by $O_r(1)$. For the derivative sum, each omitted term satisfies

$$be^{-bs/a_j} \leq Ca_j$$

uniformly for s in compact subsets of $(0, \infty)$, and hence the omitted terms change the derivative sum by $O_r(a_j) = o(a_j^2)$. Thus both displayed estimates remain valid after removing A .

The equation $\mu_j(t) = a_j - j$ has a unique solution, since μ_j is continuous and strictly decreasing from $n - r$ to 0, while $0 < a_j - j < n - r$ for all large n . To locate it, fix $\varepsilon \in (0, 1)$. The first displayed estimate gives

$$\mu_j((1 - \varepsilon)/a_j) > a_j - j, \quad \mu_j((1 + \varepsilon)/a_j) < a_j - j$$

for all sufficiently large n , uniformly in j . Therefore

$$1 - \varepsilon < a_j t_j^* < 1 + \varepsilon,$$

and $a_j t_j^* \rightarrow 1$. Substituting $t = t_j^*$ into the preceding derivative estimate, equivalently taking $s = a_j t_j^* \rightarrow 1$, gives $-\mu_j'(t_j^*) \sim a_j^2$ uniformly in j . The second-derivative bound follows from

$$\mu_j''(t) = \sum_{b \notin A} b^2 e^{-bt} = O(a_j^3) \quad (t \asymp a_j^{-1}),$$

again by the same finite-geometric estimate. Since $t_j^* \sim 1/a_j$ and $a_{j+1}/a_j \geq 1 + \eta$, the saddle times are eventually strictly ordered and uniformly separated multiplicatively. $\square$

Terminal interval notation. For the remainder of this subsection, fix an admissible tuple

$$1 \ll a_1 < \cdots < a_r \ll n, \quad a_{j+1}/a_j \geq 1 + \eta,$$

put $A = \{a_1, \dots, a_r\}$, and consider ordered times $t_1 > \cdots > t_r > 0$. Define

$$Z_j(t) = \sum_{b \notin A} \mathbf{1}_{\{T_b > t_j\}}, \quad z_j = a_j - j, \quad \mu_j(t_j) = \mathbb{E}Z_j(t).$$

Let

$$I_1 = (t_1, \infty), \quad I_j = (t_j, t_{j-1}] \quad (2 \leq j \leq r), \quad I_{r+1} = (0, t_r],$$

and write

$$Y_1(t) = Z_1(t), \quad Y_j(t) = Z_j(t) - Z_{j-1}(t) \quad (2 \leq j \leq r).$$

The corresponding interval-count target is

$$y_1 = a_1 - 1, \quad y_j = a_j - a_{j-1} - 1 \quad (2 \leq j \leq r).$$

Finally, put

$$\begin{aligned} m_j(t) &= \mathbb{E}Y_j(t), \\ H_j(t) &= z_j - \mu_j(t_j), \quad H_0(t) = 0, \\ d_j(t) &= y_j - m_j(t) = H_j(t) - H_{j-1}(t), \\ s_j &= a_j t_j, \quad s_j^* = a_j t_j^*. \end{aligned}$$

The transformation between Z and Y is triangular with determinant one, so $\mathbb{P}(Z(t) = z) = \mathbb{P}(Y(t) = y)$.

Lemma 7.7 (Coordinatewise Fourier concentration). *Let*

$$W = \sum_{\nu \in \mathcal{I}} W_\nu$$

be a sum of independent $\mathbb{Z}^r$ -valued random vectors, each supported in a fixed cube $[-M_0, M_0]^r$. Let $A_1, \dots, A_r \geq 1$. Suppose that, for every coordinate j , there is a set $\mathcal{I}_j \subset \mathcal{I}$ with $|\mathcal{I}_j| \geq c_0 A_j$, and that the sets $\mathcal{I}_1, \dots, \mathcal{I}_r$ are disjoint. Suppose also that, for every $\nu \in \mathcal{I}_j$, the law of W_ν gives mass at least c_0 to two atoms whose difference is e_j , the j -th coordinate vector. Then there are constants $c, C > 0$, depending only on r, M_0, c_0 , such that the characteristic function $\phi_W(u) = \mathbb{E}e^{iu \cdot W}$ satisfies

$$|\phi_W(u)| \leq \exp \left[-c \sum_{j=1}^r A_j \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2 \right], \quad u \in [-\pi, \pi]^r.$$

Consequently,

$$\sup_z \mathbb{P}(W = z) \leq C \prod_{j=1}^r A_j^{-1/2}.$$

The same conclusions hold uniformly after any exponential tilt

$$\mathbb{P}_\lambda(W_\nu = x) = \frac{e^{\lambda \cdot x} \mathbb{P}(W_\nu = x)}{\mathbb{E}e^{\lambda \cdot W_\nu}}, \quad \|\lambda\|_\infty \leq \Lambda,$$

with constants depending additionally on Λ .

Proof. For a single summand Y , write $p_x = \mathbb{P}(Y = x)$. If two atoms x and $x + e_j$ both have mass at least c_0 , then

$$\begin{aligned} 1 - |\mathbb{E} e^{iu \cdot Y}|^2 &= \frac{1}{2} \sum_{y,z} p_y p_z |e^{iu \cdot y} - e^{iu \cdot z}|^2 \\ &\geq c_0^2 |1 - e^{iu_j}|^2 \geq c \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2. \end{aligned}$$

After decreasing c , this gives

$$|\mathbb{E} e^{iu \cdot Y}| \leq \exp\{-c \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2\}.$$

Multiplying this estimate over the disjoint blocks $\mathcal{I}_j$ gives the displayed characteristic-function bound. Fourier inversion then gives

$$\begin{aligned} \mathbb{P}(W = z) &\leq (2\pi)^{-r} \int_{[-\pi, \pi]^r} \exp\left[-c \sum_{j=1}^r A_j \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2\right] du \\ &\leq C \prod_{j=1}^r A_j^{-1/2}. \end{aligned}$$

Under a tilt with $\|\lambda\|_\infty \leq \Lambda$, all probabilities of atoms in the fixed support are multiplied by factors bounded between positive constants depending only on M_0, r, Λ . Hence the two distinguished atom probabilities remain bounded below, with a possibly smaller constant, and the same proof applies. $\square$

Lemma 7.8 (Weighted chain comparison). *Fix $r \geq 1$ and $\eta > 0$. Let $0 < A_1 < \dots < A_r$ satisfy $A_{j+1}/A_j \geq 1 + \eta$, and put $u_{r+1} = 0$. Then*

$$\sum_{j=1}^r A_j u_j^2 \leq C_{r,\eta} \sum_{j=1}^r A_j (u_j - u_{j+1})^2.$$

Equivalently, if $d_j = H_j - H_{j-1}$ with $H_0 = 0$, then

$$\sum_{j=1}^r \frac{d_j^2}{A_j} \asymp_{r,\eta} \sum_{j=1}^r \frac{H_j^2}{A_j}.$$

Proof. Write $\Delta_j = u_j - u_{j+1}$. Then $u_j = \sum_{k=j}^r \Delta_k$. By Cauchy's inequality with the geometrically decaying weights $A_j/A_k \leq (1 + \eta)^{-(k-j)}$,

$$A_j u_j^2 \leq C_{r,\eta} \sum_{k=j}^r A_k \Delta_k^2.$$

Summing over j gives the first display. The second follows by applying the first inequality to the inverse triangular map and its reverse; since r is fixed and the ratios are bounded below, both triangular maps have uniformly bounded operator norms in the weighted norms. $\square$

Lemma 7.9 (Truncated triangular comparison). *Fix $r \geq 1$, $\eta > 0$, and $M < \infty$. Let $0 < A_1 < \dots < A_r$ satisfy $A_{j+1}/A_j \geq 1 + \eta$. If $d_j = H_j - H_{j-1}$, $H_0 = 0$, and $|d_j| \leq MA_j$ for every j , then*

$$\sum_{j=1}^r \min \left\{ \frac{d_j^2}{A_j}, A_j \right\} \geq c_{r,\eta,M} \sum_{j=1}^r \min \left\{ \frac{H_j^2}{A_j}, A_j \right\}.$$

Proof. Write $x_j = d_j/A_j$. Then $|x_j| \leq M$ and

$$\frac{H_j}{A_j} = \sum_{k \leq j} \frac{A_k}{A_j} x_k, \quad \frac{A_k}{A_j} \leq (1 + \eta)^{-(j-k)}.$$

Since $\min\{y^2, 1\} \leq y^2$, Cauchy's inequality and the geometric bound on the ratios give, for each fixed j ,

$$\begin{aligned} A_j \min \left\{ \frac{H_j^2}{A_j^2}, 1 \right\} &\leq A_j \left(\sum_{k \leq j} \frac{A_k}{A_j} x_k \right)^2 \\ &\leq C_{r,\eta} A_j \sum_{k \leq j} \frac{A_k}{A_j} x_k^2 = C_{r,\eta} \sum_{k \leq j} A_k x_k^2. \end{aligned}$$

Because $|x_k| \leq M$, we have $x_k^2 \leq C_M \min\{x_k^2, 1\}$. Summing the preceding inequality over j and using that r is fixed gives

$$\sum_{j=1}^r A_j \min \left\{ \frac{H_j^2}{A_j^2}, 1 \right\} \leq C_{r,\eta,M} \sum_{k=1}^r A_k \min\{x_k^2, 1\}.$$

This is exactly the claimed inequality after rewriting $A_j \min\{H_j^2/A_j^2, 1\} = \min\{H_j^2/A_j, A_j\}$ and $A_k \min\{x_k^2, 1\} = \min\{d_k^2/A_k, A_k\}$. $\square$

Lemma 7.10 (Boundary-aware truncated chain comparison). *Fix $r \geq 1$ and $M < \infty$. Let $0 < A_1 < \dots < A_r$, let B_- and B_+ be disjoint subsets of $\{1, \dots, r\}$, and put*

$$G = \{1, \dots, r\} \setminus (B_- \cup B_+).$$

Let $H_0 = 0$, $d_j = H_j - H_{j-1}$, and

$$\phi_j(x) = \min \left\{ \frac{x^2}{A_j}, A_j \right\}.$$

Assume

$$|d_j| \leq MA_j \quad (j \in G), \quad |H_j| \leq MA_j \quad (j \in B_+).$$

Then

$$\begin{aligned} \sum_{j \in G} \phi_j(d_j) + \sum_{j \in B_-} |H_j| + \sum_{j \in B_+} A_j \\ \geq c_{r,M} \sum_{j \in G} \phi_j(H_j). \end{aligned}$$

Proof. Decompose G into maximal consecutive runs. Let $\{\ell, \ell + 1, \dots, u\}$ be one such run and put $b = \ell - 1$. For every j in the run,

$$H_j = H_b + \sum_{k=\ell}^j d_k,$$

where $H_b = H_0 = 0$ if $\ell = 1$. For a fixed positive number A and at most $r + 1$ real numbers x_i ,

$$\min \left\{ \frac{(\sum_i x_i)^2}{A}, A \right\} \leq C_r \sum_i \min \left\{ \frac{x_i^2}{A}, A \right\}. \quad (7.1)$$

Indeed, if one term on the right equals A , the claim is immediate; otherwise it follows from Cauchy's inequality.

If $k \leq j$ belongs to the run, then $A_k \leq A_j$ and $|d_k| \leq MA_k$. Hence

$$\phi_j(d_k) \leq C_M \phi_k(d_k). \quad (7.2)$$

To see this, the assertion is immediate when $|d_k| \leq A_k$; when $|d_k| > A_k$, one has $\phi_k(d_k) = A_k$ and $\phi_j(d_k) \leq d_k^2/A_j \leq M^2 A_k$.

It remains to control the left boundary. If $b \in B_-$, then

$$\phi_j(H_b) \leq |H_b|.$$

If $b \in B_+$, then $|H_b| \leq MA_b$ and $A_b \leq A_j$, so

$$\phi_j(H_b) \leq M^2 A_b.$$

Combining these estimates with (7.1) and (7.2), and then summing first over j in the run and finally over the finitely many runs, proves the lemma. $\square$

Lemma 7.11 (Edge Fourier concentration). *Let $e_1, \dots, e_r$ be the coordinate vectors in $\mathbb{Z}^r$ and put $e_{r+1} = 0$. Let*

$$W = \sum_{\nu \in \mathcal{I}} W_\nu$$

be a sum of independent random vectors supported on $\{e_1, \dots, e_r, 0\}$. Let $A_1, \dots, A_r \geq 1$ satisfy $A_{j+1}/A_j \geq 1 + \eta$ for $j < r$. Suppose that, for every edge j , there is a set $\mathcal{I}_j \subset \mathcal{I}$ with $|\mathcal{I}_j| \geq c_0 A_j$, the sets $\mathcal{I}_j$ are disjoint, and for every $\nu \in \mathcal{I}_j$ the two atoms e_j and e_{j+1} each have probability at least c_0 . Then

$$|\mathbb{E} e^{iu \cdot W}| \leq \exp \left[-c \sum_{j=1}^r A_j \text{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2 \right], \quad u_{r+1} = 0,$$

and consequently

$$\sup_z \mathbb{P}(W = z) \leq C \prod_{j=1}^r A_j^{-1/2}.$$

The same conclusions hold uniformly after any exponential tilt $\|\lambda\|_\infty \leq \Lambda$, with constants depending additionally on Λ .

Proof. If a summand Y gives mass at least c_0 to e_j and to e_{j+1} , then

$$1 - |\mathbb{E}e^{iu \cdot Y}|^2 \geq c_0^2 |e^{iu_j} - e^{iu_{j+1}}|^2 \geq c \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2.$$

As in Lemma 7.7, this implies

$$|\mathbb{E}e^{iu \cdot Y}| \leq \exp\{-c \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2\}.$$

Multiplying over the disjoint edge blocks gives the displayed Fourier bound. For the concentration estimate, Fourier inversion and Lemma 7.8 give

$$\begin{aligned} \mathbb{P}(W = z) &\leq (2\pi)^{-r} \int_{[-\pi, \pi]^r} \exp\left[-c \sum_{j=1}^r A_j \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2\right] du \\ &\leq C \prod_{j=1}^r A_j^{-1/2}. \end{aligned}$$

Bounded tilts only multiply the finitely many atom probabilities by bounded factors, so the edge-block lower bounds persist after changing constants. $\square$

Lemma 7.12 (Forward-edge projected concentration). *Let $G \neq \emptyset$ be a subset of $\{1, \dots, r\}$ and let $0 < A_1 < \dots < A_r$ satisfy $A_{j+1}/A_j \geq 1 + \eta$. Let W be a sum of independent random vectors supported on $\{e_j : j \in G\} \cup \{0\}$ after projection onto the coordinates in G . Suppose that for each $j \in G$ there is an index $\kappa(j) \in (G \cap \{j+1, \dots, r\}) \cup \{0\}$ and a set of at least $c_0 A_j$ summands, disjoint as j varies, for which the two projected atoms e_j and $e_{\kappa(j)}$ have probabilities at least c_0 ; here $e_0 = 0$. Then, uniformly in z_G ,*

$$\mathbb{P}(W_G = z_G) \leq C \prod_{j \in G} A_j^{-1/2}.$$

The same estimate holds after any exponential tilt whose coordinates are bounded in absolute value by a fixed constant.

Proof. For a summand in the block assigned to j , the same two-atom argument used in Lemma 7.11 gives the factor

$$\exp\{-c \operatorname{dist}(u_j - u_{\kappa(j)}, 2\pi\mathbb{Z})^2\}, \quad u_0 = 0.$$

Multiplying over the disjoint blocks gives

$$|\phi(u)| \leq \exp\left[-c \sum_{j \in G} A_j \operatorname{dist}(u_j - u_{\kappa(j)}, 2\pi\mathbb{Z})^2\right].$$

Because every directed edge goes from j to a larger index, or to the anchor 0, each path reaches 0 after at most r steps. Along such a path the weights only increase. Therefore, for $u_0 = 0$,

$$\sum_{j \in G} A_j u_j^2 \leq C_{r, \eta} \sum_{j \in G} A_j (u_j - u_{\kappa(j)})^2.$$

The torus version follows from the same path argument. For each j , choose integer multiples of 2π successively along the directed path $P(j)$ from j to the anchor 0. The triangle inequality on the circle gives

$$\text{dist}(u_j, 2\pi\mathbb{Z}) \leq \sum_{\ell \in P(j)} \text{dist}(u_\ell - u_{\kappa(\ell)}, 2\pi\mathbb{Z}).$$

Cauchy's inequality, together with monotonicity of the weights along this path, gives the displayed comparison with distances to $2\pi\mathbb{Z}$. Fourier inversion then gives

$$\mathbb{P}(W_G = z_G) \leq C \int_{[-\pi, \pi]^G} \exp \left[-c \sum_{j \in G} A_j u_j^2 \right] du_G \leq C \prod_{j \in G} A_j^{-1/2}.$$

Bounded tilts distort all atoms in the finite support by bounded factors, so the two-atom lower bounds persist after changing constants. $\square$

Lemma 7.13 (Terminal interval edge blocks). *Fix $r \geq 1$, $0 < c_* < C_* < \infty$, and $c_0 > 0$. In the terminal interval notation above, fix $j \in \{1, \dots, r\}$ and suppose*

$$c_* \leq s_j = a_j t_j \leq C_*, \quad m_j(t) = \sum_{b \notin A} \mathbb{P}(T_b \in I_j) \geq c_0 a_j.$$

Then there are constants $0 < \alpha < \beta < \infty$ and $c > 0$, depending only on r, c_, C_* , and c_0 , and a set of at least ca_j rates $b \in [\alpha a_j, \beta a_j] \setminus A$ such that*

$$\mathbb{P}(T_b \in I_j) \geq c, \quad \mathbb{P}(T_b \leq t_j) \geq c.$$

Consequently, after projecting any chosen subset of the interval coordinates larger than j either to their own coordinates or to the anchor 0, a further subblock of size at least ca_j gives two projected atoms e_j and $e_{\kappa(j)}$, where $\kappa(j) > j$ is a retained coordinate or $\kappa(j) = 0$, each with probability at least c . The conclusion remains valid under every interval-coordinate exponential tilt whose coordinates are bounded by a fixed constant, after changing c .

Proof. Choose $\alpha > 0$ small and $\beta < \infty$ large. The contribution to $m_j(t)$ from $b < \alpha a_j$ is at most $\alpha a_j + 1$. Since $t_j \geq c_*/a_j$, the contribution from $b > \beta a_j$ is at most

$$\sum_{b > \beta a_j} e^{-bt_j} \leq C a_j e^{-c_* \beta}.$$

After first decreasing α and then increasing β , at least a fixed positive multiple of a_j of the total interval mass comes from $[\alpha a_j, \beta a_j]$. This block contains $O(a_j)$ rates, so at least ca_j of them satisfy $\mathbb{P}(T_b \in I_j) \geq c$. For every such rate,

$$bt_j \geq \alpha c_*, \quad \mathbb{P}(T_b \leq t_j) = 1 - e^{-bt_j} \geq 1 - e^{-\alpha c_*} > 0.$$

The event $\{T_b \leq t_j\}$ is the disjoint union of the later categories $I_{j+1}, \dots, I_{r+1}$. For each selected rate, one of these finitely many categories has probability bounded below. A second pigeonhole argument produces a common later category on a subblock of size ca_j . Projection maps that category either to a retained atom $e_{\kappa(j)}$ or to the anchor 0. Removing the fixed set A changes the block size by only $O_r(1)$. A bounded exponential tilt changes every atom probability in the finite categorical support by a bounded multiplicative factor, which proves the final assertion. $\square$

Lemma 7.14 (Projected good–bad interval bound). *Fix $r \geq 1$, $\eta > 0$, and a compact scaled chamber $K \subset (0, \infty)^r$. In the terminal interval notation above, suppose $s = (a_1 t_1, \dots, a_r t_r) \in K$ and $t_1 > \dots > t_r$. Suppose the deterministic targets satisfy*

$$c_0 a_j \leq y_j \leq C_0 a_j, \quad 1 \leq j \leq r.$$

Then, uniformly for such t and y ,

$$\mathbb{P}(Y(t) = y) \leq C \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j=1}^r \min \left\{ \frac{(y_j - m_j(t))^2}{a_j}, a_j \right\} \right].$$

The constants depend only on r, η, K, c_0 , and C_0 .

Proof. Write $d_j = y_j - m_j(t)$. Since $s \in K$, finite-geometric estimates give $m_j(t) = O_K(a_j)$ and hence $|d_j| \leq M_K a_j$. Fix a small $\varepsilon > 0$. Let

$$G = \{j : |d_j| \leq \varepsilon a_j\}, \quad B = \{1, \dots, r\} \setminus G.$$

For $j \in G$ the target lower bound gives $m_j(t) \geq (c_0 - \varepsilon)a_j$; choose $\varepsilon < c_0/2$, so Lemma 7.13 applies to every good coordinate.

Use one combined exponential tilt. Put

$$\lambda_j = \begin{cases} \delta d_j / a_j, & j \in G, \\ \delta \operatorname{sgn}(d_j), & j \in B, \end{cases}$$

where $\delta > 0$ is a sufficiently small fixed constant. Let $K_t(\lambda) = \log \mathbb{E} \exp\{\lambda \cdot (Y(t) - m(t))\}$. For every bounded tilt ξ , the categorical support and the finite-geometric bound $\operatorname{Var}_\xi Y_j(t) \leq C a_j$ give

$$\operatorname{Var}_\xi(v \cdot Y(t)) \leq C \sum_{j=1}^r a_j v_j^2.$$

Therefore

$$K_t(\lambda) \leq C \sum_{j=1}^r a_j \lambda_j^2.$$

On the other hand,

$$\lambda \cdot d \geq \delta \sum_{j \in G} \frac{d_j^2}{a_j} + \delta \varepsilon \sum_{j \in B} a_j.$$

Choosing δ small enough yields

$$-\lambda \cdot d + K_t(\lambda) \leq -c \sum_{j \in G} \frac{d_j^2}{a_j} - c \sum_{j \in B} a_j.$$

Under this bounded tilt all categorical probabilities are changed by bounded multiplicative factors. Project the tilted array onto the good coordinates and ignore the bad-coordinate constraints. For each $j \in G$, Lemma 7.13 gives a candidate block B_j of at least $c_0 a_j$ rates with two projected

atoms e_j and $e_{\kappa(j)}$, where $\kappa(j) > j$ is another good coordinate or $\kappa(j) = 0$. We now extract disjoint subblocks rigorously. Process the indices of G in increasing order. Since $a_{i+1}/a_i \geq 1 + \eta$,

$$\sum_{i < j} a_i \leq C_{r,\eta} a_j.$$

Choose $c' > 0$ so small that $c'(1 + C_{r,\eta}) \leq c_0$. After subblocks of sizes $c'a_i$ have been chosen for all earlier good indices $i < j$, at most $c' \sum_{i < j} a_i \leq c' C_{r,\eta} a_j$ elements of B_j have been used. At least $c'a_j$ unused elements of B_j therefore remain. Continuing greedily produces disjoint subblocks of size at least $c'a_j$ for every $j \in G$. Lemma 7.12 gives

$$\mathbb{P}_\lambda(Y_G(t) = y_G) \leq C \prod_{j \in G} a_j^{-1/2}.$$

Consequently

$$\begin{aligned} \mathbb{P}(Y(t) = y) &\leq \exp\{-\lambda \cdot d + K_t(\lambda)\} \mathbb{P}_\lambda(Y_G(t) = y_G) \\ &\leq C \prod_{j \in G} a_j^{-1/2} \exp \left[-c \sum_{j \in G} \frac{d_j^2}{a_j} - c \sum_{j \in B} a_j \right]. \end{aligned}$$

For $j \in B$, the factor e^{-ca_j} absorbs the missing local factor: $e^{-ca_j} \leq C a_j^{-1/2} e^{-c'a_j}$. Since for $j \in G$ we have $|d_j| \leq \varepsilon a_j$, this is the desired bound, after changing constants. $\square$

Lemma 7.15 (Actual-mean cumulative Chernoff bounds). *Fix $r \geq 1$. Let $B \subset \{1, \dots, r\}$, and put*

$$\mu_j = \mathbb{E} Z_j(t), \quad j \in B.$$

There are constants $\delta_0, C > 0$, depending only on r , such that, for every $|\theta| \leq \delta_0$,

$$\log \mathbb{E} \exp \left\{ \theta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq C \theta^2 \sum_{j \in B} \mu_j.$$

After decreasing δ_0 , if necessary, fix $0 < \delta \leq \delta_0$ and $C_0 < \infty$. There are $c = c(\delta, r, C_0) > 0$ and $L_0 = L_0(\delta, r, C_0)$ such that, whenever $L \geq L_0$, both of the following assertions hold.

If $z_j \leq C_0 L^{-1} \mu_j$ for every $j \in B$, then

$$-\sum_{j \in B} (-\delta)(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ -\delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq -c \sum_{j \in B} \mu_j.$$

If $\mu_j \leq C_0 L^{-1} z_j$ for every $j \in B$, then

$$-\sum_{j \in B} \delta(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ \delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq -c \sum_{j \in B} z_j.$$

In particular, the first exponent is at most $-cL \sum_{j \in B} a_j$ if $\mu_j \geq c_0 L a_j$, while the second is at most $-c \sum_{j \in B} a_j$ if $z_j \geq c_0 a_j$.

Proof. For a single rate $b \notin A$, define

$$R_b = \sum_{j \in B} \mathbf{1}_{\{T_b > t_j\}}, \quad m_b = \mathbb{E} R_b.$$

Then $0 \leq R_b \leq r$, and

$$\sum_{b \notin A} m_b = \sum_{j \in B} \mu_j.$$

For $|\theta| \leq 1$, let

$$K_b(\theta) = \log \mathbb{E} e^{\theta(R_b - m_b)}.$$

Then $K_b(0) = K'_b(0) = 0$, and

$$K''_b(\theta) = \text{Var}_\theta(R_b) \leq \mathbb{E}_\theta R_b^2 \leq r \mathbb{E}_\theta R_b.$$

Since $0 \leq R_b \leq r$, a bounded tilt changes the expectation of the nonnegative variable R_b by at most a factor depending only on r :

$$\mathbb{E}_\theta R_b = \frac{\mathbb{E} R_b e^{\theta R_b}}{\mathbb{E} e^{\theta R_b}} \leq e^{2r} m_b, \quad |\theta| \leq 1.$$

Therefore $K''_b(\theta) \leq C_r m_b$ for $|\theta| \leq 1$, and Taylor's formula gives $K_b(\theta) \leq C_r \theta^2 m_b$. Summing over $b \notin A$ proves the first displayed inequality for both signs of θ .

For the lower-tail assertion, use the preceding bound and $z_j \leq C_0 L^{-1} \mu_j$:

$$\begin{aligned} & - \sum_{j \in B} (-\delta)(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ -\delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \\ & \leq -\delta \left(1 - \frac{C_0}{L} \right) \sum_{j \in B} \mu_j + C \delta^2 \sum_{j \in B} \mu_j. \end{aligned}$$

Decrease δ_0 , if necessary, so that $C \delta_0 \leq 1/4$, and then choose L_0 so that $C_0/L \leq 1/4$. The last display is at most $-\delta \sum_{j \in B} \mu_j/2$.

For the upper-tail assertion, the same cumulant bound and $\mu_j \leq C_0 L^{-1} z_j$ give

$$\begin{aligned} & - \sum_{j \in B} \delta(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ \delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \\ & \leq -\delta \sum_{j \in B} z_j + (\delta + C \delta^2) \sum_{j \in B} \mu_j \\ & \leq -\delta \left(1 - \frac{2C_0}{L} \right) \sum_{j \in B} z_j, \end{aligned}$$

after decreasing δ_0 once more. Enlarging L_0 makes this at most $-\delta \sum_{j \in B} z_j/2$. The two final scale bounds follow immediately from the corresponding assumptions. $\square$

Lemma 7.16 (Mixed cumulative–interval localization). *Fix $r \geq 1$ and $\eta > 0$. There is $L_0 = L_0(r, \eta)$ such that the following holds for every $L \geq L_0$. In the terminal interval notation, let $s_j = a_j t_j$ and suppose $s \notin [L^{-1}, L]^r$. Put*

$$B_-(s) = \{j : s_j < L^{-1}\}, \quad B_+(s) = \{j : s_j > L\}, \quad G(s) = \{1, \dots, r\} \setminus (B_-(s) \cup B_+(s)).$$

On every region on which these three sets are fixed,

$$\begin{aligned} \mathbb{P}(Y(t) = y) &\leq C_L \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j \in B_-} |H_j(t)| - c(\log L) \sum_{j \in B_+} a_j \right] \\ &\times \exp \left[-c_L \sum_{j \in G(s)} \min \left\{ \frac{H_j(t)^2}{a_j}, a_j \right\} \right]. \end{aligned} \quad (7.3)$$

Here $c > 0$ is independent of L , while $C_L < \infty$ and $c_L > 0$ depend only on L, r , and η . Consequently,

$$\begin{aligned} \mathbb{P}(Y(t) = y) &\leq C_L \prod_{j=1}^r a_j^{-1/2} \exp \left[-cL \sum_{j \in B_-} a_j - c(\log L) \sum_{j \in B_+} a_j \right] \\ &\times \exp \left[-c_L \sum_{j \in G(s)} a_j \min\{(s_j - s_j^*)^2, 1\} \right]. \end{aligned} \quad (7.4)$$

Proof. Throughout the proof the offender sets are fixed. Write $B_- = B_-(s)$, $B_+ = B_+(s)$, and $G = G(s)$. Finite-geometric estimates give, uniformly in the admissible terminal range,

$$\begin{aligned} j \in B_- : \quad &\mu_j(t_j) \geq cLa_j, \quad H_j(t) = z_j - \mu_j(t_j) < 0, \\ j \in B_+ : \quad &\mu_j(t_j) \leq Ca_j/L, \quad 0 < H_j(t) \leq z_j \leq a_j, \end{aligned} \quad (7.5)$$

for L sufficiently large. In particular, $|H_j(t)| \geq cLa_j$ on B_- and $H_j(t) \geq ca_j$ on B_+ . For $j \in G$, one has $s_j \in [L^{-1}, L]$, and therefore

$$m_j(t) \leq \mu_j(t_j) \leq CLa_j, \quad |d_j(t)| \leq MLa_j. \quad (7.6)$$

Let

$$K_t(\theta) = \log \mathbb{E} \exp\{\theta \cdot (Y(t) - m(t))\}$$

be the centered interval-coordinate cumulant. If $\gamma = (\gamma_1, \dots, \gamma_r)$ is a cumulative-coordinate vector, put

$$(\mathcal{T}\gamma)_k = \sum_{j \geq k} \gamma_j.$$

Since $Z_j = Y_1 + \dots + Y_j$ and $d_j = H_j - H_{j-1}$,

$$(\mathcal{T}\gamma) \cdot d = \gamma \cdot H. \quad (7.7)$$

Choose $\delta > 0$ so small that 3δ is admissible in Lemma 7.15 and $3\delta r < 1/2$. Define

$$\gamma_j^- = -\delta \mathbf{1}_{\{j \in B_-\}}, \quad \gamma_j^+ = \delta(\log L) \mathbf{1}_{\{j \in B_+\}}.$$

We also introduce a local interval tilt on G . Ratio separation implies that there is $c_\eta > 0$ such that $y_j \geq c_\eta a_j$ for all j and all sufficiently large n . Fix $0 < \varepsilon < c_\eta/4$ and put

$$G_0 = \{j \in G : |d_j| \leq \varepsilon a_j\}, \quad G_1 = G \setminus G_0.$$

For a constant $\delta_L > 0$ to be chosen below, set

$$\lambda_j = \begin{cases} \delta_L d_j / a_j, & j \in G_0, \\ \delta_L \operatorname{sgn}(d_j), & j \in G_1, \\ 0, & j \notin G. \end{cases}$$

Finally, let

$$\theta = \mathcal{T}\gamma^- + \mathcal{T}\gamma^+ + \lambda.$$

Generalized Hölder inequality gives the precise combined-cumulant estimate

$$K_t(\theta) \leq \frac{1}{3}K_t(3\mathcal{T}\gamma^-) + \frac{1}{3}K_t(3\mathcal{T}\gamma^+) + \frac{1}{3}K_t(3\lambda). \quad (7.8)$$

Thus, by (7.7),

$$\begin{aligned} -\theta \cdot d + K_t(\theta) &\leq \frac{1}{3}\{-3\gamma^- \cdot H + K_t(3\mathcal{T}\gamma^-)\} \\ &\quad + \frac{1}{3}\{-3\gamma^+ \cdot H + K_t(3\mathcal{T}\gamma^+)\} \\ &\quad + \frac{1}{3}\{-3\lambda \cdot d + K_t(3\lambda)\}. \end{aligned} \quad (7.9)$$

We estimate the three braces separately.

For B_- , (7.5) implies $z_j \leq CL^{-1}\mu_j(t_j)$. Lemma 7.15, applied with parameter 3δ , gives

$$-3\gamma^- \cdot H + K_t(3\mathcal{T}\gamma^-) \leq -c \sum_{j \in B_-} \mu_j(t_j) \leq -c \sum_{j \in B_-} |H_j(t_j)|. \quad (7.10)$$

The same bound also retains the weaker loss $-cL \sum_{j \in B_-} a_j$.

For B_+ , put

$$R_b^+ = \sum_{j \in B_+} \mathbf{1}_{\{T_b > t_j\}}, \quad 0 \leq R_b^+ \leq r.$$

Since the summands indexed by $b \notin A$ are independent,

$$\begin{aligned} K_t(3\mathcal{T}\gamma^+) &= \sum_{b \notin A} \log \mathbb{E} \exp\{3\delta(\log L)(R_b^+ - \mathbb{E}R_b^+)\} \\ &\leq \sum_{b \notin A} \log \mathbb{E} \exp\{3\delta(\log L)R_b^+\} \\ &\leq L^{3\delta r} \sum_{b \notin A} \mathbb{P}(R_b^+ > 0) \\ &\leq L^{3\delta r} \sum_{j \in B_+} \mu_j(t_j) \leq CL^{-1/2} \sum_{j \in B_+} a_j. \end{aligned} \quad (7.11)$$

On the other hand, (7.5) gives

$$-3\gamma^+ \cdot H \leq -c(\log L) \sum_{j \in B_+} a_j.$$

After increasing L_0 , (7.11) is absorbed by this linear gain, and hence

$$-3\gamma^+ \cdot H + K_t(3\mathcal{T}\gamma^+) \leq -c(\log L) \sum_{j \in B_+} a_j. \quad (7.12)$$

For the local tilt, let v vanish outside G . Under every bounded interval-coordinate tilt ξ , a categorical summand contributes at most one coordinate, and therefore

$$\text{Var}_\xi(v \cdot Y(t)) \leq \sum_{j \in G} v_j^2 \mathbb{E}_\xi Y_j(t) \leq C_L \sum_{j \in G} a_j v_j^2, \quad (7.13)$$

where the last inequality follows from (7.6) and bounded distortion of atom probabilities. Taylor's formula for the cumulant gives

$$K_t(3\lambda) \leq C_L \sum_{j \in G} a_j \lambda_j^2.$$

Choose $\delta_L > 0$ sufficiently small. Then

$$-3\lambda \cdot d + K_t(3\lambda) \leq -c_L \sum_{j \in G} \min \left\{ \frac{d_j^2}{a_j}, a_j \right\}. \quad (7.14)$$

Indeed, on G_0 the linear term is a fixed multiple of d_j^2/a_j , whereas on G_1 it is at least a fixed multiple of a_j .

We next recover the product local factor under the single total tilt θ . For fixed L , its interval coordinates are bounded by a constant depending only on L and r . If $j \in G_0$, then

$$m_j(t) \geq y_j - \varepsilon a_j \geq (c_\eta/2)a_j.$$

Lemma 7.13, with $s_j \in [L^{-1}, L]$, supplies a candidate forward-edge block of order a_j . Process the indices of G_0 increasingly and choose small subblocks greedily. Since $a_{i+1}/a_i \geq 1 + \eta$ and r is fixed,

$$\sum_{i < j} a_i \leq C_{r,\eta} a_j,$$

so disjoint subblocks of size $c_{L,r,\eta} a_j$ can be retained for every $j \in G_0$. Boundedness of the total tilt preserves the two-atom lower bounds. Lemma 7.12, applied after projection onto G_0 , yields

$$\mathbb{P}_\theta(Y_{G_0}(t) = y_{G_0}) \leq C_L \prod_{j \in G_0} a_j^{-1/2}. \quad (7.15)$$

The exact change-of-measure identity and (7.9)–(7.15) now give

$$\begin{aligned} \mathbb{P}(Y(t) = y) &\leq C_L \prod_{j \in G_0} a_j^{-1/2} \exp \left[-c \sum_{j \in B_-} |H_j| - c(\log L) \sum_{j \in B_+} a_j \right] \\ &\quad \times \exp \left[-c_L \sum_{j \in G} \min \left\{ \frac{d_j^2}{a_j}, a_j \right\} \right]. \end{aligned} \quad (7.16)$$

For $j \in G_1$, the last exponent contains $-c_L a_j$; for $j \in B_-$ and $j \in B_+$, (7.5) and the preceding tail losses contain the same fixed-order exponential cost. These factors absorb all missing $a_j^{-1/2}$ terms, after changing constants.

It remains to pass from interval residuals to cumulative residuals without losing boundary terms. The assumptions of Lemma 7.10 hold with $A_j = a_j$ and $M = M_L$: (7.6) controls d_j on G ,

while (7.5) gives $|H_j| \leq a_j$ on B_+ . Split a fixed fraction of each offender loss in (7.16) for this comparison. Since $\log L$ is large, Lemma 7.10 yields

$$\begin{aligned} c_L \sum_{j \in G} \min \left\{ \frac{d_j^2}{a_j}, a_j \right\} + c \sum_{j \in B_-} |H_j| + c(\log L) \sum_{j \in B_+} a_j \\ \geq c'_L \sum_{j \in G} \min \left\{ \frac{H_j^2}{a_j}, a_j \right\} + c' \sum_{j \in B_-} |H_j| + c'(\log L) \sum_{j \in B_+} a_j. \end{aligned} \quad (7.17)$$

Combining (7.16) and (7.17) proves (7.3).

Finally, on G the scaled coordinate s_j belongs to $[L^{-1}, L]$. Uniform finite-geometric derivative estimates and strict monotonicity give

$$|H_j(t)| = |\mu_j(t_j^*) - \mu_j(t_j)| \geq c_L a_j \min\{|s_j - s_j^*|, 1\}. \quad (7.18)$$

Together with (7.5), this converts (7.3) into (7.4). $\square$

Lemma 7.17 (Anisotropic tilted local upper bound). *Let $W = \sum_{\nu \in \mathcal{I}} W_\nu$ be as in Lemma 7.7. Assume that the hypotheses of Lemma 7.7 hold uniformly under all tilts $\|\lambda\|_\infty \leq \Lambda$. Assume also that, for those tilted laws,*

$$\text{Var}_\lambda(v \cdot W) \leq C_0 \sum_{j=1}^r A_j v_j^2, \quad v \in \mathbb{R}^r.$$

Then there are constants $C, c > 0$, depending only on the constants in the assumptions, such that for every lattice point z , with $h = z - \mathbb{E}W$,

$$\mathbb{P}(W = z) \leq C \prod_{j=1}^r A_j^{-1/2} \exp \left[-c \sum_{j=1}^r \min \left\{ \frac{h_j^2}{A_j}, |h_j| \right\} \right].$$

In particular, if $|h_j| \leq A_j$ for every j , then the exponent may be written as $-c \sum_j h_j^2 / A_j$.

Proof. Let

$$K(\lambda) = \log \mathbb{E} e^{\lambda \cdot (W - \mathbb{E}W)}.$$

For $\|\lambda\|_\infty \leq \Lambda$, the covariance upper bound gives

$$K(\lambda) = \int_0^1 (1-s) \lambda^T \text{Var}_{s\lambda}(W) \lambda ds \leq C \sum_{j=1}^r A_j \lambda_j^2.$$

Choose a sufficiently small $\delta > 0$, depending only on the constants in the assumptions and with $\delta \leq \Lambda$, and put

$$\lambda_j = \delta \text{sgn}(h_j) \min\{|h_j|/A_j, 1\}.$$

By tilting and then applying Lemma 7.7 under the tilted law,

$$\begin{aligned} \mathbb{P}(W = z) &= \exp\{-\lambda \cdot h + K(\lambda)\} \mathbb{P}_\lambda(W = z) \\ &\leq C \prod_{j=1}^r A_j^{-1/2} \exp\{-\lambda \cdot h + K(\lambda)\}. \end{aligned}$$

For the chosen λ ,

$$\lambda \cdot h = \delta \sum_{j=1}^r |h_j| \min\{|h_j|/A_j, 1\},$$

while

$$K(\lambda) \leq C\delta^2 \sum_{j=1}^r A_j \min\{h_j^2/A_j^2, 1\}.$$

Taking δ small enough makes the second display at most one half of the first, after changing constants. This proves the claim. $\square$

Lemma 7.18 (Anisotropic terminal local CLT). *Fix $r \geq 1$, $\eta > 0$, $\kappa > 0$, and $0 < c_0 < C_0 < \infty$. Let*

$$1 \ll a_1 < \dots < a_r \ll n, \quad a_{j+1}/a_j \geq 1 + \eta,$$

where the ratio-separation condition is empty when $r = 1$. Let $t_1 > \dots > t_r$ satisfy

$$\frac{c_0}{a_j} \leq t_j \leq \frac{C_0}{a_j}, \quad \frac{t_j}{t_{j+1}} \geq 1 + \kappa \quad (1 \leq j < r).$$

Let $A = \{a_1, \dots, a_r\}$, and put

$$Z_j(t) = \sum_{b \notin A} \mathbf{1}_{\{T_b > t_j\}}, \quad Z(t) = (Z_1(t), \dots, Z_r(t)).$$

If $\Sigma(t) = \text{Var } Z(t)$, then there are constants $c, C > 0$, depending only on $r, \eta, \kappa, c_0, C_0$, such that

$$c \text{ diag}(a_1, \dots, a_r) \leq \Sigma(t) \leq C \text{ diag}(a_1, \dots, a_r)$$

in quadratic-form order. Moreover,

$$\begin{aligned} \mathbb{P}(Z(t) = z) &= (2\pi)^{-r/2} (\det \Sigma(t))^{-1/2} \\ &\times \exp \left[-\frac{1}{2} (z - \mathbb{E}Z(t))^T \Sigma(t)^{-1} (z - \mathbb{E}Z(t)) \right] + o\left((a_1 \dots a_r)^{-1/2}\right) \end{aligned}$$

uniformly whenever the Mahalanobis norm of $z - \mathbb{E}Z(t)$ is bounded. In addition, for every fixed $\gamma \in (0, 1/6)$, the coordinatewise moderate-deviation upper bound

$$\mathbb{P}(Z(t) = z) \leq C(\det \Sigma(t))^{-1/2} \exp \left[-c \sum_{j=1}^r \frac{(z_j - \mathbb{E}Z_j(t))^2}{a_j} \right]$$

holds uniformly whenever

$$|z_j - \mathbb{E}Z_j(t)| \leq a_j^{1/2+\gamma}, \quad 1 \leq j \leq r.$$

Proof. For a single rate $b \notin A$, write

$$V_b(t) = (\mathbf{1}_{\{T_b > t_1\}}, \dots, \mathbf{1}_{\{T_b > t_r\}}).$$

The vector $V_b(t)$ takes the cumulative values

$$(1, \dots, 1), \quad (0, 1, \dots, 1), \quad \dots, \quad (0, \dots, 0, 1), \quad (0, \dots, 0),$$

according as T_b falls in

$$(t_1, \infty), \quad (t_2, t_1], \quad \dots, \quad (t_r, t_{r-1}], \quad (0, t_r].$$

Adjacent support points differ by the coordinate vectors, so the lattice span is $\mathbb{Z}^r$.

Choose constants $0 < d_1 < d_2 < \infty$, depending only on $r, \eta, \kappa, c_0, C_0$, with $d_2/d_1 < 1 + \eta$. Then the rate intervals $[d_1 a_j, d_2 a_j]$ are disjoint for different j 's. For every integer rate $b \in [d_1 a_j, d_2 a_j]$ and all large n , the two adjacent time categories separated by the threshold t_j have probabilities bounded below. For instance,

$$e^{-bt_j} \geq e^{-d_2 C_0}, \quad b(t_{j-1} - t_j) \geq d_1 c_0 \frac{\kappa}{1 + \kappa}$$

for the upper adjacent interval, with the evident endpoint modifications when $j = 1$ or $j = r$. The fixed omitted set A removes only r rates, so each block still contains order a_j available rates.

These disjoint adjacent blocks imply the covariance lower bound: for every $u \in \mathbb{R}^r$, the j -th block contributes at least $ca_j u_j^2$ to $\text{Var}(u \cdot Z(t))$. Thus

$$\text{Var}(u \cdot Z(t)) \geq c \sum_{j=1}^r a_j u_j^2.$$

The upper bound follows from $\text{Var } Z_j(t) = O(a_j)$, which is a direct finite-geometric estimate, and Cauchy's inequality:

$$\text{Var}(u \cdot Z(t)) \leq r \sum_{j=1}^r u_j^2 \text{Var } Z_j(t) \leq C \sum_{j=1}^r a_j u_j^2.$$

This proves the covariance comparison.

The same blocks give the coordinatewise Fourier bound of Lemma 7.7, with $A_j = a_j$. This is the point at which anisotropy is used: rather than bounding the whole off-origin region by a scalar factor e^{-ca_1} , we retain the product-scale estimate

$$|\mathbb{E} e^{iu \cdot Z(t)}| \leq \exp \left[-c \sum_{j=1}^r a_j \text{dist}(u_j, 2\pi\mathbb{Z})^2 \right].$$

This bound is stable under bounded exponential tilts, because the support is finite and the two distinguished adjacent atom probabilities remain bounded below.

We next prove the local CLT. Fourier inversion gives

$$\mathbb{P}(Z(t) = z) = (2\pi)^{-r} \int_{[-\pi, \pi]^r} e^{-iu \cdot (z - \mathbb{E}Z(t))} \phi_t(u) du,$$

where $\phi_t(u) = \mathbb{E} e^{iu \cdot (Z(t) - \mathbb{E}Z(t))}$. Put $D_a = \text{diag}(a_1, \dots, a_r)$. On the anisotropic small box $|u_j| \leq R a_j^{-1/2}$, the finite-tail estimates and boundedness of the increments give the uniform expansion

$$\log \phi_t(u) = -\frac{1}{2} u^T \Sigma(t) u + O \left((\max_j |u_j|) \sum_{j=1}^r a_j u_j^2 \right).$$

Indeed, the third cumulant in the direction u is bounded by a constant times $\max_j |u_j|$ times the corresponding variance contribution, and the variance upper bound just proved controls the latter by $C \sum_j a_j u_j^2$. After the change of variables $u = D_a^{-1/2} v$, the error term is $O_R(a_1^{-1/2}) = o(1)$. Since the Mahalanobis norm of $z - \mathbb{E}Z(t)$ is bounded, this gives the Gaussian integral on every fixed v -ball. The contribution from the complement of such a ball is controlled uniformly by the coordinatewise Fourier bound above:

$$\int_{\max_j |u_j| > R a_j^{-1/2}} \exp \left[-c \sum_j a_j \text{dist}(u_j, 2\pi\mathbb{Z})^2 \right] du \leq C \left(\prod_j a_j^{-1/2} \right) e^{-cR^2}.$$

Letting $R \rightarrow \infty$ yields the stated local CLT and the error $o((a_1 \cdots a_r)^{-1/2})$, uniformly over the allowed parameters.

For the moderate-deviation upper bound, apply Lemma 7.17 to the terminal array. The covariance upper bound required there has already been proved and remains valid under the small tilts used below. If $|z_j - \mathbb{E}Z_j(t)| \leq a_j^{1/2+\gamma}$, then for large n this displacement is at most a_j , and Lemma 7.17 gives

$$\mathbb{P}(Z(t) = z) \leq C \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j=1}^r \frac{(z_j - \mathbb{E}Z_j(t))^2}{a_j} \right].$$

The covariance comparison implies $\prod_j a_j^{-1/2} \asymp (\det \Sigma(t))^{-1/2}$, proving the displayed form. $\square$

Lemma 7.19 (Scaled interval-count localization). *Fix $r \geq 1$, $\eta > 0$, and a compact set $K \subset (0, \infty)^r$. In the terminal interval notation, define*

$$\Phi(t) = \prod_{j=1}^r a_j e^{-a_j t_j}.$$

Uniformly for $s = (s_1, \dots, s_r) \in K$ with $t_1 > \dots > t_r$,

$$\mathbb{P}(Y(t) = y) \leq C \left(\prod_{j=1}^r a_j^{-1/2} \right) \exp \left[-c \sum_{j=1}^r a_j \min\{(s_j - s_j^*)^2, 1\} \right], \quad (7.19)$$

where $C, c > 0$ depend only on r, η , and K .

For a precise uniform tail statement, let $\mathfrak{A}_N(\eta)$ be the set of integer tuples $(n, a_1, \dots, a_r)$ satisfying

$$n \geq N, \quad N \leq a_1 < \dots < a_r \leq n/N, \quad a_{j+1}/a_j \geq 1 + \eta.$$

Then

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \sup_{(n, a) \in \mathfrak{A}_N(\eta)} \frac{1}{\prod_{j=1}^r a_j^{-1}} \int_{(a_1 t_1, \dots, a_r t_r) \notin [L^{-1}, L]^r} \Phi(t) \mathbb{P}(Z(t) = z) dt = 0. \quad (7.20)$$

In particular, for every $\varepsilon > 0$ there are $L < \infty$ and N_0 such that the integral in (7.20) is at most $\varepsilon \prod_j a_j^{-1}$ for every $(n, a) \in \mathfrak{A}_{N_0}(\eta)$.

Proof. Ratio separation gives

$$c_\eta a_j \leq y_j \leq C_\eta a_j, \quad 1 \leq j \leq r,$$

for all sufficiently large a_1 . On the fixed compact scaled chamber K , finite-geometric estimates give $m_j(t) = O_K(a_j)$ and hence $|d_j(t)| \leq M_K a_j$. Lemma 7.14 therefore yields

$$\mathbb{P}(Y(t) = y) \leq C \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j=1}^r \min \left\{ \frac{d_j(t)^2}{a_j}, a_j \right\} \right].$$

Since $d_j = H_j - H_{j-1}$ and $H_0 = 0$, Lemma 7.9 gives

$$\sum_{j=1}^r \min \left\{ \frac{d_j(t)^2}{a_j}, a_j \right\} \geq c_K \sum_{j=1}^r \min \left\{ \frac{H_j(t)^2}{a_j}, a_j \right\}.$$

On K , the derivative of $s \mapsto \mu_j(s/a_j)$ has magnitude comparable to a_j , uniformly in j . The mean-value theorem near the saddle, followed by compactness and strict monotonicity away from it, gives

$$|H_j(t)| \geq c_K a_j \min\{|s_j - s_j^*|, 1\}.$$

This proves (7.19).

We turn to (7.20). Fix $L \geq L_0(r, \eta)$ and partition the complement of $[L^{-1}, L]^r$ according to the finitely many possible pairs of offender sets (B_-, B_+) . On each such region, Lemma 7.16 gives (7.4). Since

$$\Phi(t) dt_1 \cdots dt_r = \prod_{j=1}^r e^{-s_j} ds_j,$$

and discarding the chamber constraint only enlarges the integration domain, the resulting bound factors coordinatewise. Every nonoffending coordinate contributes at most

$$C_L \int_{L^{-1}}^L e^{-c_L a_j \min\{(s-s_j^*)^2, 1\}} ds \leq C_L a_j^{-1/2}. \quad (7.21)$$

For a lower offender, the integration interval has length at most L^{-1} , and

$$a_j^{1/2} e^{-c_L a_j} \leq \alpha_{L,N}^- := \sup_{a \geq N} a^{1/2} e^{-c_L a} \rightarrow 0 \quad (N \rightarrow \infty). \quad (7.22)$$

For an upper offender,

$$\int_L^\infty e^{-s} ds = e^{-L},$$

and

$$a_j^{1/2} L^{-c a_j} \leq \alpha_{L,N}^+ := \sup_{a \geq N} a^{1/2} L^{-c a} \rightarrow 0 \quad (N \rightarrow \infty). \quad (7.23)$$

The original factor $\prod_j a_j^{-1/2}$, together with (7.21)–(7.23), therefore supplies a factor a_j^{-1} for every coordinate and an additional quantity tending to zero for every nonempty offender set. The number of possible offender sets is finite. Hence, for each fixed $L \geq L_0$,

$$\limsup_{N \rightarrow \infty} \sup_{(n,a) \in \mathfrak{A}_N(\eta)} \frac{\int_{\substack{t_1 > \dots > t_r \\ s \notin [L^{-1}, L]^r}} \Phi(t) \mathbb{P}(Z(t) = z) dt}{\prod_j a_j^{-1}} = 0.$$

This implies (7.20) and the final assertion. $\square$

Lemma 7.20 (Product-scale terminal localization). *Fix $r \geq 1$ and $\eta > 0$. In the setting of Lemma 7.6, put*

$$z = (a_1 - 1, a_2 - 2, \dots, a_r - r), \quad \Phi(t) = \prod_{j=1}^r a_j e^{-a_j t_j},$$

and let $Z(t)$ have the same definition as in Lemma 7.18. There is $\delta_0 > 0$, depending only on r and η , such that for every fixed $0 < \delta < \delta_0$ there are constants $C, c > 0$, depending only on r, η, δ , for which the following holds. If

$$\mathcal{N}_\delta = \{t_1 > \dots > t_r : |t_j/t_j^* - 1| \leq \delta \text{ for all } j\},$$

and

$$u_j = a_j^{3/2}(t_j - t_j^*), \quad 1 \leq j \leq r,$$

then, uniformly for $t \in \mathcal{N}_\delta$,

$$\Phi(t) \mathbb{P}(Z(t) = z) dt_1 \cdots dt_r \leq C \left(\prod_{j=1}^r a_j^{-1} \right) \exp \left[-c \sum_{j=1}^r \min(u_j^2, a_j) \right] du_1 \cdots du_r.$$

Moreover,

$$\begin{aligned} & \int_{\{t_1 > \dots > t_r\} \setminus \mathcal{N}_\delta} \Phi(t) \mathbb{P}(Z(t) = z) dt_1 \cdots dt_r \\ &= o \left(\prod_{j=1}^r a_j^{-1} \right), \end{aligned}$$

uniformly over the log-separated terminal range.

Proof. Choose δ_0 small enough that, whenever $|t_j/t_j^* - 1| \leq \delta_0$ for all j , the ratios t_j/t_{j+1} are bounded below by $1 + \kappa$, for some $\kappa > 0$ depending only on η . On this neighborhood, Lemma 7.6 gives

$$t_j \asymp a_j^{-1}, \quad -\mu'_j(t_j) \asymp a_j^2, \quad \mu''_j(t_j) = O(a_j^3),$$

uniformly.

For $t \in \mathcal{N}_\delta$, put $s_j = a_j t_j$. Since

$$s_j - s_j^* = a_j(t_j - t_j^*) = u_j / \sqrt{a_j},$$

Lemma 7.19, applied on the compact scaled neighborhood determined by δ , gives

$$\mathbb{P}(Z(t) = z) \leq C \left(\prod_{j=1}^r a_j^{-1/2} \right) \exp \left[-c \sum_{j=1}^r \min(u_j^2, a_j) \right].$$

Also

$$\Phi(t) dt_1 \cdots dt_r = \left(\prod_{j=1}^r a_j e^{-s_j} \right) \left(\prod_{j=1}^r a_j^{-3/2} \right) du_1 \cdots du_r \leq C \prod_{j=1}^r a_j^{-1/2} du_1 \cdots du_r,$$

because s_j remains in a compact subset of $(0, \infty)$. Multiplying the two displays proves the product-scale bound.

It remains to remove the complement of $\mathcal{N}_\delta$. In the scaled variables, Lemma 7.19 first removes the complement of a fixed compact set at cost $o(\prod_j a_j^{-1})$. On the remaining compact set, the same lemma gives

$$\mathbb{P}(Z(t) = z) \leq C \left(\prod_{j=1}^r a_j^{-1/2} \right) \exp \left[-c \sum_{j=1}^r a_j \min\{(s_j - s_j^*)^2, 1\} \right].$$

If $t \notin \mathcal{N}_\delta$, then, on this compact set and for all large n , at least one scaled coordinate satisfies $|s_j - s_j^*| \geq c_\delta$. Since

$$\Phi(t) dt_1 \cdots dt_r = \prod_{j=1}^r e^{-s_j} ds_j,$$

the compact contribution outside $\mathcal{N}_\delta$ is bounded by

$$C \left(\prod_{j=1}^r a_j^{-1/2} \right) \int_{\max_j |s_j - s_j^*| \geq c_\delta} \exp \left[-c \sum_{j=1}^r a_j \min\{(s_j - s_j^*)^2, 1\} \right] ds.$$

Changing variables $v_j = \sqrt{a_j}(s_j - s_j^*)$ gives

$$O \left(e^{-c_\delta a_1} \prod_{j=1}^r a_j^{-1} \right) = o \left(\prod_{j=1}^r a_j^{-1} \right).$$

Together with the compact-tail estimate, this proves the second assertion. $\square$

Lemma 7.21 (Covariance stability on the terminal saddle scale). *Fix $r \geq 1$, $\eta > 0$, and $R < \infty$. In the setting of Lemma 7.6, let*

$$D_a = \text{diag}(a_1, \dots, a_r), \quad H_a = \text{diag}(a_1^{-3/2}, \dots, a_r^{-3/2}),$$

and let $\Sigma(t) = \text{Var } Z(t)$. Then, uniformly over the log-separated terminal range,

$$\sup_{\|u\| \leq R} \left\| D_a^{-1/2} (\Sigma(t^* + H_a u) - \Sigma(t^*)) D_a^{-1/2} \right\|_{\text{op}} \longrightarrow 0. \quad (7.24)$$

Consequently, uniformly on the same fixed u -ball,

$$\frac{\det \Sigma(t^* + H_a u)}{\det \Sigma(t^*)} \longrightarrow 1 \quad (7.25)$$

and

$$\left\| D_a^{1/2} (\Sigma(t^* + H_a u)^{-1} - \Sigma(t^*)^{-1}) D_a^{1/2} \right\|_{\text{op}} \longrightarrow 0. \quad (7.26)$$

Proof. For $j \leq k$, independence of the nonspecified clocks gives

$$\Sigma_{jk}(t) = \sum_{b \notin A} e^{-bt_j} (1 - e^{-bt_k}). \quad (7.27)$$

The same formula with $j = k$ is the Bernoulli variance. On a fixed saddle ball $t = t^* + H_a u$, Lemma 7.6 gives $t_j \asymp a_j^{-1}$, uniformly. Differentiating (7.27) and using the finite-geometric estimates yields

$$|\partial_{t_j} \Sigma_{jk}(t)| \leq \sum_{b \geq 1} b e^{-b t_j} \leq C_R a_j^2,$$

while

$$|\partial_{t_k} \Sigma_{jk}(t)| \leq \sum_{b \geq 1} b e^{-b(t_j + t_k)} \leq C_R a_j^2.$$

For $j = k$, the derivative is bounded by the same geometric sum. Hence

$$|\Sigma_{jk}(t^* + H_a u) - \Sigma_{jk}(t^*)| \leq C_R a_j^2 (a_j^{-3/2} + a_k^{-3/2}).$$

After division by $\sqrt{a_j a_k}$, the right-hand side is at most

$$C_R \left(a_k^{-1/2} + \frac{a_j^{3/2}}{a_k^2} \right) \leq C_R a_1^{-1/2}.$$

Since r is fixed, entrywise convergence implies (7.24).

Lemma 7.18 gives uniform positive lower and upper bounds for the eigenvalues of the normalized matrices

$$D_a^{-1/2} \Sigma(t) D_a^{-1/2}$$

throughout the saddle neighborhood. The determinant and inverse are continuous, uniformly Lipschitz functions on this compact set of positive definite matrices. Applying these facts to (7.24) proves (7.25) and (7.26). $\square$

Lemma 7.22 (Terminal integrated saddle). *Fix $r \geq 1$ and $\eta > 0$. In the setting of Lemma 7.6, put*

$$z = (a_1 - 1, a_2 - 2, \dots, a_r - r), \quad \Phi(t) = \prod_{j=1}^r a_j e^{-a_j t_j}.$$

Let $A = \{a_1, \dots, a_r\}$, and put

$$Z_j(t) = \sum_{b \notin A} \mathbf{1}_{\{T_b > t_j\}}, \quad Z(t) = (Z_1(t), \dots, Z_r(t)).$$

Then

$$\begin{aligned} & \int_{t_1 > \dots > t_r} \Phi(t) \mathbb{P}(Z(t) = z) dt_1 \cdots dt_r \\ &= (1 + o(1)) \Phi(t^*) \prod_{j=1}^r (-\mu'_j(t_j^*))^{-1}, \end{aligned}$$

uniformly over the log-separated terminal range.

Proof. Let $\mathcal{N}_\delta$ be the product saddle neighborhood from Lemma 7.20, with $\delta > 0$ fixed and sufficiently small. That lemma gives

$$\int_{\{t_1 > \dots > t_r\} \setminus \mathcal{N}_\delta} \Phi(t) \mathbb{P}(Z(t) = z) dt_1 \cdots dt_r = o \left(\prod_{j=1}^r a_j^{-1} \right).$$

Since Lemma 7.6 gives

$$\Phi(t^*) \prod_{j=1}^r (-\mu'_j(t_j^*))^{-1} \asymp \prod_{j=1}^r a_j^{-1},$$

the contribution outside $\mathcal{N}_\delta$ is negligible on the required relative scale.

Inside $\mathcal{N}_\delta$, put

$$H = \text{diag}(a_1^{-3/2}, \dots, a_r^{-3/2}), \quad t = t^* + Hu.$$

This scale is forced by the balance $-\mu'_j(t_j^*) \asymp a_j^2$ and $\text{Var } Z_j(t^*) \asymp a_j$: changing t_j by $a_j^{-3/2}$ moves the mean by order $a_j^{1/2}$, the natural standard-deviation scale. Let

$$D = \text{diag}(-\mu'_1(t_1^*), \dots, -\mu'_r(t_r^*)).$$

For bounded u , Lemma 7.6 and Taylor's theorem give the normalized mean expansion

$$\left\| D_a^{-1/2} (\mathbb{E}Z(t) - z + DHu) \right\| \longrightarrow 0, \quad D_a = \text{diag}(a_1, \dots, a_r), \quad (7.28)$$

uniformly on every fixed u -ball. Indeed, the j -th Taylor remainder is

$$O(a_j^3 |t_j - t_j^*|^2) = O(u_j^2) = o(a_j^{1/2}),$$

whereas

$$\mu'_j(t_j^*)(t_j - t_j^*) = -u_j a_j^{1/2} (1 + o(1)).$$

The neighborhood $\mathcal{N}_\delta$ supplies constants κ, c_0, C_0 for the hypotheses of Lemma 7.18; hence the anisotropic local CLT applies uniformly for bounded u . Lemma 7.21, together with (7.28), gives

$$\begin{aligned} (\det \Sigma(t))^{-1/2} &= (\det \Sigma(t^*))^{-1/2} (1 + o(1)), \\ (z - \mathbb{E}Z(t))^T \Sigma(t)^{-1} (z - \mathbb{E}Z(t)) &= u^T H^T D^T \Sigma(t^*)^{-1} DHu + o(1), \end{aligned} \quad (7.29)$$

uniformly on every fixed ball. Also,

$$\Phi(t^* + Hu) = \Phi(t^*) (1 + o(1))$$

uniformly there, because $a_j(t_j - t_j^*) = u_j / \sqrt{a_j} = o(1)$. Consequently, on $\|u\| \leq R$, the integrand has the uniform asymptotic Gaussian form

$$\Phi(t^*) (2\pi)^{-r/2} (\det \Sigma(t^*))^{-1/2} \exp \left[-\frac{1}{2} u^T H^T D^T \Sigma(t^*)^{-1} DHu \right] |\det H| \, du.$$

The matrix $H^T D^T \Sigma(t^*)^{-1} DH$ is uniformly positive definite and uniformly bounded in quadratic-form order, because $D_{jj} \asymp a_j^2$, $H_{jj} = a_j^{-3/2}$, and $\Sigma(t^*) \asymp \text{diag}(a_1, \dots, a_r)$. Hence, as $n \rightarrow \infty$ with R fixed, the contribution from $\|u\| \leq R$ is

$$\begin{aligned} &\Phi(t^*) (2\pi)^{-r/2} (\det \Sigma(t^*))^{-1/2} |\det H| \\ &\quad \times \int_{\|u\| \leq R} \exp \left[-\frac{1}{2} u^T H^T D^T \Sigma(t^*)^{-1} DHu \right] \, du + o_R \left(\prod_{j=1}^r a_j^{-1} \right). \end{aligned}$$

Letting $R \rightarrow \infty$, uniformly over the allowed rate tuples, the Gaussian integral gives the determinant cancellation

$$(2\pi)^{-r/2}(\det \Sigma(t^*))^{-1/2} |\det H| \int_{\mathbb{R}^r} \exp \left[-\frac{1}{2} u^T H^T D^T \Sigma(t^*)^{-1} D H u \right] du = |\det D|^{-1}.$$

Indeed,

$$\det(H^T D^T \Sigma(t^*)^{-1} D H) = |\det H|^2 |\det D|^2 (\det \Sigma(t^*))^{-1}.$$

Thus the fixed-ball saddle contribution tends to

$$\Phi(t^*) \prod_{j=1}^r (-\mu'_j(t_j^*))^{-1}.$$

It remains only to justify the passage from fixed R to the whole neighborhood. This is precisely the product-scale bound from Lemma 7.20: on $\mathcal{N}_\delta$,

$$\Phi(t) \mathbb{P}(Z(t) = z) dt \leq C \left(\prod_{j=1}^r a_j^{-1} \right) \exp \left[-c \sum_{j=1}^r \min(u_j^2, a_j) \right] du.$$

The functions on the right have uniformly integrable tails, since

$$\sup_{a \geq 1} \int_{\mathbb{R}} e^{-c \min(u^2, a)} \mathbf{1}_{\{|u| > R\}} \mathbf{1}_{\{|u| \leq C\sqrt{a}\}} du \rightarrow 0 \quad (R \rightarrow \infty).$$

Consequently the contribution from $\mathcal{N}_\delta \cap \{\|u\| > R\}$ is

$$o_R \left(\prod_{j=1}^r a_j^{-1} \right)$$

uniformly as $R \rightarrow \infty$. Combining this tail estimate with the negligible contribution outside $\mathcal{N}_\delta$ proves the lemma. $\square$

Proposition 7.23 (Separated terminal joint estimate). *Fix $r \geq 1$ and $\eta > 0$. Uniformly over integers*

$$1 \ll a_1 < \dots < a_r \ll n$$

satisfying

$$a_{j+1}/a_j \geq 1 + \eta \quad (1 \leq j < r),$$

one has

$$\mathbb{P}(I_{n,a_1} = \dots = I_{n,a_r} = 1) \sim \frac{e^{-r}}{a_1 \dots a_r}.$$

Proof. Let $A = \{a_1, \dots, a_r\}$. On the event that all $I_{n,a_j} = 1$ occur, the time ordering is forced. Rate a is deleted at the transition at which the number of alive particles drops from a to $a - 1$. Since $a_1 < \dots < a_r$, the deletion at level a_r occurs before the deletion at level a_{r-1} , and so on. Thus necessarily

$$T_{a_1} > T_{a_2} > \dots > T_{a_r}.$$

Condition on $T_{a_j} = t_j$ in this chamber. For $b \notin A$, define

$$Z_j(t) = \sum_{b \notin A} \mathbf{1}_{\{T_b > t_j\}}, \quad 1 \leq j \leq r.$$

At time t_j , precisely $j - 1$ of the specified clocks have larger times, namely $a_1, \dots, a_{j-1}$. Therefore the joint event is equivalent to

$$Z_j(t) = a_j - j, \quad 1 \leq j \leq r.$$

Conversely, in the chamber $t_1 > \dots > t_r$, these equations imply that each specified rate a_j has exactly $a_j - 1$ clocks with larger time, and hence all the events $I_{n,a_j} = 1$ occur. Consequently the joint probability is exactly

$$\int_{t_1 > \dots > t_r} \left(\prod_{j=1}^r a_j e^{-a_j t_j} \right) \mathbb{P}(Z_j(t) = a_j - j, 1 \leq j \leq r) dt_1 \cdots dt_r.$$

By Lemma 7.22, this integral equals

$$(1 + o(1)) \prod_{j=1}^r \frac{a_j e^{-a_j t_j^*}}{-\mu'_j(t_j^*)}.$$

Lemma 7.6 gives $a_j t_j^* \rightarrow 1$ and $-\mu'_j(t_j^*) \sim a_j^2$. Hence

$$\prod_{j=1}^r \frac{a_j e^{-a_j t_j^*}}{-\mu'_j(t_j^*)} \sim \prod_{j=1}^r \frac{a_j e^{-1}}{a_j^2} = \frac{e^{-r}}{a_1 \cdots a_r}.$$

□

7.3 Terminal point process and full factorial moments

Theorem 7.24 (Local terminal Poisson point process). *Let $b_n \rightarrow \infty$ and $b_n = o(n)$, and define*

$$\Xi_n = \sum_{a=1}^n I_{n,a} \delta_{\log(a/b_n)}.$$

For every bounded interval $J \subset \mathbb{R}$,

$$\Xi_n|_J \Rightarrow \text{PPP}(e^{-1} dx)|_J$$

in the space of locally finite counting measures on J , equipped with the vague topology. Equivalently, counts in finitely many disjoint bounded intervals converge jointly to independent Poisson random variables with means e^{-1} times their lengths.

Proof. It suffices to prove convergence of mixed factorial moments. If $\log(a/b_n) \in J$, then $a \rightarrow \infty$, $a = o(n)$, and all such a 's are within a bounded multiplicative factor. Also

$$\sum_{\log(a/b_n) \in J} \frac{1}{a} \rightarrow |J|.$$

For fixed r , expand $\mathbb{E}(\Xi_n(J))_r$ over ordered distinct r -tuples. First restrict to tuples satisfying $|\log a_i - \log a_j| > \eta$ for all $i \neq j$. After sorting such a tuple increasingly, the separation condition implies

$$\frac{a_{j+1}}{a_j} \geq e^\eta = 1 + (e^\eta - 1),$$

so Proposition 7.23 applies uniformly, with separation parameter $e^\eta - 1$, and gives

$$e^{-r} \sum_{\eta\text{-sep}}^* \frac{1}{a_1 \cdots a_r} + o(1).$$

As $n \rightarrow \infty$, this tends to e^{-r} times the Lebesgue measure of the η -separated part of J^r . Letting $\eta \downarrow 0$ gives $(e^{-1}|J|)^r$.

The excluded tuples are controlled by Lemma 7.1. Indeed,

$$\begin{aligned} & \sum_{\substack{\log(a_i/b_n) \in J \ 1 \leq i \leq r \\ |\log a_i - \log a_j| \leq \eta \text{ for some } i \neq j}} \prod_{i=1}^r \frac{1}{a_i} \\ & \longrightarrow \text{Leb}\{(x_1, \dots, x_r) \in J^r : |x_i - x_j| \leq \eta \text{ for some } i \neq j\} = O_{r,J}(\eta). \end{aligned}$$

Thus the excluded contribution vanishes as $\eta \downarrow 0$. For finitely many disjoint intervals $J_1, \dots, J_m$, the mixed factorial moment expansion is the same expansion with each selected rate colored by the interval from which it comes. The separated part factors into the corresponding products of lengths, and the same non-separated estimate removes the diagonals. These limits are the mixed factorial moments of independent Poisson variables with means $e^{-1}|J_1|, \dots, e^{-1}|J_m|$, so the corresponding count vectors converge in distribution.

It remains to pass from interval counts to convergence of point measures. The death-process bound gives, for every fixed bounded interval J ,

$$\sup_n \mathbb{E} \Xi_n(J) \leq C_J < \infty. \quad (7.30)$$

Indeed, the relevant terminal coordinates lie in a bounded multiplicative window and $\mathbb{P}(I_{n,a} = 1) \leq 2/(a+1)$. Let $f \geq 0$ be continuous with compact support in J . Choose a step function $g = \sum_{\ell=1}^m c_\ell \mathbf{1}_{J_\ell}$, where the J_ℓ are disjoint half-open intervals, such that $\|f - g\|_\infty \leq \varepsilon$. Since $x \mapsto e^{-x}$ is one-Lipschitz on $[0, \infty)$,

$$\left| \mathbb{E} e^{-\langle f, \Xi_n \rangle} - \mathbb{E} e^{-\langle g, \Xi_n \rangle} \right| \leq \varepsilon \mathbb{E} \Xi_n(J) \leq C_J \varepsilon. \quad (7.31)$$

The already proved count-vector convergence gives convergence of the Laplace functional for g . The same approximation applies to the limiting Poisson process, whose expected mass on J is finite. Letting first $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ proves convergence of Laplace functionals for every nonnegative compactly supported continuous f . These Laplace functionals determine the law in the vague topology, completing the proof. $\square$

Theorem 7.25 (Sukhatme fixed factorial moments). *For every fixed $r \geq 1$,*

$$\mathbb{E}(F_n)_r \sim (e^{-1} \log n)^r.$$

Consequently,

$$\frac{F_n}{e^{-1} \log n} \rightarrow 1$$

in probability.

Proof. Let

$$L_n = \lceil \log n \rceil, \quad U_n = \lfloor n / \log n \rfloor,$$

and put

$$G_n = \sum_{L_n \leq a \leq U_n} I_{n,a}.$$

We first prove the factorial asymptotic for G_n . Fix $\eta > 0$. In the expansion of $\mathbb{E}(G_n)_r$, restrict to ordered distinct r -tuples whose logarithms are pairwise separated by at least η . For each such ordered tuple, sort the rates increasingly before applying Proposition 7.23. Pairwise logarithmic separation implies

$$\frac{a_{j+1}}{a_j} \geq e^\eta = 1 + (e^\eta - 1),$$

so Proposition 7.23 applies uniformly with separation parameter $e^\eta - 1$. Hence the separated contribution is

$$e^{-r} \sum_{\substack{L_n \leq a_1, \dots, a_r \leq U_n \\ a_i \neq a_j, |\log a_i - \log a_j| > \eta}} \frac{1}{a_1 \cdots a_r} (1 + o(1)),$$

with the error uniform for fixed r and η . For fixed η , the excluded logarithmic diagonal neighborhood has weighted lattice sum $O_{r,\eta}((\log n)^{r-1})$. Hence the separated weighted sum is

$$\left(\sum_{L_n \leq a \leq U_n} \frac{1}{a} \right)^r + o((\log n)^r).$$

The non-separated part itself is negligible as well. By Lemma 7.1, its contribution is bounded by a constant times the same logarithmic diagonal weighted sum, hence it is $O_{r,\eta}((\log n)^{r-1}) = o((\log n)^r)$. Since

$$\sum_{L_n \leq a \leq U_n} \frac{1}{a} \sim \log n,$$

we obtain

$$\mathbb{E}(G_n)_r \sim (e^{-1} \log n)^r.$$

It remains to show that the omitted ranges do not affect the leading term. By Lemma 7.1, any factorial-moment term containing at least one index outside $[L_n, U_n]$ is bounded by a constant times the corresponding product $\prod 1/a_j$. The logarithmic mass of the omitted set is

$$\sum_{a < L_n} \frac{1}{a} + \sum_{a > U_n} \frac{1}{a} = O(\log \log n) = o(\log n).$$

Therefore the total contribution of tuples with at least one omitted index is

$$O((\log n)^{r-1} \log \log n) = o((\log n)^r).$$

This proves $\mathbb{E}(F_n)_r \sim (e^{-1} \log n)^r$.

For $r = 1, 2$, the factorial moment asymptotics imply $\mathbb{E}F_n \sim e^{-1} \log n$ and

$$\mathbb{E}F_n^2 = \mathbb{E}(F_n)_2 + \mathbb{E}F_n = (e^{-1} \log n)^2 + o((\log n)^2).$$

Consequently

$$\text{Var}(F_n) = \mathbb{E}F_n^2 - (\mathbb{E}F_n)^2 = o((\log n)^2).$$

Let $\lambda_n = e^{-1} \log n$. Since $\mathbb{E}F_n/\lambda_n \rightarrow 1$ and $\text{Var}(F_n) = o(\lambda_n^2)$, Chebyshev's inequality gives $F_n/\lambda_n \rightarrow 1$ in probability. $\square$

Theorem 7.25 proves the fixed-order factorial moments and a law of large numbers for the full Sukhatme count. The global centered fluctuation theory is completed in Section 9, where a terminal-rank Palm coupling yields total-variation approximation by a Poisson law with the exact mean.

8 Endpoint-regular fixed-point universality

We now promote the endpoint fixed-point prediction to a theorem. The statement is formulated in terminal rate coordinates. Put

$$\lambda_{n,a} = \theta_{n,n+1-a}, \quad 1 \leq a \leq n,$$

so that terminal coordinate a is label $n+1-a$. Let

$$T_{n,a} \sim \text{Exp}(\lambda_{n,a}), \quad \tau_n(a) = 1 + \sum_{b \neq a} \mathbf{1}_{\{T_{n,b} > T_{n,a}\}} = n+1 - \sigma_n(n+1-a).$$

Thus τ_n is conjugate to σ_n by the reversal permutation and has the same cycle counts. In Gumbel variables

$$Y_{n,a} = \log \lambda_{n,a} + G_a,$$

the fixed-point indicator is

$$I_{n,a} = \mathbf{1} \left\{ \sum_{b \neq a} \mathbf{1}_{\{Y_{n,b} < Y_{n,a}\}} = a-1 \right\}.$$

Indeed, label $n+1-a$ is fixed exactly when there are $a-1$ labels below it in priority order. We write

$$F_n = \sum_{a=1}^n I_{n,a} = C_1(\sigma_n)$$

inside this section.

Multiplying all rates by a common deterministic factor adds the same constant to all priorities and does not change the Luce order. The following definition therefore fixes a convenient normalization.

Definition 8.1 (Uniform endpoint regular variation). The terminal rate array $(\lambda_{n,a})_{1 \leq a \leq n}$ is *uniformly endpoint regularly varying with index $\gamma > 0$* if there exist positive constants s_n , a constant $C_0 > 0$, and a positive measurable slowly varying function L at infinity such that the normalized rates

$$\bar{\lambda}_{n,a} = s_n \lambda_{n,a}$$

satisfy the following conditions.

(ER1) For every sequence $A_n \rightarrow \infty$ with $A_n = o(n)$,

$$\sup_{1 \leq a \leq A_n} \left| \frac{\bar{\lambda}_{n,a}}{C_0(a/n)^\gamma L(n/a)} - 1 \right| \rightarrow 0.$$

(ER2) For every $\delta > 0$, there are constants $0 < c_\delta < C_\delta < \infty$ such that

$$c_\delta \leq \bar{\lambda}_{n,a} \leq C_\delta, \quad \delta n \leq a \leq n,$$

for all sufficiently large n .

(ER3) The normalized terminal rates are nondecreasing:

$$\bar{\lambda}_{n,1} \leq \bar{\lambda}_{n,2} \leq \cdots \leq \bar{\lambda}_{n,n}.$$

A common multiplication of all rates does not change the Luce order. We therefore replace $\lambda_{n,a}$ by $\bar{\lambda}_{n,a}$ throughout this section and suppress the bar. We use repeatedly the uniform convergence theorem and Potter bounds for slowly varying functions in the forms of Bingham, Goldie, and Teugels [4, Theorems 1.2.1 and 1.5.6].

For example, suppose $\lambda_{n,a} = f(1 - a/n)$, where

$$f(1 - u) \sim C_0 u^\gamma L(1/u), \quad u \downarrow 0,$$

where f is bounded above and below away from zero on every compact subinterval of $[0, 1]$, and where the function $u \mapsto f(1 - u)$ is nondecreasing on $(0, 1]$. Then the sampled array satisfies Definition 8.1. The global monotonicity in this example is intentional: eventual endpoint monotonicity alone would not imply condition (ER3) for the full triangular array.

Lemma 8.2 (The endpoint partial-sum bound is automatic). *Under Definition 8.1, there is $c_\Sigma > 0$ such that, for all sufficiently large n ,*

$$\sum_{b=1}^a \lambda_{n,b} \geq c_\Sigma a \lambda_{n,a}, \quad 1 \leq a \leq n.$$

Proof. Suppose otherwise. Then there are $n_k \rightarrow \infty$ and $a_k \in [n_k]$ such that

$$\frac{\sum_{b \leq a_k} \lambda_{n_k,b}}{a_k \lambda_{n_k,a_k}} \rightarrow 0.$$

Pass to a subsequence. If (a_k) is bounded, the ratio is at least $1/a_k$, a contradiction. Assume therefore that $a_k \rightarrow \infty$.

If $a_k/n_k \rightarrow 0$, uniform endpoint regular variation and Potter bounds give

$$\frac{1}{a_k} \sum_{b=1}^{a_k} \frac{\lambda_{n_k,b}}{\lambda_{n_k,a_k}} \rightarrow \int_0^1 x^\gamma dx = \frac{1}{\gamma + 1}.$$

For completeness, on $\varepsilon a_k \leq b \leq a_k$ this is an ordinary Riemann sum by the uniform convergence theorem for slowly varying functions. On $b < \varepsilon a_k$, Potter's bound gives $\lambda_{n_k,b}/\lambda_{n_k,a_k} \leq C(b/a_k)^{\gamma/2}$, so the omitted part is $O(\varepsilon^{1+\gamma/2})$. Letting $\varepsilon \downarrow 0$ proves the display, again a contradiction.

The remaining case has $a_k \geq \delta n_k$ for some $\delta > 0$. By monotonicity and (ER2),

$$\sum_{b=1}^{a_k} \lambda_{n_k,b} \geq \sum_{b=\lceil a_k/2 \rceil}^{a_k} \lambda_{n_k,b} \geq \frac{a_k}{2} c_{\delta/2},$$

whereas $\lambda_{n_k,a_k} \leq C_\delta$. The ratio is therefore bounded below by $c_{\delta/2}/(2C_\delta) > 0$, the final contradiction. $\square$

Set

$$A_\gamma = \Gamma(1 + 1/\gamma), \quad K_\gamma = A_\gamma^\gamma, \quad c_\gamma = \gamma K_\gamma e^{-K_\gamma}.$$

For fixed $r \geq 1$ and $\eta > 0$, define the admissible family

$$\mathfrak{E}_N(r, \eta) = \left\{ (n, a_1, \dots, a_r) : N \leq a_1 < \dots < a_r \leq n/N, \quad \frac{a_{j+1}}{a_j} \geq 1 + \eta \right\}.$$

Here and below, a statement asserted uniformly for $1 \ll a_1 < \dots < a_r \ll n$ means that its supremum over $\mathfrak{E}_N(r, \eta)$ tends to zero as $N \rightarrow \infty$. This convention makes all endpoint uniformity statements independent of a particular choice of intermediate cutoff sequences.

Lemma 8.3 (Weighted regularly varying terminal sums). *Assume Definition 8.1. Let $a = a_n \rightarrow \infty$ with $a = o(n)$. For every fixed integer $m \geq 0$, put*

$$J_{\gamma, m}(\kappa) = \int_0^\infty x^{m\gamma} e^{-\kappa x^\gamma} dx = \frac{1}{\gamma} \Gamma\left(m + \frac{1}{\gamma}\right) \kappa^{-m-1/\gamma}.$$

Then, uniformly for κ in compact subintervals of $(0, \infty)$,

$$\sum_{b=1}^n \left(\frac{\lambda_{n,b}}{\lambda_{n,a}} \right)^m \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = a J_{\gamma, m}(\kappa) (1 + o(1)). \quad (8.1)$$

The same assertion holds after deleting any fixed number of summands, uniformly in the deleted indices. In particular,

$$\sum_{b=1}^n \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = a A_\gamma \kappa^{-1/\gamma} (1 + o(1)), \quad (8.2)$$

$$\sum_{b=1}^n \frac{\lambda_{n,b}}{\lambda_{n,a}} \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = \frac{a A_\gamma}{\gamma} \kappa^{-1-1/\gamma} (1 + o(1)), \quad (8.3)$$

and, with $p_{n,b}(\kappa) = \exp(-\kappa \lambda_{n,b}/\lambda_{n,a})$,

$$\sum_{b=1}^n p_{n,b}(\kappa) (1 - p_{n,b}(\kappa)) = a A_\gamma \kappa^{-1/\gamma} (1 - 2^{-1/\gamma}) (1 + o(1)). \quad (8.4)$$

Proof. Fix a compact interval $[\kappa_-, \kappa_+] \Subset (0, \infty)$, and put

$$z_n = \frac{n}{a} \rightarrow \infty, \quad R_n = (\log(e + z_n))^{4/\gamma}, \quad B_n = \lceil a R_n \rceil.$$

Since $R_n = o(z_n)$, one has $B_n = o(n)$. Definition 8.1 therefore applies uniformly to all indices at most B_n .

Fix $0 < \varepsilon < 1 < R < \infty$. On $\varepsilon a \leq b \leq Ra$, the uniform convergence theorem for slowly varying functions gives

$$\frac{\lambda_{n,b}}{\lambda_{n,a}} = \left(\frac{b}{a} \right)^\gamma (1 + o(1))$$

uniformly. Hence the contribution of this range to the left-hand side of (8.1), divided by a , converges uniformly in $\kappa \in [\kappa_-, \kappa_+]$ to

$$\int_\varepsilon^R x^{m\gamma} e^{-\kappa x^\gamma} dx.$$

The range $b < \varepsilon a$ contains at most εa terms, and

$$\sup_{x \geq 0, \kappa \in [\kappa_-, \kappa_+]} x^m e^{-\kappa x} < \infty.$$

Its normalized contribution is therefore $O_m(\varepsilon)$.

It remains to control the upper tail without applying regular variation at a fixed positive fraction of n . Potter bounds, with exponent $\gamma/2$, imply that for all sufficiently large n , uniformly for $Ra < b \leq B_n$,

$$c \left(\frac{b}{a} \right)^{\gamma/2} \leq \frac{\lambda_{n,b}}{\lambda_{n,a}} \leq C \left(\frac{b}{a} \right)^{3\gamma/2}.$$

Consequently the normalized contribution of this range is at most

$$C_m \int_R^\infty x^{3m\gamma/2} \exp(-c\kappa_- x^{\gamma/2}) dx + o(1),$$

which tends to zero as $R \rightarrow \infty$.

For $b > B_n$, monotonicity and the preceding Potter lower bound give

$$\frac{\lambda_{n,b}}{\lambda_{n,a}} \geq \frac{\lambda_{n,B_n}}{\lambda_{n,a}} \geq cR_n^{\gamma/2} = c(\log(e + z_n))^2.$$

Therefore

$$\begin{aligned} & \frac{1}{a} \sum_{b > B_n} \left(\frac{\lambda_{n,b}}{\lambda_{n,a}} \right)^m e^{-\kappa \lambda_{n,b}/\lambda_{n,a}} \\ & \leq z_n \sup_{x \geq c(\log(e+z_n))^2} x^m e^{-\kappa x} \rightarrow 0. \end{aligned}$$

Letting first $n \rightarrow \infty$, then $\varepsilon \downarrow 0$ and $R \uparrow \infty$, proves (8.1). A fixed deleted summand is bounded uniformly by $\sup_{x \geq 0} x^m e^{-\kappa x}$, so finitely many deletions change the sum by $O(1) = o(a)$. The special cases (8.2)–(8.3) follow from $J_{\gamma,0}(\kappa) = A_\gamma \kappa^{-1/\gamma}$ and $J_{\gamma,1}(\kappa) = A_\gamma \kappa^{-1-1/\gamma}/\gamma$. Finally, (8.4) is the difference of (8.2) at κ and at 2κ . $\square$

Lemma 8.4 (Endpoint product envelope). *Assume Definition 8.1. For every finite set $A \subset [n]$,*

$$\mathbb{P}(I_{n,a} = 1 \text{ for all } a \in A) \leq C^{|A|} \prod_{a \in A} \frac{1}{a},$$

where C depends only on the constant c_Σ from Lemma 8.2.

Proof. Reveal the Luce order as a death process. When the alive set is S , the next label deleted from S is $b \in S$ with probability $\lambda_{n,b}/\sum_{c \in S} \lambda_{n,c}$. The event $I_{n,a} = 1$ says that terminal coordinate a is deleted at the transition at which the number of alive labels drops from a to $a-1$. Condition on the history before that transition. If coordinate a is already dead, the conditional probability is zero. If it is alive, the alive set has size a and contains a . Since the rates are nondecreasing, the total rate of any alive set of size a is at least $\sum_{b=1}^a \lambda_{n,b}$, and Lemma 8.2 gives

$$\frac{\lambda_{n,a}}{\sum_{c \in S} \lambda_{n,c}} \leq \frac{1}{c_\Sigma a}.$$

Order the elements of A decreasingly and condition successively before the corresponding death levels. Multiplying the conditional bounds proves the claim. $\square$

8.1 Separated endpoint fixed-point asymptotics

Fix $r \geq 1$ and $\eta > 0$, and let $(n, a_1, \dots, a_r) \in \mathfrak{E}_N(r, \eta)$. Put

$$B = \{a_1, \dots, a_r\}, \quad z_j = a_j - j, \quad 1 \leq j \leq r.$$

For $g = (g_1, \dots, g_r)$, set

$$y_j(g) = \log \lambda_{n, a_j} + g_j, \quad Z_j(g) = \sum_{b \notin B} \mathbf{1}_{\{Y_{n, b} < y_j(g)\}},$$

and write

$$M_j(g_j) = \mathbb{E} Z_j(g) = \sum_{b \notin B} \exp \left(-e^{-g_j} \frac{\lambda_{n, b}}{\lambda_{n, a_j}} \right).$$

The relevant chamber is

$$\mathcal{C}_n = \{g : y_1(g) < \dots < y_r(g)\}.$$

When $g \in \mathcal{C}_n$, put $y_0 = -\infty$, $y_{r+1} = \infty$, and define interval counts

$$W_1 = Z_1, \quad W_j = Z_j - Z_{j-1} \quad (2 \leq j \leq r).$$

Thus a nonspecified priority contributes e_j to W when it lies in $[y_{j-1}, y_j)$, and contributes 0 when it lies in $[y_r, \infty)$. The target interval vector is

$$w_1 = a_1 - 1, \quad w_j = a_j - a_{j-1} - 1 \quad (2 \leq j \leq r),$$

and ratio separation gives

$$c_\eta a_j \leq w_j \leq a_j \quad (1 \leq j \leq r) \tag{8.5}$$

for all sufficiently large N .

Lemma 8.5 (Endpoint Gumbel saddle geometry). *For each j , the equation*

$$M_j(g_j^*) = z_j$$

has a unique solution. Uniformly over $\mathfrak{E}_N(r, \eta)$, as $N \rightarrow \infty$,

$$g_j^* \longrightarrow g_\gamma := -\log K_\gamma, \quad M_j'(g_j^*) \sim \frac{a_j}{\gamma}. \tag{8.6}$$

Moreover, for every fixed $L < \infty$,

$$\sup_{|g| \leq L} |M_j''(g)| \leq C_L a_j, \quad c_L a_j \leq M_j'(g) \leq C_L a_j \tag{8.7}$$

uniformly whenever $|g - g_j^| \leq 1$. There are constants $d_\eta > 0$, $\delta_\eta > 0$, and N_0 such that, for $N \geq N_0$,*

$$y_{j+1}(g_{j+1}) - y_j(g_j) \geq d_\eta \tag{8.8}$$

for every g satisfying $|g_j - g_j^| \leq \delta_\eta$, $1 \leq j \leq r$. In particular, this entire product neighborhood lies in $\mathcal{C}_n$.*

Proof. The function M_j is continuous and strictly increasing from 0 to $n - r$, so the saddle exists uniquely. With $\kappa = e^{-g}$, Lemma 8.3, including its finite-deletion assertion, gives uniformly on compact g -sets

$$M_j(g) = a_j A_\gamma e^{g/\gamma} (1 + o(1)). \quad (8.9)$$

Differentiating term by term and applying (8.1) with $m = 1$ gives

$$\begin{aligned} M'_j(g) &= \sum_{b \notin B} e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \exp \left(-e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right) \\ &= \frac{a_j A_\gamma}{\gamma} e^{g/\gamma} (1 + o(1)). \end{aligned} \quad (8.10)$$

The saddle equation and $z_j/a_j \rightarrow 1$ imply $A_\gamma e^{g_j^*/\gamma} \rightarrow 1$, which is equivalent to $g_j^* \rightarrow -\gamma \log A_\gamma = -\log K_\gamma$. Substitution in (8.10) gives the derivative asymptotic. Since

$$M''_j(g) = \sum_{b \notin B} \left[\left(e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right)^2 - e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right] e^{-e^{-g} \lambda_{n,b}/\lambda_{n,a_j}},$$

(8.1) with $m = 1, 2$ proves the second-derivative bound. The lower and upper derivative bounds then follow from (8.10) on a compact neighborhood of the saddles.

It remains to separate the thresholds. Uniform endpoint regular variation and Potter bounds imply that there is $c_\eta > 0$ such that

$$\log \frac{\lambda_{n,a_{j+1}}}{\lambda_{n,a_j}} \geq 2c_\eta \quad (8.11)$$

for all sufficiently large N . To verify this uniformly, first restrict a_{j+1}/a_j to a fixed compact interval $[1 + \eta, R]$ and use the uniform convergence theorem; for ratios larger than R , use the Potter lower bound $\lambda_{n,a_{j+1}}/\lambda_{n,a_j} \geq c(a_{j+1}/a_j)^{\gamma/2}$, and then choose R large. Because all g_j^* converge to the same constant, the saddle threshold gaps are at least c_η for large N . Taking $\delta_\eta < c_\eta/4$ proves (8.8). $\square$

Lemma 8.6 (Endpoint interval edge blocks). *Fix $L_0 > 0$, and consider any vector $g \in \mathcal{C}_n$. Write*

$$\mathcal{I}_j(g) = [y_{j-1}(g), y_j(g)) \quad (1 \leq j \leq r), \quad \mathcal{I}_0(g) = [y_r(g), \infty).$$

Let

$$m_j(g) = \mathbb{E} W_j(g).$$

For every $c_0 > 0$, there are constants $0 < \alpha < \beta < \infty$ and $c > 0$, depending only on r, η, L_0, c_0 , with the following property. If $|g_j| \leq L_0$ and $m_j(g) \geq c_0 a_j$, then at least $c a_j$ indices $b \in [\alpha a_j, \beta a_j] \setminus B$ satisfy

$$\mathbb{P}(Y_{n,b} \in \mathcal{I}_j(g)) \geq c$$

and, for one fixed $\kappa(j) \in \{j + 1, \dots, r, 0\}$,

$$\mathbb{P}(Y_{n,b} \in \mathcal{I}_{\kappa(j)}(g)) \geq c.$$

Here $\kappa(j) = 0$ denotes the anchor interval. The selected subblocks may be chosen disjoint as j varies, after reducing c .

In the saddle neighborhood from Lemma 8.5, one may instead choose disjoint blocks of size at least ca_j on which the two adjacent atoms e_j and e_{j+1} each have probability at least c , with $e_{r+1} = 0$. All these conclusions remain valid, with modified constants, under every interval-coordinate exponential tilt whose coordinates are bounded by a fixed constant.

Proof. The contribution to $m_j(g)$ from $b < \alpha a_j$ is at most αa_j . Since membership in the j -th interval implies $Y_{n,b} < y_j$, the contribution from $b > \beta a_j$ is bounded by

$$\sum_{b > \beta a_j} \exp \left(-e^{-g_j} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right).$$

The moving-cutoff proof of Lemma 8.3, with a Potter bound on the intermediate tail and monotonicity beyond the moving cutoff, shows that this is $o_\beta(a_j)$, uniformly whenever $|g_j| \leq L_0$. Choose α small and β large. A positive multiple of a_j of the expected interval mass then comes from $b \in [\alpha a_j, \beta a_j]$. This range contains $O(a_j)$ indices, so at least ca_j of them have j -th interval probability at least c .

On the same middle range, regular variation on fixed multiplicative intervals keeps $\lambda_{n,b}/\lambda_{n,a_j}$ between two positive constants. Since g_j is bounded, the probability $\mathbb{P}(Y_{n,b} \geq y_j)$ is also bounded below. Split this upper event among the finitely many later intervals and the anchor; the pigeonhole principle produces a common $\kappa(j) > j$, or the anchor 0, on a subblock of order a_j . Disjoint subblocks are extracted greedily. Indeed, $a_{i+1}/a_i \geq 1 + \eta$ and r is fixed, so $\sum_{i < j} a_i \leq C_{r,\eta} a_j$.

In the saddle neighborhood, choose a smaller multiplicative block $[(1 - \delta)a_j, (1 + \delta)a_j]$, with δ small enough that these blocks are disjoint. On this block the centered threshold $y_j - \log \lambda_{n,b}$ remains in a compact interval, and (8.8) gives uniformly positive intervals on both sides of y_j . The standard Gumbel density is positive and continuous, so the immediate lower and upper interval probabilities are both bounded below. This gives the adjacent atoms. A bounded exponential tilt multiplies all probabilities in the finite categorical support by factors bounded above and below, so every two-atom lower bound persists. $\square$

Lemma 8.7 (Endpoint Gumbel anisotropic local CLT). *Let $\mathcal{U}_n = \{g : |g_j - g_j^*| \leq \delta_\eta, 1 \leq j \leq r\}$, where δ_η is from Lemma 8.5. Let*

$$\Sigma(g) = \text{Var } Z(g), \quad D_a = \text{diag}(a_1, \dots, a_r).$$

There are constants $0 < c < C < \infty$ such that, uniformly over $\mathfrak{E}_N(r, \eta)$ and $g \in \mathcal{U}_n$,

$$cD_a \leq \Sigma(g) \leq CD_a \tag{8.12}$$

in quadratic-form order. For every fixed $R < \infty$, uniformly for lattice points $q \in \mathbb{Z}^r$ satisfying

$$\|D_a^{-1/2}(q - M(g))\| \leq R,$$

$$\begin{aligned} \mathbb{P}(Z(g) = q) &= (2\pi)^{-r/2} (\det \Sigma(g))^{-1/2} \\ &\times \exp \left[-\frac{1}{2}(q - M(g))^T \Sigma(g)^{-1} (q - M(g)) \right] + o \left(\prod_{j=1}^r a_j^{-1/2} \right). \end{aligned} \tag{8.13}$$

The error is uniform over the displayed family. Moreover, for every fixed $\nu < 1/6$,

$$\mathbb{P}(Z(g) = q) \leq C \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j=1}^r \frac{(q_j - M_j(g_j))^2}{a_j} \right] \quad (8.14)$$

uniformly whenever $|q_j - M_j(g_j)| \leq a_j^{1/2+\nu}$ for every j .

Proof. For $b \notin B$, let

$$V_b(g) = (\mathbf{1}_{\{Y_{n,b} < y_1\}}, \dots, \mathbf{1}_{\{Y_{n,b} < y_r\}}).$$

These independent vectors have the cumulative categorical support

$$(1, \dots, 1), (0, 1, \dots, 1), \dots, (0, \dots, 0, 1), 0.$$

The adjacent blocks in Lemma 8.6 imply that, for every $u \in \mathbb{R}^r$, the j -th block contributes at least $ca_j u_j^2$ to $\text{Var}(u \cdot Z(g))$. Hence the covariance lower bound in (8.12) holds. The upper bound follows from $\text{Var} Z_j(g) = O(a_j)$, which is (8.4), and Cauchy's inequality.

The same adjacent blocks give, uniformly under bounded tilts,

$$|\mathbb{E} e^{iu \cdot (Z(g) - M(g))}| \leq \exp \left[-c \sum_{j=1}^r a_j \text{dist}(u_j, 2\pi\mathbb{Z})^2 \right]. \quad (8.15)$$

Indeed, the two adjacent cumulative atoms differ by the coordinate vector e_j , and the blocks are disjoint. Thus the lattice generated by the support differences is the full lattice $\mathbb{Z}^r$.

Fourier inversion now gives the local expansion. On the anisotropic small box $|u_j| \leq Ra_j^{-1/2}$, boundedness of the categorical increments and the covariance upper bound give

$$\log \mathbb{E} e^{iu \cdot (Z - M)} = -\frac{1}{2} u^T \Sigma(g) u + O \left((\max_j |u_j|) \sum_{j=1}^r a_j u_j^2 \right).$$

After the change of variables $u = D_a^{-1/2} v$, the error is $O_R(a_1^{-1/2}) = o(1)$. The complement of a fixed v -ball is uniformly negligible on the product scale by (8.15). This proves (8.13).

For (8.14), apply Lemma 7.17 with $W = Z(g)$ and $A_j = a_j$. The coordinatewise Fourier hypothesis under bounded tilts follows from the adjacent blocks, and the required tilted covariance upper bound follows from bounded distortion of the categorical probabilities and the argument proving (8.12). Since $|q_j - M_j(g_j)| \leq a_j^{1/2+\nu} = o(a_j)$, the truncated exponent in that lemma is exactly $(q_j - M_j(g_j))^2/a_j$. This proves (8.14). $\square$

Lemma 8.8 (Endpoint mixed cumulative–interval localization). *There is $L_0 = L_0(r, \eta, \gamma)$ such that the following holds for $L \geq L_0$. For $g \in \mathcal{C}_n$, put*

$$\begin{aligned} B_R(g) &= \{j : g_j > L\}, \\ B_L(g) &= \{j : g_j < -L\}, \\ G(g) &= \{1, \dots, r\} \setminus (B_R(g) \cup B_L(g)), \end{aligned}$$

and let

$$H_j = z_j - M_j(g_j), \quad H_0 = 0, \quad d_j = H_j - H_{j-1}.$$

On every region on which the three index sets are fixed,

$$\begin{aligned} \mathbb{P}(Z(g) = z) &\leq C_L \prod_{j=1}^r a_j^{-1/2} \exp \left[-c \sum_{j \in B_R} |H_j| - c \sum_{j \in B_L} a_j \right] \\ &\times \exp \left[-c_L \sum_{j \in G} \min \left\{ \frac{H_j^2}{a_j}, a_j \right\} \right]. \end{aligned} \quad (8.16)$$

Consequently,

$$\begin{aligned} \mathbb{P}(Z(g) = z) &\leq C_L \prod_{j=1}^r a_j^{-1/2} \exp \left[-ce^{L/(2\gamma)} \sum_{j \in B_R} a_j - c \sum_{j \in B_L} a_j \right] \\ &\times \exp \left[-c_L \sum_{j \in G} a_j \min \{ (g_j - g_j^*)^2, 1 \} \right]. \end{aligned} \quad (8.17)$$

Here $c > 0$ is independent of L , while C_L, c_L may depend on L, r, η, γ .

Proof. Fix the offender sets and abbreviate them by B_R, B_L, G . From (8.9), uniformly for large L ,

$$\begin{aligned} j \in B_R : \quad &M_j(g_j) \geq ce^{L/\gamma} a_j, \quad H_j < 0, \\ j \in B_L : \quad &M_j(g_j) \leq Ce^{-L/\gamma} a_j, \quad 0 < H_j \leq a_j. \end{aligned} \quad (8.18)$$

For $j \in G$, $g_j \in [-L, L]$, so $m_j(g) = O_L(a_j)$ and $|d_j| \leq M_L a_j$.

Let

$$K_g(\theta) = \log \mathbb{E} \exp \{ \theta \cdot (W(g) - m(g)) \}$$

be the centered interval-coordinate cumulant. For a cumulative-coordinate vector α , define

$$(\mathcal{T}\alpha)_k = \sum_{j \geq k} \alpha_j.$$

Because $Z_j = W_1 + \cdots + W_j$,

$$(\mathcal{T}\alpha) \cdot d = \alpha \cdot H. \quad (8.19)$$

Choose $\delta > 0$ so small that 3δ is admissible in Lemma 7.15. Set

$$\alpha_j^- = -\delta \mathbf{1}_{\{j \in B_R\}}, \quad \alpha_j^+ = \delta \mathbf{1}_{\{j \in B_L\}}.$$

For the nonoffending interval coordinates, fix $0 < \varepsilon < c_\eta/4$, where c_η is from (8.5), and put

$$G_0 = \{j \in G : |d_j| \leq \varepsilon a_j\}, \quad G_1 = G \setminus G_0.$$

For a sufficiently small constant $\delta_L > 0$, define

$$\lambda_j = \begin{cases} \delta_L d_j / a_j, & j \in G_0, \\ \delta_L \operatorname{sgn}(d_j), & j \in G_1, \\ 0, & j \notin G. \end{cases}$$

Use the single total interval-coordinate tilt

$$\Theta = \mathcal{T}\alpha^- + \mathcal{T}\alpha^+ + \lambda.$$

Generalized Hölder inequality gives

$$K_g(\Theta) \leq \frac{1}{3}K_g(3\mathcal{T}\alpha^-) + \frac{1}{3}K_g(3\mathcal{T}\alpha^+) + \frac{1}{3}K_g(3\lambda), \quad (8.20)$$

and therefore, by (8.19),

$$\begin{aligned} -\Theta \cdot d + K_g(\Theta) &\leq \frac{1}{3}\{-3\alpha^- \cdot H + K_g(3\mathcal{T}\alpha^-)\} \\ &\quad + \frac{1}{3}\{-3\alpha^+ \cdot H + K_g(3\mathcal{T}\alpha^+)\} \\ &\quad + \frac{1}{3}\{-3\lambda \cdot d + K_g(3\lambda)\}. \end{aligned} \quad (8.21)$$

The proof of Lemma 7.15 uses only independence, nested indicator counts, and the bound that one summand contributes at most r to the cumulative linear form. It therefore applies verbatim to the present lower-priority counts. For B_R , (8.18) gives $z_j \leq Ce^{-L/\gamma}M_j(g_j)$. The lower-tail part of Lemma 7.15, applied with parameter 3δ , yields

$$-3\alpha^- \cdot H + K_g(3\mathcal{T}\alpha^-) \leq -c \sum_{j \in B_R} M_j(g_j) \leq -c \sum_{j \in B_R} |H_j|. \quad (8.22)$$

For B_L , one has $M_j(g_j) \leq Ce^{-L/\gamma}z_j$. The upper-tail part of the same lemma gives

$$-3\alpha^+ \cdot H + K_g(3\mathcal{T}\alpha^+) \leq -c \sum_{j \in B_L} z_j \leq -c \sum_{j \in B_L} a_j. \quad (8.23)$$

Under every bounded interval tilt ξ , the categorical support implies

$$\text{Var}_\xi(v \cdot W(g)) \leq C_L \sum_{j \in G} a_j v_j^2$$

for vectors supported on G . Taylor's formula for the cumulant and a sufficiently small choice of δ_L therefore give

$$-3\lambda \cdot d + K_g(3\lambda) \leq -c_L \sum_{j \in G} \min \left\{ \frac{d_j^2}{a_j}, a_j \right\}. \quad (8.24)$$

All coordinates of the total tilt Θ are bounded. For $j \in G_0$, $m_j(g) \geq w_j - \varepsilon a_j \geq (c_\eta/2)a_j$. Lemma 8.6 supplies forward-edge blocks of order a_j , which may be chosen disjoint. After projection onto G_0 , a later atom whose coordinate is not retained becomes the anchor atom. Lemma 7.12 therefore gives under the single tilted law

$$\mathbb{P}_\Theta(W_{G_0} = w_{G_0}) \leq C_L \prod_{j \in G_0} a_j^{-1/2}. \quad (8.25)$$

The exact change-of-measure identity, together with (8.21)–(8.25), now yields the same bound as (8.16), but with interval residuals d_j in place of cumulative residuals on G , and initially only the

local factors for G_0 . Every index in G_1 , B_R , or B_L carries a fixed exponential loss of order a_j , which absorbs its missing factor $a_j^{-1/2}$.

It remains to compare interval and cumulative residuals without discarding the boundary values. Apply Lemma 7.10 with $A_j = a_j$, with its set B_- equal to the present B_R , and with its set B_+ equal to the present B_L . Its assumptions hold by $|d_j| \leq M_L a_j$ on G and $|H_j| \leq a_j$ on B_L . Reserving a fixed fraction of the offender costs in (8.22)–(8.23) therefore gives (8.16).

Finally, on G , the derivative bounds in (8.7), monotonicity, and compactness imply

$$|H_j| \geq c_L a_j \min\{|g_j - g_j^*|, 1\}. \quad (8.26)$$

On B_R , (8.18) gives $|H_j| \geq ce^{L/\gamma} a_j$, which may be weakened to $ce^{L/(2\gamma)} a_j$. These observations convert (8.16) into (8.17). $\square$

Lemma 8.9 (Endpoint Gumbel product localization). *Uniformly over $\mathfrak{E}_N(r, \eta)$, the following assertions hold.*

- (i) *For every fixed compact cube $K \subset \mathbb{R}^r$ containing all saddle vectors for large N , there are $C_K, c_K > 0$ such that, throughout $K \cap \mathcal{C}_n$,*

$$\mathbb{P}(Z(g) = z) \leq C_K \prod_{j=1}^r a_j^{-1/2} \exp \left[-c_K \sum_{j=1}^r a_j \min\{(g_j - g_j^*)^2, 1\} \right]. \quad (8.27)$$

- (ii) *With $\varphi(g) = e^{-g} e^{-e^{-g}}$,*

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \sup_{\mathfrak{E}_N(r, \eta)} \frac{\int_{\mathcal{C}_n \cap \{g \notin [-L, L]^r\}} \prod_{j=1}^r \varphi(g_j) \mathbb{P}(Z(g) = z) dg}{\prod_{j=1}^r a_j^{-1}} = 0. \quad (8.28)$$

Proof. Choose L so that $K \subset [-L, L]^r$. With no offenders, the proof of Lemma 8.8 gives (8.27); equivalently, apply (8.16) with $G = \{1, \dots, r\}$.

For the tail, partition the integration region according to the finitely many sets B_R, B_L, G , and use (8.17). Every nonoffending coordinate contributes a factor $O(a_j^{-1/2})$ after integration on its Gaussian scale, in addition to the prefactor $a_j^{-1/2}$. For a left offender, e^{-ca_j} absorbs the missing local factor and

$$\int_{-\infty}^{-L} \varphi(g) dg = e^{-e^L}.$$

For a right offender, $e^{-ce^{L/(2\gamma)} a_j}$ absorbs the missing local factor and

$$\int_L^{\infty} \varphi(g) dg = 1 - e^{-e^{-L}} \leq e^{-L}.$$

At least one offender is present. Summing over the finitely many offender sets proves (8.28). $\square$

Lemma 8.10 (Endpoint covariance stability on the saddle scale). *Let*

$$H_a = \text{diag}(a_1^{-1/2}, \dots, a_r^{-1/2}), \quad \Sigma_* = \Sigma(g^*).$$

For every fixed $R < \infty$,

$$\sup_{\|u\| \leq R} \left\| D_a^{-1/2} [\Sigma(g^* + H_a u) - \Sigma_*] D_a^{-1/2} \right\|_{\text{op}} \longrightarrow 0 \quad (8.29)$$

uniformly over $\mathfrak{E}_N(r, \eta)$. Consequently, uniformly on the same sets,

$$\frac{\det \Sigma(g^* + H_a u)}{\det \Sigma_*} \longrightarrow 1 \quad (8.30)$$

and

$$\left\| D_a^{1/2} [\Sigma(g^* + H_a u)^{-1} - \Sigma_*^{-1}] D_a^{1/2} \right\|_{\text{op}} \longrightarrow 0. \quad (8.31)$$

Proof. For $j \leq k$, put

$$p_{b,j}(g_j) = \exp \left(-e^{-g_j} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right).$$

Since the threshold events are nested in the chamber,

$$\Sigma_{jk}(g) = \sum_{b \notin B} p_{b,j}(g_j)(1 - p_{b,k}(g_k)). \quad (8.32)$$

Lemma 8.3, with $m = 1$, gives uniformly on a fixed saddle neighborhood

$$|\partial_{g_j} \Sigma_{jk}(g)| \leq C a_j, \quad |\partial_{g_k} \Sigma_{jk}(g)| \leq C a_k. \quad (8.33)$$

The same estimate, with the evident interpretation, holds on the diagonal. Hence, for $\|u\| \leq R$,

$$|\Sigma_{jk}(g^* + H_a u) - \Sigma_{jk}(g^*)| \leq C_R(\sqrt{a_j} + \sqrt{a_k}).$$

After division by $\sqrt{a_j a_k}$, this is at most $C_R(a_j^{-1/2} + a_k^{-1/2}) \leq 2C_R a_1^{-1/2}$. Since r is fixed, (8.29) follows. By (8.12), the normalized matrices $D_a^{-1/2} \Sigma(g) D_a^{-1/2}$ remain in a compact subset of the positive definite matrices. Continuity of determinant and inversion on this set gives (8.30) and (8.31). $\square$

Proposition 8.11 (Separated endpoint local estimate). *Assume Definition 8.1. Fix $r \geq 1$ and $\eta > 0$. Then*

$$\lim_{N \rightarrow \infty} \sup_{(n,a) \in \mathfrak{E}_N(r, \eta)} \left| \frac{\mathbb{P}(I_{n,a_1} = \dots = I_{n,a_r} = 1)}{c_\gamma^r / (a_1 \dots a_r)} - 1 \right| = 0. \quad (8.34)$$

Proof. On the joint fixed-point event, the specified priorities are necessarily ordered as $y_1 < \dots < y_r$. Conditional on their centered Gumbels g in this chamber, exactly $j - 1$ specified priorities lie below y_j , so the event is equivalent to $Z_j(g) = a_j - j$, $1 \leq j \leq r$. Therefore

$$\begin{aligned} & \mathbb{P}(I_{n,a_1} = \dots = I_{n,a_r} = 1) \\ &= \int_{\mathcal{C}_n} \prod_{j=1}^r \varphi(g_j) \mathbb{P}(Z(g) = z) dg. \end{aligned} \quad (8.35)$$

Lemma 8.5 puts a fixed product neighborhood of g^* inside the chamber. Lemma 8.9 removes the noncompact region on the scale $\prod_j a_j^{-1}$, and its compact bound makes the part outside any fixed saddle neighborhood exponentially negligible on that scale.

Set

$$D_n = \text{diag}(M'_1(g_1^*), \dots, M'_r(g_r^*)), \quad g = g^* + H_a u.$$

For bounded u , Taylor's theorem and (8.7) give

$$M(g) - z = D_n H_a u + r_n(u), \quad \|D_n^{-1/2} r_n(u)\| \longrightarrow 0 \quad (8.36)$$

uniformly on fixed u -balls. Indeed, the j -th remainder is $O_R(1)$, hence is $o(\sqrt{a_j})$. The specified density satisfies

$$\prod_{j=1}^r \varphi(g_j^* + u_j/\sqrt{a_j}) = \prod_{j=1}^r \varphi(g_j^*) (1 + o(1)) \quad (8.37)$$

uniformly on fixed balls.

Apply Lemma 8.7, then use Lemma 8.10 to replace $\Sigma(g)$ by Σ_* . On every fixed u -ball, the integrand in (8.35), including the Jacobian $|\det H_a|$, is

$$\begin{aligned} & (1 + o(1)) \prod_{j=1}^r \varphi(g_j^*) (2\pi)^{-r/2} (\det \Sigma_*)^{-1/2} |\det H_a| \\ & \times \exp \left[-\frac{1}{2} u^T H_a D_n \Sigma_*^{-1} D_n H_a u \right] du. \end{aligned} \quad (8.38)$$

The matrices in the exponent are uniformly positive definite and uniformly bounded. The compact localization estimate (8.27), after the change of variables, provides the integrable majorant

$$C \left(\prod_{j=1}^r a_j^{-1} \right) \exp \left[-c \sum_{j=1}^r \min\{u_j^2, a_j\} \right].$$

Thus the fixed-ball limit may be extended uniformly to $\mathbb{R}^r$. The Gaussian determinant cancels exactly:

$$\begin{aligned} & (2\pi)^{-r/2} (\det \Sigma_*)^{-1/2} |\det H_a| \int_{\mathbb{R}^r} e^{-\frac{1}{2} u^T H_a D_n \Sigma_*^{-1} D_n H_a u} du \\ & = |\det D_n|^{-1}. \end{aligned} \quad (8.39)$$

Consequently,

$$\mathbb{P}(I_{n,a_1} = \dots = I_{n,a_r} = 1) = (1 + o(1)) \prod_{j=1}^r \frac{\varphi(g_j^*)}{M'_j(g_j^*)}. \quad (8.40)$$

By (8.6),

$$\varphi(g_j^*) \longrightarrow K_\gamma e^{-K_\gamma}, \quad M'_j(g_j^*) \sim a_j/\gamma.$$

Each factor in (8.40) is therefore asymptotic to c_γ/a_j , uniformly over the admissible family. This proves (8.34). $\square$

Theorem 8.12 (Endpoint fixed-point universality). *Assume the terminal rate array is uniformly endpoint regularly varying with index $\gamma > 0$ in the sense of Definition 8.1. Let*

$$\Xi_n = \sum_{a=1}^n I_{n,a} \delta_{\log(a/b_n)},$$

where $b_n \rightarrow \infty$ and $b_n = o(n)$. For every bounded interval $J \subset \mathbb{R}$, the restriction $\Xi_n|_J$, regarded as a random locally finite counting measure with the vague topology, satisfies

$$\Xi_n|_J \Rightarrow \text{PPP}(c_\gamma dx)|_J. \quad (8.41)$$

Equivalently, counts in finitely many disjoint bounded intervals converge jointly to independent Poisson random variables with means c_γ times their lengths. Moreover, for every fixed $r \geq 1$,

$$\mathbb{E}(F_n)_r \sim (c_\gamma \log n)^r. \quad (8.42)$$

In particular,

$$\mathbb{E}C_1(\sigma_n) \sim c_\gamma \log n, \quad \frac{C_1(\sigma_n)}{c_\gamma \log n} \longrightarrow 1 \quad \text{in probability.} \quad (8.43)$$

Proof. We first prove the local point-process assertion. For every bounded interval $I = (u, v]$, the elementary logarithmic Riemann sum gives

$$\sum_{\log(a/b_n) \in I} \frac{1}{a} \longrightarrow v - u = |I|. \quad (8.44)$$

All coordinates in this sum tend to infinity and are $o(n)$. Fix disjoint bounded intervals $I_1, \dots, I_m$ and nonnegative integers $k_1, \dots, k_m$, and put $k = \sum_\ell k_\ell$. Expand

$$\mathbb{E} \prod_{\ell=1}^m (\Xi_n(I_\ell))_{k_\ell}$$

over ordered distinct terminal coordinates, colored according to their intervals. Restrict first to tuples whose logarithms are pairwise more than η apart. After sorting the coordinates, Proposition 8.11 applies with ratio-separation parameter $e^\eta - 1$, uniformly over the entire restricted sum. Its contribution is therefore

$$c_\gamma^k (1 + o(1)) \sum_{\eta\text{-separated}}^* \frac{1}{a_1 \cdots a_k}. \quad (8.45)$$

The logarithmic diagonal neighborhood has negligible harmonic weight as $\eta \downarrow 0$. Indeed, uniformly for a in a fixed multiplicative window,

$$\sum_{b: |\log b - \log a| \leq \eta} \frac{1}{b} \leq C\eta + o(1), \quad (8.46)$$

and hence a union bound over pairs gives an $O_{k, I_1, \dots, I_m}(\eta) + o(1)$ bound for the weighted sum of excluded tuples. Lemma 8.4 gives the same bound, up to a fixed constant, for their probability contribution. Using (8.44) in (8.45) and then letting $\eta \downarrow 0$, we obtain

$$\mathbb{E} \prod_{\ell=1}^m (\Xi_n(I_\ell))_{k_\ell} \longrightarrow \prod_{\ell=1}^m (c_\gamma |I_\ell|)^{k_\ell}. \quad (8.47)$$

These are the mixed factorial moments of independent Poisson variables, so the interval-count vectors converge to the asserted independent Poisson vector.

We now upgrade count convergence to convergence of point measures. By the one-point case of Lemma 8.4,

$$\sup_n \mathbb{E} \Xi_n(J) \leq C \sup_n \sum_{\log(a/b_n) \in J} \frac{1}{a} < \infty \quad (8.48)$$

for every bounded J , after ignoring finitely many initial n . Let $f \geq 0$ be continuous with compact support in J . Choose a finite partition of J into half-open intervals and a step function f_π , constant on the partition cells, with $\|f - f_\pi\|_\infty \rightarrow 0$ as the mesh tends to zero. Since $|e^{-x} - e^{-y}| \leq |x - y|$ for $x, y \geq 0$,

$$\left| \mathbb{E} e^{-\langle \Xi_n, f \rangle} - \mathbb{E} e^{-\langle \Xi_n, f_\pi \rangle} \right| \leq \|f - f_\pi\|_\infty \mathbb{E} \Xi_n(J), \quad (8.49)$$

uniformly in n . For fixed π , the interval-count convergence just proved gives

$$\mathbb{E} e^{-\langle \Xi_n, f_\pi \rangle} \longrightarrow \exp \left[-c_\gamma \int_J (1 - e^{-f_\pi(x)}) dx \right].$$

Letting the mesh tend to zero in view of (8.49) yields

$$\mathbb{E} e^{-\langle \Xi_n, f \rangle} \longrightarrow \exp \left[-c_\gamma \int_J (1 - e^{-f(x)}) dx \right], \quad (8.50)$$

the Laplace functional of $\text{PPP}(c_\gamma dx)|_J$. Laplace functionals on nonnegative compactly supported continuous functions determine the law of a locally finite counting measure and are convergence determining for the vague topology. This proves (8.41); the uniform first-moment bound (8.48) also gives tightness directly.

It remains to prove the full factorial moments. Put

$$L_n = \lceil \log n \rceil, \quad U_n = \lfloor n / \log n \rfloor, \quad H_n = \sum_{L_n \leq a \leq U_n} \frac{1}{a}.$$

Then $H_n \sim \log n$. Fix once and for all a positive separation η_0 . The harmonic weight of ordered r -tuples in $[L_n, U_n]$ containing a pair with $|\log a_i - \log a_j| \leq \eta_0$ is $O_r(H_n^{r-1})$. To see this, fix one coordinate a_i , use

$$\sum_{b: |\log b - \log a_i| \leq \eta_0} \frac{1}{b} \leq C_{\eta_0},$$

and sum the remaining $r - 2$ coordinates freely. Repeated coordinates are included in this bound. Therefore the separated weighted sum is

$$H_n^r + O_r(H_n^{r-1}). \quad (8.51)$$

On the separated tuples, Proposition 8.11 applies uniformly and gives the contribution

$$c_\gamma^r H_n^r + o(H_n^r).$$

The nonseparated probability contribution is $O_r(H_n^{r-1})$ by Lemma 8.4 and the preceding harmonic estimate. Thus, for $G_n = \sum_{L_n \leq a \leq U_n} I_{n,a}$,

$$\mathbb{E}(G_n)_r \sim (c_\gamma \log n)^r. \quad (8.52)$$

Finally, the harmonic mass outside $[L_n, U_n]$ is

$$H_n^{\text{out}} := \sum_{a < L_n} \frac{1}{a} + \sum_{a > U_n} \frac{1}{a} = O(\log \log n).$$

By the product envelope, the total contribution to $\mathbb{E}(F_n)_r$ from ordered tuples containing at least one omitted coordinate is at most

$$C_r H_n^{\text{out}} \left(\sum_{a=1}^n \frac{1}{a} \right)^{r-1} = O_r((\log n)^{r-1} \log \log n) = o((\log n)^r).$$

Combining this with (8.52) proves (8.42).

Taking $r = 1, 2$ gives $\mathbb{E}F_n \sim c_\gamma \log n$ and $\text{Var}(F_n) = o((\log n)^2)$. Chebyshev's inequality yields (8.43). $\square$

8.2 Further endpoint limits

The same regular-variation calculation also gives two boundary limits that are useful for interpreting the endpoint process. The first is a mesoscopic one-coordinate displacement law; the second is a microscopic infinite random permutation.

Theorem 8.13 (Endpoint displacement law). *Assume Definition 8.1. Let $a_{n,1}, \dots, a_{n,m}$ be distinct terminal coordinates, where m is fixed, $a_{n,j} \rightarrow \infty$, and $a_{n,j} = o(n)$ for every j . Then*

$$\left(\frac{\tau_n(a_{n,1})}{a_{n,1}}, \dots, \frac{\tau_n(a_{n,m})}{a_{n,m}} \right) \Rightarrow \left(A_\gamma E_1^{-1/\gamma}, \dots, A_\gamma E_m^{-1/\gamma} \right),$$

where $E_1, \dots, E_m$ are independent $\text{Exp}(1)$ variables and $A_\gamma = \Gamma(1 + 1/\gamma)$. Equivalently,

$$\log \frac{\tau_n(a_{n,j})}{a_{n,j}} \Rightarrow X_\gamma,$$

where X_γ has density

$$k_\gamma(x) = \gamma K_\gamma e^{-\gamma x} \exp\{-K_\gamma e^{-\gamma x}\}, \quad K_\gamma = A_\gamma^\gamma.$$

Proof. Write

$$Y_{n,a} = \log \lambda_{n,a} + G_a,$$

with independent standard Gumbels. Conditional on $G_a = g$,

$$\tau_n(a) - 1 = \sum_{b \neq a} \mathbf{1}_{\{Y_{n,b} < Y_{n,a}\}}$$

is a sum of independent Bernoulli variables with mean

$$M_{n,a}(g) = \sum_{b \neq a} \exp\left(-e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a}}\right).$$

Lemma 8.3 gives, uniformly for g in compact sets,

$$\frac{M_{n,a}(g)}{a} \longrightarrow A_\gamma e^{g/\gamma}.$$

The conditional variance is at most the conditional mean, hence is $O(a)$ on compact g -sets. Chebyshev's inequality therefore gives, uniformly on such sets,

$$\frac{\tau_n(a)}{a} - A_\gamma e^{G_a/\gamma} \longrightarrow 0 \quad \text{in probability.}$$

Truncating the Gumbel variables to a fixed compact set and then letting the truncation grow proves the one-coordinate assertion. For finitely many specified coordinates, delete them from the Bernoulli sums. Lemma 8.3 is unchanged after finitely many deletions, and a union bound over the conditional Chebyshev estimates proves the joint convergence. The limiting variables are independent because the specified Gumbels are independent.

Finally, if $E = e^{-G} \sim \text{Exp}(1)$, then $A_\gamma e^{G/\gamma} = A_\gamma E^{-1/\gamma}$. The change of variables $x = \log A_\gamma + G/\gamma$ gives the displayed density k_γ . $\square$

8.3 Mesoscopic endpoint tangent measure

The microscopic limit in Theorem 8.19 is an infinite random permutation. On every mesoscopic scale containing $b_n \rightarrow \infty$ terminal coordinates, the correct object is instead a dense normalized graph measure. The unnormalized graph cannot have a locally finite point-process limit, because a compact source window contains order b_n points.

Define

$$\mathcal{K}_\gamma(x, y) = \frac{1}{y} k_\gamma\left(\log \frac{y}{x}\right) = \gamma K_\gamma \frac{x^\gamma}{y^{\gamma+1}} \exp\left\{-K_\gamma \left(\frac{x}{y}\right)^\gamma\right\}, \quad x, y > 0.$$

Theorem 8.14 (Mesoscopic endpoint tangent measure). *Assume Definition 8.1. Let $b_n \rightarrow \infty$ with $b_n = o(n)$, and put*

$$\mathcal{M}_n^{(b_n)} = \frac{1}{b_n} \sum_{a=1}^n \delta_{(a/b_n, \tau_n(a)/b_n)}.$$

Each $\mathcal{M}_n^{(b_n)}$ is a Radon measure on compact subsets of $(0, \infty)^2$, and the density $\mathcal{K}_\gamma$ is locally integrable there. Then, vaguely in probability on $(0, \infty)^2$,

$$\mathcal{M}_n^{(b_n)} \longrightarrow \mathcal{K}_\gamma(x, y) dx dy.$$

More precisely, for every $\varphi \in C_c((0, \infty)^2)$,

$$\mathbb{E} \left| \langle \mathcal{M}_n^{(b_n)}, \varphi \rangle - \int_0^\infty \int_0^\infty \varphi(x, y) \mathcal{K}_\gamma(x, y) dy dx \right| \longrightarrow 0.$$

The limiting kernel is bistochastic with respect to Lebesgue measure.

Lemma 8.15 (Uniform mesoscopic Laplace sums). *Fix $0 < x_- < x_+ < \infty$ and $0 < u_- < u_+ < \infty$. Uniformly for $x_- b_n \leq a \leq x_+ b_n$ and $u \in [u_-, u_+]$,*

$$\frac{1}{b_n} \sum_{c=1}^n \exp\left\{-u \frac{\lambda_{n,c}}{\lambda_{n,a}}\right\} = \frac{a}{b_n} A_\gamma u^{-1/\gamma} + o(1).$$

For every fixed $u_- > 0$, the left-hand side is uniformly bounded for all $u \geq u_-$.

Proof. Put $X_n = n/b_n \rightarrow \infty$ and choose

$$R_n = (\log(e + X_n))^{4/\gamma}, \quad B_n = \lceil 2x_+ b_n R_n \rceil.$$

Then $B_n = o(n)$. Uniform endpoint regular variation applies to all $a \in [x_- b_n, x_+ b_n]$ and $c \leq B_n$. On every fixed multiplicative interval $\delta a \leq c \leq Ra$, the uniform convergence theorem for slowly varying functions gives

$$\lambda_{n,c}/\lambda_{n,a} = (c/a)^\gamma(1 + o(1))$$

uniformly. Hence

$$\frac{1}{b_n} \sum_{\delta a \leq c \leq Ra} e^{-u\lambda_{n,c}/\lambda_{n,a}} = \frac{a}{b_n} \int_\delta^R e^{-uv^\gamma} dv + o(1).$$

The range $c < \delta a$ contributes at most $x_+ \delta$. Potter's bound with exponent $\gamma/2$ gives, for $Ra < c \leq B_n$,

$$\lambda_{n,c}/\lambda_{n,a} \geq c(c/a)^{\gamma/2},$$

so the normalized contribution is bounded by

$$C \frac{a}{b_n} \int_R^\infty e^{-cu - v^{\gamma/2}} dv.$$

For $c > B_n$, monotonicity and Potter's bound give

$$\lambda_{n,c}/\lambda_{n,a} \geq cR_n^{\gamma/2},$$

and therefore

$$\frac{1}{b_n} \sum_{c > B_n} e^{-u\lambda_{n,c}/\lambda_{n,a}} \leq X_n \exp\{-cu - R_n^{\gamma/2}\} = o(1).$$

Let first $n \rightarrow \infty$, then $\delta \downarrow 0$ and $R \uparrow \infty$. Since

$$\int_0^\infty e^{-uv^\gamma} dv = A_\gamma u^{-1/\gamma},$$

the first assertion follows. Monotonicity in u gives the final bound. $\square$

Let $E_{n,a} = \lambda_{n,a} T_{n,a}$. These variables are independent and standard exponential. Conditional on $E_{n,a} = u$,

$$\tau_n(a) = 1 + \sum_{c \neq a} \xi_{n,a,c}(u), \quad \mathbb{P}(\xi_{n,a,c}(u) = 1) = e^{-u\lambda_{n,c}/\lambda_{n,a}},$$

with independent summands. Put

$$r_{n,a}(u) = 1 + \sum_{c \neq a} e^{-u\lambda_{n,c}/\lambda_{n,a}}.$$

Lemma 8.16 (Conditional rank self-averaging). *Fix $0 < x_- < x_+ < \infty$. For every $\varepsilon, \delta > 0$,*

$$\sup_{x_- b_n \leq a \leq x_+ b_n} \mathbb{P}(|\tau_n(a) - r_{n,a}(E_{n,a})| > \delta b_n) \leq \varepsilon + \frac{C_\varepsilon}{\delta^2 b_n}$$

for all large n . Moreover, uniformly for $a/b_n \in [x_-, x_+]$ and u in a compact subset of $(0, \infty)$,

$$\frac{r_{n,a}(u)}{b_n} \longrightarrow \frac{a}{b_n} A_\gamma u^{-1/\gamma}.$$

Proof. Conditionally on $E_{n,a} = u$,

$$\text{Var}(\tau_n(a) \mid E_{n,a} = u) \leq r_{n,a}(u).$$

On $\{E_{n,a} \geq \varepsilon\}$, Lemma 8.15 and monotonicity in u give $r_{n,a}(E_{n,a}) \leq C_\varepsilon b_n$. Conditional Chebyshev therefore bounds the deviation probability on this event by $C_\varepsilon/(\delta^2 b_n)$. The complementary event has probability at most ε . The second assertion follows immediately from Lemma 8.15; omitting the $c = a$ summand and adding the leading one changes the sum by $O(1)$. $\square$

Proof of Theorem 8.14. Fix $\varphi \in C_c((0, \infty)^2)$, and choose $0 < x_- < x_+ < \infty$ containing the first-coordinate projection of its support. Set

$$Z_n(\varphi) = \frac{1}{b_n} \sum_a \varphi(a/b_n, \tau_n(a)/b_n)$$

and

$$\widehat{Z}_n(\varphi) = \frac{1}{b_n} \sum_a \varphi(a/b_n, r_{n,a}(E_{n,a})/b_n).$$

Only $O(b_n)$ summands are nonzero. Uniform continuity of φ and Lemma 8.16 give

$$\mathbb{E}|Z_n(\varphi) - \widehat{Z}_n(\varphi)| \longrightarrow 0.$$

Indeed, split according as the two second coordinates differ by at most δ , use the modulus of continuity on that event, and then let $n \rightarrow \infty$, $\delta \downarrow 0$, and $\varepsilon \downarrow 0$.

The summands in $\widehat{Z}_n(\varphi)$ are independent functions of the independent variables $E_{n,a}$. Hence

$$\text{Var}(\widehat{Z}_n(\varphi)) \leq C_\varphi/b_n \longrightarrow 0.$$

It remains to compute the expectations. If $U \sim \text{Exp}(1)$, Lemma 8.16, uniformly on compact u -sets, gives

$$\mathbb{E}\widehat{Z}_n(\varphi) \longrightarrow \int_0^\infty \int_0^\infty \varphi(x, xA_\gamma u^{-1/\gamma}) e^{-u} du dx.$$

To justify the passage, truncate u to $[\varepsilon, M]$, use uniform convergence and an ordinary Riemann sum, and then use the exponential mass of $(0, \varepsilon) \cup (M, \infty)$.

For fixed x , make the change of variables $y = xA_\gamma u^{-1/\gamma}$. Then $u = K_\gamma(x/y)^\gamma$ and

$$e^{-u} \left| \frac{du}{dy} \right| = \mathcal{K}_\gamma(x, y).$$

This proves the L^1 convergence of test-function integrals. A countable convergence-determining subset of $C_c((0, \infty)^2)$ then gives vague convergence in probability.

Finally, the first marginal identity follows from the same change of variables: $\mathcal{K}_\gamma(x, \cdot)$ is the density of $xA_\gamma U^{-1/\gamma}$. For the second marginal, put $v = K_\gamma(x/y)^\gamma$ to obtain

$$\int_0^\infty \mathcal{K}_\gamma(x, y) dx = K_\gamma^{-1/\gamma} \int_0^\infty v^{1/\gamma} e^{-v} dv = 1.$$

Thus the kernel is bistochastic. $\square$

Corollary 8.17 (Rectangle law). *For bounded intervals $I, J \subseteq (0, \infty)$,*

$$\frac{1}{b_n} \# \{a : a/b_n \in I, \tau_n(a)/b_n \in J\} \xrightarrow{\mathbb{P}} \int_I \int_J \mathcal{K}_\gamma(x, y) dy dx.$$

Proof. The boundary of $I \times J$ has zero mass under the continuous limiting density, so the assertion follows from vague convergence and approximation by compactly supported continuous functions. $\square$

Proposition 8.18 (Multiplicative semigroup and diagonal traces). *For every $r \geq 1$,*

$$\mathcal{K}_\gamma^r(x, y) = \frac{1}{y} k_\gamma^{*r} \left(\log \frac{y}{x} \right).$$

In particular,

$$\mathcal{K}_\gamma^r(x, x) = \frac{1}{x} k_\gamma^{*r}(0).$$

Proof. For $r = 2$, substitute $z = xe^w$ in the composition integral:

$$\int_0^\infty \mathcal{K}_\gamma(x, z) \mathcal{K}_\gamma(z, y) dz = \frac{1}{y} \int_{\mathbb{R}} k_\gamma(w) k_\gamma \left(\log \frac{y}{x} - w \right) dw.$$

This is $y^{-1} k_\gamma^{*2}(\log(y/x))$. Iterate. $\square$

8.4 Microscopic endpoint permutation

Theorem 8.19 (Microscopic endpoint permutation). *Assume Definition 8.1. Extend the finite permutation $\tau_n \in S_n$ to a permutation $\bar{\tau}_n$ of $\mathbb{N}$ by*

$$\bar{\tau}_n(a) = \begin{cases} \tau_n(a), & a \leq n, \\ a, & a > n. \end{cases}$$

Then, in the product topology on $\mathbb{N}^{\mathbb{N}}$,

$$\bar{\tau}_n \Rightarrow \tau^{(\gamma)}, \tag{8.53}$$

where, for independent $E_1, E_2, \dots \sim \text{Exp}(1)$,

$$\tau^{(\gamma)}(a) = 1 + \# \left\{ b \neq a : \frac{E_b}{b^\gamma} > \frac{E_a}{a^\gamma} \right\}.$$

Almost surely, $\tau^{(\gamma)}$ is a permutation of $\mathbb{N}$.

Proof. Normalize by the first terminal rate and put

$$r_{n,b} = \frac{\lambda_{n,b}}{\lambda_{n,1}}, \quad Q_{n,b} = \lambda_{n,1} T_{n,b} = \frac{E_b}{r_{n,b}},$$

using independent $E_b \sim \text{Exp}(1)$. For every fixed B , uniform endpoint regular variation and the uniform convergence theorem for slowly varying functions give

$$(r_{n,1}, \dots, r_{n,B}) \longrightarrow (1, 2^\gamma, \dots, B^\gamma).$$

Hence, under this coupling,

$$(Q_{n,1}, \dots, Q_{n,B}) \longrightarrow (E_1, E_2/2^\gamma, \dots, E_B/B^\gamma)$$

almost surely.

We remove the tail uniformly. Fix m and $\varepsilon > 0$. For fixed $a \leq m$, the ratios $r_{n,a}$ remain bounded, so

$$\mathbb{P}(Q_{n,a} < \varepsilon) \leq C_m \varepsilon.$$

Choose $W_n = \lfloor n/\log n \rfloor$. Potter's bound, applied with reference coordinate 1, gives

$$r_{n,b} \geq cb^{\gamma/2}, \quad B < b \leq W_n,$$

for all sufficiently large n . By monotonicity, the same lower bound at $b = W_n$ applies to every $b > W_n$. Consequently,

$$\begin{aligned} \sum_{b>B} \mathbb{P}(Q_{n,b} > \varepsilon) &= \sum_{b>B} e^{-\varepsilon r_{n,b}} \\ &\leq \sum_{b>B} e^{-c\varepsilon b^{\gamma/2}} + ne^{-c\varepsilon W_n^{\gamma/2}}. \end{aligned} \tag{8.54}$$

The second term tends to zero, and the first tends to zero as $B \rightarrow \infty$, uniformly for all sufficiently large n . Thus, outside an event of probability at most $C_m \varepsilon + o_B(1)$, every comparison determining the ranks of the first m coordinates is decided by the first B clocks. The same argument applies to the limiting clocks E_b/b^γ . Finite collections of all these clocks have no ties. Taking first $n \rightarrow \infty$, then $B \rightarrow \infty$, and finally $\varepsilon \downarrow 0$, proves finite-dimensional convergence of $(\bar{\tau}_n(1), \dots, \bar{\tau}_n(m))$ for every fixed m , which is (8.53).

For every $t > 0$,

$$\sum_{b=1}^{\infty} \mathbb{P}(E_b/b^\gamma > t) = \sum_{b=1}^{\infty} e^{-b^\gamma t} < \infty.$$

Borel–Cantelli implies that only finitely many limiting clocks exceed any positive threshold. Moreover,

$$\mathbb{P}\left(\sup_{b \geq 1} \frac{E_b}{b^\gamma} > t\right) \leq \sum_{b \geq 1} e^{-b^\gamma t} \longrightarrow 0 \quad (t \rightarrow \infty),$$

so the clock set has a finite maximum. The clocks are almost surely distinct, and zero is their only accumulation point. Their decreasing enumeration therefore contains every label exactly once and assigns every positive integer rank exactly once. Thus $\tau^{(\gamma)}$ is almost surely a permutation of $\mathbb{N}$. $\square$

Remark 8.20 (The Gumbel origin of the constant). The preceding proof makes the constant transparent. Conditional on $Y_{n,a} = \log \lambda_{n,a} + g$, the expected number of lower terminal priorities is

$$\sum_b \exp\left(-e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) \sim a A_\gamma e^{g/\gamma}.$$

The fixed-point saddle is therefore $A_\gamma e^{g/\gamma} = 1$, so $e^{-g} = K_\gamma = A_\gamma^\gamma$. The density of the specified centered Gumbel at the saddle is $K_\gamma e^{-K_\gamma}$, while the derivative of the mean with respect to g is a/γ . Their quotient is exactly c_γ/a , with

$$c_\gamma = \gamma K_\gamma e^{-K_\gamma}.$$

For $\gamma = 1$, $A_\gamma = K_\gamma = 1$, and the constant reduces to the Sukhatme value e^{-1} .

9 Global endpoint Poisson approximation

The local point-process theorem and the fixed-order factorial moments do not, by themselves, imply a Poisson approximation for the total fixed-point count, whose mean diverges like $c_\gamma \log n$. We now prove the global approximation in total variation. The argument uses an approximate Palm coupling: to force a specified terminal coordinate a to be fixed, we interchange its clock with the clock currently occupying terminal rank a . This changes the total number of fixed points only through the intended point and, exceptionally, a two-cycle.

Retain the notation of Section 8. Thus $T_{n,a} \sim \text{Exp}(\lambda_{n,a})$ are independent,

$$\tau_n(a) = 1 + \sum_{b \neq a} \mathbf{1}_{\{T_{n,b} > T_{n,a}\}}, \quad I_{n,a} = \mathbf{1}_{\{\tau_n(a)=a\}}, \quad F_n = \sum_{a=1}^n I_{n,a}.$$

Put

$$p_{n,a} = \mathbb{P}(I_{n,a} = 1), \quad m_n = \mathbb{E}F_n = \sum_{a=1}^n p_{n,a}.$$

Theorem 9.1 (Global endpoint Poisson approximation). *Assume Definition 8.1. Then*

$$d_{\text{TV}}(\mathcal{L}(F_n), \text{Pois}(m_n)) \longrightarrow 0.$$

More precisely,

$$d_{\text{TV}}(\mathcal{L}(F_n), \text{Pois}(m_n)) = O\left(\frac{\log \log n}{\sqrt{\log n}}\right).$$

Since $m_n \sim c_\gamma \log n$, it follows that

$$\frac{F_n - m_n}{\sqrt{m_n}} \Rightarrow N(0, 1).$$

For the Sukhatme rates, this gives

$$d_{\text{TV}}(\mathcal{L}(F_n), \text{Pois}(\mathbb{E}F_n)) \rightarrow 0, \quad \mathbb{E}F_n \sim e^{-1} \log n.$$

9.1 An approximate-Palm Stein bound

We first isolate the form of the Stein–Chen argument that will be used. The standard solution $f = f_B$ of the Poisson Stein equation satisfies

$$\|\Delta f\|_\infty \leq \frac{1 - e^{-\lambda}}{\lambda}, \quad \|f\|_\infty \leq \min\{1, \lambda^{-1/2}\};$$

see Barbour, Holst, and Janson [2, Lemma I.1.1].

Lemma 9.2 (Stein bound with approximate Palm laws). *Let $X_1, \dots, X_N$ be indicators,*

$$W = \sum_{i=1}^N X_i, \quad p_i = \mathbb{E}X_i, \quad \lambda = \mathbb{E}W > 0.$$

For every i with $p_i > 0$, let $Y^{(i)}$ be an integer-valued random variable constructed on the same probability space as W , and put

$$\varepsilon_i = d_{\text{TV}}(\mathcal{L}(Y^{(i)}), \mathcal{L}(W \mid X_i = 1)).$$

Indices with $p_i = 0$ are omitted. Then

$$\begin{aligned} d_{\text{TV}}(\mathcal{L}(W), \text{Pois}(\lambda)) &\leq \frac{1 - e^{-\lambda}}{\lambda} \sum_{i=1}^N p_i \mathbb{E}|W + 1 - Y^{(i)}| \\ &\quad + 2 \min\{1, \lambda^{-1/2}\} \sum_{i=1}^N p_i \varepsilon_i. \end{aligned}$$

Proof. Fix $B \subset \mathbb{Z}_{\geq 0}$, and let $f = f_B$ solve

$$\lambda f(k+1) - kf(k) = \mathbf{1}_{\{k \in B\}} - \mathbb{P}(\text{Pois}(\lambda) \in B).$$

Using

$$\mathbb{E}[Wf(W)] = \sum_i p_i \mathbb{E}[f(W) \mid X_i = 1],$$

we obtain

$$\begin{aligned} &\mathbb{P}(W \in B) - \mathbb{P}(\text{Pois}(\lambda) \in B) \\ &= \sum_i p_i \left(\mathbb{E}\{f(W+1) - f(Y^{(i)})\} + \mathbb{E}f(Y^{(i)}) - \mathbb{E}[f(W) \mid X_i = 1] \right). \end{aligned}$$

The first difference is bounded by $\|\Delta f\|_{\infty} \mathbb{E}|W + 1 - Y^{(i)}|$, and the second by $2\|f\|_{\infty} \varepsilon_i$. Apply the displayed Stein factors and take the supremum over B . $\square$

9.2 The terminal rank transposition

Fix n temporarily and suppress it from the notation. The clock-tie set has product-exponential measure zero. On its complement, let $b_a(t)$ be the unique label occupying terminal rank a , so that $\tau_t(b_a(t)) = a$, and define $\mathcal{T}_a t$ by interchanging t_a and $t_{b_a(t)}$. Define $\mathcal{T}_a$ arbitrarily, say as the identity, on the tie set. Then label a has terminal rank a in $\mathcal{T}_a t$.

Let $\nu_a = \mathcal{L}(\mathcal{T}_a T)$, and let $\mu_a^* = \mathcal{L}(T \mid I_a = 1)$ be the exact Palm law.

Lemma 9.3 (Exact swap likelihood ratio). *For a configuration t satisfying $I_a(t) = 1$, put*

$$R_a(t) = \sum_{b=1}^n \exp\{-(\lambda_a - \lambda_b)(t_b - t_a)\}.$$

If

$$f(t) = \prod_{j=1}^n \lambda_j e^{-\lambda_j t_j}$$

is the product-exponential density, then

$$\frac{d\nu_a}{dt} = \mathbf{1}_{\{I_a=1\}} f(t) R_a(t), \quad \frac{d\mu_a^*}{dt} = \frac{\mathbf{1}_{\{I_a=1\}}}{p_a} f(t).$$

Consequently,

$$\mathbb{E}[R_a \mid I_a = 1] = p_a^{-1}$$

and

$$d_{\text{TV}}(\nu_a, \mu_a^*) = \frac{1}{2} \mathbb{E}[|p_a R_a(T) - 1| \mid I_a = 1].$$

Proof. For $1 \leq b \leq n$, let

$$\mathcal{R}_{a,b} = \{t : b_a(t) = b\}$$

inside the no-tie region, and let $S_{a,b}$ interchange coordinates a and b . The sets $\mathcal{R}_{a,b}$ form a measurable partition. For every t with $I_a(t) = 1$, the unique preimage of t lying in $\mathcal{R}_{a,b}$ is $S_{a,b}t$: in that preimage label b occupies rank a , and the defining transposition returns t . Coordinate transpositions have Jacobian one, and

$$\frac{f(S_{a,b}t)}{f(t)} = \exp\{-(\lambda_a - \lambda_b)(t_b - t_a)\}.$$

Changing variables separately on the rank chambers and summing over b gives the density of ν_a . The Palm density is immediate. Integrating the first density gives $p_a \mathbb{E}[R_a \mid I_a = 1] = 1$, and integrating the absolute difference of the two densities gives the total-variation identity. $\square$

9.3 Uniform endpoint estimates

Put $\beta = \gamma/2$. Fix a constant $K_0 > 4/\gamma$, and define

$$h_n = K_0 \log \log n, \quad U_n = \lfloor ne^{-2h_n} \rfloor, \quad V_n = \lfloor ne^{-h_n} \rfloor.$$

Thus $U_n = n(\log n)^{-2K_0+o(1)}$ and $V_n = n(\log n)^{-K_0+o(1)}$. For $a \leq U_n$, write

$$x_{a,b} = \frac{\lambda_{n,b}}{\lambda_{n,a}}.$$

We use the uniform convergence theorem and Potter bounds for slowly varying functions in the forms recorded in Bingham, Goldie, and Teugels [4, Theorems 1.2.1 and 1.5.6].

Lemma 9.4 (Uniform endpoint package). *There are constants $A_0 < \infty$, $0 < \kappa_- < \kappa_+ < \infty$, and $0 < c < C < \infty$ such that, for all sufficiently large n and $A_0 \leq a \leq U_n$, the following hold.*

(i) For $a \leq b \leq V_n$,

$$c(b/a)^\beta \leq x_{a,b} \leq C(b/a)^{\gamma+\beta},$$

whereas for $1 \leq b \leq a$,

$$c(b/a)^{\gamma+\beta} \leq x_{a,b} \leq C(b/a)^\beta.$$

(ii) For $x \geq 1$,

$$\#\{b \leq V_n : x_{a,b} \leq x\} \leq Ca(1 + x^{1/\beta}).$$

(iii) Let

$$\mu_a(t) = \sum_{b \neq a} e^{-\lambda_{n,b}t}.$$

There is a unique $t_a^* > 0$ satisfying $\mu_a(t_a^*) = a - 1$. If $\kappa_a^* = \lambda_{n,a}t_a^*$, then

$$\kappa_- \leq \kappa_a^* \leq \kappa_+.$$

(iv) Uniformly for $\kappa \in [\kappa_-/4, 2\kappa_+]$ and $j = 0, 1, 2, 3, 4$,

$$\sum_{b \neq a} x_{a,b}^j e^{-\kappa x_{a,b}} \leq Ca.$$

For $j = 0, 1, 2$, the same sums are bounded below by ca .

(v) At $t = t_a^*$,

$$v_a(t) = \sum_{b \neq a} e^{-\lambda_{n,b}t} (1 - e^{-\lambda_{n,b}t}) \asymp a,$$

$$D_a(t) = -\mu'_a(t) = \sum_{b \neq a} \lambda_{n,b} e^{-\lambda_{n,b}t} \asymp a \lambda_{n,a}.$$

The same comparisons hold whenever $\lambda_{n,a}t \in [\kappa_-/2, 2\kappa_+]$. Throughout this scaled neighborhood, at least ca Bernoulli parameters $e^{-\lambda_{n,b}t}$ belong to a fixed compact subinterval of $(0, 1)$.

Proof. Uniform endpoint regular variation gives, uniformly for $a, b \leq V_n = o(n)$,

$$x_{a,b} = \left(\frac{b}{a}\right)^\gamma \frac{L(n/b)}{L(n/a)} (1 + o(1)).$$

Potter's bound with exponent $\beta = \gamma/2$ gives both inequalities in (i). If $b \geq a$ and $x_{a,b} \leq x$, the lower bound in (i) implies $b \leq Cax^{1/\beta}$; adding the at most a indices below a proves (ii).

We next bracket the saddle. Since $V_n/U_n = e^{h_n} \rightarrow \infty$, the interval $[[a/2], [2a]]$ lies below V_n for all large n . On this interval the ratios $x_{a,b}$ are bounded above by a fixed constant C_0 . Choose $\kappa_- > 0$ so small that $e^{-\kappa_- C_0} \geq 3/4$. The interval contains $3a/2 + O(1)$ indices, and hence, after increasing A_0 ,

$$\sum_{b \neq a} e^{-\kappa_- x_{a,b}} > a - 1.$$

For the opposite inequality, the reverse Potter bound gives, for $\kappa \geq 1$,

$$\sum_{b < a} e^{-\kappa x_{a,b}} \leq Ca \int_0^1 e^{-c\kappa u^{\gamma+\beta}} du \leq Ca\kappa^{-1/(\gamma+\beta)},$$

$$\sum_{a \leq b \leq V_n} e^{-\kappa x_{a,b}} \leq Ca \int_1^\infty e^{-c\kappa u^\beta} du \leq Ca\kappa^{-1/\beta}.$$

For $b > V_n$, monotonicity and (i) give

$$x_{a,b} \geq x_{a,V_n} \geq c(V_n/U_n)^\beta = ce^{\beta h_n} = c(\log n)^{K_0\beta}.$$

Because $K_0\beta > 2$,

$$\sum_{b > V_n} e^{-\kappa x_{a,b}} \leq n \exp\{-c\kappa(\log n)^{K_0\beta}\} = o(a)$$

uniformly for $a \geq A_0$ and fixed $\kappa > 0$. Choose a sufficiently large fixed κ_+ so that the first two sums at κ_+ are together at most $a/2$, and then take n large enough that the last sum is at most $a/4$. After increasing A_0 once more, the total is less than $a - 1$. The choices above may be made with fixed margins: for some $c_0 > 0$,

$$\mu_a(\kappa_-/\lambda_{n,a}) \geq (1 + c_0)a, \quad \mu_a(\kappa_+/\lambda_{n,a}) \leq (1 - c_0)a.$$

Since the sum is continuous and strictly decreasing in κ , (iii) follows.

For (iv), the contribution of $b < a$ is at most a , since monotonicity gives $x_{a,b} \leq 1$. For $a \leq b \leq V_n$, dyadic summation using (ii) gives

$$\begin{aligned} \sum_{a \leq b \leq V_n} x_{a,b}^j e^{-\kappa x_{a,b}} \\ \leq C a \sum_{m \geq 0} 2^{m(j+1/\beta)} e^{-c\kappa 2^m} \leq C_j a \end{aligned}$$

uniformly for $\kappa \geq \kappa_-/4$. On $b > V_n$, the preceding lower bound on $x_{a,b}$ and the elementary inequality $x^j e^{-\kappa x} \leq C_{j,\kappa_-} e^{-(\kappa/2)x}$ for large x give an $o(a)$ contribution. This proves the upper bounds. Restricting to a fixed subinterval of $[a, 2a]$ gives ca terms for which $x_{a,b}$ stays in a fixed compact subset of $(0, \infty)$, proving the lower bounds for $j = 0, 1, 2$.

Part (v) follows from (iv) after putting $\kappa = \lambda_{n,a}t$. The lower variance bound and the final nondegeneracy assertion follow by restricting to the same fixed subinterval of $[a, 2a]$. $\square$

Lemma 9.5 (Likelihood-ratio tail sum). *There are constants $C, c > 0$ such that, uniformly under the hypotheses of Lemma 9.4, for every $M \geq e^2$ and every $\kappa \in [\kappa_-/2, 2\kappa_+]$,*

$$\sum_{b: \lambda_{n,b} > \lambda_{n,a}} (x_{a,b} - 1) e^{-\kappa x_{a,b}} M^{-1/(x_{a,b}-1)} \leq C a e^{-c\sqrt{\log M}}.$$

Proof. Put $L = \log M$. For $x > 1$ and $\kappa \in [\kappa_-/2, 2\kappa_+]$,

$$\begin{aligned} \kappa x + \frac{L}{x-1} &= \frac{\kappa x}{2} + \left(\frac{\kappa x}{2} + \frac{L}{x-1} \right) \\ &\geq \frac{\kappa x}{2} + 2\sqrt{\frac{\kappa L x}{2(x-1)}} \geq \frac{\kappa x}{2} + c\sqrt{L}. \end{aligned}$$

Therefore

$$(x-1) e^{-\kappa x} M^{-1/(x-1)} \leq C e^{-c\sqrt{\log M}} x e^{-(\kappa/2)x}.$$

Summing this inequality and applying Lemma 9.4(iv) with $j = 1$ and $\kappa/2 \in [\kappa_-/4, \kappa_+]$ proves the claim. This argument includes the entire rate range and is uniform in M . $\square$

9.4 A quantitative one-dimensional saddle

For fixed a , let

$$S_a(t) = \sum_{b \neq a} \xi_b(t), \quad \xi_b(t) \sim \text{Bernoulli}(e^{-\lambda_{n,b}t})$$

independently. Thus $I_{n,a} = 1$ is obtained by integrating the event $S_a(t) = a-1$ against the density of $T_{n,a}$.

Lemma 9.6 (Quantitative endpoint saddle). *There are constants $\rho, c, C > 0$ such that, uniformly for $A_0 \leq a \leq U_n$,*

$$p_{n,a} = \frac{\lambda_{n,a} e^{-\lambda_{n,a} t_a^*}}{D_a(t_a^*)} \left(1 + O \left(a^{-\rho} + e^{-c\sqrt{\log(a+2)}} \right) \right).$$

Put

$$u = \lambda_{n,a} \sqrt{a} (t - t_a^*), \quad L_a = (\log(a+2))^{1/4}.$$

Under the Palm law $\mathcal{L}(T_{n,a} \mid I_{n,a} = 1)$,

$$\mathbb{P}(|u| > L_a \mid I_{n,a} = 1) \leq C e^{-c\sqrt{\log(a+2)}}.$$

Finally, whenever $|u| \leq L_a$, uniformly in $b \neq a$ and $s \in \{0, 1\}$,

$$\frac{\mathbb{P}(S_a(t) - \xi_b(t) = a - 1 - s)}{\mathbb{P}(S_a(t) = a - 1)} = 1 + O(a^{-\rho}).$$

Proof. Suppress n and write

$$q_b(t) = e^{-\lambda_b t}, \quad \mu(t) = \sum_{b \neq a} q_b(t), \quad v(t) = \sum_{b \neq a} q_b(t)(1 - q_b(t)).$$

In the scaled neighborhood $\lambda_a t \in [\kappa_-/2, 2\kappa_+]$, Lemma 9.4 gives

$$v(t) \asymp a, \quad -\mu'(t) \asymp a\lambda_a, \quad |\mu^{(j)}(t)| \leq C a \lambda_a^j \quad (j = 2, 3, 4),$$

and at least ca of the $q_b(t)$ lie in a fixed compact subinterval of $(0, 1)$.

We first prove a uniform local expansion. Let

$$\psi_t(\theta) = \mathbb{E} e^{i\theta(S_a(t) - \mu(t))}.$$

The nondegenerate Bernoulli block gives, for $|\theta| \leq \pi$,

$$|\psi_t(\theta)| \leq e^{-ca\theta^2}.$$

Indeed, $|1 - q + qe^{i\theta}|^2 = 1 - 4q(1 - q)\sin^2(\theta/2)$, and $\sin^2(\theta/2) \geq c\theta^2$ on $[-\pi, \pi]$. Uniformly for $|\theta| \leq \theta_0$,

$$\log \psi_t(\theta) = -\frac{1}{2}v(t)\theta^2 + O(a|\theta|^3).$$

Fourier inversion, split at $|\theta| = a^{-2/5}$, therefore gives

$$\mathbb{P}(S_a(t) = \ell) = \frac{1}{\sqrt{2\pi v(t)}} \exp \left\{ -\frac{(\ell - \mu(t))^2}{2v(t)} \right\} + O(a^{-1}) \quad (28)$$

uniformly for $|\ell - \mu(t)| \leq L_a \sqrt{a}$ in the central window. On the inner Fourier interval, the integral of the cubic error is $O(a \int |\theta|^3 e^{-ca\theta^2} d\theta) = O(a^{-1})$; the complementary Fourier integral is exponentially small. The same estimate holds after deleting any one summand, with its own mean and variance.

The variance derivative satisfies

$$|v'(t)| \leq \sum_{b \neq a} \lambda_b e^{-\lambda_b t} = \lambda_a \sum_{b \neq a} x_{a,b} e^{-\kappa x_{a,b}} \leq C a \lambda_a, \quad \kappa = \lambda_a t,$$

by Lemma 9.4(iv). Hence, if $|u| \leq L_a$, Taylor's theorem gives

$$\frac{a - 1 - \mu(t)}{\sqrt{v(t)}} = c_a u + O\left(\frac{1 + u^2}{\sqrt{a}}\right),$$

where c_a is bounded above and below by positive constants, and

$$\frac{v(t)}{v(t_a^*)} = 1 + O\left(\frac{1 + |u|}{\sqrt{a}}\right).$$

The Gaussian main term in (28) is at least $a^{-1/2}e^{-CL_a^2}$. Hence its absolute $O(a^{-1})$ error is a relative $O(a^{-\rho})$ error for some fixed $\rho > 0$, because $L_a^2 = \sqrt{\log(a+2)} = o(\log a)$. Deleting one summand changes the mean and variance by at most one, so the ratio of the two Gaussian main terms differs from one by

$$O\left(\frac{1 + |a - 1 - \mu(t)|}{a}\right) = O\left(\frac{L_a}{\sqrt{a}}\right).$$

After decreasing ρ , this proves the final assertion of the lemma.

We also need a local upper bound. Put $h = a - 1 - \mu(t)$ and let

$$K_t(\theta) = \log \mathbb{E} \exp\{\theta(S_a(t) - \mu(t))\}.$$

For $|\theta| \leq \theta_0$, bounded Bernoulli summands and Lemma 9.4(iv) give

$$K_t''(\theta) = \text{Var}_\theta S_a(t) \leq Ca, \quad K_t(0) = K_t'(0) = 0,$$

and therefore $K_t(\theta) \leq Ca\theta^2$. If $|h| \leq c_0a$, choose $\theta = \delta h/a$, with $\delta > 0$ sufficiently small. Exponential tilting gives

$$\begin{aligned} \mathbb{P}(S_a(t) = a - 1) &= e^{-\theta h + K_t(\theta)} \mathbb{P}_\theta(S_a(t) = a - 1) \\ &\leq Ca^{-1/2} \exp\left\{-c\frac{h^2}{a}\right\}. \end{aligned}$$

Here the maximal atom under $\mathbb{P}_\theta$ is $O(a^{-1/2})$: the tilt is bounded, so the block of ca Bernoulli parameters in a fixed compact subset of $(0, 1)$ remains nondegenerate, and the same Fourier estimate used above applies. If $c_0a < |h| \leq Ca$, use instead a fixed tilt $\theta = \delta \text{sgn}(h)$; choosing δ sufficiently small gives exponent at most $-c|h| \leq -c'a$, which is bounded above by $-c'h^2/a$. Thus, throughout the fixed scaled neighborhood,

$$\mathbb{P}(S_a(t) = a - 1) \leq Ca^{-1/2} \exp\left\{-c\frac{h^2}{a}\right\}. \quad (29)$$

The construction of κ_- and κ_+ in Lemma 9.4 in fact gives fixed positive margins at the two bracket points. Monotonicity of μ therefore implies $|h| \geq ca$ outside the scaled neighborhood, and Bernstein's inequality makes that integrated contribution exponentially small.

Put

$$v_* = v(t_a^*), \quad D_* = D_a(t_a^*), \quad w = \frac{D_*}{\sqrt{v_*}}(t - t_a^*).$$

On $|u| \leq L_a$, equivalently $|w| \leq CL_a$, Taylor's theorem and (28) give

$$\begin{aligned} &\lambda_a e^{-\lambda_a t} \mathbb{P}(S_a(t) = a - 1) dt \\ &= \frac{\lambda_a e^{-\lambda_a t_a^*} e^{-w^2/2}}{D_* \sqrt{2\pi}} (1 + O(a^{-\rho})) dw, \end{aligned}$$

after decreasing ρ once more. The error includes the Taylor remainders, which are polynomially small uniformly on the chosen polylogarithmic window. By (29), the remainder of the fixed scaled neighborhood contributes at most

$$C \frac{\lambda_a e^{-\lambda_a t_a^*}}{D_*} e^{-cL_a^2},$$

and the region outside that neighborhood is exponentially smaller. Therefore

$$\begin{aligned} p_{n,a} &= \int_0^\infty \lambda_a e^{-\lambda_a t} \mathbb{P}(S_a(t) = a-1) dt \\ &= \frac{\lambda_a e^{-\lambda_a t_a^*}}{D_*} \left(1 + O\left(a^{-\rho} + e^{-cL_a^2}\right) \right). \end{aligned}$$

Since $L_a^2 = \sqrt{\log(a+2)}$, this is the first assertion. Dividing the same central-tail estimate by the saddle asymptotic proves the Palm concentration bound. $\square$

The preceding lemma and Lemma 8.3 give, uniformly on endpoint ranges whose lower endpoint tends to infinity and whose upper endpoint is $o(n)$,

$$p_{n,a} = \frac{c_\gamma + o(1)}{a}.$$

We shall also use the global bound

$$p_{n,a} \leq \frac{C}{a}, \quad 1 \leq a \leq n,$$

which follows by exposing the death process and applying Lemma 8.2 at the transition with a alive labels.

Lemma 9.7 (Logarithmic order of the mean). *There are constants $0 < c < C < \infty$ such that, for all sufficiently large n ,*

$$c \log n \leq m_n \leq C \log n.$$

Proof. The global point bound gives the upper estimate. For the lower estimate, Lemma 9.4 and Lemma 9.6 imply, after increasing a fixed lower cutoff, that

$$p_{n,a} \geq c/a, \quad A_1 \leq a \leq U_n.$$

Indeed, the saddle prefactor is comparable to $1/a$, and the relative error tends to zero. Since $\log U_n = \log n - O(\log \log n)$, summing over this interval proves the lower bound. $\square$

9.5 The swap law is asymptotically Palm

Proposition 9.8 (Quantitative swap–Palm approximation). *There are constants $C, c, \rho > 0$ such that, for all sufficiently large n and $A_0 \leq a \leq U_n$,*

$$\varepsilon_{n,a} := d_{\text{TV}}(\nu_{n,a}, \mu_{n,a}^*) \leq C \left(a^{-\rho} + e^{-c\sqrt{\log(a+2)}} \right).$$

In particular,

$$\sum_{a=1}^{U_n} \frac{\varepsilon_{n,a}}{a} \leq C.$$

Proof. Fix a and suppress n . Put $t_* = t_a^*$, $\kappa_* = \lambda_a t_*$, and $k = a - 1$. Conditional on $T_a = t$, define

$$B_b = \mathbf{1}_{\{T_b > t\}}, \quad N(t) = \sum_{b \neq a} B_b.$$

Thus $I_a = 1$ is the event $N(t) = k$. For $b \neq a$, put

$$H_b(t, T_b) = \exp\{-(\lambda_a - \lambda_b)(T_b - t)\}, \quad H_a = 1,$$

so that $R_a = \sum_b H_b$.

A direct integration gives

$$\mathbb{E}[H_b \mid T_a = t] = \frac{\lambda_b}{\lambda_a} e^{(\lambda_a - \lambda_b)t}.$$

Therefore

$$\begin{aligned} M_a(t) &:= \mathbb{E}[R_a \mid T_a = t] = \frac{e^{\lambda_a t}}{\lambda_a} \sum_{b=1}^n \lambda_b e^{-\lambda_b t} \\ &= 1 + \frac{e^{\lambda_a t}}{\lambda_a} D_a(t). \end{aligned}$$

Let

$$\eta_a = a^{-\rho_0} + e^{-c_0 \sqrt{\log(a+2)}},$$

where ρ_0, c_0 are chosen from Lemma 9.6. That lemma and the identity

$$\frac{\lambda_a e^{-\lambda_a t_*}}{D_a(t_*)} M_a(t_*) = 1 + \frac{\lambda_a e^{-\lambda_a t_*}}{D_a(t_*)}$$

give

$$p_a M_a(t_*) = 1 + O(\eta_a).$$

Moreover,

$$\frac{d}{dt} \log M_a(t) = \lambda_a - \frac{\sum_b \lambda_b^2 e^{-\lambda_b t}}{\sum_b \lambda_b e^{-\lambda_b t}},$$

so Lemma 9.4(iv) gives $|\frac{d}{dt} \log M_a(t)| \leq C\lambda_a$ in the saddle neighborhood. Consequently, on

$$\mathcal{C}_a(t) = \{|\lambda_a \sqrt{a}(t - t_*)| \leq L_a\},$$

we have

$$p_a M_a(t) = 1 + O\left(\eta_a + \frac{L_a}{\sqrt{a}}\right).$$

We next compare conditioning on the exact Bernoulli count with the unconditioned center. For $s \in \{0, 1\}$ and $b \neq a$, let

$$A_{b,s} = \mathbb{E}[H_b \mathbf{1}_{\{B_b = s\}} \mid T_a = t].$$

Then

$$\begin{aligned} &\mathbb{E}[H_b \mid T_a = t, N(t) = k] \\ &= A_{b,1} \frac{\mathbb{P}(N(t) - B_b = k - 1)}{\mathbb{P}(N(t) = k)} + A_{b,0} \frac{\mathbb{P}(N(t) - B_b = k)}{\mathbb{P}(N(t) = k)}. \end{aligned}$$

The deleted-summand assertion of Lemma 9.6 makes both ratios $1 + O(a^{-\rho_1})$, uniformly in b , throughout the central window. Since all terms are nonnegative and $H_a = 1$ is deterministic,

$$\mathbb{E}[R_a \mid T_a = t, N(t) = k] = M_a(t)(1 + O(a^{-\rho_1})).$$

We now control fluctuations of R_a . Let

$$B_a = a^{1/8}, \quad \overline{H}_b = H_b \wedge B_a, \quad \overline{R}_a = \sum_b \overline{H}_b.$$

If $\lambda_b \leq \lambda_a$, then $H_b \leq e^{\lambda_a t}$, which is bounded in the central window, so such an index does not contribute to $(H_b - B_a)_+$ for large a . If $x_b = \lambda_b/\lambda_a > 1$ and $\kappa = \lambda_a t$, direct integration gives

$$\mathbb{E}[(H_b - B_a)_+ \mid T_a = t] = (x_b - 1)e^{-\kappa x_b} B_a^{-1/(x_b-1)}.$$

Lemma 9.5 therefore yields

$$\sum_b \mathbb{E}[(H_b - B_a)_+ \mid T_a = t] \leq Cae^{-c\sqrt{\log a}}.$$

This estimate survives conditioning on $N(t) = k$. Indeed, for $Y_b = (H_b - B_a)_+$,

$$\begin{aligned} \mathbb{E}[Y_b \mid T_a = t, N(t) = k] \\ = \sum_{s=0}^1 \mathbb{E}[Y_b \mathbf{1}_{\{B_b=s\}} \mid T_a = t] \frac{\mathbb{P}(N(t) - B_b = k - s)}{\mathbb{P}(N(t) = k)} \leq C\mathbb{E}[Y_b \mid T_a = t], \end{aligned}$$

by the same deleted-summand ratio bound. Summing over b gives

$$\mathbb{E}[R_a - \overline{R}_a \mid T_a = t, N(t) = k] \leq Cae^{-c\sqrt{\log a}}. \quad (30)$$

Conditional on $T_a = t$ and $N(t) = k$, the random set

$$\mathcal{S} = \{b \neq a : B_b = 1\}$$

has law

$$\mathbb{P}(\mathcal{S} = S \mid |\mathcal{S}| = k) \propto \prod_{b \in S} \frac{q_b}{1 - q_b}, \quad q_b = e^{-\lambda_b t}.$$

This is a product Bernoulli measure conditioned on one cardinality. Product Bernoulli measures are strongly Rayleigh, and single-cardinality conditioning preserves this property [5, Corollary 4.18]. Strongly Rayleigh measures have the stochastic covering property [13, Proposition 2.2], so the concentration theorem [13, Theorem 3.1] applies.

Given $\mathcal{S}$, the clocks are independent with the corresponding one-sided truncated exponential laws. Define

$$g_b^+(t) = \mathbb{E}[\overline{H}_b \mid T_a = t, B_b = 1], \quad g_b^-(t) = \mathbb{E}[\overline{H}_b \mid T_a = t, B_b = 0]$$

for $b \neq a$, and

$$G(S) = 1 + \sum_{b \in S} g_b^+(t) + \sum_{\substack{b \neq a \\ b \notin S}} g_b^-(t).$$

Both g_b^+ and g_b^- lie in $[0, B_a]$. Hence, for arbitrary subsets S, S' ,

$$|G(S) - G(S')| \leq B_a |S \Delta S'|.$$

Thus G/B_a is Lipschitz one for Hamming distance. Pemantle–Peres concentration gives

$$\mathbb{P}(|G - \mathbb{E}G| > u \mid T_a = t, N(t) = k) \leq 2 \exp \left\{ -\frac{u^2}{8k B_a^2} \right\},$$

and therefore

$$\mathbb{E}[|G - \mathbb{E}G| \mid T_a = t, N(t) = k] \leq C B_a \sqrt{a}. \quad (31)$$

Conditional on $\mathcal{S}$, the truncated clock contributions are independent, and $0 \leq \bar{H}_b \leq B_a$. Consequently,

$$\begin{aligned} \text{Var}(\bar{R}_a \mid \mathcal{S}, T_a = t) &\leq \sum_b \mathbb{E}[\bar{H}_b^2 \mid \mathcal{S}, T_a = t] \\ &\leq B_a G(\mathcal{S}). \end{aligned}$$

Since $\mathbb{E}G = \mathbb{E}[\bar{R}_a \mid T_a = t, N(t) = k] = O(a)$, conditional Cauchy–Schwarz gives

$$\mathbb{E}[|\bar{R}_a - G| \mid T_a = t, N(t) = k] \leq C \sqrt{B_a a}. \quad (32)$$

Combining the conditional mean comparison, (30), (31), and (32), and using $B_a = a^{1/8}$, yields

$$\begin{aligned} \mathbb{E}[|R_a - M_a(t)| \mid T_a = t, N(t) = k] \\ \leq C a \left(a^{-\rho_2} + a^{-3/8} + e^{-c\sqrt{\log a}} \right). \end{aligned}$$

Since $p_a = O(a^{-1})$, the last estimate and the central estimate for $p_a M_a(t)$ imply, for some $\rho_3 > 0$,

$$\mathbb{E}[|p_a R_a - 1| \mid T_a = t, I_a = 1] \leq C \left(a^{-\rho_3} + e^{-c\sqrt{\log a}} \right)$$

throughout the central window.

It remains to remove its complement. Define the central event on an output configuration by

$$\mathcal{C}_a = \{|\lambda_a \sqrt{a}(t_a - t_*)| \leq L_a\}.$$

Lemma 9.6 gives

$$\mathbb{P}(\mathcal{C}_a^c \mid I_a = 1) \leq C e^{-cL_a^2}.$$

Under ν_a , the output coordinate t_a is the a -th largest input clock; write this order statistic as O_a . At $t_{\pm} = t_* \pm L_a/(\lambda_a \sqrt{a})$, Taylor’s theorem and Lemma 9.4(v) show that the expected number of clocks above t_+ is at most $a - cL_a \sqrt{a}$, while the expected number above t_- is at least $a + cL_a \sqrt{a}$. Bernstein’s inequality therefore gives

$$\nu_a(\mathcal{C}_a^c) = \mathbb{P}(|\lambda_a \sqrt{a}(O_a - t_*)| > L_a) \leq C e^{-cL_a^2}.$$

The likelihood-ratio identity gives

$$p_a \mathbb{E}[R_a \mathbf{1}_{\mathcal{C}_a^c} \mid I_a = 1] = \nu_a(\mathcal{C}_a^c).$$

Using $|p_a R_a - 1| \leq p_a R_a + 1$ on $\mathcal{C}_a^c$, we conclude that

$$\mathbb{E}[|p_a R_a - 1| \mid I_a = 1] \leq C \left(a^{-\rho_3} + e^{-c\sqrt{\log(a+2)}} \right).$$

Lemma 9.3 proves the first assertion. Finally,

$$\sum_{a \geq 2} a^{-1-\rho_3} < \infty, \quad \int_2^\infty \frac{e^{-c\sqrt{\log x}}}{x} dx < \infty,$$

and the finitely many indices below A_0 are absorbed into the constant. $\square$

9.6 The swap changes the count only through a two-cycle

Let $J_{n,a}$ be the indicator that a belongs to a nontrivial two-cycle of τ_n , and put $q_{n,a} = \mathbb{P}(J_{n,a} = 1)$.

Lemma 9.9 (Exact effect of the rank swap). *For every no-tie clock configuration,*

$$F_n(\mathcal{T}_{n,a}T) = F_n + 1 - I_{n,a} + J_{n,a}.$$

In particular,

$$|F_n + 1 - F_n(\mathcal{T}_{n,a}T)| = I_{n,a} + J_{n,a}.$$

Proof. If a is already fixed, the map does nothing, $I_{n,a} = 1$, and $J_{n,a} = 0$. Suppose a is not fixed, and let b occupy terminal rank a . Interchanging the clocks of a and b exchanges their ranks and leaves every other rank unchanged. It fixes a . It fixes b as well precisely when the old rank of a was b , namely when a and b formed a two-cycle. This gives the signed identity. Since $I_{n,a}$ and $J_{n,a}$ are mutually exclusive, the absolute-value identity follows. $\square$

Lemma 9.10 (Two-cycle transition bound). *There is $C < \infty$ such that, for $1 \leq a < b \leq n$,*

$$\mathbb{P}(\tau_n(a) = b, \tau_n(b) = a) \leq C \min \left\{ \frac{1}{ab}, \frac{\lambda_{n,a}}{b\lambda_{n,b}} \right\}.$$

Proof. Expose the increasing exponential race as a pure-death process. When the alive set is S , the next deleted coordinate is $j \in S$ with probability $\lambda_{n,j} / \sum_{c \in S} \lambda_{n,c}$. The two-cycle event requires that a be deleted at the transition from b alive labels to $b - 1$, and that b subsequently be deleted at the transition from a alive labels to $a - 1$.

At level b , any alive set of size b has total rate at least $\sum_{j=1}^b \lambda_{n,j} \geq c_\Sigma b \lambda_{n,b}$ by Lemma 8.2. Hence the conditional probability of deleting a is at most

$$C \frac{\lambda_{n,a}}{b\lambda_{n,b}}.$$

At level a , if b is still alive, the total rate is at least

$$\lambda_{n,b} + \sum_{j=1}^{a-1} \lambda_{n,j}.$$

Moreover,

$$\sum_{j=1}^{a-1} \lambda_{n,j} \geq (c_\Sigma a - 1) \lambda_{n,a} \geq c' a \lambda_{n,a}$$

for all sufficiently large a ; the finitely many smaller values are absorbed into the constant. Thus the conditional probability of deleting b at level a is at most

$$C \min \left\{ 1, \frac{\lambda_{n,b}}{a \lambda_{n,a}} \right\}.$$

Multiplying the two conditional bounds proves the claim. $\square$

Lemma 9.11 (Weighted two-cycle summability). *With U_n, V_n as above,*

$$\sum_{a \leq U_n} p_{n,a} q_{n,a} = O(1), \quad \sum_{a > U_n} p_{n,a} q_{n,a} = O(\log \log n).$$

Proof. For an unordered pair $s < \ell$, write

$$P_{s,\ell} = \mathbb{P}(\tau_n(s) = \ell, \tau_n(\ell) = s).$$

Since $p_{n,a} \leq C/a$,

$$\begin{aligned} \sum_{a \leq U_n} p_{n,a} q_{n,a} &= \sum_{s < \ell} (p_{n,s} \mathbf{1}_{\{s \leq U_n\}} + p_{n,\ell} \mathbf{1}_{\{\ell \leq U_n\}}) P_{s,\ell} \\ &\leq C \sum_{s \leq U_n} \frac{1}{s} \sum_{\ell > s} P_{s,\ell}. \end{aligned}$$

For $\ell \leq V_n$, Potter's bound gives

$$\frac{\lambda_{n,s}}{\lambda_{n,\ell}} \leq C(s/\ell)^\beta.$$

Lemma 9.10 therefore bounds the summand after the factor $1/s$ by

$$C \min \left\{ \frac{1}{s^2 \ell}, \frac{s^{\beta-1}}{\ell^{1+\beta}} \right\}.$$

For fixed s , split at $\ell = s^{1+1/\beta}$. Below this point the first term sums to $O(\log(s+1)/s^2)$; above it the second term sums to $O(s^{-2})$. These bounds remain valid if the splitting point lies outside $(s, V_n]$. Summing over s gives a finite constant.

For $\ell > V_n$, monotonicity and Potter's bound give

$$\frac{\lambda_{n,s}}{\lambda_{n,\ell}} \leq \frac{\lambda_{n,s}}{\lambda_{n,V_n}} \leq C(s/V_n)^\beta.$$

Hence these pairs contribute at most

$$C \sum_{s \leq U_n} \frac{1}{s} \left(\frac{s}{V_n} \right)^\beta \sum_{\ell > V_n} \frac{1}{\ell} \leq C \log(n/V_n) \left(\frac{U_n}{V_n} \right)^\beta = o(1).$$

This proves the first estimate. The second follows from $q_{n,a} \leq 1$:

$$\sum_{a > U_n} p_{n,a} q_{n,a} \leq C \sum_{a > U_n} \frac{1}{a} = O(\log(n/U_n)) = O(\log \log n).$$

$\square$

9.7 Proof of the global theorem

Proof of Theorem 9.1. Apply Lemma 9.2 with $W = F_n$, $X_a = I_{n,a}$, and

$$Y^{(a)} = F_n(\mathcal{T}_{n,a}T).$$

By Lemma 9.9,

$$\mathbb{E}|F_n + 1 - Y^{(a)}| = p_{n,a} + q_{n,a}.$$

The first Stein term is therefore

$$\frac{1 - e^{-m_n}}{m_n} \sum_{a=1}^n p_{n,a}(p_{n,a} + q_{n,a}).$$

Since $p_{n,a} \leq C/a$,

$$\sum_a p_{n,a}^2 = O(1).$$

Lemma 9.11 gives

$$\sum_a p_{n,a}q_{n,a} = O(\log \log n).$$

By Lemma 9.7, the first Stein term is $O(\log \log n / \log n)$.

For the approximate-Palm term, data processing for total variation gives

$$d_{\text{TV}}(\mathcal{L}(F_n(\mathcal{T}_{n,a}T)), \mathcal{L}(F_n \mid I_{n,a} = 1)) \leq d_{\text{TV}}(\nu_{n,a}, \mu_{n,a}^*) = \varepsilon_{n,a}.$$

Therefore Proposition 9.8 gives

$$\begin{aligned} \sum_{a \leq U_n} p_{n,a} \varepsilon_{n,a} &\leq C \sum_{a \leq U_n} \frac{1}{a} \left(a^{-\rho} + e^{-c\sqrt{\log(a+2)}} \right) \\ &\leq C. \end{aligned}$$

On the remaining range, use $\varepsilon_{n,a} \leq 1$ and $p_{n,a} \leq C/a$:

$$\sum_{a > U_n} p_{n,a} \varepsilon_{n,a} = O(\log \log n).$$

Lemma 9.7 therefore bounds the second Stein term by

$$O\left(m_n^{-1/2} \log \log n\right) = O\left(\frac{\log \log n}{\sqrt{\log n}}\right).$$

Combining the two estimates proves the total-variation bound.

The exact asymptotic $m_n \sim c_\gamma \log n$ is Theorem 8.12. If $Z_n \sim \text{Pois}(m_n)$, then $(Z_n - m_n)/\sqrt{m_n} \Rightarrow N(0, 1)$. Total-variation convergence transfers this limit from Z_n to F_n . For the Sukhatme profile, $\gamma = 1$ and $c_1 = e^{-1}$. $\square$

Remark 9.12 (No growing-dimensional dissociation is needed). The proof does not require uniform factorization of joint fixed-point probabilities for sets of cardinality comparable to $\log n$. The rank transposition produces an approximate Palm coupling one coordinate at a time. The only direct interaction that can change the fixed-point count is a two-cycle, and its weighted contribution is summable. Thus the formerly conjectured high-order dissociation statement is strictly stronger than what is needed for global Poisson approximation.

10 Endpoint short cycles

Recall that the terminal-coordinate permutation τ_n is conjugate to σ_n by reversal. In particular,

$$C_r(\tau_n) = C_r(\sigma_n), \quad r \geq 1.$$

The multiplicative kernel in Proposition 8.18 determines the asymptotic number of every fixed endpoint cycle length. Unlike the regular positive-profile regime, each of these expected cycle counts diverges logarithmically.

For $r \geq 1$, put

$$\alpha_{\gamma,r} = \frac{1}{r} k_{\gamma}^{*r}(0).$$

Thus $\alpha_{\gamma,1} = c_{\gamma}$.

Theorem 10.1 (Endpoint short-cycle law). *Assume Definition 8.1. For every fixed $R \geq 1$ and fixed nonnegative integers $m_1, \dots, m_R$,*

$$\mathbb{E} \prod_{r=1}^R (C_r(\sigma_n))_{m_r} \sim \prod_{r=1}^R (\alpha_{\gamma,r} \log n)^{m_r}.$$

Consequently, for every fixed r ,

$$\mathbb{E} C_r(\sigma_n) \sim \frac{k_{\gamma}^{*r}(0)}{r} \log n,$$

and, jointly for every fixed R ,

$$\left(\frac{C_1}{\alpha_{\gamma,1} \log n}, \dots, \frac{C_R}{\alpha_{\gamma,R} \log n} \right) \longrightarrow (1, \dots, 1)$$

in probability and in every fixed L^p , $p < \infty$.

The proof has three inputs. First, we prove a source-to-target endpoint entry estimate by an endpoint-specific anisotropic local central limit theorem, mixed noncompact localization under one tilted law, and covariance stability on the saddle scale. Second, a death-process envelope localizes every fixed cycle component in logarithmic coordinates, including components crossing the mesoscopic cutoff. Third, a logarithmic trace Riemann sum identifies the coefficient $k_{\gamma}^{*r}(0)/r$.

10.1 General endpoint point constraints

For fixed $\ell \geq 1$, $\eta > 0$, and $M < \infty$, define

$$\begin{aligned} \mathfrak{Q}_N(\ell, \eta, M) = \{ & (n, p_1, \dots, p_{\ell}, q_1, \dots, q_{\ell}) : N \leq p_j, q_j \leq n/N, \\ & p_1, \dots, p_{\ell} \text{ distinct}, \quad q_1 < \dots < q_{\ell}, \quad q_{j+1}/q_j \geq e^{\eta}, \\ & e^{-M} \leq p_j/q_j \leq e^M \quad (1 \leq j \leq \ell) \}. \end{aligned} \quad (10.1)$$

A uniform assertion below means that the corresponding supremum over $\mathfrak{Q}_N(\ell, \eta, M)$ tends to zero as $N \rightarrow \infty$. The coordinates p_j are called *source terminal coordinates*, and the q_j are the target terminal positions.

Let $P = \{p_1, \dots, p_\ell\}$ and put

$$R_{n,j} = \frac{\lambda_{n,p_j}}{\lambda_{n,q_j}}, \quad z_j = q_j - j.$$

For $h = (h_1, \dots, h_\ell) \in \mathbb{R}^\ell$, define thresholds

$$y_j(h_j) = \log \lambda_{n,q_j} + h_j$$

and cumulative lower-priority counts

$$Z_j(h) = \sum_{b \notin P} \mathbf{1}_{\{Y_{n,b} < y_j(h_j)\}}, \quad 1 \leq j \leq \ell.$$

The relevant chamber is

$$\mathcal{C}_{n,q} = \{h : y_1(h_1) < \dots < y_\ell(h_\ell)\}.$$

Finally, for $R > 0$ let

$$\varphi_R(h) = R e^{-h} \exp\{-R e^{-h}\}. \quad (10.2)$$

This is the density of $\log R + G$ when G is standard Gumbel.

Lemma 10.2 (Source-to-target endpoint saddle geometry). *Uniformly over $\mathfrak{Q}_N(\ell, \eta, M)$, the ratios $R_{n,j}$ remain in a fixed compact subinterval of $(0, \infty)$. For each j , the equation*

$$M_j(h_j^*) = z_j, \quad M_j(h) = \mathbb{E} Z_j(h), \quad (10.3)$$

has a unique solution, and

$$h_j^* \longrightarrow g_\gamma := -\log K_\gamma, \quad d_{n,j} := M_j'(h_j^*) \sim \frac{q_j}{\gamma}. \quad (10.4)$$

For every fixed $L < \infty$,

$$\sup_{|h| \leq L} |M_j''(h)| \leq C_L q_j, \quad c_L q_j \leq M_j'(h) \leq C_L q_j \quad (10.5)$$

whenever $|h - h_j^*| \leq 1$. Moreover, there are $\delta_{\eta,M} > 0$, $c_{\eta,M} > 0$, and N_0 such that, for $N \geq N_0$,

$$y_{j+1}(h_{j+1}) - y_j(h_j) \geq c_{\eta,M} \quad (10.6)$$

whenever $|h_j - h_j^*| \leq \delta_{\eta,M}$ for every j . Thus this product neighborhood is contained in $\mathcal{C}_{n,q}$.

Proof. Uniform regular variation on the fixed multiplicative range $e^{-M} \leq p_j/q_j \leq e^M$ gives

$$R_{n,j} = \left(\frac{p_j}{q_j}\right)^\gamma (1 + o(1)) \quad (10.7)$$

uniformly, proving the compactness assertion. By Lemma 8.3, including its finite-deletion clause, uniformly on compact h -sets,

$$M_j(h) = q_j A_\gamma e^{h/\gamma} (1 + o(1)), \quad (10.8)$$

$$M'_j(h) = \frac{q_j A_\gamma}{\gamma} e^{h/\gamma} (1 + o(1)). \quad (10.9)$$

The function M_j is continuous and strictly increasing from 0 to $n - \ell$, so the saddle exists uniquely. Since $z_j/q_j \rightarrow 1$, (10.8) gives $A_\gamma e^{h_j^*/\gamma} \rightarrow 1$, equivalently $h_j^* \rightarrow -\log K_\gamma$. Substitution into (10.9) gives $d_{n,j} \sim q_j/\gamma$.

Termwise differentiation expresses M_j'' as a linear combination of the weighted sums in Lemma 8.3 with powers one and two. This gives (10.5).

Finally, the target separation and the Potter lower bound imply that, for some $c_\eta > 0$ and all large N ,

$$\log \frac{\lambda_{n,q_{j+1}}}{\lambda_{n,q_j}} \geq 2c_\eta. \quad (10.10)$$

Indeed, on a fixed compact range of q_{j+1}/q_j this follows from the uniform convergence theorem, while larger ratios are handled by $\lambda_{n,q_{j+1}}/\lambda_{n,q_j} \geq c(q_{j+1}/q_j)^{\gamma/2}$. Since all h_j^* converge to the same constant, a sufficiently small fixed $\delta_{\eta,M}$ gives (10.6). $\square$

Lemma 10.3 (Source-to-target endpoint anisotropic local CLT). *Let*

$$D_q = \text{diag}(q_1, \dots, q_\ell), \quad \Sigma(h) = \text{Var } Z(h),$$

and let $\mathcal{U}_n$ be the saddle neighborhood in Lemma 10.2. Uniformly over $\mathfrak{Q}_N(\ell, \eta, M)$ and $h \in \mathcal{U}_n$,

$$cD_q \leq \Sigma(h) \leq CD_q \quad (10.11)$$

in quadratic-form order. For every fixed $A < \infty$, uniformly for lattice points $v \in \mathbb{Z}^\ell$ satisfying

$$\|D_q^{-1/2}(v - M(h))\| \leq A,$$

we have

$$\begin{aligned} \mathbb{P}(Z(h) = v) &= (2\pi)^{-\ell/2} (\det \Sigma(h))^{-1/2} \\ &\times \exp \left[-\frac{1}{2} (v - M(h))^T \Sigma(h)^{-1} (v - M(h)) \right] + o \left(\prod_{j=1}^{\ell} q_j^{-1/2} \right). \end{aligned} \quad (10.12)$$

For every fixed $\nu < 1/6$, there are $C, c > 0$ such that

$$\mathbb{P}(Z(h) = v) \leq C \prod_{j=1}^{\ell} q_j^{-1/2} \exp \left[-c \sum_{j=1}^{\ell} \frac{(v_j - M_j(h_j))^2}{q_j} \right] \quad (10.13)$$

whenever $|v_j - M_j(h_j)| \leq q_j^{1/2+\nu}$ for every j .

Proof. For $b \notin P$, the vector

$$V_b(h) = (\mathbf{1}_{\{Y_{n,b} < y_1(h_1)\}}, \dots, \mathbf{1}_{\{Y_{n,b} < y_\ell(h_\ell)\}})$$

has the cumulative categorical support

$$(1, \dots, 1), (0, 1, \dots, 1), \dots, (0, \dots, 0, 1), 0.$$

By (10.6), the adjacent threshold gaps are uniformly positive in $\mathcal{U}_n$. Choose a small multiplicative interval $[(1 - \delta)q_j, (1 + \delta)q_j]$ for each j , with δ so small that the ℓ intervals are disjoint. Uniform regular variation and (10.4) imply that, for every index b in the j -th block, the two categories immediately below and above the threshold $y_j(h_j)$ both have probability bounded below by a positive constant. Removing the fixed set P changes each block size by at most ℓ . Consequently, for every j , there are at least cq_j independent summands having two atoms whose difference is the coordinate vector e_j , with both atom probabilities at least c . These bounds persist under every bounded exponential tilt.

The disjoint blocks give the covariance lower bound in (10.11). The upper bound follows from

$$\text{Var } Z_j(h) = O(q_j),$$

which is Lemma 8.3 applied at the relevant compact values of e^{-h_j} , followed by Cauchy's inequality. The same blocks and Lemma 7.7 give, uniformly under bounded tilts,

$$\left| \mathbb{E} e^{iu \cdot (Z(h) - M(h))} \right| \leq \exp \left[-c \sum_{j=1}^{\ell} q_j \text{dist}(u_j, 2\pi\mathbb{Z})^2 \right]. \quad (10.14)$$

On the anisotropic box $|u_j| \leq Aq_j^{-1/2}$, bounded categorical increments and the covariance upper bound give

$$\log \mathbb{E} e^{iu \cdot (Z - M)} = -\frac{1}{2} u^T \Sigma(h) u + O \left((\max_j |u_j|) \sum_j q_j u_j^2 \right).$$

After the change of variables $u = D_q^{-1/2} w$, the remainder is $O_A(q_1^{-1/2}) = o(1)$. Fourier inversion and (10.14) control the complementary region on the product scale. This proves (10.12). The tilted local upper bound (10.13) follows from Lemma 7.17: the tilted covariance upper bound follows from bounded distortion of the categorical probabilities, and the edge blocks supply the required tilted concentration estimate. $\square$

Lemma 10.4 (Source-to-target mixed localization). *There is $L_0 = L_0(\ell, \eta, M, \gamma)$ such that the following holds for $L \geq L_0$. For $h \in \mathcal{C}_{n,q}$, set*

$$\begin{aligned} B_R(h) &= \{j : h_j > L\}, \\ B_L(h) &= \{j : h_j < -L\}, \\ G(h) &= \{1, \dots, \ell\} \setminus (B_R(h) \cup B_L(h)), \end{aligned}$$

and put

$$H_j = z_j - M_j(h_j), \quad H_0 = 0.$$

On every region on which these three sets are fixed,

$$\begin{aligned} \mathbb{P}(Z(h) = z) &\leq C_L \prod_{j=1}^{\ell} q_j^{-1/2} \exp \left[-c \sum_{j \in B_R} |H_j| - c \sum_{j \in B_L} q_j \right] \\ &\quad \times \exp \left[-c_L \sum_{j \in G} \min \left\{ \frac{H_j^2}{q_j}, q_j \right\} \right]. \end{aligned} \quad (10.15)$$

Consequently,

$$\begin{aligned} \mathbb{P}(Z(h) = z) &\leq C_L \prod_{j=1}^{\ell} q_j^{-1/2} \exp \left[-ce^{L/(2\gamma)} \sum_{j \in B_R} q_j - c \sum_{j \in B_L} q_j \right] \\ &\times \exp \left[-c_L \sum_{j \in G} q_j \min\{(h_j - h_j^*)^2, 1\} \right]. \end{aligned} \quad (10.16)$$

Moreover,

$$\lim_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \sup_{\Omega_N(\ell, \eta, M)} \frac{\int_{\mathcal{C}_{n,q} \cap \{h \notin [-L, L]^\ell\}} \prod_{j=1}^{\ell} \varphi_{R_{n,j}}(h_j) \mathbb{P}(Z(h) = z) dh}{\prod_{j=1}^{\ell} q_j^{-1}} = 0. \quad (10.17)$$

On every fixed compact cube containing the saddle vectors,

$$\mathbb{P}(Z(h) = z) \leq C \prod_{j=1}^{\ell} q_j^{-1/2} \exp \left[-c \sum_{j=1}^{\ell} q_j \min\{(h_j - h_j^*)^2, 1\} \right]. \quad (10.18)$$

Proof. Define interval counts

$$W_1 = Z_1, \quad W_j = Z_j - Z_{j-1} \quad (2 \leq j \leq \ell),$$

with targets

$$w_1 = q_1 - 1, \quad w_j = q_j - q_{j-1} - 1 \quad (2 \leq j \leq \ell).$$

The target separation gives $c_\eta q_j \leq w_j \leq q_j$. Let $m_j = \mathbb{E}W_j$ and $d_j = w_j - m_j$; then $d_j = H_j - H_{j-1}$. For $j \in G$, one has $m_j \leq M_j(h_j) = O_L(q_j)$, and hence $|d_j| = O_L(q_j)$.

Fix the offender sets. Let

$$K_h(\theta) = \log \mathbb{E} \exp\{\theta \cdot (W(h) - m(h))\}$$

be the centered interval-coordinate cumulant, and let

$$(\mathcal{T}\alpha)_k = \sum_{j \geq k} \alpha_j.$$

Then $(\mathcal{T}\alpha) \cdot d = \alpha \cdot H$. Choose $\delta > 0$ sufficiently small and set

$$\alpha_j^- = -\delta \mathbf{1}_{\{j \in B_R\}}, \quad \alpha_j^+ = \delta \mathbf{1}_{\{j \in B_L\}}.$$

On G , use the local interval tilt

$$\lambda_j = \begin{cases} \delta_L d_j / q_j, & |d_j| \leq \varepsilon q_j, \\ \delta_L \operatorname{sgn}(d_j), & |d_j| > \varepsilon q_j, \end{cases}$$

where $\varepsilon < c_\eta/4$ and $\delta_L > 0$ is sufficiently small; put $\lambda_j = 0$ off G . Use the single total interval-coordinate tilt

$$\Theta = \mathcal{T}\alpha^- + \mathcal{T}\alpha^+ + \lambda. \quad (10.19)$$

Generalized Hölder inequality yields

$$K_h(\Theta) \leq \frac{1}{3}K_h(3\mathcal{T}\alpha^-) + \frac{1}{3}K_h(3\mathcal{T}\alpha^+) + \frac{1}{3}K_h(3\lambda). \quad (10.20)$$

By (10.8), for large L ,

$$\begin{aligned} j \in B_R: \quad & M_j(h_j) \geq ce^{L/\gamma}q_j, \quad H_j < 0, \\ j \in B_L: \quad & M_j(h_j) \leq Ce^{-L/\gamma}q_j, \quad 0 < H_j \leq q_j. \end{aligned} \quad (10.21)$$

The proof of Lemma 7.15 applies to the present nested lower-priority counts without change. Applied with parameter 3δ , it gives

$$-3\alpha^- \cdot H + K_h(3\mathcal{T}\alpha^-) \leq -c \sum_{j \in B_R} |H_j|, \quad (10.22)$$

$$-3\alpha^+ \cdot H + K_h(3\mathcal{T}\alpha^+) \leq -c \sum_{j \in B_L} q_j. \quad (10.23)$$

Under every bounded interval tilt, the weighted-sum estimate gives

$$\text{Var}_\xi(v \cdot W(h)) \leq C_L \sum_{j \in G} q_j v_j^2$$

for v supported on G . Taylor's formula for the cumulant and the choice of δ_L therefore give

$$-3\lambda \cdot d + K_h(3\lambda) \leq -c_L \sum_{j \in G} \min \left\{ \frac{d_j^2}{q_j}, q_j \right\}. \quad (10.24)$$

All coordinates of Θ are bounded for fixed L . For every locally regular coordinate $j \in G$ with $|d_j| \leq \varepsilon q_j$, the interval mean is at least $(c_\eta/2)q_j$. The middle-block argument in Lemma 8.6, with q_j in place of a_j and with the fixed deletion set P , supplies forward two-atom blocks of order q_j . They can be chosen disjointly, and persist under the total tilt. After projection onto the locally regular coordinates, Lemma 7.12 gives

$$\mathbb{P}_\Theta(W_{G_0} = w_{G_0}) \leq C_L \prod_{j \in G_0} q_j^{-1/2}. \quad (10.25)$$

The exact change-of-measure identity, together with (10.20)–(10.25), gives the desired exponential estimate first in terms of the interval residuals d_j . Indices lacking a local factor carry a fixed exponential loss of order q_j , which absorbs the missing factor $q_j^{-1/2}$.

To pass from interval to cumulative residuals without losing boundary terms, apply Lemma 7.10 with $A_j = q_j$, with its B_- equal to the present B_R , and its B_+ equal to the present B_L . The assumptions follow from $|d_j| = O_L(q_j)$ on G and $|H_j| \leq q_j$ on B_L . Reserving a fixed fraction of the offender costs proves (10.15). On G , the derivative bounds and monotonicity imply

$$|H_j| \geq c_L q_j \min\{|h_j - h_j^*|, 1\}.$$

On B_R , (10.21) gives $|H_j| \geq ce^{L/\gamma}q_j$. This proves (10.16); the compact estimate (10.18) is the case with no offenders.

It remains to integrate the tail. By (10.7), $R_{n,j}$ lies in a fixed compact subinterval of $(0, \infty)$. Uniformly in such R ,

$$\varphi_R(h) \leq Ce^{-h} \quad (h \geq 0), \quad \varphi_R(h) \leq Ce^{-h}e^{-ce^{-h}} \quad (h \leq 0). \quad (10.26)$$

Each nonoffending coordinate contributes $O(q_j^{-1/2})$ after integration on its Gaussian scale, in addition to the prefactor $q_j^{-1/2}$. For every offender, the exponential loss in (10.16) absorbs the missing local factor, and the remaining density tail tends to zero as $L \rightarrow \infty$. Summing over the finitely many offender sets proves (10.17). $\square$

Lemma 10.5 (Source-to-target covariance stability). *Let*

$$H_q = \text{diag}(q_1^{-1/2}, \dots, q_\ell^{-1/2}), \quad \Sigma_* = \Sigma(h^*).$$

For every fixed $A < \infty$,

$$\sup_{\|u\| \leq A} \left\| D_q^{-1/2} [\Sigma(h^* + H_q u) - \Sigma_*] D_q^{-1/2} \right\|_{\text{op}} \longrightarrow 0 \quad (10.27)$$

uniformly over $\mathfrak{Q}_N(\ell, \eta, M)$. Consequently,

$$\frac{\det \Sigma(h^* + H_q u)}{\det \Sigma_*} \longrightarrow 1 \quad (10.28)$$

and

$$\left\| D_q^{1/2} [\Sigma(h^* + H_q u)^{-1} - \Sigma_*^{-1}] D_q^{1/2} \right\|_{\text{op}} \longrightarrow 0 \quad (10.29)$$

uniformly on the same sets.

Proof. For $j \leq k$, put

$$p_{b,j}(h_j) = \exp \left(-e^{-h_j} \frac{\lambda_{n,b}}{\lambda_{n,q_j}} \right).$$

The threshold events are nested in the chamber, so

$$\Sigma_{jk}(h) = \sum_{b \notin P} p_{b,j}(h_j) (1 - p_{b,k}(h_k)). \quad (10.30)$$

Lemma 8.3, with power one, gives on a fixed saddle neighborhood

$$|\partial_{h_j} \Sigma_{jk}(h)| \leq Cq_j, \quad |\partial_{h_k} \Sigma_{jk}(h)| \leq Cq_k.$$

Hence, for $\|u\| \leq A$,

$$|\Sigma_{jk}(h^* + H_q u) - \Sigma_{jk}(h^*)| \leq C_A(\sqrt{q_j} + \sqrt{q_k}).$$

After division by $\sqrt{q_j q_k}$ this is at most $C_A(q_j^{-1/2} + q_k^{-1/2}) \leq 2C_A q_1^{-1/2}$. Since ℓ is fixed, (10.27) follows. The normalized covariance matrices remain in a compact subset of the positive-definite matrices by (10.11); continuity of determinant and inversion gives (10.28)–(10.29). $\square$

Lemma 10.6 (Endpoint integrated point-constraint saddle). *Uniformly over $\mathfrak{Q}_N(\ell, \eta, M)$,*

$$\begin{aligned} \int_{\mathcal{C}_{n,q}} \prod_{j=1}^{\ell} \varphi_{R_{n,j}}(h_j) \mathbb{P}(Z(h) = z) dh \\ = (1 + o(1)) \prod_{j=1}^{\ell} \frac{\varphi_{R_{n,j}}(h_j^*)}{d_{n,j}}, \end{aligned} \quad (10.31)$$

where $d_{n,j} = M'_j(h_j^*)$.

Proof. By Lemma 10.4, it suffices first to integrate over a fixed compact cube and then over a shrinking saddle neighborhood; the complement is $o(\prod_j q_j^{-1})$. Put

$$h = h^* + H_q u, \quad D_n = \text{diag}(d_{n,1}, \dots, d_{n,\ell}).$$

For bounded u , Taylor's theorem and (10.5) give

$$M(h) - z = D_n H_q u + o(D_q^{1/2}) \quad (10.32)$$

uniformly. The density factors satisfy

$$\prod_j \varphi_{R_{n,j}}(h_j) = \prod_j \varphi_{R_{n,j}}(h_j^*) (1 + o(1))$$

on bounded u -balls. Lemma 10.3 and Lemma 10.5 therefore give, uniformly on such balls, the Gaussian integrand with covariance Σ_* and mean shift $D_n H_q u$.

Since $dh = |\det H_q| du$, the Gaussian integral contributes

$$\begin{aligned} (2\pi)^{-\ell/2} (\det \Sigma_*)^{-1/2} |\det H_q| \\ \times \int_{\mathbb{R}^\ell} \exp \left[-\frac{1}{2} u^T H_q^T D_n^T \Sigma_*^{-1} D_n H_q u \right] du = |\det D_n|^{-1}. \end{aligned} \quad (10.33)$$

Indeed,

$$\det(H_q^T D_n^T \Sigma_*^{-1} D_n H_q) = |\det H_q|^2 |\det D_n|^2 (\det \Sigma_*)^{-1}.$$

The compact localization bound (10.18) is an integrable Gaussian majorant in u , and (10.17) handles the outer region. Dominated convergence proves (10.31). $\square$

Proposition 10.7 (Separated endpoint entry estimate). *Uniformly over $\mathfrak{Q}_N(\ell, \eta, M)$,*

$$\mathbb{P}(\tau_n(p_j) = q_j, \ 1 \leq j \leq \ell) = (1 + o(1)) \prod_{j=1}^{\ell} \frac{1}{q_j} k_\gamma \left(\log \frac{q_j}{p_j} \right). \quad (10.34)$$

Proof. Condition on the specified centered Gumbels G_{p_j} and put

$$h_j = \log R_{n,j} + G_{p_j}.$$

Then the specified priority is $Y_{n,p_j} = \log \lambda_{n,q_j} + h_j$, and h_j has density $\varphi_{R_{n,j}}$. Since $q_1 < \dots < q_\ell$, the point constraints force $Y_{n,p_1} < \dots < Y_{n,p_\ell}$. In this chamber exactly $j - 1$ specified priorities

lie below the j -th threshold, so the event is equivalent to $Z_j(h) = q_j - j$ for all j . Thus the probability is exactly the integral in (10.31).

By Lemma 10.2, $h_j^* \rightarrow -\log K_\gamma$ and $d_{n,j} \sim q_j/\gamma$. Therefore

$$\begin{aligned} \frac{\varphi_{R_{n,j}}(h_j^*)}{d_{n,j}} &= \frac{\gamma K_\gamma}{q_j} R_{n,j} e^{-K_\gamma R_{n,j}} (1 + o(1)) \\ &= \frac{1}{q_j} k_\gamma \left(\log \frac{q_j}{p_j} \right) (1 + o(1)), \end{aligned}$$

where the second equality uses (10.7). Multiply over j . □

10.2 A global cycle envelope

The next lemma makes precise the monotonicity used when forced deletions are interspersed with ordinary Luce deletions.

Lemma 10.8 (Scheduled-deletion survival). *Let S be a finite set of labels with positive rates, and let $p \in S$. Start from S and run a schedule in which each step is either an ordinary Luce deletion from the current set or a forced removal of a chosen competitor $c \neq p$. The forced choices may depend on the previously exposed history. For $m \geq 0$, the probability that p is still present after m ordinary deletions is at most*

$$H_m(S, p) := \mathbb{P}(\#\{b \in S \setminus \{p\} : T_b < T_p\} \geq m) \quad (10.35)$$

for an unmodified exponential race on S .

Proof. The function $H_m(S, p)$ is increasing under addition of competitors, by coupling the clocks. It also satisfies the exact recursion

$$H_m(S, p) = \sum_{b \in S \setminus \{p\}} \frac{\lambda_b}{\sum_{c \in S} \lambda_c} H_{m-1}(S \setminus \{b\}, p), \quad m \geq 1, \quad (10.36)$$

with $H_0 = 1$: to survive m ordinary deletions, the first ordinary deletion must be a competitor b , after which memorylessness leaves the same problem with $m - 1$ deletions.

Induct on the remaining schedule. If the next step is a forced removal of c , the induction hypothesis and monotonicity give a bound by $H_m(S \setminus \{c\}, p) \leq H_m(S, p)$. If the next step is ordinary, condition on the deleted label and use (10.36). This also covers adaptive forced choices after conditioning on the current history. □

Lemma 10.9 (Forced-deletion bound). *Consider compatible point constraints $\tau_n(p_i) = q_i$, $1 \leq i \leq L$, with distinct sources and distinct targets. Then*

$$\mathbb{P}(\tau_n(p_i) = q_i, 1 \leq i \leq L) \leq C^L \prod_{i=1}^L \frac{\lambda_{n,p_i}}{q_i \lambda_{n,q_i}}. \quad (10.37)$$

If (p_*, q_*) is one distinguished constraint, the right-hand side may be multiplied by

$$\mathbb{P}(\tau_n(p_*) \leq q_* + L). \quad (10.38)$$

Proof. Expose the increasing exponential race as a pure-death chain. When q labels are alive, every alive set has total rate at least

$$\sum_{b \leq q} \lambda_{n,b} \geq c_{\Sigma} q \lambda_{n,q}$$

by Lemma 8.2. Hence the conditional probability of deleting source p_i at level q_i is at most $C \lambda_{n,p_i} / (q_i \lambda_{n,q_i})$. Sum over compatible death histories and pull out these bounds at the forced levels. The remaining transition weights have total mass at most one, proving (10.37).

For the refinement, retain the condition that p_* survives until level q_* . Before that level at most $L - 1$ deletions are forced. Thus the residual schedule contains at least $n - q_* - L$ ordinary deletions before p_* may be removed. Lemma 10.8 bounds the residual path mass by

$$\begin{aligned} H_{n-q_*-L}([n], p_*) &= \mathbb{P}(\#\{b \neq p_* : T_b < T_{p_*}\} \geq n - q_* - L) \\ &= \mathbb{P}(\tau_n(p_*) \leq q_* + L), \end{aligned}$$

which proves (10.38). $\square$

Lemma 10.10 (Endpoint survival tail). *Put*

$$U_n = \left\lfloor \frac{n}{(\log n)^2} \right\rfloor, \quad V_n = 2U_n.$$

There are constants $C, c, \delta > 0$ such that, uniformly for $1 \leq q < p \leq n$, $q \leq U_n$, and every fixed L ,

$$\mathbb{P}(\tau_n(p) \leq q + L) \leq C \exp \left\{ -c \left(\frac{p \wedge V_n}{q + L} \right)^\delta \right\}. \quad (10.39)$$

Proof. Put $m = q + L$, $\tilde{p} = p \wedge V_n$, and choose $\beta \in (0, \gamma)$ and $\delta = \beta / (1 + \beta)$. If $d = \tilde{p} / m$ is bounded, enlarge C . Otherwise let $s = \lfloor m d^\delta \rfloor$. Then $Cm \leq s \leq \tilde{p} / 2$, so in particular the block $\{1, \dots, s\}$ does not contain p . Take $t = 1 / \lambda_{n,s}$. For every $b \leq s$,

$$\mathbb{P}(T_{n,b} > t) = e^{-\lambda_{n,b} / \lambda_{n,s}} \geq e^{-1}.$$

If $T_{n,p} < t$ and at least m of these first s clocks exceed t , then at least m clocks exceed $T_{n,p}$, so $\tau_n(p) \geq m + 1$. Therefore

$$\mathbb{P}(\tau_n(p) \leq m) \leq e^{-\lambda_{n,p} / \lambda_{n,s}} + e^{-cs}$$

by a binomial Chernoff bound. Potter's lower bound and monotonicity give

$$\lambda_{n,p} / \lambda_{n,s} \geq c(\tilde{p} / s)^\beta.$$

By the choice of s , both s and $(\tilde{p} / s)^\beta$ are bounded below by a constant times d^δ . This proves (10.39). $\square$

Let Γ be a prescribed collection of pairwise vertex-disjoint directed cycles in terminal coordinates, and let E_Γ be the event that all those cycles occur. For a cycle component γ define its capped logarithmic oscillation by

$$\omega_{U_n}(\gamma) = \text{osc}_{a \in \gamma} \log(a \wedge U_n). \quad (10.40)$$

Lemma 10.11 (Capped path drop). *Let γ be a nontrivial directed cycle containing at least one vertex not exceeding U_n . There is a downward edge $p \mapsto q$ of γ such that $q \leq U_n$ and*

$$\frac{p \wedge V_n}{q} \geq \exp\{\omega_{U_n}(\gamma)/|\gamma|\}. \quad (10.41)$$

Proof. If $\omega_{U_n}(\gamma) = 0$, then the component must contain a vertex above U_n and a vertex equal to U_n ; any directed entrance from the upper part to a vertex not exceeding U_n gives a downward edge satisfying (10.41). Assume henceforth that the oscillation is positive. Choose vertices whose capped logarithms attain the maximum and minimum, and follow the directed cycle from the former to the latter. Put $z_i = v_i \wedge U_n$ along this path. The product of the ratios z_i/z_{i+1} over the downward capped steps is at least the net ratio between the endpoints, namely $e^{\omega_{U_n}(\gamma)}$. There are at most $|\gamma|$ steps, so one downward capped step has ratio at least the right-hand side of (10.41). For that step, $z_{i+1} = q \leq U_n$ and $p \wedge V_n \geq p \wedge U_n = z_i$, which proves the claim. $\square$

Corollary 10.12 (Cycle envelope). *If L is the total number of vertices of Γ , then*

$$\mathbb{P}(E_\Gamma) \leq \frac{C_L}{\prod_{a \in V(\Gamma)} a}. \quad (10.42)$$

If γ is a nontrivial component meeting $[1, U_n]$, then

$$\mathbb{P}(E_\Gamma) \leq \frac{C_L}{\prod_{a \in V(\Gamma)} a} \Psi_L(\omega_{U_n}(\gamma)), \quad (10.43)$$

where

$$\Psi_L(x) = C_L \exp\{-c_L e^{c_L x}\}. \quad (10.44)$$

Consequently, if

$$\Omega_{U_n}(\Gamma) = \max_{\gamma \text{ nontrivial } \gamma \cap [1, U_n] \neq \emptyset} \omega_{U_n}(\gamma),$$

then, whenever the maximum is over a nonempty set,

$$\mathbb{P}(E_\Gamma) \leq \frac{C_L}{\prod_{a \in V(\Gamma)} a} \Psi_L(\Omega_{U_n}(\Gamma)). \quad (10.45)$$

Proof. Apply Lemma 10.9. Around every directed cycle the rate ratios telescope:

$$\prod_{a \mapsto b} \frac{\lambda_{n,a}}{b \lambda_{n,b}} = \frac{1}{\prod_{a \in \gamma} a}.$$

This proves (10.42). For a component meeting $[1, U_n]$, choose the edge supplied by Lemma 10.11 as the distinguished constraint in Lemma 10.9. Lemma 10.10 gives (10.43). Choosing a component attaining the maximum gives (10.45). $\square$

10.3 The logarithmic trace

Lemma 10.13 (Continuum multiplicative trace). *For every fixed $r \geq 1$,*

$$I_r(T) := \int_{[0,T]^r} \prod_{j=1}^r k_\gamma(x_{j+1} - x_j) dx_1 \cdots dx_r = T k_\gamma^{*r}(0) + O_r(1), \quad (10.46)$$

where $x_{r+1} = x_1$.

Proof. For $r = 1$ the identity is exact. For $r \geq 2$, put $z_j = x_{j+1} - x_j$, $1 \leq j < r$, and $z_r = -\sum_{j < r} z_j$. If $s_0 = 0$ and $s_j = z_1 + \cdots + z_j$, let $\mathcal{R}(z) = \max_j s_j - \min_j s_j$. The admissible translations x_1 have length $[T - \mathcal{R}(z)]_+$, so

$$I_r(T) = \int_{\mathbb{R}^{r-1}} [T - \mathcal{R}(z)]_+ \prod_{j=1}^r k_\gamma(z_j) dz_1 \cdots dz_{r-1}.$$

The product integral without the bracket is $k_\gamma^{*r}(0)$. The density k_γ has an exponential right tail, a superexponential left tail, and a finite first absolute moment. Since $\mathcal{R}(z) \leq 2 \sum_{j < r} |z_j|$, the error integral is finite uniformly in T . $\square$

We shall use the elementary harmonic-coordinate bounds

$$\sum_{A \leq a \leq B} \frac{1}{a} \leq 1 + \log(B/A), \quad (10.47)$$

and, for $J_a = [\log a, \log(a+1))$,

$$|J_a| = \frac{1}{a} + O(a^{-2}). \quad (10.48)$$

Lemma 10.14 (Discrete endpoint trace). *Let $L_n \rightarrow \infty$, $U_n = o(n)$, and $T_n = \log(U_n/L_n) \rightarrow \infty$. Then*

$$\begin{aligned} \sum_{\substack{L_n \leq a_1, \dots, a_r \leq U_n \\ a_1, \dots, a_r \text{ distinct}}} \prod_{j=1}^r \frac{1}{a_{j+1}} k_\gamma \left(\log \frac{a_{j+1}}{a_j} \right) \\ = k_\gamma^{*r}(0) T_n + o(T_n), \end{aligned} \quad (10.49)$$

where $a_{r+1} = a_1$.

Proof. Because $\prod_j a_{j+1}^{-1} = \prod_j a_j^{-1}$, the sum is a logarithmic Riemann sum. First restrict to $\text{osc}(\log a_1, \dots, \log a_r) \leq M$. On this region the cyclic kernel product and its first derivatives are bounded by a constant depending on r, M . Using the cells J_a in (10.48), cell integration and (10.47) show that the difference from the continuum integral over $[\log L_n, \log U_n]^r$ is

$$O_{r,M}(T_n/L_n) = o(T_n). \quad (10.50)$$

The same bound controls the replacement of $1/a$ by $|J_a|$.

We next record a uniform discrete tail bound. The local supremum of k_γ over an interval of length one is bounded by an integrable density $\bar{k}_\gamma$ having the same exponential right tail, a superexponential left tail, and a finite first moment. Enlarging every logarithmic cell by one and using (10.47), the discrete cyclic sum with oscillation larger than M is bounded by a constant times the corresponding continuum cyclic integral with k_γ replaced by $\bar{k}_\gamma$ and oscillation larger than $M - r$. The increment representation in Lemma 10.13 therefore gives

$$\lim_{M \rightarrow \infty} \sup_n \frac{1}{T_n} \sum_{\substack{L_n \leq a_1, \dots, a_r \leq U_n \\ \text{osc}(\log a) > M}} \prod_{j=1}^r \frac{1}{a_{j+1}} k_\gamma \left(\log \frac{a_{j+1}}{a_j} \right) = 0. \quad (10.51)$$

Finally, an equality $a_i = a_j$ replaces one harmonic factor by a square: $\sum_{a \geq L_n} a^{-2} = O(L_n^{-1})$. The remaining localized relative sums are uniformly bounded by the same convolution majorant, so all equality diagonals contribute $O_r(L_n^{-1})$. Combine (10.50)–(10.51) with Lemma 10.13, and then let $M \rightarrow \infty$. $\square$

10.4 Factorial-moment summation

The next elementary observation is used repeatedly. If a component has r vertices, one may choose one logarithmic translation coordinate and $r - 1$ relative coordinates. By (10.47), if its logarithmic oscillation is at most x , the total harmonic mass of the relative coordinates is at most $C_r(1+x)^{r-1}$. For an ordered collection of m components and L vertices, the corresponding bound is

$$C_L(1+x)^{L-m}(1+T_n)^m. \quad (10.52)$$

If one component is required to meet $[1, L_n]$, one of the T_n translation factors in (10.52) is replaced by $O(1 + \log L_n + x)$. If a component crosses U_n and its capped oscillation is at most x , its lower anchor belongs to $[U_n e^{-x}, U_n]$, so its translation factor is $O(1+x)$ rather than T_n ; each vertex above U_n contributes the harmonic factor

$$H_n := \sum_{a > U_n} \frac{1}{a} = O(\log \log n). \quad (10.53)$$

These bounds follow by summing the nonanchor vertices successively over multiplicative intervals and will be used without further comment.

Lemma 10.15 (Negligible endpoint cycle configurations). *Fix an ordered collection of m cycle components with total number of vertices L . Put*

$$L_n = \lceil \log n \rceil, \quad U_n = \left\lfloor \frac{n}{(\log n)^2} \right\rfloor, \quad T_n = \log(U_n/L_n).$$

Then:

- (i) *among collections with all vertices in $[L_n, U_n]$, the total probability of those for which some nontrivial component has logarithmic oscillation greater than M is at most $\varepsilon_M T_n^m$, where $\varepsilon_M \downarrow 0$;*
- (ii) *if all component oscillations are at most M , the total probability of collections containing two vertices at logarithmic distance at most η is at most*

$$C_{L,M} \eta T_n^m + O_{L,M}(T_n^{m-1}); \quad (10.54)$$

- (iii) *collections having at least one vertex outside $[L_n, U_n]$ contribute $o(T_n^m)$.*

Proof. For (i), let Ω be the maximum logarithmic oscillation among the nontrivial components. Since all vertices are at most U_n , this is the same as $\Omega_{U_n}(\Gamma)$. Corollary 10.12 and (10.52) give, after decomposing $\Omega \in [k, k+1)$,

$$\begin{aligned} & \sum_{\substack{\Gamma: V(\Gamma) \subset [L_n, U_n] \\ \Omega(\Gamma) > M}} \mathbb{P}(E_\Gamma) \\ & \leq C_L T_n^m \sum_{k \geq M} (2+k)^{L-m} \Psi_L(k) =: \varepsilon_M T_n^m. \end{aligned}$$

The superexponential decay in (10.44) gives $\varepsilon_M \downarrow 0$.

For (ii), parameterize each component by one translation coordinate and its bounded relative coordinates. If the close pair lies in one component, it cuts out a band of width $O(\eta)$ in a bounded relative-coordinate domain. The logarithmic lattice error is $O(L_n^{-1})$ by (10.47). Summing the remaining coordinates gives $O_{L,M}((\eta + L_n^{-1})T_n^m)$. If the close pair lies in two different components, it links two translation coordinates within a bounded interval whose length depends only on M , and therefore removes one free translation factor. Its contribution is $O_{L,M}(T_n^{m-1})$. This proves (10.54).

For (iii), first consider collections with no vertex above U_n but with a vertex below L_n . Use the maximum-oscillation envelope as in (i). By the observation preceding the lemma, the component meeting the lower range has translation mass $O(1 + \log L_n + \Omega)$ rather than T_n . Hence the total is bounded by

$$C_L T_n^{m-1} \sum_{k \geq 0} (1 + \log L_n + k)(2 + k)^{L-m} \Psi_L(k) = O_L((1 + \log L_n)T_n^{m-1}) = o(T_n^m). \quad (10.55)$$

The case in which all components are fixed points follows directly from $\sum_{a < L_n} a^{-1} = O(\log \log n)$ and the same conclusion.

Now suppose that some vertex exceeds U_n . A component lying wholly above U_n has harmonic mass at most $H_n^r = o(T_n)$ for its fixed length r . For the remaining components, if at least one nontrivial component meets $[1, U_n]$, choose the one of maximal capped oscillation and use (10.45); the summation in (i), together with (10.55), is then $O_L(T_n^{m-1})$. If no such component exists, every remaining component is either a fixed point or wholly above U_n , and its harmonic mass is $O(T_n)$. Thus any collection containing a wholly upper component contributes $o(T_n^m)$.

It remains to treat crossing components. Let $h \geq 1$ be the total number of vertices above U_n . For every component meeting $[1, U_n]$, use the capped oscillation (10.40), and let Ω be their maximum. Corollary 10.12 supplies the factor $\Psi_L(\Omega)$. A crossing component has its lower anchor in $[U_n e^{-\Omega}, U_n]$, so it has no free T_n translation factor; its upper vertices contribute H_n^h . The harmonic-volume bound therefore gives

$$\begin{aligned} & \sum_{\Gamma: \Gamma \text{ has a crossing component}} \mathbb{P}(E_\Gamma) \\ & \leq C_L H_n^h T_n^{m-1} \sum_{k \geq 0} (2 + k)^{L-m+1} \Psi_L(k) = O_L(H_n^h T_n^{m-1}) = o(T_n^m), \end{aligned} \quad (10.56)$$

because h and L are fixed and $(\log \log n)^h = o(\log n)$. This completes (iii). $\square$

Proof of Theorem 10.1. Let

$$m = \sum_{r=1}^R m_r, \quad L = \sum_{r=1}^R r m_r.$$

If $m = 0$, the assertion is trivial. Expanding the mixed factorial moment gives a sum over ordered collections of distinct directed cycles. Collections sharing a vertex have probability zero, so only vertex-disjoint collections contribute. The falling factorial orders cycles of the same length; hence no $m_r!$ divisor appears.

Fix $M < \infty$ and $\eta > 0$. Call a collection good if all vertices lie in $[L_n, U_n]$, every component has logarithmic oscillation at most M , and all L vertices are pairwise separated by more than η on logarithmic scale. For a good collection, every source-to-target ratio lies in $[e^{-M}, e^M]$, and

the sorted targets are separated by ratios at least e^η . Proposition 10.7 therefore gives, uniformly,

$$\mathbb{P}(E_\Gamma) = (1 + o(1)) \prod_{a \rightarrow b \in \Gamma} \frac{1}{b} k_\gamma \left(\log \frac{b}{a} \right). \quad (10.57)$$

The total leading kernel mass of such collections is $O_L(T_n^m)$ by Lemma 10.14 and the nonnegativity of the kernel, so the accumulated uniform relative error in (10.57) is $o(T_n^m)$.

For one directed cycle of length r , Lemma 10.14 gives

$$k_\gamma^{*r}(0)T_n + o(T_n)$$

when a starting vertex is distinguished. Division by r removes cyclic rotations. For an ordered collection of components, the unrestricted leading kernel sum factors over components. Equality restrictions and cross-component close pairs remove $O_L(T_n^{m-1})$, while same-component η -bands remove $O_{L,M}(\eta T_n^m)$. This follows from the harmonic-coordinate proof of Lemma 10.15; on the bounded oscillation region the leading kernel is bounded, so the same volume estimates apply verbatim. Hence the good contribution is

$$\prod_{r=1}^R \left(\frac{k_\gamma^{*r}(0)}{r} T_n \right)^{m_r} + o(T_n^m) + O_{L,M}(\eta T_n^m). \quad (10.58)$$

The same lemma controls all nongood collections. Let first $n \rightarrow \infty$, then $\eta \downarrow 0$, and finally $M \rightarrow \infty$. Since

$$T_n = \log(U_n/L_n) = \log n - 3 \log \log n + O(1) \sim \log n,$$

(10.58) proves the mixed factorial-moment asymptotic.

For the L^p conclusion, put

$$X_{n,r} = \frac{C_r}{\alpha_{\gamma,r} \log n}.$$

The first two factorial moments give $X_{n,r} \rightarrow 1$ in L^2 , hence in probability. Given any fixed $p < \infty$, choose an integer $s > p$. The factorial-moment theorem and the identities expressing ordinary powers as finite linear combinations of falling factorials give

$$\sup_n \mathbb{E} X_{n,r}^s < \infty.$$

Thus $|X_{n,r} - 1|^p$ is uniformly integrable, and convergence in probability gives $X_{n,r} \rightarrow 1$ in L^p . Since R is fixed, the same argument applies to the vector in any norm on $\mathbb{R}^R$. $\square$

Remark 10.16 (Transpositions). For $r = 2$,

$$\begin{aligned} k_\gamma^{*2}(0) &= \int_{\mathbb{R}} k_\gamma(x) k_\gamma(-x) dx \\ &= \gamma K_\gamma^2 \int_{\mathbb{R}} e^{-2K_\gamma \cosh t} dt = 2\gamma K_\gamma^2 K_0(2K_\gamma), \end{aligned}$$

where $t = \gamma x$ and $K_0(z) = \int_0^\infty e^{-z \cosh t} dt$. Thus

$$\mathbb{E} C_2 \sim \gamma K_\gamma^2 K_0(2K_\gamma) \log n.$$

In the Sukhatme case this coefficient is $K_0(2)$.

11 Conclusions and theorem map

The results establish a coherent hierarchy of exact, microscopic, mesoscopic, and macroscopic laws for Luce permutations.

At the two ends of the ordering, Theorems 3.1 and 3.2 give exact top-window comparison identities, while Theorem 4.2 identifies the total mass and the defect of the limiting bottom-window law. The vanishing-block construction in Proposition 2.6 shows that these microscopic statistics cannot in general be inferred from a permuton limit alone.

For uniformly positive continuous profiles, Theorem 5.1 and Corollary 5.16 give joint Poisson convergence, including total variation on each fixed cycle-count vector. Theorem 5.19 proves that, for some $q = q(f) \in (0, 1)$,

$$\lambda_r = \frac{1}{r} + O_f(q^r/r).$$

The diagonal-band, sparse-entry, thin-graph, and deterministic-overlap results then extend the same microscopic density to broad deterministic collections of point constraints. Under uniformly comparable weights, Theorem 6.1 supplies the complementary macroscopic conclusion: the ranked cycle lengths converge to PD(1), and their size-biased order converges to GEM(1).

For the Sukhatme and endpoint-regular regimes, Theorems 7.5, 7.24, 7.25, and 8.12 establish the logarithmic fixed-point mean, local terminal Poisson point processes, and all fixed factorial moments. Theorem 9.1 upgrades the full endpoint fixed-point count to total-variation approximation by a Poisson law with its exact mean. Theorems 8.13, 8.14, and 8.19 identify, respectively, the finite-dimensional endpoint displacement law, the deterministic mesoscopic tangent measure, and the microscopic infinite endpoint permutation. Finally, Theorem 10.1 determines the logarithmic means and all fixed mixed factorial moments of the endpoint cycle counts of every fixed length, and hence their joint law of large numbers.

12 Proof mechanisms

The proofs use two complementary mechanisms.

12.1 Local-entry and terminal-condition analysis

The boundary, regular-profile, and endpoint results begin with an exact event representation and then identify its incoming mass.

- (i) In the top-window problem, the relevant set is the distinct-word support. The total-variation distance is exactly the independent-sampling collision probability.
- (ii) In the bottom-window problem, the candidate infinite law has total mass

$$\mathbb{P}(N_0 \geq k),$$

and its missing mass is exactly $\mathbb{P}(N_0 < k)$. The defect is therefore caused by too few clocks surviving beyond the terminal endpoint.

- (iii) In the regular positive-profile regime, the kernel $\rho(x, y)$ is both the Luce permuton density and the microscopic point-entry density for $\sigma_n(i) = j$. Separated local estimates are completed by endpoint, collision, and target-crowding bounds before factorial moments are summed.

- (iv) In the Sukhatme profile, the relevant layer is logarithmic in the terminal coordinate $a = n + 1 - i$. The one-point probability is asymptotic to e^{-1}/a , and logarithmic summation yields both the mean and the local Poisson point process.
- (v) For endpoint-regular arrays, after an irrelevant common normalization, the rates behave as

$$C_0(a/n)^\gamma L(n/a).$$

The shifted-Gumbel saddle produces the local coefficient c_γ/a , while the multiplicative transition kernel on logarithmic coordinates has diagonal return density $x^{-1}k_\gamma^{*r}(0)$.

- (vi) The global endpoint Poisson theorem uses an approximate-Palm rank transposition. Swapping the clock of coordinate a with the clock occupying terminal rank a forces $I_{n,a} = 1$; the pushforward law is close to the exact Palm law, and the fixed-point count changes only through the intended point and a possible two-cycle.

The shifted-Gumbel coordinates are especially effective because the specified priority density and the derivative of the threshold mean cancel in the integrated saddle. This cancellation explains the same local coefficients in the regular, Sukhatme, and endpoint-regular regimes.

12.2 Clock-slab mixing and regeneration

The macroscopic theorem is not a local saddle calculation. Short clock-time slabs produce conditional permanent matchings with hereditary edge flatness. A common minorization yields a uniform L^2 contraction, a quantitative prefix law gives relative balance in each target category, and a two-scale smoothing argument upgrades global balance to the dependently selected closure slab. Deletion of an interior macroscopic cycle preserves regular histories, so the successive residual fractions become asymptotically independent uniforms. This regeneration mechanism yields the GEM(1) and PD(1) limits.

13 Open problems

The results above settle every fixed cycle length in the regular and endpoint regimes and the macroscopic limit under uniformly comparable weights. The main remaining questions concern the scales between these two extremes.

Conjecture 13.1 (Mesoscopic cycle process). *Under the hypotheses of Theorem 6.1, if $r_n \rightarrow \infty$ and $r_n = o(n)$, then*

$$\mathbb{E}C_{r_n}(\sigma_n) \sim \frac{1}{r_n}.$$

More strongly, let $a_n \rightarrow \infty$ with $a_n = o(n)$, and let $[u_j, v_j] \Subset (0, \infty)$, $1 \leq j \leq m$, be pairwise disjoint intervals. If $L_{n,1}, L_{n,2}, \dots$ are the cycle lengths, then the vector

$$(\#\{i : u_j a_n \leq L_{n,i} \leq v_j a_n\})_{j=1}^m$$

should converge to independent Poisson variables with respective means $\log(v_j/u_j)$. Equivalently, the submacroscopic cycle lengths should converge locally, after scaling by a_n , to the scale-invariant Poisson process with intensity dx/x .

The scale-invariant process is the corresponding benchmark for uniform permutations and the Ewens sampling formula; see Arratia, Barbour, and Tavaré [1].

Conjecture 13.2 (Total number of cycles). *Assume the exact sampled-profile model*

$$\theta_{n,i} = f(i/n), \quad f \in C[0,1], \quad 0 < m \leq f \leq M < \infty,$$

and put

$$\mathcal{N}_n^{\text{cyc}} = \sum_{r=1}^n C_r(\sigma_n).$$

Then

$$\mathbb{E} \mathcal{N}_n^{\text{cyc}} = \log n + \kappa_f + o(1), \quad \frac{\mathcal{N}_n^{\text{cyc}} - \mathbb{E} \mathcal{N}_n^{\text{cyc}}}{\sqrt{\log n}} \Rightarrow N(0,1),$$

where

$$\kappa_f = \gamma_E + (\lambda_1 - 1) + \sum_{r=2}^{\infty} \left(\lambda_r - \frac{1}{r} \right)$$

and γ_E is the Euler–Mascheroni constant.

The trace correction is absolutely convergent by Theorem 5.19. The missing input is control uniform in growing cycle length. Whether the same constant is stable under every uniformly vanishing triangular-array perturbation allowed in Section 5 is a separate universality question.

Conjecture 13.3 (Endpoint-singular macroscopic cycles). *Under Definition 8.1, the ranked macroscopic cycle lengths converge to PD(1).*

Theorem 6.1 does not apply because the ratio of the largest to the smallest weight diverges. Theorem 10.1 nevertheless shows that, for every fixed R ,

$$\sum_{r=1}^R r C_r(\sigma_n) = O_{\mathbb{P}}(\log n) = o_{\mathbb{P}}(n).$$

A proof of Conjecture 13.3 must additionally control the total mass in cycle lengths growing with n , particularly throughout the mesoscopic range.

The bulk–endpoint independence and longest-increasing-subsequence conjectures are recorded in Appendix A. Proposition 2.6 shows that the specific sparse-cycle and longest-increasing-subsequence conclusions formulated here cannot follow from permuton convergence alone without additional microscopic control.

A Additional conjectures

The conjectures below combine the bulk and endpoint mechanisms or concern a statistic not controlled by the cycle arguments. They are not used elsewhere in the paper.

Conjecture A.1 (Bulk–endpoint independence). *Suppose that there are normalizing constants $s_n > 0$ and a continuous function $f : [0,1) \rightarrow (0,\infty)$ such that, for every compact $K \Subset [0,1)$,*

$$\sup_{i/n \in K} |s_n \theta_{n,i} - f(i/n)| \longrightarrow 0,$$

that

$$f(1-u) \sim Cu^\gamma L(1/u), \quad u \downarrow 0,$$

and that the normalized terminal arrays $s_n \lambda_{n,a}$ satisfy Definition 8.1 with the same index γ . Let ρ be the associated Luce permuton density. For a compact set $K \Subset (0, 1)$, a bounded interval $J \Subset \mathbb{R}$, and $b_n \rightarrow \infty$, $b_n = o(n)$, define

$$\Xi_n^{\text{bulk}} = \sum_{i=1}^n \mathbf{1}_{\{\sigma_n(i)=i\}} \delta_{i/n}, \quad \Xi_n^{\text{term}} = \sum_{a=1}^n I_{n,a} \delta_{\log(a/b_n)}.$$

Then, in the product vague topology,

$$\left(\Xi_n^{\text{bulk}}|_K, \Xi_n^{\text{term}}|_J \right) \Rightarrow \left(\Xi^{\text{bulk}}|_K, \Xi^{\text{term}}|_J \right),$$

where the two limiting point processes are independent Poisson point processes with respective intensities

$$\rho(x, x) dx \quad \text{and} \quad c_\gamma dx.$$

This prediction combines the separated bulk saddle with the endpoint regular-variation saddle. A proof should require a mixed-scale integrated local limit together with uniform domination across the region between the two scales.

Conjecture A.2 (Longest increasing subsequences under regularity). *Under the uniformly positive continuous-profile hypotheses of Section 5,*

$$\frac{\text{LIS}(\sigma_n)}{\sqrt{n}} \xrightarrow{\mathbb{P}} \mathcal{L}(\rho),$$

where the inhomogeneous Hammersley functional is

$$\mathcal{L}(\rho) = 2 \sup_{(x,y) \in \mathcal{A}} \int_0^1 \sqrt{\rho(x(t), y(t)) x'(t) y'(t)} dt.$$

Here $\mathcal{A}$ is the class of absolutely continuous, coordinatewise nondecreasing paths $(x(t), y(t))$ from $(0, 0)$ to $(1, 1)$.

This is the natural nonuniform-permutation analogue of the inhomogeneous Hammersley variational laws; see Deuschel and Zeitouni [9] and Sjöstrand [18]. Additional microscopic regularity is indispensable. Indeed, in Proposition 2.6 choose the planted permutation to be increasing and choose

$$\sqrt{n} \ll m_n \ll n.$$

The limiting permuton is unchanged, while the planted block forces $\text{LIS}(\sigma_n) \geq m_n$ with probability tending to one, thereby changing the $\sqrt{n}$ scale.

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